

AQUIRE

Awesome Quantum Information Research Explorer

1,600 unique resources | 30 major categories | 164 topics

792 fine subcategories | Snapshot date 8 October 2026

AQUIRE is a research and learning reference for quantum computing, quantum machine learning, hardware, fault tolerance, software, and related technologies. It combines a structured subject map with a bibliography of retrieved scholarly records and official resources. Use it to identify a question, choose a reading route, and trace the evidence back to its original source.

The catalog confirms bibliographic identity through retrieved metadata or an official page. Topic classification is title screened or curated and does not replace reading the paper. Bibliographic authenticity is not a guarantee of scientific validity or practical quantum advantage.

The interactive HTML provides combined search and filters. This DOCX, the PDF, and the Markdown contain the same stable resource identifiers and the complete selected catalog. CSV and JSON provide reusable metadata.

Catalog composition

Resource type	Unique records
Book	20
Book chapter	148
Conference paper	296
Course	6
Documentation	13
Edited book	3
Institutional resource	4
Journal article	1005
Lecture notes	1
Posted content	56
Reference book	4
Report	14
Software repository	20
Standard	3

Resource type	Unique records
Tutorial	7

1,546 records have registered DOI metadata. 54 official pages were retrieved. Supplied publication years span 1966 to 2026; undated pages remain undated.

Sources and verification

Scope and taxonomy

This is a broad research and learning map, not a claim to enumerate every possible topic or every available resource. The hierarchy contains major categories, topics, and fine subcategories. Networking, sensing, and classical post quantum cryptography are explicitly included as adjacent areas. These areas should not be conflated with quantum computation.

Authentic source records

Scholarly entries were retrieved from the Crossref REST API using publisher deposited metadata and identifiable DOIs. Official web entries were retrieved from first party institutions, software projects, or providers and required a successful HTTP response and an identifiable page title. Failed or unrelated sources were excluded. The catalog is a metadata based bibliography, not a full text systematic review or a ranking of research quality.

Subject assignments and finer tags

Topic searches were screened using title wording and subject context. Ambiguous results received editorial corrections or exclusion. Multiple supported discovery paths are retained as related topics. Fine subcategory tags are conservative lexical assignments from titles. Topic level only means that no precise fine tag was established; the full vocabulary shown in the taxonomy is not automatically assigned to every resource.

Dates and publication types

Years and publication types follow retrieved bibliographic metadata. Posted content is kept separate from journal articles because peer review cannot be inferred from a DOI. Books and book chapters are distinct resource types, and separate editions can be retained when their years differ. Undated official pages stay undated; the retrieval year is not substituted for a publication year. Conference names are shown only when the source supplies an event or a proceedings venue.

Deduplication and selection

DOIs and canonical web URLs were deduplicated. Identical normalized titles with matching lead author and publication year were also merged, preferring a journal or conference record over posted content when a match was clear. The selection balances topic coverage and retains curated foundational material. The same stable Q identifiers and resource records are used in HTML, DOCX, Markdown, PDF, CSV, and JSON.

What the verification status means

Registered DOI means that a corresponding bibliographic record was retrieved from Crossref. It does not mean that every DOI resolver or full text download was individually tested. Page retrieved means that the official web page and its title were successfully retrieved on the snapshot date. Links can redirect, change, require authentication, or lead to paywalled content after delivery. No publication year, author, venue, or review status was invented to fill missing fields.

Coverage and limitations

Coverage is selective and varies by topic, especially standards, economics, and specialized engineering. Bibliographic identity is stronger evidence than an unverified reference string, but it is not a guarantee of scientific validity, peer review quality, or practical advantage. Metadata can contain publisher errors or incomplete fields. Papers were not systematically screened for retractions and corrections across all external databases. Use the source record and paper itself when a claim is important.

Offline use and exports

The HTML is self contained and uses no external scripts or fonts. Search, topic browsing, sorting, saved resources, and exports operate locally. External resource links need internet access. Browser saved resources use local storage and are not sent to a server. Search supports words combined with AND and quoted phrases. Journal, conference, author, publisher, year, type, topic, and verification filters can be combined. Exports contain all filtered results, while printing prints the currently displayed page of cards.

Primary source documentation: [Crossref REST API](#); [NIST quantum information science](#); [IBM Quantum Learning](#); [MIT OpenCourseWare](#)

Reading paths

Foundations to algorithms

Build the mathematical and algorithmic vocabulary needed to read quantum computing research.

1. Review Hilbert spaces, tensor products, and unitary operators. Topic D01T01.
2. Work with states, density matrices, and partial traces. Topic D01T02.
3. Understand gates, circuits, and universal models. Topic D03T01.
4. Implement a small quantum Fourier transform. Topic D05T01.
5. Study phase estimation with explicit precision and overlap assumptions. Topic D05T02.
6. Compare oracle query cost with complete implementation cost. Topic D04T02.

Suggested output: A tested notebook for small algorithms, with input assumptions and classical comparisons.

Quantum machine learning and hybrid AI

Connect classical ML experience to defensible quantum learning experiments.

1. Account for the cost of encoding the dataset. Topic D10T01.
2. Compare a quantum kernel with classical kernels on the same split. Topic D10T02.
3. Build a variational classifier with a reproducible training loop. Topic D10T03.
4. Measure gradients and investigate trainability. Topic D09T04.
5. Read generalization and sample complexity results. Topic D10T06.
6. Evaluate GPU simulation cost and memory before scaling experiments. Topic D24T04.

Suggested output: A benchmark report with held out metrics, uncertainty, encoding cost, and strong classical baselines.

Quantum chemistry and simulation

Follow a physical problem from model definition to algorithm and resource estimate.

1. Choose the basis, active space, observable, and accuracy. Topic D08T01.
2. Map fermionic operators to qubits and use valid symmetries. Topic D08T02.

AQUIRE

3. Run a small VQE study with reference energies. Topic D09T01.
4. Compare digital simulation methods and error budgets. Topic D07T01.
5. Study block encoding and qubitization access assumptions. Topic D06T01.
6. Translate the algorithm into logical and physical resources. Topic D08T04.

Suggested output: A small molecule or lattice demonstration and a transparent resource estimate.

Error correction to fault tolerant architecture

Connect code properties to a complete logical computing system.

1. Construct stabilizer checks and logical operators. Topic D19T01.
2. Simulate syndrome extraction under an explicit noise model. Topic D19T02.
3. Measure decoder accuracy and latency. Topic D20T06.
4. Map logical operations through lattice surgery or another stated method. Topic D20T03.
5. Include non Clifford resources and factory scheduling. Topic D20T02.
6. Combine distance, hardware, and runtime assumptions in an estimate. Topic D20T05.

Suggested output: Logical error scaling plots and an architecture aware cost model.

Hardware and experimental reliability

Evaluate one hardware platform using control and characterization evidence.

1. Select a platform and understand its physical qubit model. Topic D12T01.
2. Study control pulses and parameter constraints. Topic D18T01.
3. Review calibration and drift tracking. Topic D18T05.
4. Interpret benchmarking with its model assumptions. Topic D22T02.
5. Investigate leakage, crosstalk, and correlated errors. Topic D18T04.
6. Assess mitigation with sampling overhead and bias. Topic D21T01.

Suggested output: A platform comparison that distinguishes device metrics from logical performance.

Quantum software and GPU systems

Build an executable workflow and understand its full software stack.

1. Learn one supported programming framework. Topic D23T01.
2. Inspect intermediate representation and classical control semantics. Topic D23T02.
3. Measure the effect of mapping and routing. Topic D23T04.
4. Estimate state vector and density matrix memory. Topic D24T01.
5. Benchmark CPU, GPU, and distributed simulation where available. Topic D24T04.
6. Add semantic and statistical checks to the workflow. Topic D23T06.

Suggested output: A versioned project with circuit definitions, environment details, and performance measurements.

Networking and security

Separate quantum networking protocols from classical post quantum security.

1. Understand teleportation and the required classical communication. Topic D26T01.

AQUIRE

2. Study repeaters and memory and loss constraints. Topic D26T02.
 3. Analyze entanglement distribution and routing. Topic D26T03.
 4. Read QKD security with finite size and device assumptions. Topic D27T01.
 5. Compare delegation, blindness, and verifiability. Topic D27T03.
 6. Study classical post quantum cryptography as a distinct deployment path. Topic D27T05.
- Suggested output: A protocol comparison with explicit trust models, costs, and deployment requirements.

Sensing and scientific applications

Choose a scientific target and distinguish computational from sensing claims.

1. Define an estimation task and attainable measurement strategy. Topic D28T01.
2. Read sensing performance with bandwidth and spatial resolution. Topic D28T02.
3. Identify a concrete scientific or engineering observable. Topic D29T04.
4. Evaluate numerical algorithm assumptions and conditioning. Topic D06T06.
5. Specify what can be verified with the available measurements. Topic D22T06.
6. Prepare a reproducible comparison with complete uncertainty reporting. Topic D30T04.

Suggested output: An application study with a specified observable, validation plan, and practical cost accounting.

Questions to take to every resource

1. What task, input model, observable, and accuracy are specified?
2. Is this a theorem, simulation, experiment, review, tutorial, or roadmap?
3. Does the cost include preparation, oracles, compilation, shots, correction, and classical processing?
4. What strong classical baseline is appropriate?
5. Which noise and trust assumptions support the conclusion?
6. Can the result be reproduced from the available record?

Category index

Category	Topics	Primary resources
D01 Foundations and mathematical tools	5	42
D02 Entanglement and quantum resources	5	42
D03 Models of quantum computation	6	71
D04 Quantum complexity and limitations	5	34
D05 Core quantum algorithms	6	72
D06 Advanced algorithm design	6	76
D07 Hamiltonian simulation and many body physics	6	56

Category	Topics	Primary resources
D08 Quantum chemistry and materials	5	39
D09 Variational and hybrid quantum methods	6	74
D10 Quantum machine learning	7	88
D11 Optimization and operations research	5	55
D12 Superconducting qubit hardware	6	59
D13 Trapped ion quantum computing	5	44
D14 Neutral atom and Rydberg processors	5	38
D15 Spin qubits and semiconductor platforms	5	48
D16 Photonic quantum computing	6	57
D17 Other qubit platforms and transduction	5	57
D18 Control calibration and noise characterization	5	46
D19 Quantum error correction codes	6	54
D20 Fault tolerance and logical operations	6	60
D21 Error mitigation and near term reliability	5	39
D22 Characterization verification and benchmarks	6	50
D23 Software programming and compilation	6	75
D24 Classical simulation and hybrid systems	5	45
D25 Architecture cloud and distributed computing	5	65
D26 Quantum networks and communication	5	52
D27 Quantum cryptography and security	6	61
D28 Quantum sensing and metrology	5	44
D29 Specialized scientific applications	5	31
D30 Education research practice and ecosystem	5	26

D01 Foundations and mathematical tools

The language of quantum information connects linear algebra, probability, physical states, and measurements. These resources build the prerequisites needed to read algorithm and hardware papers.

Prerequisites: Linear algebra, probability, and introductory quantum mechanics.

Assessment focus: State normalization, operator properties, measurement definitions, and physical interpretation.

D01T01 Linear algebra and Hilbert spaces

Hilbert spaces provide the state space for quantum systems; tensor products describe composition. Linear algebra is also the implementation language of gates and many simulators.

Fine subcategories: State vectors; inner products; tensor products; eigenvalues; spectral decomposition; unitary operators.

Primary catalog resources: 8. Related topic assignments are searchable in the HTML.

Q0001 Hilbert space in quantum mechanics

Wenbing MENG, Pan ZHOU.

Journal article | 2024 | Region - Educational Research and Reviews | vol. 6 | no. 7 | pp. 93.

[DOI 10.32629/rerr.v6i7.2344](#) | [Source record](#) | Registered DOI metadata

Q0002 Algebra and Hilbert space structures induced by quantum probes

Go Kato, Masaki Owari, Koji Maruyama.

Journal article | 2020 | Annals of Physics | vol. 412 | pp. 168046 | article 168046.

[DOI 10.1016/j.aop.2019.168046](#) | [Source record](#) | Registered DOI metadata

Q0003 Quantum Information Science I | Physics | MIT OpenCourseWare

Author metadata not supplied.

Course | 2018 | MIT OpenCourseWare.

[Official resource](#) | [Source record](#) | Official page retrieved

Q0004 Hilbert space representations of the quantum $\ast$ -algebra $\mathcal{U}_q(\mathfrak{su}_{1,1})$

David Dubray.

Journal article | 2015 | Banach Journal of Mathematical Analysis | vol. 9 | no. 3 | pp. 261-277.

[DOI 10.15352/bjma/09-3-19](#) | [Source record](#) | Registered DOI metadata

Q0005 Quantum Mechanics in Hilbert Space

Author metadata not supplied.

Book chapter | 2015 | Hilbert Space and Quantum Mechanics | pp. 611-696.

[DOI 10.1142/9789814635844_0019](#) | [Source record](#) | Registered DOI metadata

Q0006 A class of Hilbert space representations of the quantum plane and the quantum algebra $U_q(\mathfrak{sl}_2)$

Asao Arai.

Journal article | 2008 | Reports on Mathematical Physics | vol. 61 | no. 2 | pp. 171-180.

[DOI 10.1016/s0034-4877\(08\)00013-x](#) | [Source record](#) | Registered DOI metadata

Q0007 Hilbert space representation of an algebra of observables for q -deformed relativistic quantum mechanics

W. Zippold.

Journal article | 1995 | Zeitschrift für Physik C Particles and Fields | vol. 67 | no. 4 | pp. 681-686.

[DOI 10.1007/bf01553995](#) | [Source record](#) | Registered DOI metadata

Q0008 Quantum diffusions on the algebra of all bounded operators on a hilbert space

RL Hudson.

Book chapter | 1989 | Lecture Notes in Mathematics | pp. 256-269.

[DOI 10.1007/bfb0083556](#) | [Source record](#) | Registered DOI metadata

D01T02 Quantum states and density operators

Density operators represent both pure states and statistical mixtures. Reduced states are essential when only part of a larger system is measured or controlled.

Fine subcategories: Pure states; mixed states; Bloch sphere; partial trace; reduced states; purification.

Primary catalog resources: 9. Related topic assignments are searchable in the HTML.

Q0009 Geometry of Quantum States

Ingemar Bengtsson, Karol Życzkowski.

Book | 2017 | Cambridge University Press.

[DOI 10.1017/9781139207010](#) | [Source record](#) | Registered DOI metadata

Q0010 Quantum Computation and Quantum Information

Michael A. Nielsen, Isaac L. Chuang.

Book | 2012 | Cambridge University Press.

[DOI 10.1017/cbo9780511976667](#) | [Source record](#) | Registered DOI metadata

Q0011 Geometry of Quantum States

Ingemar Bengtsson, Karol Życzkowski.

Book | 2006 | Cambridge University Press.

[DOI 10.1017/cbo9780511535048](#) | [Source record](#) | Registered DOI metadata

Q0012 Ph219/CS219 Quantum Computation

Author metadata not supplied.

Course | 2026 | Caltech John Preskill.

[Official resource](#) | [Source record](#) | Official page retrieved

Q0013 Feynman Path Integral and Landau Density Matrix in Probability Representation of Quantum States

Olga V. Man'ko.

Journal article | 2025 | Physics | vol. 7 | no. 4 | pp. 66.

[DOI 10.3390/physics7040066](#) | [Source record](#) | Registered DOI metadata

Q0014 Measuring the Density Matrix of Quantum-Modeled Cognitive States

Wendy Xiomara Chavarría-Garza, Osvaldo Aquines-Gutiérrez, Ajax Santos-Guevara, Humberto Martínez-Huerta, Jose Ruben Morones-Ibarra, Jonathan Rincon Saucedo.

Journal article | 2024 | Quantum Reports | vol. 6 | no. 2 | pp. 156-171.

[DOI 10.3390/quantum6020013](#) | [Source record](#) | Registered DOI metadata

Q0015 Density matrix renormalization group algorithm for mixed quantum states

Chu Guo.

Journal article | 2022 | Physical Review B | vol. 105 | no. 19 | article 195152.

[DOI 10.1103/physrevb.105.195152](#) | [Source record](#) | Registered DOI metadata

Fine tags: mixed states

Q0016 Neutral excitations of quantum Hall states: A density matrix renormalization group study

Prashant Kumar, F. D. M. Haldane.

Journal article | 2022 | Physical Review B | vol. 106 | no. 7 | article 075116.

[DOI 10.1103/physrevb.106.075116](#) | [Source record](#) | Registered DOI metadata

Q0017 Physics 219 Course Information

Author metadata not supplied.

Lecture notes | Undated | Caltech John Preskill.

[Official resource](#) | [Source record](#) | Official page retrieved

D01T03 Quantum measurements

Measurements convert quantum states into classical outcomes according to a specified measurement model. General measurements require POVMs and instruments rather than only orthogonal projectors.

Fine subcategories: Projective measurements; POVMs; weak measurement; quantum instruments; measurement disturbance.

Primary catalog resources: 8. Related topic assignments are searchable in the HTML.

Q0018 Implementing positive-operator-valued-measurement elements in photonic circuits for performing minimum quantum state tomography of path qudits

W. R. Cardoso, D. F. Barros, M. R. Barros, S. Pádua.

Journal article | 2019 | Physical Review A | vol. 99 | no. 6 | article 062324.

[DOI 10.1103/physreva.99.062324](#) | [Source record](#) | Registered DOI metadata

Q0019 A Measure of Quantum Correlation Based on von Neumann Entropy and Positive Operator-Valued Measurement

Yang Leng, Wen-Juan Li.

Journal article | 2018 | International Journal of Theoretical Physics | vol. 57 | no. 11 | pp. 3480-3484.

[DOI 10.1007/s10773-018-3862-8](#) | [Source record](#) | Registered DOI metadata

Q0020 Research on Deep Space Optical Quantum OFDM System Based on Positive Operator Valued Measurement Detection

Xiao Zhao, Xiaolin Zhou, Chongbin Xu, Xin Wang.

Book chapter | 2018 | Communications in Computer and Information Science | pp. 307-317.

[DOI 10.1007/978-981-10-7877-4_28](#) | [Source record](#) | Registered DOI metadata

Q0021 Realization of Single-Qubit Positive-Operator-Valued Measurement via a One-Dimensional Photonic Quantum Walk

Zhihao Bian, Jian Li, Hao Qin, Xiang Zhan, Rong Zhang, Barry C. Sanders, Peng Xue.

Journal article | 2015 | Physical Review Letters | vol. 114 | no. 20 | article 203602.

[DOI 10.1103/physrevlett.114.203602](#) | [Source record](#) | Registered DOI metadata

Q0022 QUANTUM THREE-QUBIT W-STATE SHARING VIA POSITIVE OPERATOR-VALUED MEASUREMENT AND PROJECTIVE MEASUREMENT

MING-QIANG BAI, ZHI-WEN MO.

Journal article | 2012 | Modern Physics Letters B | vol. 26 | no. 31 | pp. 1250208.

[DOI 10.1142/s0217984912502089](#) | [Source record](#) | Registered DOI metadata

Q0023 Quantum Tasks with Non-maximally Quantum Channels via Positive Operator-Valued Measurement

Jia-Yin Peng, Ming-Xing Luo, Zhi-Wen Mo.

Journal article | 2012 | International Journal of Theoretical Physics | vol. 52 | no. 1 | pp. 253-265.

[DOI 10.1007/s10773-012-1328-y](#) | [Source record](#) | Registered DOI metadata

Q0024 Symmetric Remote Single-Qubit State Preparation via Positive Operator-Valued Measurement

Zhang-Yin Wang.

Journal article | 2010 | International Journal of Theoretical Physics | vol. 49 | no. 6 | pp. 1357-1369.

[DOI 10.1007/s10773-010-0316-3](#) | [Source record](#) | Registered DOI metadata

Q0025 Positive-operator and projection valued measurements in quantum key distribution

Howard E. Brandt.

Journal article | 2007 | Journal of Modern Optics | vol. 54 | no. 16-17 | pp. 2357-2363.

[DOI 10.1080/09500340701639557](#) | [Source record](#) | Registered DOI metadata

D01T04 Quantum channels and open systems

Quantum channels describe physical transformations including noise and discarded subsystems. Open system models connect microscopic dynamics to experimentally observed decoherence.

Fine subcategories: Kraus representations; completely positive maps; Lindblad equations; Stinespring dilation; non Markovian dynamics.

Primary catalog resources: 8. Related topic assignments are searchable in the HTML.

Q0026 An Open-Quantum-Systems Theory of Quantum-Biological Communication Channels

Liang Dong.

Journal article | 2026 | IEEE Transactions on Molecular, Biological, and Multi-Scale Communications | vol. 12 | pp. 858-874.

[DOI 10.1109/tmbmc.2026.3716243](#) | [Source record](#) | Registered DOI metadata

Q0027 Efficient simulation of open quantum systems coupled to a reservoir through multiple channels

Hanggai Nuomin, Jiaxi Wu, Peng Zhang, David N. Beratan.

Journal article | 2024 | The Journal of Chemical Physics | vol. 161 | no. 12 | article 124114.

[DOI 10.1063/5.0226183](#) | [Source record](#) | Registered DOI metadata

Q0028 Transport in open quantum systems in the presence of lossy channels

Katha Ganguly, Manas Kulkarni, Bijay Kumar Agarwalla.

Journal article | 2024 | Physical Review B | vol. 110 | no. 23 | article 235425.

[DOI 10.1103/physrevb.110.235425](#) | [Source record](#) | Registered DOI metadata

Q0029 Quantum Systems, Channels, Information

Alexander S. Holevo.

Edited book | 2019 | De Gruyter.

[DOI 10.1515/9783110642490](#) | [Source record](#) | Registered DOI metadata

Q0030 Quantum speed-up capacity in different types of quantum channels for two-qubit open systems

Wei Wu, Xin Liu, Chao Wang.

Journal article | 2018 | Chinese Physics B | vol. 27 | no. 6 | pp. 060302.

[DOI 10.1088/1674-1056/27/6/060302](#) | [Source record](#) | Registered DOI metadata

Q0031 The decay of quantum systems with a small number of open channels

F Dittes.

Journal article | 2000 | Physics Reports | vol. 339 | no. 4 | pp. 215-316.

[DOI 10.1016/s0370-1573\(00\)00065-x](#) | [Source record](#) | Registered DOI metadata

Q0032 GitHub - qutip/qutip: QuTiP: Quantum Toolbox in Python · GitHub

Author metadata not supplied.

Software repository | Undated | QuTiP.

[Official resource](#) | [Source record](#) | Official page retrieved

Q0033 QuTiP: Quantum Toolbox in Python — QuTiP 5.3 Documentation

Author metadata not supplied.

Documentation | Undated | QuTiP.

[Official resource](#) | [Source record](#) | Official page retrieved

D01T05 Quantum entropy and information measures

Information measures quantify uncertainty, correlations, and distinguishability. Operational meanings depend on the coding, estimation, or security task being studied.

Fine subcategories: Von Neumann entropy; relative entropy; mutual information; min entropy; data processing inequalities.

Primary catalog resources: 9. Related topic assignments are searchable in the HTML.

Q0034 The Theory of Quantum Information

John Watrous.

Book | 2018 | Cambridge University Press.

[DOI 10.1017/9781316848142](#) | [Source record](#) | Registered DOI metadata

Q0035 Quantum Information Theory

Mark M. Wilde.

Book | 2016 | Cambridge University Press.

[DOI 10.1017/9781316809976](#) | [Source record](#) | Registered DOI metadata

Q0036 Quantum Information Theory

Mark M. Wilde.

Book | 2013 | Cambridge University Press.

[DOI 10.1017/cbo9781139525343](#) | [Source record](#) | Registered DOI metadata

Q0037 Quantum Accessible Information and Classical Entropy Inequalities

A. S. Holevo, A. V. Utkin.

Journal article | 2026 | Lobachevskii Journal of Mathematics | vol. 47 | no. 6 | pp. 2499-2527.

[DOI 10.1134/s1995080226617704](#) | [Source record](#) | Registered DOI metadata

Q0038 Quantum dynamical entropy and dissipative information flows

Anonymous.

Journal article | 2026 | APS Open Science.

[DOI 10.1103/6hlj-9t5r](#) | [Source record](#) | Registered DOI metadata

Q0039 Quantum-inspired information entropy in multifield turbulence

Go Yatomi, Motoki Nakata.

Journal article | 2025 | Physical Review Research | vol. 7 | no. 2 | article 023212.

[DOI 10.1103/physrevresearch.7.023212](#) | [Source record](#) | Registered DOI metadata

Q0040 Quantum Entropy and Correlations in Quantum Information

Reinhold A. Bertlmann, Nicolai Friis.

Book chapter | 2023 | Modern Quantum Theory | pp. 659-703.

[DOI 10.1093/oso/9780199683338.003.0020](#) | [Source record](#) | Registered DOI metadata

Q0041 Quantum Information Dimension and Geometric Entropy

Fabio Anza, James P. Crutchfield.

Journal article | 2022 | PRX Quantum | vol. 3 | no. 2 | article 020355.

[DOI 10.1103/prxquantum.3.020355](#) | [Source record](#) | Registered DOI metadata

AQUIRE

Q0042 Quantum Information Science | Media Arts and Sciences | MIT OpenCourseWare

Author metadata not supplied.

Course | 2006 | MIT OpenCourseWare.

[Official resource](#) | [Source record](#) | Official page retrieved

D02 Entanglement and quantum resources

Entanglement and other nonclassical resources supply capabilities that must be characterized separately from hardware size. Resource theory studies what operations can generate or consume each resource.

Prerequisites: Density operators, composite systems, and basic information measures.

Assessment focus: Allowed operations, resource monotones, witness validity, and operational task.

D02T01 Entanglement detection and quantification

Entanglement criteria test whether a state is separable, while monotones quantify particular resources. Different measures need not order mixed states in the same way.

Fine subcategories: Witnesses; separability criteria; negativity; concurrence; entanglement entropy.

Primary catalog resources: 8. Related topic assignments are searchable in the HTML.

Q0043 Data-Driven Quantification of Quantum k-Entanglement via Machine Learning

Jie Guo, Jinchuan Hou, Xiaofei Qi, Kan He.

Journal article | 2026 | Entropy | vol. 28 | no. 7 | pp. 832.

[DOI 10.3390/e28070832](https://doi.org/10.3390/e28070832) | [Source record](#) | Registered DOI metadata

Q0044 Detection of quantum entanglement across the event horizon

Patryk Michalski, Andrzej Dragan.

Journal article | 2026 | Physical Review A | vol. 114 | no. 1 | article 012420.

[DOI 10.1103/1mch-w3zx](https://doi.org/10.1103/1mch-w3zx) | [Source record](#) | Registered DOI metadata

Q0045 Entanglement detection via subsystem quantum Fisher information

Jiawei Zang, Wei Wu, Pingxing Chen.

Journal article | 2026 | Physical Review A | vol. 114 | no. 3 | article 032443.

[DOI 10.1103/kxxy-llb2](https://doi.org/10.1103/kxxy-llb2) | [Source record](#) | Registered DOI metadata

Q0046 Entanglement detection with quantum-inspired kernels and SVMs

Ana Martínez-Sabiote, Michalis Skotiniotis, Jara J. Bermejo-Vega, Daniel Manzano, Carlos Cano.

Journal article | 2026 | The Journal of Supercomputing | vol. 82 | no. 2 | article 100.

[DOI 10.1007/s11227-026-08229-7](https://doi.org/10.1007/s11227-026-08229-7) | [Source record](#) | Registered DOI metadata

Q0047 Reliable entanglement detection via quantum generative adversarial model

Li Xu, Fengting Zhou, Jing Wang, Jin-Min Liang, Zongqiang Chen, Ming Li.

Journal article | 2026 | Physica Scripta | vol. 101 | no. 26 | pp. 265103.

[DOI 10.1088/1402-4896/ae7e63](https://doi.org/10.1088/1402-4896/ae7e63) | [Source record](#) | Registered DOI metadata

Q0048 Quantification of Entanglement and Coherence with Purity Detection

He Lu, Ting Zhang, Graem Smith, John Smolin, Lu Liu, Xu-Jie Peng, Qi Zhao, Davide Girolami et al..

Posted content | 2024 | Springer Science and Business Media LLC.

[DOI 10.21203/rs.3.rs-3844047/v1](https://doi.org/10.21203/rs.3.rs-3844047/v1) | [Source record](#) | Registered DOI metadata

Q0049 Quantification of entanglement and coherence with purity detection

Ting Zhang, Graeme Smith, John A. Smolin, Lu Liu, Xu-Jie Peng, Qi Zhao, Davide Girolami, Xiongfeng Ma et al..

Journal article | 2024 | npj Quantum Information | vol. 10 | no. 1 | article 60.

[DOI 10.1038/s41534-024-00857-2](https://doi.org/10.1038/s41534-024-00857-2) | [Source record](#) | Registered DOI metadata

Q0050 Detection and quantification of entanglement with measurement-device-independent and universal entanglement witness*

Zhi-Jin Ke, Yi-Tao Wang, Shang Yu, Wei Liu, Yu Meng, Zhi-Peng Li, Hang Wang, Qiang Li et al..

Journal article | 2020 | Chinese Physics B | vol. 29 | no. 8 | pp. 080301.

[DOI 10.1088/1674-1056/ab9288](https://doi.org/10.1088/1674-1056/ab9288) | [Source record](#) | Registered DOI metadata

D02T02 Bell nonlocality and contextuality

Bell nonlocality concerns correlations incompatible with local hidden variable models. Contextuality concerns dependence on measurement context and requires its own experimental assumptions.

Fine subcategories: Bell inequalities; loopholes; device independence; contextuality inequalities; self testing.

Primary catalog resources: 8. Related topic assignments are searchable in the HTML.

Q0051 Revealing hidden nonlocality and preparation contextuality for an arbitrary input Bell inequality

Asmita Kumari, Saikat Patra.

Journal article | 2026 | Physica Scripta | vol. 101 | no. 34 | pp. 345101.

[DOI 10.1088/1402-4896/ae9993](https://doi.org/10.1088/1402-4896/ae9993) | [Source record](#) | Registered DOI metadata

Q0052 Trade-off relations between Bell nonlocality and local Kochen–Specker contextuality in generalized Bell scenarios

Lucas E A Porto, Gabriel Ruffolo, Rafael Rabelo, Marcelo Terra Cunha, Paweł Kurzyński.

Journal article | 2024 | New Journal of Physics | vol. 26 | no. 8 | pp. 083028.

[DOI 10.1088/1367-2630/ad7167](https://doi.org/10.1088/1367-2630/ad7167) | [Source record](#) | Registered DOI metadata

Q0053 Contextuality or Nonlocality: What Would John Bell Choose Today?

Marian Kupczynski.

Journal article | 2023 | Entropy | vol. 25 | no. 2 | pp. 280.

[DOI 10.3390/e25020280](https://doi.org/10.3390/e25020280) | [Source record](#) | Registered DOI metadata

Q0054 Synchronous Observation of Bell Nonlocality and State-Dependent Contextuality

Peng Xue, Lei Xiao, G. Ruffolo, A. Mazzari, T. Temistocles, M. Terra Cunha, R. Rabelo.

Journal article | 2023 | Physical Review Letters | vol. 130 | no. 4 | article 040201.

[DOI 10.1103/physrevlett.130.040201](https://doi.org/10.1103/physrevlett.130.040201) | [Source record](#) | Registered DOI metadata

Q0055 Assumption-Free Derivation of the Bell-Type Criteria of Contextuality/Nonlocality

Ehtibar N. Dzhafarov.

Journal article | 2021 | Entropy | vol. 23 | no. 11 | pp. 1543.

[DOI 10.3390/e23111543](https://doi.org/10.3390/e23111543) | [Source record](#) | Registered DOI metadata

Q0056 Classical causal models cannot faithfully explain Bell nonlocality or Kochen-Specker contextuality in arbitrary scenarios

J. C. Pearl, E. G. Cavalcanti.

Journal article | 2021 | Quantum | vol. 5 | pp. 518 | article 518.

[DOI 10.22331/q-2021-08-05-518](https://doi.org/10.22331/q-2021-08-05-518) | [Source record](#) | Registered DOI metadata

Q0057 Experimental observation of quantum contextuality beyond Bell nonlocality

Zheng-Hao Liu, Hui-Xian Meng, Zhen-Peng Xu, Jie Zhou, Sheng Ye, Qiang Li, Kai Sun, Hong-Yi Su et al..

Journal article | 2019 | Physical Review A | vol. 100 | no. 4 | article 042118.

[DOI 10.1103/physreva.100.042118](https://doi.org/10.1103/physreva.100.042118) | [Source record](#) | Registered DOI metadata

Q0058 Sharing nonlocality and nontrivial preparation contextuality using the same family of Bell expressions

Asmita Kumari, A. K. Pan.

Journal article | 2019 | Physical Review A | vol. 100 | no. 6 | article 062130.

[DOI 10.1103/physreva.100.062130](https://doi.org/10.1103/physreva.100.062130) | [Source record](#) | Registered DOI metadata

D02T03 Coherence and resource theories

Resource theories specify free states and operations before defining a resource. Coherence, asymmetry, and entanglement are related but distinct examples.

Fine subcategories: Incoherent operations; asymmetry; conversion rates; resource monotones; coherence distillation.

Primary catalog resources: 8. Related topic assignments are searchable in the HTML.

Q0059 One-shot manipulation of coherence in dynamic quantum resource theory

Yu Luo.

Journal article | 2025 | Physical Review A | vol. 111 | no. 2 | article 022447.

[DOI 10.1103/physreva.111.022447](https://doi.org/10.1103/physreva.111.022447) | [Source record](#) | Registered DOI metadata

Q0060 Generic aspects of the resource theory of quantum coherence

Fabio Deelan Cunden, Paolo Facchi, Giuseppe Florio, Giovanni Gramegna.

Journal article | 2021 | Physical Review A | vol. 103 | no. 2 | article 022401.

[DOI 10.1103/physreva.103.022401](https://doi.org/10.1103/physreva.103.022401) | [Source record](#) | Registered DOI metadata

Q0061 Resource theory of quantum coherence with probabilistically nondistinguishable pointers and corresponding wave-particle duality

Chirag Srivastava, Sreetama Das, Ujjwal Sen.

Journal article | 2021 | Physical Review A | vol. 103 | no. 2 | article 022417.

[DOI 10.1103/physreva.103.022417](https://doi.org/10.1103/physreva.103.022417) | [Source record](#) | Registered DOI metadata

Q0062 Dynamical resource theory of quantum coherence

Gaurav Saxena, Eric Chitambar, Gilad Gour.

Journal article | 2020 | Physical Review Research | vol. 2 | no. 2 | article 023298.

[DOI 10.1103/physrevresearch.2.023298](https://doi.org/10.1103/physrevresearch.2.023298) | [Source record](#) | Registered DOI metadata

Q0063 The resource theory of coherence for quantum channels

F. H. Kamin, F. T. Tabesh, S. Salimi, F. Kheirandish.

Journal article | 2020 | Quantum Information Processing | vol. 19 | no. 7 | article 210.

[DOI 10.1007/s11128-020-02702-9](https://doi.org/10.1007/s11128-020-02702-9) | [Source record](#) | Registered DOI metadata

Q0064 General resource theory of quantum coherence in multipartite system

Feng Liu, Dong-Mei Gao, Xiao-Qiu Cai.

Journal article | 2019 | Acta Physica Sinica | vol. 68 | no. 23 | pp. 230301.

[DOI 10.7498/aps.68.20190966](https://doi.org/10.7498/aps.68.20190966) | [Source record](#) | Registered DOI metadata

Q0065 Operational resource theory of total quantum coherence

Si-ren Yang, Chang-shui Yu.

Journal article | 2018 | Annals of Physics | vol. 388 | pp. 305-314.

[DOI 10.1016/j.aop.2017.11.028](https://doi.org/10.1016/j.aop.2017.11.028) | [Source record](#) | Registered DOI metadata

Q0066 Structure of the Resource Theory of Quantum Coherence

Alexander Streltsov, Swapan Rana, Paul Boes, Jens Eisert.

Journal article | 2017 | Physical Review Letters | vol. 119 | no. 14 | article 140402.

[DOI 10.1103/physrevlett.119.140402](https://doi.org/10.1103/physrevlett.119.140402) | [Source record](#) | Registered DOI metadata

D02T04 Magic and stabilizer resources

Magic characterizes resources outside stabilizer computation. It matters both for non Clifford computation and for the cost of classical simulation.

Fine subcategories: Nonstabilizer states; mana; robustness of magic; stabilizer rank; simulation costs.

Primary catalog resources: 7. Related topic assignments are searchable in the HTML.

Q0067 No-cost Bell nonlocality certification from quantum tomography and its applications in quantum-magic-resource witnessing

Paweł Cieśliński, Lukas Knips, Harald Weinfurter, Wiesław Laskowski.

Journal article | 2026 | Physical Review A | vol. 113 | no. 5 | article 052404.

[DOI 10.1103/1nkl-sphd](#) | [Source record](#) | Registered DOI metadata

Q0068 Qualia from quantum magic: a quantum resource approach to phenomenal consciousness

Gandhimohan M. Viswanathan.

Journal article | 2026 | Frontiers in Physics | vol. 13 | article 1715825.

[DOI 10.3389/fphy.2025.1715825](#) | [Source record](#) | Registered DOI metadata

Q0069 Stabilizer Ranks, Barnes Wall Lattices and Magic Monotones

Amolak Ratan Kalra, Pulkit Sinha.

Journal article | 2026 | Quantum | vol. 10 | pp. 2179 | article 2179.

[DOI 10.22331/q-2026-07-29-2179](#) | [Source record](#) | Registered DOI metadata

Fine tags: stabilizer rank

Q0070 Stabilizer entanglement enhances magic injection

Zong-Yue Hou, ChunJun Cao, Zhi-Cheng Yang.

Journal article | 2026 | npj Quantum Information | vol. 12 | no. 1 | article 113.

[DOI 10.1038/s41534-026-01265-4](#) | [Source record](#) | Registered DOI metadata

Q0071 Leveraging magic resources for quantum channel capacity enhancement under stabilizer convolution

Wenlong Sun, Yuanfeng Jin.

Journal article | 2025 | Physical Review A | vol. 112 | no. 5 | article 052412.

[DOI 10.1103/qnz3-4fsc](#) | [Source record](#) | Registered DOI metadata

Q0072 Stabilizer Testing and Magic Entropy via Quantum Fourier Analysis

Kaifeng Bu, Weichen Gu, Arthur Jaffe.

Journal article | 2025 | Communications in Mathematical Physics | vol. 406 | no. 10 | article 236.

[DOI 10.1007/s00220-025-05421-3](#) | [Source record](#) | Registered DOI metadata

Q0073 Stabilizer entropies are monotones for magic-state resource theory

Lorenzo Leone, Lennart Bittel.

Journal article | 2024 | Physical Review A | vol. 110 | no. 4 | article L040403.

[DOI 10.1103/physreva.110.l040403](#) | [Source record](#) | Registered DOI metadata

D02T05 Entanglement distillation

Distillation converts noisy shared resources into fewer high quality ones. Rates and achievable fidelities depend on allowed communication and operation classes.

Fine subcategories: Purification; hashing; bound entanglement; one shot protocols; network resource conversion.

Primary catalog resources: 11. Related topic assignments are searchable in the HTML.

Q0074 Improving entanglement resilience in quantum memories with error-detection-based distillation

Huidan Zheng, Gunsik Min, Ilkwon Sohn, Jun Heo.

Journal article | 2026 | Scientific Reports.

[DOI 10.1038/s41598-026-72375-4](#) | [Source record](#) | Registered DOI metadata

Q0075 Nonlocality Distillation Can Outperform Entanglement Distillation

Peter Høyer, Jibran Rashid, Razeen Ud Din.

Conference paper | 2026 | 2026 International Conference on Quantum Communications, Networking, and Computing (QNCN) | pp. 291-297.

[DOI 10.1109/qcnc69040.2026.00049](#) | [Source record](#) | Registered DOI metadata

Q0076 Constant-Rate Entanglement Distillation for Fast Quantum Interconnects

Christopher Pattison, Gefen Baranes, Juan Pablo Bonilla Ataides, Mikhail D. Lukin, Hengyun Zhou.

Conference paper | 2025 | Proceedings of the 52nd Annual International Symposium on Computer Architecture | pp. 257-270.

[DOI 10.1145/3695053.3731069](#) | [Source record](#) | Registered DOI metadata

Q0077 Entanglement Distribution Delay Optimization in Quantum Networks With Distillation

Mahdi Chehimi, Kenneth Goodenough, Walid Saad, Don Towsley, Tony X. Zhou.

Journal article | 2025 | IEEE Journal on Selected Areas in Communications | vol. 43 | no. 5 | pp. 1871-1886.

[DOI 10.1109/jsac.2025.3543485](#) | [Source record](#) | Registered DOI metadata

Q0078 Quantum Code-Assisted Entanglement Distillation Under Bit-Flip Noise

Huidan Zheng, Gunsik Min, Yujin Kang, Youshin Chung, Hyelyn Kwak, Jun Heo.

Conference paper | 2025 | 2025 International Conference on Mobile, Military, Maritime IT Convergence (ICMIC) | pp. 317-320.

[DOI 10.1109/icmic66299.2025.11257738](#) | [Source record](#) | Registered DOI metadata

Q0079 Quantum change point and entanglement distillation

Abhishek Banerjee, Pratapaditya Bej, Somshubhro Bandyopadhyay.

Journal article | 2024 | Physical Review A | vol. 109 | no. 4 | article 042407.

[DOI 10.1103/physreva.109.042407](#) | [Source record](#) | Registered DOI metadata

Q0080 Entanglement Distillation Optimization Using Fuzzy Relations for Quantum State Tomography

Timothy Ganesan, Irraivan Elamvazuthi.

Journal article | 2023 | Algorithms | vol. 16 | no. 7 | pp. 313.

[DOI 10.3390/a16070313](#) | [Source record](#) | Registered DOI metadata

Q0081 Learning Quantum Entanglement Distillation With Noisy Classical Communications

Hari Hara Suthan Chittoor, Osvaldo Simeone.

Conference paper | 2023 | ICASSP 2023 - 2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP) | pp. 1-5.

[DOI 10.1109/icassp49357.2023.10097202](#) | [Source record](#) | Registered DOI metadata

Q0082 Optimal Entanglement Distillation Policies for Quantum Switches

Vivek Kumar, Nitish K. Chandra, Kaushik P. Seshadreesan, Alan Scheller-Wolf, Sridhar Tayur.

Conference paper | 2023 | 2023 IEEE International Conference on Quantum Computing and Engineering (QCE) | pp. 1198-1204.

[DOI 10.1109/qce57702.2023.00135](#) | [Source record](#) | Registered DOI metadata

Q0083 Trapped Ion Quantum Repeaters with Entanglement Distillation based on Quantum LDPC Codes

Ann Kang, Saikat Guha, Narayanan Rengaswamy, Kaushik P. Seshadreesan.

Conference paper | 2023 | 2023 IEEE International Conference on Quantum Computing and Engineering (QCE) | pp. 1165-1171.

[DOI 10.1109/qce57702.2023.00131](https://doi.org/10.1109/qce57702.2023.00131) | [Source record](#) | Registered DOI metadata

Q0084 Multi-User Distillation of Common Randomness and Entanglement From Quantum States

Farzin Salek, Andreas Winter.

Journal article | 2022 | IEEE Transactions on Information Theory | vol. 68 | no. 2 | pp. 976-988.

[DOI 10.1109/tit.2021.3124965](https://doi.org/10.1109/tit.2021.3124965) | [Source record](#) | Registered DOI metadata

D03 Models of quantum computation

Different models specify the available operations and how an algorithm is executed. Equivalence of models does not imply identical engineering costs or noise tolerance.

Prerequisites: Qubits, unitary operations, and measurement.

Assessment focus: Model assumptions, universality, resource generation, and noise constraints.

D03T01 Circuit and universal gate models

A circuit is an ordered composition of gates and measurements. Universality describes what transformations can be approximated, while compilation determines their actual cost.

Fine subcategories: Universal gate sets; reversible circuits; qudit gates; multi controlled gates; universality proofs.

Primary catalog resources: 8. Related topic assignments are searchable in the HTML.

Q0085 Quantum Computing for Computer Scientists

Noson S. Yanofsky, Mirco A. Mannucci.

Book | 2008 | Cambridge University Press.

[DOI 10.1017/cbo9780511813887](https://doi.org/10.1017/cbo9780511813887) | [Source record](#) | Registered DOI metadata

Q0086 Quantum Computer Science

N. David Mermin.

Book | 2007 | Cambridge University Press.

[DOI 10.1017/cbo9780511813870](https://doi.org/10.1017/cbo9780511813870) | [Source record](#) | Registered DOI metadata

Q0087 Quantum theory, the Church–Turing principle and the universal quantum computer

David Deutsch.

Journal article | 1985 | Proceedings of the Royal Society of London. A. Mathematical and Physical Sciences | vol. 400 | no. 1818 | pp. 97-117.

[DOI 10.1098/rspa.1985.0070](https://doi.org/10.1098/rspa.1985.0070) | [Source record](#) | Registered DOI metadata

Q0088 Time-optimal universal quantum gates on superconducting circuits

Ze Li, Ming-Jie Liang, Zheng-Yuan Xue.

Journal article | 2023 | Physical Review A | vol. 108 | no. 4 | article 042617.

[DOI 10.1103/physreva.108.042617](https://doi.org/10.1103/physreva.108.042617) | [Source record](#) | Registered DOI metadata

Q0089 Quantum-Dot Cellular Automata-Based Encoder Circuit Using Layered Universal Gates

Birinderjit Singh Kalyan, Balwinder Singh.

Book chapter | 2021 | Nanoelectronic Devices for Hardware and Software Security | pp. 217-230.

[DOI 10.1201/9781003126645-11](https://doi.org/10.1201/9781003126645-11) | [Source record](#) | Registered DOI metadata

Q0090 Electric circuits for universal quantum gates and quantum Fourier transformation

Motohiko Ezawa.

Journal article | 2020 | Physical Review Research | vol. 2 | no. 2 | article 023278.

[DOI 10.1103/physrevresearch.2.023278](https://doi.org/10.1103/physrevresearch.2.023278) | [Source record](#) | Registered DOI metadata

Q0091 Experimental Implementation of Universal Nonadiabatic Geometric Quantum Gates in a Superconducting Circuit

Y. Xu, Z. Hua, Tao Chen, X. Pan, X. Li, J. Han, W. Cai, Y. Ma et al..

Journal article | 2020 | Physical Review Letters | vol. 124 | no. 23 | article 230503.

[DOI 10.1103/physrevlett.124.230503](https://doi.org/10.1103/physrevlett.124.230503) | [Source record](#) | Registered DOI metadata

Q0092 Fast universal quantum gates on microwave photons with all-resonance operations in circuit QED

Ming Hua, Ming-Jie Tao, Fu-Guo Deng.

Journal article | 2015 | Scientific Reports | vol. 5 | no. 1 | article 9274.

[DOI 10.1038/srep09274](#) | [Source record](#) | Registered DOI metadata

D03T02 Measurement based computing

Measurement based computation processes an entangled resource state through adaptive local measurements. Classical feed forward and resource generation are part of the computation.

Fine subcategories: Cluster states; graph states; adaptive measurements; feed forward; one way computing.

Primary catalog resources: 8. Related topic assignments are searchable in the HTML.

Q0093 MCBGD: Multi-Path Cluster-Based Graph Distribution for Scalable Measurement-Based Quantum Computation

Guanting Yu, Yongzhe Lin, Yiyang Yu, Zisen Wei, Xiaofeng Jiang.

Conference paper | 2026 | 2026 28th International Conference on Advanced Communications Technology (ICACT) | pp. 309-315.

[DOI 10.23919/icact68090.2026.11431344](#) | [Source record](#) | Registered DOI metadata

Q0094 One- and two-dimensional cluster states for topological phase simulation and measurement-based quantum computation

Tao Jiang, Jianbin Cai, Junxiang Huang, Naibin Zhou, Yukun Zhang, Jiahao Bei, Guoqing Cai, Sirui Cao et al..

Journal article | 2026 | Nature Physics | vol. 22 | no. 3 | pp. 430-438.

[DOI 10.1038/s41567-026-03179-6](#) | [Source record](#) | Registered DOI metadata

Fine tags: Cluster states

Q0095 FMCC: Flexible Measurement-based Quantum Computation over Cluster State

Yingheng Li, Aditya Pawar, Zewei Mo, Youtao Zhang, Jun Yang, Xulong Tang.

Conference paper | 2024 | Proceedings of the 29th ACM International Conference on Architectural Support for Programming Languages and Operating Systems, Volume 4 | pp. 111-126.

[DOI 10.1145/3622781.3674185](#) | [Source record](#) | Registered DOI metadata

Q0096 Universal hardware-efficient topological measurement-based quantum computation via color-code-based cluster states

Seok-Hyung Lee, Hyunseok Jeong.

Journal article | 2022 | Physical Review Research | vol. 4 | no. 1 | article 013010.

[DOI 10.1103/physrevresearch.4.013010](#) | [Source record](#) | Registered DOI metadata

Fine tags: Cluster states

Q0097 Temporal-mode continuous-variable three-dimensional cluster state for topologically protected measurement-based quantum computation

Kosuke Fukui, Warit Asavanant, Akira Furusawa.

Journal article | 2020 | Physical Review A | vol. 102 | no. 3 | article 032614.

[DOI 10.1103/physreva.102.032614](#) | [Source record](#) | Registered DOI metadata

Q0098 Constraints on Measurement-Based Quantum Computation in Effective Cluster States

Daniel Klagges, Kai Phillip Schmidt.

Journal article | 2012 | Physical Review Letters | vol. 108 | no. 23 | article 230508.

[DOI 10.1103/physrevlett.108.230508](#) | [Source record](#) | Registered DOI metadata

Fine tags: Cluster states

Q0099 MEASUREMENT-BASED QUANTUM COMPUTATION WITH CLUSTER STATES

ROBERT RAUßENDORF.

Journal article | 2009 | International Journal of Quantum Information | vol. 07 | no. 06 | pp. 1053-1203.

[DOI 10.1142/s0219749909005699](https://doi.org/10.1142/s0219749909005699) | [Source record](#) | Registered DOI metadata

Fine tags: Cluster states

Q0100 Efficient measurement-based quantum computation with cluster states in quantum-bit fixed systems

Da-Sheng Diao, Yong-Sheng Zhang, Xiang-Fa Zhou, Guang-Can Guo.

Journal article | 2008 | Physical Review A | vol. 77 | no. 4 | article 044301.

[DOI 10.1103/physreva.77.044301](https://doi.org/10.1103/physreva.77.044301) | [Source record](#) | Registered DOI metadata

Fine tags: Cluster states

D03T03 Adiabatic quantum computing

Adiabatic computation evolves a Hamiltonian along a path whose low energy states encode the answer. Gap behavior and schedule assumptions determine the required runtime.

Fine subcategories: Spectral gaps; adiabatic theorems; schedule design; universality; diabatic effects.

Primary catalog resources: 16. Related topic assignments are searchable in the HTML.

Q0101 Adiabatic Quantum Computing and Quantum Annealing

Bastien Chopard, Marco Tomassini.

Book chapter | 2026 | Natural Computing Series | pp. 265-275.

[DOI 10.1007/978-981-95-6216-9_15](https://doi.org/10.1007/978-981-95-6216-9_15) | [Source record](#) | Registered DOI metadata

Q0102 Quantum Zeno effect versus adiabatic quantum computing and quantum annealing

Naser Ahmadianiaz, Dennis Kraft, Gernot Schaller, Ralf Schützhold.

Journal article | 2026 | New Journal of Physics | vol. 28 | no. 6 | pp. 064502.

[DOI 10.1088/1367-2630/ae6e68](https://doi.org/10.1088/1367-2630/ae6e68) | [Source record](#) | Registered DOI metadata

Q0103 Adiabatic Quantum Computing

Christian Bauckhage, Rafet Sifa.

Book chapter | 2025 | Cognitive Technologies | pp. 253-276.

[DOI 10.1007/978-3-031-99402-9_12](https://doi.org/10.1007/978-3-031-99402-9_12) | [Source record](#) | Registered DOI metadata

Q0104 Adiabatic Quantum Computing

Ray LaPierre.

Book chapter | 2025 | The Materials Research Society Series | pp. 341-344.

[DOI 10.1007/978-3-031-90731-9_23](https://doi.org/10.1007/978-3-031-90731-9_23) | [Source record](#) | Registered DOI metadata

Q0105 Bayesian Update Step Using Adiabatic Quantum Computing

Rheinische Friedrich- Wilhelms-Universität, Norman Bunk, Felix Govaers, Fraunhofer FKIE.

Conference paper | 2025 | Computer Science Research Notes.

[DOI 10.24132/csrn.2025-a23](https://doi.org/10.24132/csrn.2025-a23) | [Source record](#) | Registered DOI metadata

Q0106 Determining probability density functions with adiabatic quantum computing

Matteo Robbiati, Juan M. Cruz-Martinez, Stefano Carrazza.

Journal article | 2025 | Quantum Machine Intelligence | vol. 7 | no. 1 | article 5.

[DOI 10.1007/s42484-024-00228-2](https://doi.org/10.1007/s42484-024-00228-2) | [Source record](#) | Registered DOI metadata

Q0107 Exploiting adiabatic quantum computing in deep space missions

Rebecca Casati, Enrico Prati.

Journal article | 2025 | Journal of Physics: Conference Series | vol. 3017 | no. 1 | pp. 012042.

[DOI 10.1088/1742-6596/3017/1/012042](https://doi.org/10.1088/1742-6596/3017/1/012042) | [Source record](#) | Registered DOI metadata

Q0108 Filter-enhanced adiabatic quantum computing on a digital quantum processor

Erenay Karacan, Conor Mc Keever, Michael Foss-Feig, David Hayes, Michael Lubasch.

Journal article | 2025 | Physical Review Research | vol. 7 | no. 3 | article 033153.

[DOI 10.1103/x2v8-jx1h](#) | [Source record](#) | Registered DOI metadata

Q0109 Non-adiabatic holonomies for photonic quantum computing

Vera Neef, Julien Pinske, Tom A. W. Wolterink, Karo Becker, Matthias Heinrich, Stefan Scheel, Alexander Szameit.

Journal article | 2025 | Optica Quantum | vol. 3 | no. 1 | pp. 93.

[DOI 10.1364/opticaq.530855](#) | [Source record](#) | Registered DOI metadata

Q0110 Solving Portfolio Optimization Problems using Adiabatic Quantum Computing

Bartłomiej Kolodziejczyk.

Posted content | 2025 | Springer Science and Business Media LLC.

[DOI 10.21203/rs.3.rs-6986668/v1](#) | [Source record](#) | Registered DOI metadata

Q0111 Adiabatic Quantum Computing

Bernard Zygelman.

Book chapter | 2024 | Undergraduate Topics in Computer Science | pp. 241-273.

[DOI 10.1007/978-3-031-66425-0_10](#) | [Source record](#) | Registered DOI metadata

Q0112 Quantum annealing and adiabatic quantum computing

Charles R. Giardina.

Book chapter | 2024 | Many-Sorted Algebras for Deep Learning and Quantum Technology | pp. 103-129.

[DOI 10.1016/b978-0-443-13697-9.00021-7](#) | [Source record](#) | Registered DOI metadata

Q0113 Quantum-Assisted Machine Learning by Means of Adiabatic Quantum Computing

Leonardo Tomazeli Duarte, Yannick Deville.

Conference paper | 2024 | 2024 IEEE Mediterranean and Middle-East Geoscience and Remote Sensing Symposium (M2GARSS) | pp. 371-375.

[DOI 10.1109/m2garss57310.2024.10537323](#) | [Source record](#) | Registered DOI metadata

Q0114 Simulating adiabatic quantum computing with parameterized quantum circuits

Ioannis Kolotouros, Ioannis Petrongonas, Miloš Prokop, Petros Wallden.

Journal article | 2024 | Quantum Science and Technology | vol. 10 | no. 1 | pp. 015003.

[DOI 10.1088/2058-9565/ad80c0](#) | [Source record](#) | Registered DOI metadata

Q0115 Towards Adiabatic Quantum Computing Using Compressed Quantum Circuits

Conor Mc Keever, Michael Lubasch.

Journal article | 2024 | PRX Quantum | vol. 5 | no. 2 | article 020362.

[DOI 10.1103/prxquantum.5.020362](#) | [Source record](#) | Registered DOI metadata

Q0116 Training neural networks with universal adiabatic quantum computing

Steve Abel, Juan Carlos Criado, Michael Spannowsky.

Journal article | 2024 | Frontiers in Artificial Intelligence | vol. 7 | article 1368569.

[DOI 10.3389/frai.2024.1368569](#) | [Source record](#) | Registered DOI metadata

D03T04 Quantum annealing

Annealing seeks low energy solutions of an encoded optimization problem. Embedding, temperature, open system dynamics, and solution sampling distinguish hardware annealers from ideal adiabatic algorithms.

Fine subcategories: Ising encoding; QUBO; minor embedding; anneal offsets; reverse annealing.

Primary catalog resources: 14. Related topic assignments are searchable in the HTML.

Q0117 QUANTUM ANNEALING

STEVE ABEL, LUCA A. NUTRICATI.

Book chapter | 2026 | Machine Learning Tutorials for Pure Mathematics and Theoretical Physics | pp. 59-79.

[DOI 10.1142/9781807290016_0004](#) | [Source record](#) | Registered DOI metadata

Q0118 Quantum Annealing

Michael Tsesmelis.

Book chapter | 2025 | Quantum Technologies | pp. 65-72.

[DOI 10.1007/978-3-031-90727-2_7](#) | [Source record](#) | Registered DOI metadata

Q0119 Quantum Annealing

Ryo Maezono.

Book chapter | 2025 | Essentials for Deeper Understanding of Quantum Computing | pp. 199-214.

[DOI 10.1007/978-981-96-5646-2_8](#) | [Source record](#) | Registered DOI metadata

Q0120 Quantum error mitigation in quantum annealing

Jack Raymond, Mohammad H. Amin, Andrew D. King, Richard Harris, William Bernoudy, Andrew J. Berkley, Kelly Boothby, Anatoly Smirnov et al..

Journal article | 2025 | npj Quantum Information | vol. 11 | no. 1 | article 38.

[DOI 10.1038/s41534-025-00977-3](#) | [Source record](#) | Registered DOI metadata

Q0121 Quantum Annealing

Chuck Easttom.

Book chapter | 2024 | Hardware for Quantum Computing | pp. 101-112.

[DOI 10.1007/978-3-031-66477-9_9](#) | [Source record](#) | Registered DOI metadata

Q0122 Quantum Annealing

Carleton Coffrin, Marc Vuffray.

Book chapter | 2024 | Encyclopedia of Optimization | pp. 1-8.

[DOI 10.1007/978-3-030-54621-2_855-1](#) | [Source record](#) | Registered DOI metadata

Q0123 Quantum annealing and computation

Bikas K Chakrabarti, Sudip Mukherjee.

Book chapter | 2024 | Encyclopedia of Condensed Matter Physics | pp. 536-542.

[DOI 10.1016/b978-0-323-90800-9.00057-3](#) | [Source record](#) | Registered DOI metadata

Q0124 Reverse Quantum Annealing Assisted by Forward Annealing

Manpreet Singh Jattana.

Journal article | 2024 | Quantum Reports | vol. 6 | no. 3 | pp. 452-464.

[DOI 10.3390/quantum6030030](#) | [Source record](#) | Registered DOI metadata

Fine tags: reverse annealing

Q0125 Quantum Annealing with Ocean

Rafael Pereira da Silva.

Posted content | 2023 | Elsevier BV.

[DOI 10.2139/ssrn.4526152](#) | [Source record](#) | Registered DOI metadata

Q0126 Quantum Topology Optimization via Quantum Annealing

Zisheng Ye, Xiaoping Qian, Wenxiao Pan.

Journal article | 2023 | IEEE Transactions on Quantum Engineering | vol. 4 | pp. 1-15.

[DOI 10.1109/tqe.2023.3266410](#) | [Source record](#) | Registered DOI metadata

Q0127 Customized Quantum Annealing Schedules

Mostafa Khezri, Xi Dai, Rui Yang, Tameem Albash, Adrian Lupascu, Daniel A. Lidar.

Journal article | 2022 | Physical Review Applied | vol. 17 | no. 4 | article 044005.

[DOI 10.1103/physrevapplied.17.044005](https://doi.org/10.1103/physrevapplied.17.044005) | [Source record](#) | Registered DOI metadata

Q0128 Guaranteed-accuracy quantum annealing

Takashi Imoto, Yuya Seki, Yuichiro Matsuzaki, Shiro Kawabata.

Journal article | 2022 | Physical Review A | vol. 106 | no. 4 | article 042615.

[DOI 10.1103/physreva.106.042615](https://doi.org/10.1103/physreva.106.042615) | [Source record](#) | Registered DOI metadata

Q0129 Parallel quantum annealing

Elijah Pelofske, Georg Hahn, Hristo N. Djidjev.

Journal article | 2022 | Scientific Reports | vol. 12 | no. 1 | article 4499.

[DOI 10.1038/s41598-022-08394-8](https://doi.org/10.1038/s41598-022-08394-8) | [Source record](#) | Registered DOI metadata

Q0130 Quantum annealing: an overview

Atanu Rajak, Sei Suzuki, Amit Dutta, Bikas K. Chakrabarti.

Journal article | 2022 | Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences | vol. 381 | no. 2241 | article 20210417.

[DOI 10.1098/rsta.2021.0417](https://doi.org/10.1098/rsta.2021.0417) | [Source record](#) | Registered DOI metadata

D03T05 Continuous variable computing

Continuous variable systems encode information in modes with continuous quadratures. Universal computation generally needs suitable non Gaussian resources as well as Gaussian control.

Fine subcategories: Quadratures; Gaussian states; non Gaussian operations; squeezing; continuous variable universality.

Primary catalog resources: 17. Related topic assignments are searchable in the HTML.

Q0131 Continuous-variable quantum communication

Vladyslav C. Usenko, Antonio Acín, Romain Alléaume, Ulrik L. Andersen, Eleni Diamanti, Tobias Gehring, Adnan A. E. Hajomer, Florian Kanitschar et al..

Journal article | 2026 | Reviews of Modern Physics | vol. 98 | no. 1 | article 015003.

[DOI 10.1103/mgj7-t6d3](https://doi.org/10.1103/mgj7-t6d3) | [Source record](#) | Registered DOI metadata

Q0132 Experimental memory control in continuous-variable optical quantum reservoir computing

Iris Paparelle, Johan Henaff, Jorge García-Beni, Émilie Gillet, Daniel Montesinos, Gian Luca Giorgi, Miguel C. Soriano, Roberta Zambrini et al..

Journal article | 2026 | Nature Photonics | vol. 20 | no. 4 | pp. 413-420.

[DOI 10.1038/s41566-026-01880-9](https://doi.org/10.1038/s41566-026-01880-9) | [Source record](#) | Registered DOI metadata

Q0133 Interplay of resources for universal continuous-variable quantum computing

Varun Upreti, Ulysse Chabaud.

Journal article | 2026 | Physical Review Research | vol. 8 | no. 3 | article 033379.

[DOI 10.1103/2str-dhrg](https://doi.org/10.1103/2str-dhrg) | [Source record](#) | Registered DOI metadata

Q0134 Neonatal EEG Seizure Prediction with Quantum-Inspired Continuous-Variable (Bosonic) Reservoir Computing

Krishna Bhatia, Chun-Yu Lin.

Conference paper | 2026 | ICASSP 2026 - 2026 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP) | pp. 22057-22061.

[DOI 10.1109/icassp55912.2026.11460556](https://doi.org/10.1109/icassp55912.2026.11460556) | [Source record](#) | Registered DOI metadata

Q0135 Continuous Variable Quantum Distance Bounding

Kevin Bogner, Aysajan Abidin, Dave Singelée.

Conference paper | 2025 | IEEE INFOCOM 2025 - IEEE Conference on Computer Communications Workshops (INFOCOM WKSHPS) | pp. 1-6.

[DOI 10.1109/infocomwkshps65812.2025.11153001](https://doi.org/10.1109/infocomwkshps65812.2025.11153001) | [Source record](#) | Registered DOI metadata

Q0136 Continuous-variable quantum computing on a trapped ion: neural network applications

Alexandre C Ricardo, Gubio G de Lima, Amanda G Valério, Tiago de S Farias, Celso J Villas-Boas.

Journal article | 2025 | Physica Scripta | vol. 100 | no. 12 | pp. 125109.

[DOI 10.1088/1402-4896/ae2164](https://doi.org/10.1088/1402-4896/ae2164) | [Source record](#) | Registered DOI metadata

Q0137 Integrated continuous-variable quantum light sources for quantum computing

Xuezhi Zhu, Siyu Ren, Yunyun Cao, Xiaolong Su.

Journal article | 2025 | Advanced Photonics | vol. 7 | no. 06 | pp. 1-8.

[DOI 10.1117/1.ap.7.6.064005](https://doi.org/10.1117/1.ap.7.6.064005) | [Source record](#) | Registered DOI metadata

Q0138 Measurement-based continuous-variable quantum reservoir computing

Iris Paparelle, Johan Henaff, Jorge Garcia-Beni, Roberta Zambrini, Valentina Parigi.

Conference paper | 2025 | Quantum Communications and Quantum Imaging XXIII | pp. 21.

[DOI 10.1117/12.3063922](https://doi.org/10.1117/12.3063922) | [Source record](#) | Registered DOI metadata

Q0139 Programmable Continuous-Variable Photonic Quantum Computing in the Time Domain

Shuntaro Takeda.

Conference paper | 2025 | 2025 European Conference on Optical Communications (ECOC) | pp. 1-2.

[DOI 10.1109/ecoc66593.2025.11263362](https://doi.org/10.1109/ecoc66593.2025.11263362) | [Source record](#) | Registered DOI metadata

Q0140 Quantum reservoir computing with nonlinear optics: experimental continuous variable memory control

Iris Paparelle, Johan Henaff, Jorge Garcia-Beni, Roberta Zambrini, Valentina Parigi.

Conference paper | 2025 | Emerging Topics in Artificial Intelligence (ETAI) 2025 | pp. 15.

[DOI 10.1117/12.3063910](https://doi.org/10.1117/12.3063910) | [Source record](#) | Registered DOI metadata

Q0141 Towards a Measurement-Based Quantum Computing Architecture Using Continuous-Variable Optical Qubits

Ryosuke Matsuo, Junichiro Kadamoto.

Conference paper | 2025 | 2025 IEEE International Conference on Quantum Computing and Engineering (QCE) | pp. 657-658.

[DOI 10.1109/qce65121.2025.10493](https://doi.org/10.1109/qce65121.2025.10493) | [Source record](#) | Registered DOI metadata

Q0142 Continuous-variable Quantum Boltzmann Machine

Shikha Bangar, Leanto Sunny, Kubra Yeter-Aydeniz, George Siopsis.

Posted content | 2024 | Springer Science and Business Media LLC.

[DOI 10.21203/rs.3.rs-4485601/v1](https://doi.org/10.21203/rs.3.rs-4485601/v1) | [Source record](#) | Registered DOI metadata

Q0143 Programmable Continuous-Variable Photonic Quantum Computing in the Time Domain

Shuntaro Takeda.

Conference paper | 2024 | 2024 Conference on Lasers and Electro-Optics Pacific Rim (CLEO-PR) | pp. 1-1.

[DOI 10.1109/cleo-pr60912.2024.10676435](https://doi.org/10.1109/cleo-pr60912.2024.10676435) | [Source record](#) | Registered DOI metadata

Q0144 Time-Multiplexed Programmable Continuous-Variable Photonic Quantum Computing

Shuntaro Takeda.

Conference paper | 2024 | Advanced Photonics Congress 2024 | pp. SpW2H.1.

[DOI 10.1364/sppcom.2024.spw2h.1](https://doi.org/10.1364/sppcom.2024.spw2h.1) | [Source record](#) | Registered DOI metadata

Q0145 Flow conditions for continuous variable measurement-based quantum computing

Robert I. Booth, Damian Markham.

Journal article | 2023 | Quantum | vol. 7 | pp. 1146 | article 1146.

[DOI 10.22331/q-2023-10-19-1146](#) | [Source record](#) | Registered DOI metadata

Q0146 Time-multiplexed programmable continuous-variable photonic quantum computing

Shuntaro Takeda.

Conference paper | 2023 | Quantum Communications and Quantum Imaging XXI | pp. 27.

[DOI 10.1117/12.2672528](#) | [Source record](#) | Registered DOI metadata

Q0147 Xanadu | Welcome to Xanadu

Author metadata not supplied.

Documentation | Undated | Xanadu Strawberry Fields.

[Official resource](#) | [Source record](#) | Official page retrieved

D03T06 Topological computation and anyons

Topological schemes encode and manipulate information using anyonic degrees of freedom. Mathematical universality and experimental identification of the required excitations are separate questions.

Fine subcategories: Braiding; fusion rules; non Abelian statistics; topological gates; Majorana proposals.

Primary catalog resources: 8. Related topic assignments are searchable in the HTML.

Q0148 Fault-tolerant quantum computation by anyons

A.Yu. Kitaev.

Journal article | 2003 | Annals of Physics | vol. 303 | no. 1 | pp. 2-30.

[DOI 10.1016/s0003-4916\(02\)00018-0](#) | [Source record](#) | Registered DOI metadata

Q0149 Fluctuating Charge Disproportionation in Doped $\text{La}_3\text{Ni}_2\text{O}_7$: Ambient-Pressure Fibonacci Anyons for Universal Topological Quantum Computation

Fanyuy Fai, Lukong Fai, Afungchui David.

Posted content | 2026 | Springer Science and Business Media LLC.

[DOI 10.21203/rs.3.rs-10108629/v1](#) | [Source record](#) | Registered DOI metadata

Q0150 Universal quantum computation using Ising anyons from a non-semisimple topological quantum field theory

Filippo Iulianelli, Sung Kim, Joshua Sussan, Aaron D. Lauda.

Journal article | 2025 | Nature Communications | vol. 16 | no. 1 | article 6408.

[DOI 10.1038/s41467-025-61342-8](#) | [Source record](#) | Registered DOI metadata

Q0151 Anyons and Topological Quantum Computation

Tudor D. Stanescu.

Book chapter | 2024 | Introduction to Topological Quantum Matter & Quantum Computation | pp. 365-392.

[DOI 10.1201/9781003226048-12](#) | [Source record](#) | Registered DOI metadata

Q0152 Compiling single-qubit braiding gate for Fibonacci anyons topological quantum computation

M T Rouabah, N E Belaloui, A Tounsi.

Journal article | 2021 | Journal of Physics: Conference Series | vol. 1766 | no. 1 | pp. 012029.

[DOI 10.1088/1742-6596/1766/1/012029](#) | [Source record](#) | Registered DOI metadata

Fine tags: Braiding

Q0153 Isolating Kondo anyons for topological quantum computation

Yashar Komijani.

Journal article | 2020 | Physical Review B | vol. 101 | no. 23 | article 235131.

[DOI 10.1103/physrevb.101.235131](#) | [Source record](#) | Registered DOI metadata

Q0154 Introduction to topological quantum computation with non-Abelian anyons

Bernard Field, Tapio Simula.

Journal article | 2018 | Quantum Science and Technology | vol. 3 | no. 4 | pp. 045004.

[DOI 10.1088/2058-9565/aacad2](https://doi.org/10.1088/2058-9565/aacad2) | [Source record](#) | Registered DOI metadata

Q0155 Topological Quantum Computation with Non-Abelian Anyons in Fractional Quantum Hall States

Lachezar S. Georgiev.

Book chapter | 2017 | Progress in Theoretical Chemistry and Physics | pp. 75-94.

[DOI 10.1007/978-3-319-50255-7_5](https://doi.org/10.1007/978-3-319-50255-7_5) | [Source record](#) | Registered DOI metadata

D04 Quantum complexity and limitations

Complexity theory identifies conditional speedups and lower bounds. It also explains why a problem being quantum does not automatically make it efficiently solvable.

Prerequisites: Algorithms, asymptotic analysis, and discrete mathematics.

Assessment focus: Oracle model, complexity assumptions, lower bounds, and input access.

D04T01 Quantum complexity classes

Complexity classes formalize efficiently solvable and verifiable quantum problems. A class inclusion or separation should be read together with its error and oracle model.

Fine subcategories: BQP; QMA; QCMA; interactive proofs; promise problems; oracle separations.

Primary catalog resources: 8. Related topic assignments are searchable in the HTML.

Q0156 Quantum Computing since Democritus

Scott Aaronson.

Book | 2013 | Cambridge University Press.

[DOI 10.1017/cbo9780511979309](https://doi.org/10.1017/cbo9780511979309) | [Source record](#) | Registered DOI metadata

Q0157 Quantum computational complexity

Daowen Qiu.

Book chapter | 2026 | Theoretical Foundations of Quantum Computing | pp. 207-237.

[DOI 10.1016/b978-0-44-327704-7.00012-8](https://doi.org/10.1016/b978-0-44-327704-7.00012-8) | [Source record](#) | Registered DOI metadata

Q0158 Comparing quantum complexity and quantum fidelity

Anonymous.

Journal article | 2025 | Physical Review B.

[DOI 10.1103/qccp-sgvb](https://doi.org/10.1103/qccp-sgvb) | [Source record](#) | Registered DOI metadata

Q0159 On quantum complexity

Mohsen Alishahiha.

Journal article | 2023 | Physics Letters B | vol. 842 | pp. 137979 | article 137979.

[DOI 10.1016/j.physletb.2023.137979](https://doi.org/10.1016/j.physletb.2023.137979) | [Source record](#) | Registered DOI metadata

Q0160 Quantum Complexity

Helmut Satz.

Book chapter | 2022 | More than the Sum of the Parts | pp. 127-135.

[DOI 10.1093/oso/9780192864178.003.0015](https://doi.org/10.1093/oso/9780192864178.003.0015) | [Source record](#) | Registered DOI metadata

Q0161 Quantum Complexity and Quantum Chaos

Author metadata not supplied.

Book chapter | 2022 | Quantum Mechanics | pp. 384-392.

[DOI 10.1017/9781108976299.040](https://doi.org/10.1017/9781108976299.040) | [Source record](#) | Registered DOI metadata

Q0162 Quantum Hamiltonian Complexity

Sevag Gharibian, Yichen Huang, Zeph Landau, Seung Woo Shin.

Journal article | 2015 | Foundations and Trends® in Theoretical Computer Science | vol. 10 | no. 3 | pp. 159-282.

[DOI 10.1561/04000000066](https://doi.org/10.1561/04000000066) | [Source record](#) | Registered DOI metadata

Q0163 Quantum Computational Complexity of the N -Representability Problem: QMA Complete

Yi-Kai Liu, Matthias Christandl, F. Verstraete.

Journal article | 2007 | Physical Review Letters | vol. 98 | no. 11 | article 110503.

[DOI 10.1103/physrevlett.98.110503](https://doi.org/10.1103/physrevlett.98.110503) | [Source record](#) | Registered DOI metadata

Fine tags: QMA

D04T02 Query complexity and lower bounds

Query complexity counts accesses to an input oracle. This isolates useful theoretical questions but does not include the physical cost of constructing that oracle.

Fine subcategories: Polynomial method; adversary method; oracle models; decision trees; bounded error queries.

Primary catalog resources: 8. Related topic assignments are searchable in the HTML.

Q0164 THEORETICAL FOUNDATIONS OF CLASSICAL AND QUANTUM QUERY MODELS: COMPLEXITY, LOWER BOUNDS, AND POLYNOMIAL-SPECTRAL INTERPLAY

Medak Srinivas.

Book chapter | 2026 | IIP Series | pp. 139-150.

[DOI 10.58532/nbennur3146c10](https://doi.org/10.58532/nbennur3146c10) | [Source record](#) | Registered DOI metadata

Q0165 Permutation Superposition Oracles for Quantum Query Lower Bounds

Christian Majenz, Giulio Malavolta, Michael Walter.

Conference paper | 2025 | Proceedings of the 57th Annual ACM Symposium on Theory of Computing | pp. 1508-1519.

[DOI 10.1145/3717823.3718266](https://doi.org/10.1145/3717823.3718266) | [Source record](#) | Registered DOI metadata

Q0166 Quantum Lower Bounds by Sample-to-Query Lifting

Qisheng Wang, Zhicheng Zhang.

Journal article | 2025 | SIAM Journal on Computing | vol. 54 | no. 5 | pp. 1294-1334.

[DOI 10.1137/24m1638616](https://doi.org/10.1137/24m1638616) | [Source record](#) | Registered DOI metadata

Q0167 Quantum Query Complexity for Nonstandard Oracles with Tight Bounds and Tradeoffs Beyond Grover

Yalla Jnan Devi Satya Prasad.

Posted content | 2025 | Wiley.

[DOI 10.22541/au.176616158.89278653/v1](https://doi.org/10.22541/au.176616158.89278653/v1) | [Source record](#) | Registered DOI metadata

Q0168 Explicit relation between all lower bound techniques for quantum query complexity

Loïck Magnin, Jérémie Roland.

Journal article | 2015 | International Journal of Quantum Information | vol. 13 | no. 04 | pp. 1350059.

[DOI 10.1142/s0219749913500597](https://doi.org/10.1142/s0219749913500597) | [Source record](#) | Registered DOI metadata

Q0169 Lower Bounds on Quantum Query Complexity for Read-Once Formulas with XOR and MUX Operators

Hideaki FUKUHARA, Eiji TAKIMOTO.

Journal article | 2010 | IEICE Transactions on Information and Systems | vol. E93-D | no. 2 | pp. 280-289.

[DOI 10.1587/transinf.e93.d.280](https://doi.org/10.1587/transinf.e93.d.280) | [Source record](#) | Registered DOI metadata

Q0170 Lower Bounds for Randomized and Quantum Query Complexity Using Kolmogorov Arguments

Sophie Laplante, Frédéric Magniez.

Journal article | 2008 | SIAM Journal on Computing | vol. 38 | no. 1 | pp. 46-62.

[DOI 10.1137/050639090](https://doi.org/10.1137/050639090) | [Source record](#) | Registered DOI metadata

Q0171 An Algorithmic Argument for Nonadaptive Query Complexity Lower Bounds on Advised Quantum Computation

Harumichi Nishimura, Tomoyuki Yamakami.

Book chapter | 2004 | Lecture Notes in Computer Science | pp. 827-838.

[DOI 10.1007/978-3-540-28629-5_65](#) | [Source record](#) | Registered DOI metadata

D04T03 Quantum communication complexity

Communication complexity asks how much information must be exchanged to solve a distributed task. Prior entanglement and classical side channels change the model.

Fine subcategories: One way communication; simultaneous messages; entanglement assisted protocols; lower bounds.

Primary catalog resources: 8. Related topic assignments are searchable in the HTML.

Q0172 Quantum Communication Complexity of Regularized Linear Regression Protocols

Sayaki Matsushita.

Journal article | 2026 | IEEE Transactions on Quantum Engineering | vol. 7 | pp. 3103012-3103012.

[DOI 10.1109/tqe.2026.3687237](#) | [Source record](#) | Registered DOI metadata

Q0173 Hybrid Quantum Cryptography from Communication Complexity

Francesco Mazzoncini, Balthazar Bauer, Peter Brown, Romain Alléaume.

Journal article | 2025 | Quantum | vol. 9 | pp. 1862 | article 1862.

[DOI 10.22331/q-2025-09-24-1862](#) | [Source record](#) | Registered DOI metadata

Q0174 Quantum Communication Complexity and Raz's Problem

Nipun Agarwal.

Posted content | 2024 | Institute of Electrical and Electronics Engineers (IEEE).

[DOI 10.36227/techrxiv.171779455.51487026/v1](#) | [Source record](#) | Registered DOI metadata

Q0175 Quantum Communication Complexity of Linear Regression

Ashley Montanaro, Changpeng Shao.

Journal article | 2024 | ACM Transactions on Computation Theory | vol. 16 | no. 1 | pp. 1-30.

[DOI 10.1145/3625225](#) | [Source record](#) | Registered DOI metadata

Q0176 Quantum Contextuality Provides Communication Complexity Advantage

Shashank Gupta, Debashis Saha, Zhen-Peng Xu, Adán Cabello, A. S. Majumdar.

Journal article | 2023 | Physical Review Letters | vol. 130 | no. 8 | article 080802.

[DOI 10.1103/physrevlett.130.080802](#) | [Source record](#) | Registered DOI metadata

Q0177 Bounds on Oblivious Multiparty Quantum Communication Complexity

François Le Gall, Daiki Suruga.

Book chapter | 2022 | Lecture Notes in Computer Science | pp. 641-657.

[DOI 10.1007/978-3-031-20624-5_39](#) | [Source record](#) | Registered DOI metadata

Q0178 Entanglement-based quantum communication complexity beyond Bell nonlocality

Joseph Ho, George Moreno, Samurá Brito, Francesco Graffitti, Christopher L. Morrison, Ranieri Nery, Alexander Pickston, Massimiliano Proietti et al..

Journal article | 2022 | npj Quantum Information | vol. 8 | no. 1 | article 13.

[DOI 10.1038/s41534-022-00520-8](#) | [Source record](#) | Registered DOI metadata

Q0179 Quantum versus Randomized Communication Complexity, with Efficient Players

Uma Girish, Ran Raz, Avishay Tal.

Journal article | 2022 | computational complexity | vol. 31 | no. 2 | article 17.

[DOI 10.1007/s00037-022-00232-7](#) | [Source record](#) | Registered DOI metadata

D04T04 Quantum advantage and sampling hardness

Sampling advantage compares specified quantum distributions with classical algorithms under stated assumptions. A sampling demonstration is not by itself evidence of useful application advantage.

Fine subcategories: Circuit sampling; complexity assumptions; classical baselines; verification gaps; asymptotic advantage.

Primary catalog resources: 8. Related topic assignments are searchable in the HTML.

Q0180 Quantum supremacy using a programmable superconducting processor

Frank Arute, Kunal Arya, Ryan Babbush, Dave Bacon, Joseph C. Bardin, Rami Barends, Rupak Biswas, Sergio Boixo et al..

Journal article | 2019 | Nature | vol. 574 | no. 7779 | pp. 505-510.

[DOI 10.1038/s41586-019-1666-5](#) | [Source record](#) | Registered DOI metadata

Q0181 Predicting sampling advantage of stochastic Ising machines for quantum simulations

Rutger J.L.F. Berns, Davi R. Rodrigues, Giovanni Finocchio, Johan H. Mentink.

Journal article | 2026 | Physical Review Applied | vol. 25 | no. 2 | article 024085.

[DOI 10.1103/xl6n-dlq6](#) | [Source record](#) | Registered DOI metadata

Q0182 Quantum energetic advantage before computational advantage in Boson Sampling

Anonymous.

Journal article | 2026 | Physical Review Letters.

[DOI 10.1103/8jyg-m72v](#) | [Source record](#) | Registered DOI metadata

Q0183 Unconditional Quantum Advantage for Sampling with Shallow Circuits

Adam Bene Watts, Natalie Parham.

Journal article | 2026 | Quantum | vol. 10 | pp. 2188 | article 2188.

[DOI 10.22331/q-2026-08-12-2188](#) | [Source record](#) | Registered DOI metadata

Fine tags: Circuit sampling

Q0184 Instantaneous Quantum Polynomial-Time Sampling and Verifiable Quantum Advantage: Stabilizer Scheme and Classical Security

Michael J. Bremner, Bin Cheng, Zhengfeng Ji.

Journal article | 2025 | PRX Quantum | vol. 6 | no. 2 | article 020315.

[DOI 10.1103/prxquantum.6.020315](#) | [Source record](#) | Registered DOI metadata

Q0185 Quantum Computational Advantage of Noisy Boson Sampling with Partially Distinguishable Photons

Byeongseon Go, Changhun Oh, Hyunseok Jeong.

Journal article | 2025 | PRX Quantum | vol. 6 | no. 3 | article 030362.

[DOI 10.1103/rflv-gc66](#) | [Source record](#) | Registered DOI metadata

Q0186 Quantum-Advantage Cryptography via Lemniscate-Orbit Sampling and Distributional Hardness

Ralph Vince.

Posted content | 2025 | Elsevier BV.

[DOI 10.2139/ssrn.5488706](#) | [Source record](#) | Registered DOI metadata

Q0187 Robust quantum computational advantage with programmable 3050-photon Gaussian boson sampling

Chao-Yang Lu, Hua-Liang Liu, Hao Su, Si-Qiu Gong, Yi-Chao Gu, Hao-Yang Tang, Meng-Hao Jia, Qian Wei et al..

Posted content | 2025 | Springer Science and Business Media LLC.

[DOI 10.21203/rs.3.rs-7435859/v1](#) | [Source record](#) | Registered DOI metadata

D04T05 Dequantization and classical limitations

Dequantization identifies classical algorithms that reproduce a claimed benefit under comparable access assumptions. Input preparation and low rank structure often determine the comparison.

Fine subcategories: Sample query access; low rank assumptions; classical emulation; input access costs; quantum inspired algorithms.

Primary catalog resources: 2. Related topic assignments are searchable in the HTML.

Q0188 Quantum Algorithms for Trading: A Survey of Speedups, Thresholds, and Dequantization

Luigi Laura, Marco Parrillo, Alessio Pascucci, Marco Rossi, Valerio Rughetti.

Journal article | 2026 | Information | vol. 17 | no. 9 | pp. 908.

[DOI 10.3390/info17090908](https://doi.org/10.3390/info17090908) | [Source record](#) | Registered DOI metadata

Q0189 Quantum-inspired career guidance without quantum advantage: A dequantization-aware hybrid architecture, critical benchmarking, and readiness assessment

Kondeti Vinay Kumar, Chandra Shekar N, Meeravali Shaik, Savanam Chandra Sekhar.

Journal article | 2026 | International Journal of Computing and Artificial Intelligence | vol. 7 | no. 8 | pp. 52-64.

[DOI 10.33545/27076571.2026.v7.i8a.371](https://doi.org/10.33545/27076571.2026.v7.i8a.371) | [Source record](#) | Registered DOI metadata

D05 Core quantum algorithms

Primitive algorithms form reusable components of larger workflows. Their complexity must include data preparation, oracle construction, accuracy, and measurement costs.

Prerequisites: Quantum circuits, probability, and eigenvalue problems.

Assessment focus: Oracle construction, precision, state overlap, and output readout.

D05T01 Quantum Fourier transform

The quantum Fourier transform changes between computational and Fourier bases. It is a primitive in phase estimation and order finding rather than a general replacement for classical FFT output.

Fine subcategories: Phase kickback; approximate transforms; semiclassical transforms; arithmetic circuits; Fourier sampling.

Primary catalog resources: 15. Related topic assignments are searchable in the HTML.

Q0190 A review on quantum Fourier transform

Tomás Barros, Pablo Álvarez, Bárbara Vidal, Mauricio Solar.

Journal article | 2026 | Quantum Information Processing | vol. 25 | no. 2 | article 41.

[DOI 10.1007/s11128-026-05067-7](https://doi.org/10.1007/s11128-026-05067-7) | [Source record](#) | Registered DOI metadata

Q0191 Algorithms Using Quantum Fourier Transform

Author metadata not supplied.

Book chapter | 2025 | Quantum Computing for Programmers | pp. 244-292.

[DOI 10.1017/9781009548519.012](https://doi.org/10.1017/9781009548519.012) | [Source record](#) | Registered DOI metadata

Q0192 Quantum Fourier Transform

Osama M. Raisuddin, Suvranu De.

Book chapter | 2025 | Quantum Computing for Engineers | pp. 155-158.

[DOI 10.1007/978-3-032-03325-3_19](https://doi.org/10.1007/978-3-032-03325-3_19) | [Source record](#) | Registered DOI metadata

Q0193 Quantum Fourier Transform

Cheng Hsiao Wu.

Book chapter | 2025 | Nonlocal Quantum Computing Theory | pp. 25-35.

[DOI 10.1201/9781003542032-2](https://doi.org/10.1201/9781003542032-2) | [Source record](#) | Registered DOI metadata

Q0194 Quantum Fourier Transform and Applications

Eduardo R. Mucciolo.

Book chapter | 2025 | Introduction to Quantum Information Processing | pp. 97-118.

[DOI 10.1201/9781003485124-8](https://doi.org/10.1201/9781003485124-8) | [Source record](#) | Registered DOI metadata

Q0195 Quantum Fourier Transform with Leakage Control

US Department of Energy, D. Kurkcuoglu, Fermi National Accelerator Laboratory (FNAL), Batavia, IL (United States), M. Alam, J. Job, A. Li, A. Macridin, G. Perdue et al..

Report | 2025 | Office of Scientific and Technical Information (OSTI).

[DOI 10.2172/3019156](https://doi.org/10.2172/3019156) | [Source record](#) | Registered DOI metadata

Q0196 Quantum Fourier transform

Author metadata not supplied.

Book chapter | 2025 | Quantum Algorithms | pp. 225-227.

[DOI 10.1017/9781009639651.015](https://doi.org/10.1017/9781009639651.015) | [Source record](#) | Registered DOI metadata

Q0197 Short-time quantum Fourier transform processing

Sreeraj Rajindran Nair, Benjamin Southwell, Christopher Ferrie.

Conference paper | 2025 | ICASSP 2025 - 2025 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP) | pp. 1-5.

[DOI 10.1109/icassp49660.2025.10890646](https://doi.org/10.1109/icassp49660.2025.10890646) | [Source record](#) | Registered DOI metadata

Q0198 Quantum Fourier Transform Using Dynamic Circuits

Elisa Bäumer, Vinay Tripathi, Alireza Seif, Daniel Lidar, Derek S. Wang.

Journal article | 2024 | Physical Review Letters | vol. 133 | no. 15 | article 150602.

[DOI 10.1103/physrevlett.133.150602](https://doi.org/10.1103/physrevlett.133.150602) | [Source record](#) | Registered DOI metadata

Q0199 Circuit of Quantum Fractional Fourier Transform

Tieyu Zhao, Yingying Chi.

Journal article | 2023 | Fractal and Fractional | vol. 7 | no. 10 | pp. 743.

[DOI 10.3390/fractalfract7100743](https://doi.org/10.3390/fractalfract7100743) | [Source record](#) | Registered DOI metadata

Q0200 Entanglement Parallelization via Quantum Fourier Transform

Mario Mastriani.

Journal article | 2023 | Advanced Quantum Technologies | vol. 6 | no. 10 | article 2300022.

[DOI 10.1002/qute.202300022](https://doi.org/10.1002/qute.202300022) | [Source record](#) | Registered DOI metadata

Q0201 Quantum Fourier Transform

Andreas Wichert.

Book chapter | 2023 | Quantum Artificial Intelligence with Qiskit | pp. 246-254.

[DOI 10.1201/9781003374404-19](https://doi.org/10.1201/9781003374404-19) | [Source record](#) | Registered DOI metadata

Q0202 Quantum Fourier Transform Has Small Entanglement

Jielun Chen, E.M. Stoudenmire, Steven R. White.

Journal article | 2023 | PRX Quantum | vol. 4 | no. 4 | article 040318.

[DOI 10.1103/prxquantum.4.040318](https://doi.org/10.1103/prxquantum.4.040318) | [Source record](#) | Registered DOI metadata

Q0203 Quantum Fourier Transform I

Hui Yung Wong.

Book chapter | 2023 | Introduction to Quantum Computing | pp. 243-253.

[DOI 10.1007/978-3-031-36985-8_25](https://doi.org/10.1007/978-3-031-36985-8_25) | [Source record](#) | Registered DOI metadata

Q0204 Quantum Fourier Transform II

Hui Yung Wong.

Book chapter | 2023 | Introduction to Quantum Computing | pp. 255-264.

[DOI 10.1007/978-3-031-36985-8_26](https://doi.org/10.1007/978-3-031-36985-8_26) | [Source record](#) | Registered DOI metadata

D05T02 Phase estimation

Phase estimation extracts eigenphase information from controlled dynamics. Precision, coherent evolution time, and state overlap control its cost and success probability.

Fine subcategories: Iterative estimation; Bayesian estimation; robust estimation; spectral resolution; controlled evolution.

Primary catalog resources: 14. Related topic assignments are searchable in the HTML.

Q0205 Filtered quantum phase estimation

Gwonhak Lee, Minhyeok Kang, Jungsoo Hong, Stepan Fomichev, Joonsuk Huh.

Journal article | 2026 | Quantum Science and Technology | vol. 11 | no. 4 | pp. 045053.

[DOI 10.1088/2058-9565/aea127](https://doi.org/10.1088/2058-9565/aea127) | [Source record](#) | Registered DOI metadata

Q0206 Measuring the non-abelian quantum phase with the algorithm of quantum phase estimation

Seng Ghee Tan, Son-Hsien Chen, Ying-Cheng Yang, Yen-Fu Chen, Yen-Lin Chen, Jia-Hsiu Hsieh.

Journal article | 2026 | Physics Letters A | vol. 593 | pp. 132019 | article 132019.

[DOI 10.1016/j.physleta.2026.132019](https://doi.org/10.1016/j.physleta.2026.132019) | [Source record](#) | Registered DOI metadata

Q0207 Optimal Coherent Quantum Phase Estimation [Slides]

USDOE National Nuclear Security Administration (NNSA), Yigit Subasi, USDOE Office of Science (SC), Advanced Scientific Computing Research (ASCR), USDOE Laboratory Directed Research and Development (LDRD) Program, Los Alamos National Laboratory (LANL), Los Alamos, NM (United States), National Quantum Information Science (QIS) Research Centers (United States). The Quantum Science Center (QSC), Andrew Sornborger.

Report | 2026 | Office of Scientific and Technical Information (OSTI).

[DOI 10.2172/3097329](https://doi.org/10.2172/3097329) | [Source record](#) | Registered DOI metadata

Q0208 Quantum Phase Estimation Without Controlled Unitaries

Laura Clinton, Toby S. Cubitt, Raul Garcia-Patron, Ashley Montanaro, Stasja Stanisic, Maarten Stroeck.

Journal article | 2026 | PRX Quantum | vol. 7 | no. 1 | article 010345.

[DOI 10.1103/7qcr-znl2](https://doi.org/10.1103/7qcr-znl2) | [Source record](#) | Registered DOI metadata

Q0209 Quantumness of quantum phase estimation algorithms

Ugo Marzolino.

Journal article | 2026 | Annals of Physics | vol. 493 | pp. 170599 | article 170599.

[DOI 10.1016/j.aop.2026.170599](https://doi.org/10.1016/j.aop.2026.170599) | [Source record](#) | Registered DOI metadata

Q0210 Agnostic phase estimation for quantum sensing and quantum computation

Flavio Salvati, Xingrui Song, Chandrashekhar Gaikwad, Nicole Y. Halpern, David R. M. Arvidsson-Shukur, Kater Murch.

Conference paper | 2025 | Quantum Sensing, Imaging, and Precision Metrology III | pp. 178.

[DOI 10.1117/12.3053679](https://doi.org/10.1117/12.3053679) | [Source record](#) | Registered DOI metadata

Q0211 Nonlinear Spectroscopy via Generalized Quantum Phase Estimation

Ignacio Loaiza, Danial Motlagh, Kasra Hejazi, Modjtaba Shokrian Zini, Alain Delgado, Juan Miguel Arrazola.

Journal article | 2025 | Quantum | vol. 9 | pp. 1822 | article 1822.

[DOI 10.22331/q-2025-08-07-1822](https://doi.org/10.22331/q-2025-08-07-1822) | [Source record](#) | Registered DOI metadata

Q0212 Quantum Phase Estimation

Osama M. Raisuddin, Suvranu De.

Book chapter | 2025 | Quantum Computing for Engineers | pp. 159-163.

[DOI 10.1007/978-3-032-03325-3_20](https://doi.org/10.1007/978-3-032-03325-3_20) | [Source record](#) | Registered DOI metadata

Q0213 Quantum phase estimation

Author metadata not supplied.

Book chapter | 2025 | Quantum Algorithms | pp. 228-234.

[DOI 10.1017/9781009639651.016](https://doi.org/10.1017/9781009639651.016) | [Source record](#) | Registered DOI metadata

Q0214 Quantum phase estimation-enabled phase tracking for coexisting quantum/classical metropolitan networks

Jabir Marakkarakath Vadakkepurayil, Daehyun H. Ahn, Nur Fajar R. Annafianto, Ivan Burenkov, Abdella Battou, Sergey Polyakov.

Conference paper | 2025 | Quantum Sensing, Imaging, and Precision Metrology III | pp. 144.

[DOI 10.1117/12.3043163](https://doi.org/10.1117/12.3043163) | [Source record](#) | Registered DOI metadata

Q0215 Special Functions in Quantum Phase Estimation

Masahito Hayashi.

Book chapter | 2025 | CRM Series in Mathematical Physics | pp. 81-92.

[DOI 10.1007/978-3-031-90135-5_5](https://doi.org/10.1007/978-3-031-90135-5_5) | [Source record](#) | Registered DOI metadata

Q0216 A differentiable quantum phase estimation algorithm

Davide Castaldo, Soran Jahangiri, Agostino Migliore, Juan Miguel Arrazola, Stefano Corni.

Journal article | 2024 | Quantum Science and Technology | vol. 9 | no. 4 | pp. 045026.

[DOI 10.1088/2058-9565/ad69bc](#) | [Source record](#) | Registered DOI metadata

Q0217 Learning Quantum Phase Estimation by Variational Quantum Circuits

Chen-Yu Liu, Kuan-Cheng Chen, Chu-Hsuan Abraham Lin.

Conference paper | 2024 | 2024 International Joint Conference on Neural Networks (IJCNN) | pp. 1-6.

[DOI 10.1109/ijcnn60899.2024.10651206](#) | [Source record](#) | Registered DOI metadata

Q0218 Reductive quantum phase estimation

Nicholas J. C. Papadopoulos, Jarrod T. Reilly, John Drew Wilson, Murray J. Holland.

Journal article | 2024 | Physical Review Research | vol. 6 | no. 3 | article 033051.

[DOI 10.1103/physrevresearch.6.033051](#) | [Source record](#) | Registered DOI metadata

D05T03 Amplitude amplification and search

Amplitude amplification raises the probability of a marked outcome. Search speedups are measured in oracle queries and must include the implementation of the marking operation.

Fine subcategories: Grover search; fixed point amplification; search lower bounds; unknown success probability.

Primary catalog resources: 8. Related topic assignments are searchable in the HTML.

Q0219 A fast quantum mechanical algorithm for database search

Lov K. Grover.

Conference paper | 1996 | Proceedings of the twenty-eighth annual ACM symposium on Theory of computing - STOC '96 | pp. 212-219.

[DOI 10.1145/237814.237866](#) | [Source record](#) | Registered DOI metadata

Q0220 Distributed exact quantum amplitude amplification algorithm for arbitrary quantum states

Xu Zhou, Wenxuan Tao, Keren Li, Shenggen Zheng.

Journal article | 2026 | Science China Information Sciences | vol. 69 | no. 8 | article 180506.

[DOI 10.1007/s11432-026-5020-7](#) | [Source record](#) | Registered DOI metadata

Q0221 QUANTUM SEARCH DYNAMICS: A COMPREHENSIVE ANALYSIS OF GROVER'S ALGORITHM AND AMPLITUDE AMPLIFICATION

Sripada Saipriya, Dr. Mahender Veshala.

Book chapter | 2026 | IIP Series | pp. 126-138.

[DOI 10.58532/nbennur3146c9](#) | [Source record](#) | Registered DOI metadata

Fine tags: Grover search

Q0222 The Single-Rotation Quantum Search Algorithm: Amplitude Amplification with One Oracle Call

Ying Liu.

Journal article | 2026 | Journal of Quantum Information Science | vol. 16 | no. 02 | pp. 161-190.

[DOI 10.4236/jqis.2026.162006](#) | [Source record](#) | Registered DOI metadata

Q0223 Limitations of quantum counting, nested quantum search and amplitude amplification and their potential to solve discrete optimization problems

Stefan Creemers, Luis Fernando Pérez Armas.

Journal article | 2025 | Philosophical Transactions of the Royal Society A Mathematical Physical and Engineering Sciences | vol. 383 | no. 2310 | article 20240561.

[DOI 10.1098/rsta.2024.0561](#) | [Source record](#) | Registered DOI metadata

Q0224 Optimized Amplitude Amplification for Quantum State Preparation

A. A. Chernikov, K. R. Zakharova, S. S. Sysoev.

Journal article | 2025 | Lobachevskii Journal of Mathematics | vol. 46 | no. 7 | pp. 3522-3526.

[DOI 10.1134/s1995080225606666](#) | [Source record](#) | Registered DOI metadata

Q0225 Quantum amplitude amplification operators: exact quantum search

Hyeokjea Kwon, Joonwoo Bae.

Conference paper | 2022 | Quantum Communications and Quantum Imaging XX | pp. 11.

[DOI 10.1117/12.2626449](#) | [Source record](#) | Registered DOI metadata

Q0226 Search by Quantum Walks on Two-Dimensional Grid without Amplitude Amplification

Andris Ambainis, Artūrs Bačkurs, Nikolajs Nahimovs, Raitis Ozols, Alexander Rivosh.

Book chapter | 2013 | Lecture Notes in Computer Science | pp. 87-97.

[DOI 10.1007/978-3-642-35656-8_7](#) | [Source record](#) | Registered DOI metadata

D05T04 Amplitude estimation

Amplitude estimation estimates a probability or expectation using structured quantum access. Near term variants trade coherent depth for sampling and inference.

Fine subcategories: Monte Carlo estimation; maximum likelihood estimation; iterative methods; confidence intervals.

Primary catalog resources: 16. Related topic assignments are searchable in the HTML.

Q0227 Component-wise Validation of Amplitude Encoding within Quantum Amplitude Estimation

Farhat Zargoun, Marwa Swidan.

Journal article | 2026 | AlQalam Journal of Medical and Applied Sciences.

[DOI 10.54361/ajmas.269746](#) | [Source record](#) | Registered DOI metadata

Q0228 Entropic optimal transport with quantum amplitude estimation

Francisco Orts.

Journal article | 2026 | Future Generation Computer Systems | vol. 183 | pp. 108570 | article 108570.

[DOI 10.1016/j.future.2026.108570](#) | [Source record](#) | Registered DOI metadata

Q0229 Hardware-Efficient Quantum Grid Sampling for Low-Latency Amplitude Estimation

Yu-Ting Kao, Hao-Yu Lu, Yeong-Jar Chang, Jason Gemsun Young, Chao-Hung Wang, Darsen D. Lu.

Conference paper | 2026 | 2026 IEEE 19th Dallas Circuits and Systems Conference (DCAS) | pp. 1-5.

[DOI 10.1109/dcas69364.2026.11544432](#) | [Source record](#) | Registered DOI metadata

Q0230 Harnessing Bayesian Statistics to Accelerate Iterative Quantum Amplitude Estimation

Qilin Li, Atharva Vidwans, Yazhen Wang, Micheline B. Soley.

Journal article | 2026 | Quantum | vol. 10 | pp. 1962 | article 1962.

[DOI 10.22331/q-2026-01-14-1962](#) | [Source record](#) | Registered DOI metadata

Q0231 Quantum Amplitude Estimation in Practice: A Case Study in Option Pricing

Nouhaila Innan, Muhammad Kashif, Alberto Marchisio, Muhammad Moonis Usman, Muhammad Shafique.

Book chapter | 2026 | Communications in Computer and Information Science | pp. 348-358.

[DOI 10.1007/978-3-032-13852-1_35](#) | [Source record](#) | Registered DOI metadata

Q0232 An efficient quantum computing based structural reliability analysis method using quantum amplitude estimation

Jingran He.

Journal article | 2025 | Structural Safety | vol. 114 | pp. 102555 | article 102555.

[DOI 10.1016/j.strusafe.2024.102555](#) | [Source record](#) | Registered DOI metadata

Q0233 Bayesian Quantum Amplitude Estimation

Alexandra Ramôa, Luis Paulo Santos.

Journal article | 2025 | Quantum | vol. 9 | pp. 1856 | article 1856.

[DOI 10.22331/q-2025-09-11-1856](#) | [Source record](#) | Registered DOI metadata

Q0234 Classical postprocessing approach for quantum amplitude estimation

Yongdan Yang, Ruyu Yang.

Journal article | 2025 | Physical Review A | vol. 112 | no. 1 | article 012625.

[DOI 10.1103/z9hk-f37h](#) | [Source record](#) | Registered DOI metadata

Q0235 Evaluating Quantum Amplitude Estimation for Pricing Multi-Asset Basket Options

Muhammad Kashif, Shaf Khalid, Nouhaila Innan, Alberto Marchisio, Muhammad Shafique.

Conference paper | 2025 | 2025 IEEE International Conference on Quantum Artificial Intelligence (QAI) | pp. 451-458.

[DOI 10.1109/qai63978.2025.00076](#) | [Source record](#) | Registered DOI metadata

Q0236 Quantum phase estimation and quantum amplitude estimation

Steven Herbert.

Book chapter | 2025 | Quantum Computing | pp. 155-175.

[DOI 10.1093/9780191964381.003.0009](#) | [Source record](#) | Registered DOI metadata

Q0237 Adaptive measurement strategy for noisy quantum amplitude estimation with variational quantum circuits

Kohei Oshio, Yohichi Suzuki, Kaito Wada, Keigo Hisanaga, Shumpei Uno, Naoki Yamamoto.

Journal article | 2024 | Physical Review A | vol. 110 | no. 6 | article 062423.

[DOI 10.1103/physreva.110.062423](#) | [Source record](#) | Registered DOI metadata

Q0238 An Efficient Quantum Computing Based Reliability Analysis Method Using Quantum Amplitude Estimation

Jingran He.

Posted content | 2024 | Elsevier BV.

[DOI 10.2139/ssrn.4870890](#) | [Source record](#) | Registered DOI metadata

Q0239 Carbon market risk estimation using quantum conditional generative adversarial network and amplitude estimation

Xiyuan Zhou, Huan Zhao, Yuji Cao, Xiang Fei, Gaoqi Liang, Junhua Zhao.

Journal article | 2024 | Energy Conversion and Economics | vol. 5 | no. 4 | pp. 193-210.

[DOI 10.1049/enc2.12122](#) | [Source record](#) | Registered DOI metadata

Q0240 Noise-Aware Quantum Amplitude Estimation

Steven Herbert, Ifan Williams, Roland Guichard, Darren Ng.

Journal article | 2024 | IEEE Transactions on Quantum Engineering | vol. 5 | pp. 1-23.

[DOI 10.1109/tqe.2024.3476929](#) | [Source record](#) | Registered DOI metadata

Q0241 On the bias in iterative quantum amplitude estimation

Koichi Miyamoto.

Journal article | 2024 | EPJ Quantum Technology | vol. 11 | no. 1 | article 42.

[DOI 10.1140/epjqt/s40507-024-00253-x](#) | [Source record](#) | Registered DOI metadata

Q0242 Quantum multi-programming for maximum likelihood amplitude estimation

Pooja Rao, Sua Choi, Kwangmin Yu.

Conference paper | 2024 | Quantum Computing, Communication, and Simulation IV | pp. 33.

[DOI 10.1117/12.3002854](#) | [Source record](#) | Registered DOI metadata

Fine tags: maximum likelihood estimation

D05T05 Quantum walks

Quantum walks use interference in graph or configuration space to design algorithms. Walk dynamics, graph access, and hitting or detection tasks need separate definitions.

Fine subcategories: Discrete time walks; continuous time walks; hitting times; graph search; spatial search.

Primary catalog resources: 11. Related topic assignments are searchable in the HTML.

Q0243 Boomerang quantum walks

A. R. C. Buarque, W. S. Dias, Ernesto P. Raposo.

Journal article | 2026 | Quantum Information Processing | vol. 25 | no. 6 | article 199.

[DOI 10.1007/s11128-026-05222-0](#) | [Source record](#) | Registered DOI metadata

Q0244 Quantum Computing and Quantum Walks

Fei Yan, Wen Liang, Fangyan Dong.

Book chapter | 2025 | Exploring Complex Networks with Quantum Walks | pp. 3-32.

[DOI 10.1201/9781003683902-2](#) | [Source record](#) | Registered DOI metadata

Q0245 Quantum Walks

Peng Xue.

Book chapter | 2025 | Reference Module in Materials Science and Materials Engineering.

[DOI 10.1016/b978-0-443-33965-3.00003-1](#) | [Source record](#) | Registered DOI metadata

Q0246 Asynchronous Quantum Random Walks

Manfred Harringer.

Posted content | 2024 | Springer Science and Business Media LLC.

[DOI 10.21203/rs.3.rs-3372215/v5](#) | [Source record](#) | Registered DOI metadata

Q0247 A Crossover Between Open Quantum Random Walks to Quantum Walks

Norio Konno, Kaname Matsue, Etsuo Segawa.

Journal article | 2023 | Journal of Statistical Physics | vol. 190 | no. 12 | article 202.

[DOI 10.1007/s10955-023-03211-6](#) | [Source record](#) | Registered DOI metadata

Q0248 Irrational Quantum Walks

Gabriel Coutinho, Pedro Ferreira Baptista, Chris Godsil, Thomás Jung Spier, Reinhard Werner.

Journal article | 2023 | SIAM Journal on Applied Algebra and Geometry | vol. 7 | no. 3 | pp. 567-584.

[DOI 10.1137/22m1521262](#) | [Source record](#) | Registered DOI metadata

Q0249 Multidimensional Quantum Walks

Stacey Jeffery, Sebastian Zur.

Conference paper | 2023 | Proceedings of the 55th Annual ACM Symposium on Theory of Computing | pp. 1125-1130.

[DOI 10.1145/3564246.3585158](#) | [Source record](#) | Registered DOI metadata

Q0250 Quantum walks

Steven Duplij, Raimund Vogl.

Book chapter | 2023 | Innovative Quantum Computing | pp. 6-1-6-29.

[DOI 10.1088/978-0-7503-5281-9ch6](#) | [Source record](#) | Registered DOI metadata

Q0251 Geodesic quantum walks

Giuseppe Di Molfetta, Victor Deng.

Journal article | 2022 | Physical Review A | vol. 105 | no. 6 | article 062420.

[DOI 10.1103/physreva.105.062420](#) | [Source record](#) | Registered DOI metadata

Q0252 On Quantum Algorithms for Random Walks in the Nonnegative Quarter Plane

Vasileios Kalantzis, Mark S. Squillante, Shashanka Ubaru, Lior Horesh.

Journal article | 2022 | ACM SIGMETRICS Performance Evaluation Review | vol. 50 | no. 2 | pp. 42-44.

[DOI 10.1145/3561074.3561089](#) | [Source record](#) | Registered DOI metadata

Q0253 QUANTUM AI FRAMEWORKS: ENHANCING MACHINE LEARNING AND OPTIMIZATION THROUGH QUANTUM NEURAL NETWORKS, VARIATIONAL ALGORITHMS, AND QUANTUM WALKS

Narayana Gaddam.

Journal article | 2022 | INTERNATIONAL JOURNAL OF ADVANCED RESEARCH IN ENGINEERING AND TECHNOLOGY | vol. 13 | no. 6 | pp. 59-72.

[DOI 10.34218/ijaret_13_06_006](#) | [Source record](#) | Registered DOI metadata

D05T06 Hidden subgroup and factoring algorithms

Order finding underlies Shor factoring and discrete logarithm algorithms. Hidden subgroup methods generalize the structure, but non abelian cases have additional obstacles.

Fine subcategories: Shor algorithm; order finding; discrete logarithms; abelian subgroups; non abelian challenges.

Primary catalog resources: 8. Related topic assignments are searchable in the HTML.

Q0254 Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer

Peter W. Shor.

Journal article | 1997 | SIAM Journal on Computing | vol. 26 | no. 5 | pp. 1484-1509.

[DOI 10.1137/s0097539795293172](#) | [Source record](#) | Registered DOI metadata

Q0255 Shor Factoring Algorithm and the Quantum Fourier Transform

Alice Flarend, Robert Hilborn.

Book chapter | 2026 | Quantum Computing and Quantum Physics | pp. 272-293.

[DOI 10.1093/oso/9780197904930.003.0015](#) | [Source record](#) | Registered DOI metadata

Fine tags: Shor algorithm

Q0256 Contributions of Quantum Factoring on Quantum Research

Clayton Ferner.

Journal article | 2022 | Computer | vol. 55 | no. 8 | pp. 5-7.

[DOI 10.1109/mc.2022.3178307](#) | [Source record](#) | Registered DOI metadata

Q0257 Factoring Using Quantum Computers

Kristian Gjølsteen.

Book chapter | 2022 | Practical Mathematical Cryptography | pp. 137-146.

[DOI 10.1201/9781003149422-5](#) | [Source record](#) | Registered DOI metadata

Q0258 Variational Quantum Factoring

Eric Anschuetz, Jonathan Olson, Alán Aspuru-Guzik, Yudong Cao.

Book chapter | 2019 | Lecture Notes in Computer Science | pp. 74-85.

[DOI 10.1007/978-3-030-14082-3_7](#) | [Source record](#) | Registered DOI metadata

Q0259 Quantum Algorithm for Factoring

Sean Hallgren.

Book chapter | 2016 | Encyclopedia of Algorithms | pp. 1651-1652.

[DOI 10.1007/978-1-4939-2864-4_307](#) | [Source record](#) | Registered DOI metadata

AQUIRE

Q0260 Quantum Computation | Mathematics | MIT OpenCourseWare

Author metadata not supplied.

Course | 2003 | MIT OpenCourseWare.

[Official resource](#) | [Source record](#) | Official page retrieved

Q0261 Quantum factoring, discrete logarithms, and the hidden subgroup problem

R. Jozsa.

Journal article | 2001 | Computing in Science & Engineering | vol. 3 | no. 2 | pp. 34-43.

[DOI 10.1109/5992.909000](#) | [Source record](#) | Registered DOI metadata

Fine tags: discrete logarithms

D06 Advanced algorithm design

Modern algorithm frameworks organize linear algebra, simulation, and optimization around carefully controlled access models. Resource estimates depend strongly on how operators are encoded.

Prerequisites: Linear algebra, core quantum algorithms, and approximation theory.

Assessment focus: Block encoding normalization, conditioning, polynomial degree, and total cost.

D06T01 Block encoding and qubitization

Block encodings embed an operator into a larger unitary, and qubitization builds controlled spectral transformations. Normalization and oracle construction are central resource assumptions.

Fine subcategories: Signal oracles; normalization factors; quantum walks; Hamiltonian access; projected unitary encodings.

Primary catalog resources: 8. Related topic assignments are searchable in the HTML.

Q0262 Quantum Möbius Transform via Block Encoding

Cezary Pilaszewicz.

Conference paper | 2026 | 2026 IEEE Conference on Artificial Intelligence (CAI) | pp. 1968-1972.

[DOI 10.1109/cai68641.2026.11536229](https://doi.org/10.1109/cai68641.2026.11536229) | [Source record](#) | Registered DOI metadata

Q0263 Quantum block encoding for one-pair semiseparable matrices

Giacomo Antonioli, Paola Boito, Gianna Maria Del Corso, Margherita Porcelli.

Journal article | 2026 | Numerical Algorithms.

[DOI 10.1007/s11075-026-02479-5](https://doi.org/10.1007/s11075-026-02479-5) | [Source record](#) | Registered DOI metadata

Q0264 Quantum simulation of wave optics in weakly inhomogeneous media using block-encoding

Anonymous.

Journal article | 2026 | Physical Review Research.

[DOI 10.1103/snzq-fp39](https://doi.org/10.1103/snzq-fp39) | [Source record](#) | Registered DOI metadata

Q0265 An efficient quantum circuit for block encoding a pairing Hamiltonian

Diyi Liu, Weijie Du, Lin Lin, James P. Vary, Chao Yang.

Journal article | 2025 | Journal of Computational Science | vol. 85 | pp. 102480 | article 102480.

[DOI 10.1016/j.jocs.2024.102480](https://doi.org/10.1016/j.jocs.2024.102480) | [Source record](#) | Registered DOI metadata

Q0266 Block encoding bosons by signal processing

Christopher F. Kane, Siddharth Hariprakash, Neel S. Modi, Michael Kreshchuk, Christian W Bauer.

Journal article | 2025 | Quantum | vol. 9 | pp. 1747 | article 1747.

[DOI 10.22331/q-2025-05-15-1747](https://doi.org/10.22331/q-2025-05-15-1747) | [Source record](#) | Registered DOI metadata

Q0267 Classical optimization with imaginary-time block encoding on quantum computers: The MaxCut problem

Dawei Zhong, Akhil Francis, Ermal Rrapaj.

Journal article | 2025 | Physical Review A | vol. 112 | no. 4 | article 042420.

[DOI 10.1103/physreva.112.042420](https://doi.org/10.1103/physreva.112.042420) | [Source record](#) | Registered DOI metadata

Q0268 Exact block encoding of imaginary time evolution with universal quantum neural networks

Ermal Rrapaj, Evan Rule.

Journal article | 2025 | Physical Review Research | vol. 7 | no. 1 | article 013306.

[DOI 10.1103/physrevresearch.7.013306](https://doi.org/10.1103/physrevresearch.7.013306) | [Source record](#) | Registered DOI metadata

Q0269 GitHub - quantumlib/Qualtran: QUALTRAN is a Python library for expressing and analyzing Fault Tolerant Quantum algorithms. · GitHub

Author metadata not supplied.

Software repository | Undated | Google Quantum AI Qualtran.

[Official resource](#) | [Source record](#) | Official page retrieved

D06T02 Quantum signal processing

Quantum signal processing implements constrained polynomial transformations through phase sequences. Polynomial degree, phase synthesis, and approximation error determine algorithm costs.

Fine subcategories: Polynomial transformations; phase factor synthesis; parity constraints; approximation error.

Primary catalog resources: 16. Related topic assignments are searchable in the HTML.

Q0270 An adversary bound for quantum signal processing

Lorenzo Laneve.

Journal article | 2026 | Quantum | vol. 10 | pp. 2025 | article 2025.

[DOI 10.22331/q-2026-03-13-2025](#) | [Source record](#) | Registered DOI metadata

Q0271 Fast Phase Factor Finding for Quantum Signal Processing

Hongkang Ni, Lexing Ying.

Journal article | 2026 | SIAM Journal on Scientific Computing | vol. 48 | no. 4 | pp. B651-B670.

[DOI 10.1137/24m1705214](#) | [Source record](#) | Registered DOI metadata

Q0272 Mathematical and Numerical Analysis of Quantum Signal Processing

Lin Lin.

Conference paper | 2026 | Proceedings of the International Congress of Mathematicians 2026 - Volume 7: Invited Lectures (Sections 15–20) | pp. 57-74.

[DOI 10.1137/25m1806120](#) | [Source record](#) | Registered DOI metadata

Q0273 Arbeitsgemeinschaft: Quantum Signal Processing and Nonlinear Fourier Analysis

András Gilyén, Lin Lin, Christoph Thiele.

Journal article | 2025 | Oberwolfach Reports | vol. 21 | no. 4 | pp. 2671-2750.

[DOI 10.4171/owr/2024/46](#) | [Source record](#) | Registered DOI metadata

Q0274 Complementary Polynomials in Quantum Signal Processing

Bjorn K. Berntson, Christoph Sünderhauf.

Journal article | 2025 | Communications in Mathematical Physics | vol. 406 | no. 7 | article 161.

[DOI 10.1007/s00220-025-05302-9](#) | [Source record](#) | Registered DOI metadata

Q0275 Modular quantum signal processing in many variables

Zane M. Rossi, Jack L. Ceroni, Isaac L. Chuang.

Journal article | 2025 | Quantum | vol. 9 | pp. 1776 | article 1776.

[DOI 10.22331/q-2025-06-18-1776](#) | [Source record](#) | Registered DOI metadata

Q0276 Parallel Quantum Signal Processing Via Polynomial Factorization

John M. Martyn, Zane M. Rossi, Kevin Z. Cheng, Yuan Liu, Isaac L. Chuang.

Journal article | 2025 | Quantum | vol. 9 | pp. 1834 | article 1834.

[DOI 10.22331/q-2025-08-27-1834](#) | [Source record](#) | Registered DOI metadata

Q0277 Private Quantum Signal Processing

Akinori KAWACHI, Yuto NISHIKUBO.

Journal article | 2025 | IEICE Transactions on Fundamentals of Electronics, Communications and Computer Sciences | vol. E108.A | no. 9 | pp. 1105-1113 | article 2024DMP0017.

[DOI 10.1587/transfun.2024dmp0017](https://doi.org/10.1587/transfun.2024dmp0017) | [Source record](#) | Registered DOI metadata

Q0278 Prospects of quantum error mitigation for quantum signal processing

Ugnė Liaubaitė, S E Skelton.

Journal article | 2025 | Physica Scripta | vol. 100 | no. 12 | pp. 125119.

[DOI 10.1088/1402-4896/ae216a](https://doi.org/10.1088/1402-4896/ae216a) | [Source record](#) | Registered DOI metadata

Q0279 Quantum Machine Learning for Enhancing Signal Processing Applications

Kamineni Sairam, Shashikant Deepak, Rekha Chakravarthi, Saumendra Ku. Mohanty, P.S. Raghavendra Rao, Varsha Choudhary, Ankit Punia.

Journal article | 2025 | International Journal of Engineering, Science and Information Technology | vol. 5 | no. 2 | pp. 508-512.

[DOI 10.52088/ijesty.v5i2.1375](https://doi.org/10.52088/ijesty.v5i2.1375) | [Source record](#) | Registered DOI metadata

Q0280 Quantum signal processing and the quantum singular value transformation

Steven Herbert.

Book chapter | 2025 | Quantum Computing | pp. 253-276.

[DOI 10.1093/9780191964381.003.0014](https://doi.org/10.1093/9780191964381.003.0014) | [Source record](#) | Registered DOI metadata

Q0281 Qubitization and Quantum Signal Processing

Osama M. Raisuddin, Suvranu De.

Book chapter | 2025 | Quantum Computing for Engineers | pp. 179-183.

[DOI 10.1007/978-3-032-03325-3_23](https://doi.org/10.1007/978-3-032-03325-3_23) | [Source record](#) | Registered DOI metadata

Q0282 Statistical Signal Processing for Quantum Error Mitigation

Kausthubh Chandramouli, Kelly Mae Allen, Christopher Mori, Dror Baron, Mário A. T. Figueiredo.

Conference paper | 2025 | 2025 IEEE International Conference on Quantum Computing and Engineering (QCE) | pp. 269-275.

[DOI 10.1109/qce65121.2025.00038](https://doi.org/10.1109/qce65121.2025.00038) | [Source record](#) | Registered DOI metadata

Q0283 Derivative Pricing using Quantum Signal Processing

Nikitas Stamatopoulos, William J. Zeng.

Journal article | 2024 | Quantum | vol. 8 | pp. 1322 | article 1322.

[DOI 10.22331/q-2024-04-30-1322](https://doi.org/10.22331/q-2024-04-30-1322) | [Source record](#) | Registered DOI metadata

Q0284 Generalized Quantum Signal Processing

Danial Motlagh, Nathan Wiebe.

Journal article | 2024 | PRX Quantum | vol. 5 | no. 2 | article 020368.

[DOI 10.1103/prxquantum.5.020368](https://doi.org/10.1103/prxquantum.5.020368) | [Source record](#) | Registered DOI metadata

Q0285 Infinite quantum signal processing

Yulong Dong, Lin Lin, Hongkang Ni, Jiasu Wang.

Journal article | 2024 | Quantum | vol. 8 | pp. 1558 | article 1558.

[DOI 10.22331/q-2024-12-10-1558](https://doi.org/10.22331/q-2024-12-10-1558) | [Source record](#) | Registered DOI metadata

D06T03 Quantum singular value transformation

Singular value transformation applies polynomial functions to the singular values of a block encoded matrix. It unifies many algorithms while making access assumptions explicit.

Fine subcategories: Matrix functions; singular value thresholds; pseudoinverses; spectral filtering; polynomial degree.

Primary catalog resources: 18. Related topic assignments are searchable in the HTML.

Q0286 Quantum singular value transformation and beyond: exponential improvements for quantum matrix arithmetics

András Gilyén, Yuan Su, Guang Hao Low, Nathan Wiebe.

Conference paper | 2019 | Proceedings of the 51st Annual ACM SIGACT Symposium on Theory of Computing | pp. 193-204.

[DOI 10.1145/3313276.3316366](#) | [Source record](#) | Registered DOI metadata

Q0287 Avoiding Singular Value Thresholding in Quantum Low-Rank Recovery

Angshul Majumdar.

Journal article | 2026 | IEEE Signal Processing Letters | vol. 33 | pp. 2820-2824.

[DOI 10.1109/lsp.2026.3710103](#) | [Source record](#) | Registered DOI metadata

Q0288 Improved Performance of Quantum Search via Quantum Singular Value Transformation

Zheng Zhang, Minzhong Luo.

Conference paper | 2026 | Proceedings of the 2026 7th International Conference on Computer Information and Big Data Applications | pp. 240-244.

[DOI 10.1145/3813822.3813860](#) | [Source record](#) | Registered DOI metadata

Q0289 SIGNAL DENOISING BASED ON QUANTUM ADAPTIVE TRANSFORMATION AND SINGULAR VALUE DECOMPOSITION

YingYing Sun, MengYao Lv, ChenLin Wu, WeiXian Xie, ZiYuan Fang, RuiTong Zhao.

Journal article | 2026 | Journal of Computer Science and Electrical Engineering | vol. 8 | no. 3 | pp. 27-33.

[DOI 10.61784/jcsee3134](#) | [Source record](#) | Registered DOI metadata

Q0290 Simple Harmonic Motion Analysis of Elastic Structures Using Hamiltonian Simulation via Quantum Singular Value Transformation

Koya Wagatsuma, Katsuhiko Endo, Kenjiro Terada.

Journal article | 2026 | International Journal for Numerical Methods in Engineering | vol. 127 | no. 16 | article e70407.

[DOI 10.1002/nme.70407](#) | [Source record](#) | Registered DOI metadata

Q0291 Singular value transformation for unknown quantum channels

Ryotaro Niwa, Zane Marius Rossi, Philip Taranto, Mio Murao.

Journal article | 2026 | Physical Review A | vol. 114 | no. 3 | article L030401.

[DOI 10.1103/321q-k5dh](#) | [Source record](#) | Registered DOI metadata

Q0292 Correction: Robust dequantization of the quantum singular value transformation and quantum machine learning algorithms

François Le Gall.

Journal article | 2025 | computational complexity | vol. 34 | no. 1 | article 5.

[DOI 10.1007/s00037-025-00265-8](#) | [Source record](#) | Registered DOI metadata

Q0293 Implementing Credit Risk Analysis with Quantum Singular Value Transformation

Davide Veronelli, Francesca Cibrario, Emanuele Dri, Valeria Zaffaroni, Giacomo Ranieri, Davide Corbellotto, Bartolomeo Montrucchio.

Conference paper | 2025 | 2025 IEEE International Conference on Quantum Computing and Engineering (QCE) | pp. 2228-2237.

[DOI 10.1109/qce65121.2025.00243](#) | [Source record](#) | Registered DOI metadata

Q0294 Improved quantum power method and numerical integration using a quantum singular-value transformation

Nhat A. Nghiem, Hiroki Sukeno, Shuyu Zhang, Tzu-Chieh Wei.

Journal article | 2025 | Physical Review A | vol. 111 | no. 1 | article 012434.

[DOI 10.1103/physreva.111.012434](#) | [Source record](#) | Registered DOI metadata

Q0295 Matrix Encoding Method in Variational Quantum Singular Value Decomposition

Alexander I. Zenchuk, Wentao Qi, Junde Wu.

Journal article | 2025 | Quantum Information & Computation | vol. 25 | no. 4 | pp. 356-368.

[DOI 10.2478/qic-2025-0020](#) | [Source record](#) | Registered DOI metadata

Q0296 Robust Dequantization of the Quantum Singular Value Transformation and Quantum Machine Learning Algorithms

François Le Gall.

Journal article | 2025 | computational complexity | vol. 34 | no. 1 | article 2.

[DOI 10.1007/s00037-024-00262-3](#) | [Source record](#) | Registered DOI metadata

Q0297 A CS guide to the quantum singular value transformation

Ewin Tang, Kevin Tian.

Book chapter | 2024 | 2024 Symposium on Simplicity in Algorithms (SOSA) | pp. 121-143.

[DOI 10.1137/1.9781611977936.13](#) | [Source record](#) | Registered DOI metadata

Q0298 An Improved Classical Singular Value Transformation for Quantum Machine Learning

Ainesh Bakshi, Ewin Tang.

Book chapter | 2024 | Proceedings of the 2024 Annual ACM-SIAM Symposium on Discrete Algorithms (SODA) | pp. 2398-2453.

[DOI 10.1137/1.9781611977912.86](#) | [Source record](#) | Registered DOI metadata

Q0299 Nonlinear transformation of complex amplitudes via quantum singular value transformation

Naixu Guo, Kosuke Mitarai, Keisuke Fujii.

Journal article | 2024 | Physical Review Research | vol. 6 | no. 4 | article 043227.

[DOI 10.1103/physrevresearch.6.043227](#) | [Source record](#) | Registered DOI metadata

Q0300 Recursive quantum eigenvalue and singular-value transformation: Analytic construction of matrix sign function by Newton iteration

Kaoru Mizuta, Keisuke Fujii.

Journal article | 2024 | Physical Review Research | vol. 6 | no. 1 | article L012007.

[DOI 10.1103/physrevresearch.6.l012007](#) | [Source record](#) | Registered DOI metadata

Q0301 Singular Value Decomposition Quantum Algorithm for Quantum Biology

Emily K. Oh, Timothy J. Krogmeier, Anthony W. Schlimgen, Kade Head-Marsden.

Journal article | 2024 | ACS Physical Chemistry Au | vol. 4 | no. 4 | pp. 393-399.

[DOI 10.1021/acspyschemau.4c00018](#) | [Source record](#) | Registered DOI metadata

Q0302 Dequantizing the Quantum Singular Value Transformation: Hardness and Applications to Quantum Chemistry and the Quantum PCP Conjecture

Sevag Gharibian, François Le Gall.

Journal article | 2023 | SIAM Journal on Computing | vol. 52 | no. 4 | pp. 1009-1038.

[DOI 10.1137/22m1513721](#) | [Source record](#) | Registered DOI metadata

Q0303 Simulation of Linear Non-Hermitian Boundary-Value Problems with Quantum Singular-Value Transformation

I. Novikau, I.Y. Dodin, E.A. Startsev.

Journal article | 2023 | Physical Review Applied | vol. 19 | no. 5 | article 054012.

[DOI 10.1103/physrevapplied.19.054012](https://doi.org/10.1103/physrevapplied.19.054012) | [Source record](#) | Registered DOI metadata

D06T04 Linear combination of unitaries

Linear combinations of unitaries implement operators using coherent selection and state preparation. Success probability and amplification overhead affect the final resource estimate.

Fine subcategories: PREPARE and SELECT oracles; oblivious amplification; Taylor series simulation; success probability.

Primary catalog resources: 8. Related topic assignments are searchable in the HTML.

Q0304 Efficient Quantum Circuit Implementations of The qDRIFT Algorithm via Linear Combinations of Unitaries and Quantum Forking

I. J. David, I. Sinayskiy, F. Petruccione.

Journal article | 2025 | Journal of Physics: Conference Series | vol. 2970 | no. 1 | pp. 012004.

[DOI 10.1088/1742-6596/2970/1/012004](https://doi.org/10.1088/1742-6596/2970/1/012004) | [Source record](#) | Registered DOI metadata

Q0305 Hamiltonian dynamics simulation using linear combination of unitaries on an ion trap quantum computer

Michelle Wynne Sze, Yao Tang, Silas Dilkes, David Muñoz Ramo, Ross Duncan, Nathan Fitzpatrick.

Journal article | 2025 | Quantum Science and Technology | vol. 11 | no. 1 | pp. 015023.

[DOI 10.1088/2058-9565/ae2292](https://doi.org/10.1088/2058-9565/ae2292) | [Source record](#) | Registered DOI metadata

Q0306 Majorana tensor decomposition: a unifying framework for decompositions of fermionic Hamiltonians to linear combination of unitaries

Ignacio Loaiza, Aritra Sankar Brahmachari, Artur F Izmaylov.

Journal article | 2025 | Quantum Science and Technology | vol. 10 | no. 3 | pp. 035035.

[DOI 10.1088/2058-9565/add9c1](https://doi.org/10.1088/2058-9565/add9c1) | [Source record](#) | Registered DOI metadata

Q0307 An Optimal Linear-combination-of-unitaries-based Quantum Linear System Solver

Sander Gribling, Iordanis Kerenidis, Dániel Szilágyi.

Journal article | 2024 | ACM Transactions on Quantum Computing | vol. 5 | no. 2 | pp. 1-23.

[DOI 10.1145/3649320](https://doi.org/10.1145/3649320) | [Source record](#) | Registered DOI metadata

Q0308 Implementing any Linear Combination of Unitaries on Intermediate-term Quantum Computers

Shantanav Chakraborty.

Journal article | 2024 | Quantum | vol. 8 | pp. 1496 | article 1496.

[DOI 10.22331/q-2024-10-10-1496](https://doi.org/10.22331/q-2024-10-10-1496) | [Source record](#) | Registered DOI metadata

Q0309 Programmable silicon-photon quantum simulator based on a linear combination of unitaries

Yue Yu, Yulin Chi, Chonghao Zhai, Jieshan Huang, Qihuang Gong, Jianwei Wang.

Journal article | 2024 | Photonics Research | vol. 12 | no. 8 | pp. 1760.

[DOI 10.1364/prj.517294](https://doi.org/10.1364/prj.517294) | [Source record](#) | Registered DOI metadata

Q0310 Efficient Application of the Factorized form of the Unitary Coupled-Cluster Ansatz for the Variational Quantum Eigensolver Algorithm by Using Linear Combination of Unitaries

Luogen Xu, James K. Freericks.

Journal article | 2023 | Symmetry | vol. 15 | no. 7 | pp. 1429.

[DOI 10.3390/sym15071429](https://doi.org/10.3390/sym15071429) | [Source record](#) | Registered DOI metadata

Q0311 Fast black-box quantum state preparation based on linear combination of unitaries

Shengbin Wang, Zhimin Wang, Guolong Cui, Shangshang Shi, Ruimin Shang, Lixin Fan, Wendong Li, Zhiqiang Wei et al..

Journal article | 2021 | Quantum Information Processing | vol. 20 | no. 8 | article 270.

[DOI 10.1007/s11128-021-03203-z](https://doi.org/10.1007/s11128-021-03203-z) | [Source record](#) | Registered DOI metadata

D06T05 Quantum linear systems

Quantum linear systems algorithms prepare a state related to a linear system solution. Extracting every solution component can remove the intended advantage.

Fine subcategories: HHL; condition numbers; sparse access; solution observables; preconditioning.

Primary catalog resources: 10. Related topic assignments are searchable in the HTML.

Q0312 A VARIATIONAL QUANTUM ALGORITHM FOR SOLVING LINEAR SYSTEMS FOR DECISION SUPPORT SYSTEMS

Author metadata not supplied.

Journal article | 2026 | Telecommunication and information technologies | vol. 92 | no. 3 | pp. 46-51.

[DOI 10.31673/2412-4338.2026.039705](https://doi.org/10.31673/2412-4338.2026.039705) | [Source record](#) | Registered DOI metadata

Q0313 Quantum algorithm for linear systems

Yunah Choi, Jeihee Cho, Shiho Kim.

Book chapter | 2026 | Advances in Computers | pp. 1-26.

[DOI 10.1016/bs.adcom.2025.11.001](https://doi.org/10.1016/bs.adcom.2025.11.001) | [Source record](#) | Registered DOI metadata

Q0314 Resource-efficient quantum algorithm for linear systems of equations

Francesco Ghisoni, Francesco Scala, Daniele Bajoni, Dario Gerace.

Journal article | 2026 | Physical Review A | vol. 113 | no. 3 | article 032405.

[DOI 10.1103/gwzg-4pls](https://doi.org/10.1103/gwzg-4pls) | [Source record](#) | Registered DOI metadata

Q0315 A mixed-precision quantum-classical algorithm for solving linear systems

Océane Koska, Marc Baboulin, Arnaud Gazda.

Conference paper | 2025 | 2025 IEEE International Parallel and Distributed Processing Symposium Workshops (IPDPSW) | pp. 501-508.

[DOI 10.1109/ipdpsw66978.2025.00081](https://doi.org/10.1109/ipdpsw66978.2025.00081) | [Source record](#) | Registered DOI metadata

Q0316 Quantum Iterative Algorithm for Linear Systems of Equation

Debasish Roy, Sambo Raj Chandra.

Book chapter | 2024 | Lecture Notes in Networks and Systems | pp. 560-575.

[DOI 10.1007/978-3-031-62281-6_38](https://doi.org/10.1007/978-3-031-62281-6_38) | [Source record](#) | Registered DOI metadata

Q0317 Quantum multirow iteration algorithm for linear systems with nonsquare coefficient matrices

Weitao Lin, Guojing Tian, Xiaoming Sun.

Journal article | 2024 | Physical Review A | vol. 110 | no. 2 | article 022438.

[DOI 10.1103/physreva.110.022438](https://doi.org/10.1103/physreva.110.022438) | [Source record](#) | Registered DOI metadata

Q0318 Coherence dynamics in quantum algorithm for linear systems of equations

Linlin Ye, Zhaoqi Wu, Shao-Ming Fei.

Journal article | 2023 | Physica Scripta | vol. 98 | no. 12 | pp. 125104.

[DOI 10.1088/1402-4896/ad0584](https://doi.org/10.1088/1402-4896/ad0584) | [Source record](#) | Registered DOI metadata

Q0319 Enhancing the Quantum Linear Systems Algorithm Using Richardson Extrapolation

Almudena Carrera Vazquez, Ralf Hiptmair, Stefan Woerner.

Journal article | 2022 | ACM Transactions on Quantum Computing | vol. 3 | no. 1 | pp. 1-37.

[DOI 10.1145/3490631](https://doi.org/10.1145/3490631) | [Source record](#) | Registered DOI metadata

Q0320 Quantum linear system algorithm applied to communication systems

Jeonghoon Park, Jun Heo.

Journal article | 2022 | Quantum Information Processing | vol. 21 | no. 7 | article 267.

[DOI 10.1007/s11128-022-03598-3](#) | [Source record](#) | Registered DOI metadata

Q0321 Solving Large-Scale Linear Systems of Equations by a Quantum Hybrid Algorithm

M. R. Perelshtein, A. I. Pakhomchik, A. A. Melnikov, A. A. Novikov, A. Glatz, G. S. Paraoanu, V. M. Vinokur, G. B. Lesovik.

Journal article | 2022 | Annalen der Physik | vol. 534 | no. 7 | article 2200082.

[DOI 10.1002/andp.202200082](#) | [Source record](#) | Registered DOI metadata

D06T06 Quantum differential equations

Differential equation algorithms encode time evolution or discretized solution vectors. Stability, conditioning, boundary conditions, and output observables all enter the analysis.

Fine subcategories: Ordinary differential equations; partial differential equations; stability; boundary conditions; solution readout.

Primary catalog resources: 16. Related topic assignments are searchable in the HTML.

Q0322 Fast-forwarding quantum algorithms for linear dissipative differential equations

Dong An, Akwum Onwunta, Gengzhi Yang.

Journal article | 2026 | Quantum | vol. 10 | pp. 1986 | article 1986.

[DOI 10.22331/q-2026-01-27-1986](#) | [Source record](#) | Registered DOI metadata

Q0323 H-DES: a quantum–classical hybrid differential equation solver

Hamza Jaffali, Jonas Bastos de Araujo, Nadia Milazzo, Marta Reina, Henri de Boutray, Karla Baumann, Frédéric Holweck, Youcef Mohdeb et al..

Journal article | 2026 | Physica Scripta | vol. 101 | no. 21 | pp. 215107.

[DOI 10.1088/1402-4896/ae6404](#) | [Source record](#) | Registered DOI metadata

Q0324 Compact quantum algorithms for time-dependent differential equations

Sachin S. Bharadwaj, Katepalli R. Sreenivasan.

Journal article | 2025 | Physical Review Research | vol. 7 | no. 2 | article 023262.

[DOI 10.1103/physrevresearch.7.023262](#) | [Source record](#) | Registered DOI metadata

Q0325 Quantum Algorithms for Stochastic Differential Equations: A Schrödingerisation Approach

Shi Jin, Nana Liu, Wei Wei.

Journal article | 2025 | Journal of Scientific Computing | vol. 104 | no. 2 | article 56.

[DOI 10.1007/s10915-025-02970-6](#) | [Source record](#) | Registered DOI metadata

Q0326 Quantum Differential Equation Solvers: Limitations and Fast-Forwarding

Dong An, Jin-Peng Liu, Daochen Wang, Qi Zhao.

Journal article | 2025 | Communications in Mathematical Physics | vol. 406 | no. 8 | article 189.

[DOI 10.1007/s00220-025-05358-7](#) | [Source record](#) | Registered DOI metadata

Q0327 Quantum Ordinary Differential Equation Algorithms: Block-Matrix Algorithms

Osama M. Raisuddin, Suvaranu De.

Book chapter | 2025 | Quantum Computing for Engineers | pp. 259-265.

[DOI 10.1007/978-3-032-03325-3_32](#) | [Source record](#) | Registered DOI metadata

Q0328 Quantum Ordinary Differential Equation Algorithms: Time-Marching Algorithms

Osama M. Raisuddin, Suvaranu De.

Book chapter | 2025 | Quantum Computing for Engineers | pp. 267-269.

[DOI 10.1007/978-3-032-03325-3_33](#) | [Source record](#) | Registered DOI metadata

Q0329 Quantum Partial Differential Equation Algorithms

Osama M. Raisuddin, Suvaranu De.

Book chapter | 2025 | Quantum Computing for Engineers | pp. 271-275.

[DOI 10.1007/978-3-032-03325-3_34](#) | [Source record](#) | Registered DOI metadata

Q0330 Variational Quantum Algorithms for Differential Equations on a Noisy Quantum Computer

Niclas Schillo, Andreas Sturm.

Journal article | 2025 | IEEE Transactions on Quantum Engineering | vol. 6 | pp. 1-16.

[DOI 10.1109/tqe.2025.3532017](#) | [Source record](#) | Registered DOI metadata

Q0331 Variational Quantum Framework for Partial Differential Equation Constrained Optimization

Amit Surana, Abeynaya Gnanasekaran.

Journal article | 2025 | ACM Transactions on Quantum Computing | vol. 7 | no. 1 | pp. 1-36.

[DOI 10.1145/3762671](#) | [Source record](#) | Registered DOI metadata

Q0332 Diffusion-Based Quantum Error Mitigation Using Stochastic Differential Equation

Joo Yong Shim, Joongheon Kim.

Conference paper | 2024 | 2024 IEEE VTS Asia Pacific Wireless Communications Symposium (APWCS) | pp. 1-5.

[DOI 10.1109/apwcs61586.2024.10679306](#) | [Source record](#) | Registered DOI metadata

Q0333 Quantum Algorithms for Multiscale Partial Differential Equations

Junpeng Hu, Shi Jin, Lei Zhang.

Journal article | 2024 | Multiscale Modeling & Simulation | vol. 22 | no. 3 | pp. 1030-1067.

[DOI 10.1137/23m1566340](#) | [Source record](#) | Registered DOI metadata

Fine tags: partial differential equations

Q0334 Quantum Computing Solution to Sturm-Liouville Differential Equation

Author metadata not supplied.

Journal article | 2024 | International Journal of Innovative Research in Physics | vol. 5 | no. 2.

[DOI 10.15864/ijip.5203](#) | [Source record](#) | Registered DOI metadata

Q0335 Quantum algorithm for dynamic mode decomposition integrated with a quantum differential equation solver

Yuta Mizuno, Tamiki Komatsuzaki.

Journal article | 2024 | Physical Review Research | vol. 6 | no. 4 | article 043031.

[DOI 10.1103/physrevresearch.6.043031](#) | [Source record](#) | Registered DOI metadata

Q0336 Quantum algorithms for nonlinear partial differential equations

Shi Jin, Nana Liu.

Journal article | 2024 | Bulletin des Sciences Mathématiques | vol. 194 | pp. 103457 | article 103457.

[DOI 10.1016/j.bulsci.2024.103457](#) | [Source record](#) | Registered DOI metadata

Fine tags: partial differential equations

Q0337 Quantum Algorithms for Nonlinear Partial Differential Equations

Shi Jin, Nana Liu.

Posted content | 2023 | Elsevier BV.

[DOI 10.2139/ssrn.4353562](#) | [Source record](#) | Registered DOI metadata

Fine tags: partial differential equations

D07 Hamiltonian simulation and many body physics

Quantum simulation targets the evolution and observables of physical systems. Digital, analog, and hybrid approaches have different error budgets and verification methods.

Prerequisites: Quantum mechanics, Hamiltonians, and many body physics.

Assessment focus: Simulation error, model truncation, observables, and classical verification.

D07T01 Digital Hamiltonian simulation

Digital simulation approximates Hamiltonian evolution with a programmable sequence of gates. The comparison should include accuracy, sparsity, simulation time, and compilation cost.

Fine subcategories: Trotter formulas; error bounds; interaction picture; sparse Hamiltonians; time dependent evolution.

Primary catalog resources: 8. Related topic assignments are searchable in the HTML.

Q0338 Diagonal-Budgeted Trotterization for Efficient Quantum Hamiltonian Simulation

Srikar Chundury, Blake Burgstahler, Jiajia Li, In-Saeng Suh, Frank Mueller.

Conference paper | 2026 | Proceedings of the 40th ACM International Conference on Supercomputing | pp. 1350-1362.

[DOI 10.1145/3797905.3807869](#) | [Source record](#) | Registered DOI metadata

Q0339 Practical limitations on Hamiltonian simulation using quantum processors

Abhishek Sadhu, Mandira Dey, Debashree Ghosh.

Journal article | 2026 | Electronic Structure | vol. 8 | no. 3 | pp. 035006.

[DOI 10.1088/2516-1075/ae97a8](#) | [Source record](#) | Registered DOI metadata

Q0340 Snapshot-QAOA: Extending QAOA to Quantum Hamiltonian Simulation

Reuben Tate, Quinn Langfitt, Elijah Pelofske, Ammar Kirmani, Andreas Bärtzchi, John Golden, Stephan Eidenbenz.

Posted content | 2026 | Springer Science and Business Media LLC.

[DOI 10.21203/rs.3.rs-9171925/v1](#) | [Source record](#) | Registered DOI metadata

Q0341 Quantum Hamiltonian Simulation of Time-Domain Electromagnetic Fields

E. Colella, L. Bastianelli, V. Mariani Primiani, F. Moglie, G. Gradoni.

Conference paper | 2025 | 2025 International Conference on Electromagnetics in Advanced Applications (ICEAA) | pp. 1-1.

[DOI 10.1109/iceaa65662.2025.11305881](#) | [Source record](#) | Registered DOI metadata

Q0342 A Mathematical Framework for Quantum Hamiltonian Simulation and Duality

Harriet Apel, Toby Cubitt.

Journal article | 2024 | Annales Henri Poincaré | vol. 26 | no. 1 | pp. 317-364.

[DOI 10.1007/s00023-024-01432-3](#) | [Source record](#) | Registered DOI metadata

Q0343 Parallel Quantum Algorithm for Hamiltonian Simulation

Zhicheng Zhang, Qisheng Wang, Mingsheng Ying.

Journal article | 2024 | Quantum | vol. 8 | pp. 1228 | article 1228.

[DOI 10.22331/q-2024-01-15-1228](#) | [Source record](#) | Registered DOI metadata

Q0344 Quantum computing of reacting flows via Hamiltonian simulation

Zhen Lu, Yue Yang.

Journal article | 2024 | Proceedings of the Combustion Institute | vol. 40 | no. 1-4 | pp. 105440 | article 105440.

[DOI 10.1016/j.proci.2024.105440](#) | [Source record](#) | Registered DOI metadata

Q0345 Riemannian quantum circuit optimization for Hamiltonian simulation

Ayşe Kotil, Rahul Banerjee, Qunsheng Huang, Christian B Mendl.

Journal article | 2024 | Journal of Physics A: Mathematical and Theoretical | vol. 57 | no. 13 | pp. 135303.

[DOI 10.1088/1751-8121/ad2d6e](https://doi.org/10.1088/1751-8121/ad2d6e) | [Source record](#) | Registered DOI metadata

D07T02 Analog quantum simulation

Analog simulators engineer a physical Hamiltonian to approximate a target model. Calibration and validation are especially important when universal digital corrections are unavailable.

Fine subcategories: Hamiltonian engineering; cold atoms; trapped ions; spin models; simulator calibration.

Primary catalog resources: 16. Related topic assignments are searchable in the HTML.

Q0346 Simulating physics with computers

Richard P. Feynman.

Journal article | 1982 | International Journal of Theoretical Physics | vol. 21 | no. 6-7 | pp. 467-488.

[DOI 10.1007/bf02650179](https://doi.org/10.1007/bf02650179) | [Source record](#) | Registered DOI metadata

Q0347 Analog quantum simulation of chiral magnetic dynamics using optical superlattices

Sabhyata Gupta, Luis Santos.

Journal article | 2026 | Physical Review A | vol. 114 | no. 3 | article 033321.

[DOI 10.1103/hr8d-ttj8](https://doi.org/10.1103/hr8d-ttj8) | [Source record](#) | Registered DOI metadata

Q0348 Hamiltonian simulation with explicit formulas for digital-analog quantum computing

Mikel Garcia de Andoin, Thorger Müller, Gonzalo Camacho.

Journal article | 2026 | Physical Review A | vol. 113 | no. 6 | article 062607.

[DOI 10.1103/nzxg-5lbg](https://doi.org/10.1103/nzxg-5lbg) | [Source record](#) | Registered DOI metadata

Q0349 Accuracy Guarantees and Quantum Advantage in Analog Open Quantum Simulation with and without Noise

Vikram Kashyap, Georgios Styliaris, Sara Mouradian, J. Ignacio Cirac, Rahul Trivedi.

Journal article | 2025 | Physical Review X | vol. 15 | no. 2 | article 021017.

[DOI 10.1103/physrevx.15.021017](https://doi.org/10.1103/physrevx.15.021017) | [Source record](#) | Registered DOI metadata

Q0350 Analog Quantum Simulation via Rydberg Dressing

Arinjoy De, Majd Hamdan, Milan Kornjaca, Alexei Bylinskii.

Conference paper | 2025 | 2025 IEEE International Conference on Quantum Computing and Engineering (QCE) | pp. 520-521.

[DOI 10.1109/qce65121.2025.10425](https://doi.org/10.1109/qce65121.2025.10425) | [Source record](#) | Registered DOI metadata

Q0351 Analog quantum simulation of coupled electron-nuclear dynamics in molecules

Jong-Kwon Ha, Ryan J. MacDonell.

Journal article | 2025 | Chemical Science | vol. 16 | no. 41 | pp. 19423-19435.

[DOI 10.1039/d5sc04076k](https://doi.org/10.1039/d5sc04076k) | [Source record](#) | Registered DOI metadata

Q0352 Analog simulation of noisy quantum circuits

Etienne Granet, Kévin Hémerly, Henrik Dreyer.

Journal article | 2025 | Physical Review Research | vol. 7 | no. 1 | article 013213.

[DOI 10.1103/physrevresearch.7.013213](https://doi.org/10.1103/physrevresearch.7.013213) | [Source record](#) | Registered DOI metadata

Q0353 Compact quantum dot models for analog microwave co-simulation

Lorenzo Peri, Alberto Gomez-Saiz, Christopher J. B. Ford, M. Fernando Gonzalez-Zalba.

Journal article | 2025 | npj Quantum Information | vol. 11 | no. 1 | article 194.

[DOI 10.1038/s41534-025-01140-8](https://doi.org/10.1038/s41534-025-01140-8) | [Source record](#) | Registered DOI metadata

Q0354 Progress towards neutron-scattering simulation on an analog quantum processor

Nora Bauer, Victor Ale, Pontus Laurell, Serena Huang, Seth Watabe, David Alan Tennant, George Siopsis.

Journal article | 2025 | Physical Review A | vol. 111 | no. 2 | article 022442.

[DOI 10.1103/physreva.111.022442](https://doi.org/10.1103/physreva.111.022442) | [Source record](#) | Registered DOI metadata

Q0355 QTurbo: A Robust and Efficient Compiler for Analog Quantum Simulation

Junyu Zhou, Yuhao Liu, Shize Che, Anupam Mitra, Efehan Kökcü, Ermal Rrapaj, Costin Iancu, Gushu Li.

Conference paper | 2025 | Proceedings of the 31st ACM International Conference on Architectural Support for Programming Languages and Operating Systems, Volume 1 | pp. 203-218.

[DOI 10.1145/3760250.3762227](https://doi.org/10.1145/3760250.3762227) | [Source record](#) | Registered DOI metadata

Q0356 Analog quantum simulation of partial differential equations

Shi Jin, Nana Liu.

Journal article | 2024 | Quantum Science and Technology | vol. 9 | no. 3 | pp. 035047.

[DOI 10.1088/2058-9565/ad49cf](https://doi.org/10.1088/2058-9565/ad49cf) | [Source record](#) | Registered DOI metadata

Q0357 Multipurpose platform for analog quantum simulation

Shuwei Jin, Kunlun Dai, Joris Verstraten, Maxime Dixmieras, Ragheed Alhyder, Christophe Salomon, Bruno Peaudecerf, Tim de Jongh et al..

Journal article | 2024 | Physical Review Research | vol. 6 | no. 1 | article 013158.

[DOI 10.1103/physrevresearch.6.013158](https://doi.org/10.1103/physrevresearch.6.013158) | [Source record](#) | Registered DOI metadata

Q0358 SimuQ: A Framework for Programming Quantum Hamiltonian Simulation with Analog Compilation

Yuxiang Peng, Jacob Young, Pengyu Liu, Xiaodi Wu.

Journal article | 2024 | Proceedings of the ACM on Programming Languages | vol. 8 | no. POPL | pp. 2425-2455.

[DOI 10.1145/3632923](https://doi.org/10.1145/3632923) | [Source record](#) | Registered DOI metadata

Q0359 Digital-Analog Quantum Simulation of Fermionic Models

Lucas C. Céleri, Daniel Huerga, Francisco Albarrán-Arriagada, Enrique Solano, Mikel Garcia de Andoin, Mikel Sanz.

Journal article | 2023 | Physical Review Applied | vol. 19 | no. 6 | article 064086.

[DOI 10.1103/physrevapplied.19.064086](https://doi.org/10.1103/physrevapplied.19.064086) | [Source record](#) | Registered DOI metadata

Q0360 Measuring Arbitrary Physical Properties in Analog Quantum Simulation

Minh C. Tran, Daniel K. Mark, Wen Wei Ho, Soonwon Choi.

Journal article | 2023 | Physical Review X | vol. 13 | no. 1 | article 011049.

[DOI 10.1103/physrevx.13.011049](https://doi.org/10.1103/physrevx.13.011049) | [Source record](#) | Registered DOI metadata

Q0361 Predicting molecular vibronic spectra using time-domain analog quantum simulation

Ryan J. MacDonell, Tomas Navickas, Tim F. Wohlers-Reichel, Christophe H. Valahu, Arjun D. Rao, Maverick J. Millican, Michael A. Currington, Michael J. Biercuk et al..

Journal article | 2023 | Chemical Science | vol. 14 | no. 35 | pp. 9439-9451.

[DOI 10.1039/d3sc02453a](https://doi.org/10.1039/d3sc02453a) | [Source record](#) | Registered DOI metadata

D07T03 Quantum lattice and spin models

Lattice and spin models provide structured many body targets. Geometry, interactions, symmetry, and observable selection specify the actual simulation problem.

Fine subcategories: Ising models; Heisenberg models; Hubbard models; phase diagrams; frustration.

Primary catalog resources: 8. Related topic assignments are searchable in the HTML.

Q0362 Probing confinement through dynamical quantum phase transitions: From quantum spin models to lattice gauge theories

Jesse J. Osborne, Ian P. McCulloch, Jad C. Halimeh.

Journal article | 2025 | Physical Review Research | vol. 7 | no. 4 | article 043076.

[DOI 10.1103/rnv5-f32k](#) | [Source record](#) | Registered DOI metadata

Q0363 Quantum Simulating Continuum Field Theories with Large-Spin Lattice Models

Gabriele Calliari, Marco Di Liberto, Hannes Pichler, Torsten V. Zache.

Journal article | 2025 | PRX Quantum | vol. 6 | no. 3 | article 030304.

[DOI 10.1103/nt76-ttmj](#) | [Source record](#) | Registered DOI metadata

Q0364 Quantum simulation of spin-boson models with structured bath

Ke Sun, Mingyu Kang, Hanggai Nuomin, George Schwartz, David N. Beratan, Kenneth R. Brown, Jungsang Kim.

Journal article | 2025 | Nature Communications | vol. 16 | no. 1 | article 4042.

[DOI 10.1038/s41467-025-59296-y](#) | [Source record](#) | Registered DOI metadata

Q0365 Recipes for the digital quantum simulation of lattice spin systems

Guido Burkard.

Journal article | 2025 | SciPost Physics Core | vol. 8 | no. 1 | article 030.

[DOI 10.21468/scipostphyscore.8.1.030](#) | [Source record](#) | Registered DOI metadata

Q0366 Real-space spectral simulation of quantum spin models: Application to generalized Kitaev models

Francisco M. O. Brito, Aires Ferreira.

Journal article | 2024 | SciPost Physics Core | vol. 7 | no. 1 | article 006.

[DOI 10.21468/scipostphyscore.7.1.006](#) | [Source record](#) | Registered DOI metadata

Q0367 Thermal tensor renormalization group simulations of square-lattice quantum spin models

Han Li, Bin-Bin Chen, Ziyu Chen, Jan von Delft, Andreas Weichselbaum, Wei Li.

Journal article | 2019 | Physical Review B | vol. 100 | no. 4 | article 045110.

[DOI 10.1103/physrevb.100.045110](#) | [Source record](#) | Registered DOI metadata

Q0368 Quantum Simulation of SU(4) Symmetric Spin Lattice Models

Bruno Uchoa.

Conference paper | 2018 | The First International Conference on Symmetry | pp. 39.

[DOI 10.3390/proceedings2010039](#) | [Source record](#) | Registered DOI metadata

Q0369 Quantum simulation of spin models on an arbitrary lattice with trapped ions

S Korenblit, D Kafri, W C Campbell, R Islam, E E Edwards, Z-X Gong, G-D Lin, L-M Duan et al..

Journal article | 2012 | New Journal of Physics | vol. 14 | no. 9 | pp. 095024.

[DOI 10.1088/1367-2630/14/9/095024](#) | [Source record](#) | Registered DOI metadata

D07T04 Quantum thermal states

Thermal state algorithms prepare or characterize Gibbs states and finite temperature observables. Mixing, equilibration, and energy normalization can dominate costs.

Fine subcategories: Gibbs sampling; thermalization; imaginary time evolution; partition functions; finite temperature observables.

Primary catalog resources: 8. Related topic assignments are searchable in the HTML.

Q0370 Optimal Quantum Algorithm for Gibbs State Preparation

Cambyse Rouzé, Daniel Stilck França, Álvaro M. Alhambra.

Journal article | 2026 | Physical Review Letters | vol. 136 | no. 6 | article 060601.

[DOI 10.1103/lhht-svmn](#) | [Source record](#) | Registered DOI metadata

Q0371 Dissipative Variational Quantum Algorithms for Gibbs State Preparation

Yigal Ilin, Itai Arad.

Journal article | 2025 | IEEE Transactions on Quantum Engineering | vol. 6 | pp. 1-12.

[DOI 10.1109/tqe.2024.3511419](https://doi.org/10.1109/tqe.2024.3511419) | [Source record](#) | Registered DOI metadata

Q0372 Lindblad engineering for quantum Gibbs state preparation under the eigenstate thermalization hypothesis

Eric Brunner, Luuk Coopmans, Gabriel Matos, Matthias Rosenkranz, Frederic Sauvage, Yuta Kikuchi.

Journal article | 2025 | Quantum | vol. 9 | pp. 1843 | article 1843.

[DOI 10.22331/q-2025-08-29-1843](https://doi.org/10.22331/q-2025-08-29-1843) | [Source record](#) | Registered DOI metadata

Fine tags: thermalization

Q0373 Resource Estimates for Lindbladian-Based Gibbs State Preparation on Fault-Tolerant Quantum Computers

Sandia National Laboratories (SNL-NM), Albuquerque, NM (United States), Eric Bobrow, Advanced Simulation and Computing, USDOE National Nuclear Security Administration (NNSA), Riley Chien, Alina Kononov, Lucas Kovalsky, Jacob Nelson et al..

Report | 2025 | Office of Scientific and Technical Information (OSTI).

[DOI 10.2172/3023927](https://doi.org/10.2172/3023927) | [Source record](#) | Registered DOI metadata

Q0374 Variational Quantum Algorithms for Gibbs State Preparation

Mirko Consiglio.

Book chapter | 2025 | Lecture Notes in Computer Science | pp. 56-70.

[DOI 10.1007/978-3-031-81247-7_5](https://doi.org/10.1007/978-3-031-81247-7_5) | [Source record](#) | Registered DOI metadata

Q0375 Variational Gibbs state preparation on noisy intermediate-scale quantum devices

Mirko Consiglio, Jacopo Settimo, Andrea Giordano, Carlo Mastroianni, Francesco Plastina, Salvatore Lorenzo, Sabrina Maniscalco, John Gould et al..

Journal article | 2024 | Physical Review A | vol. 110 | no. 1 | article 012445.

[DOI 10.1103/physreva.110.012445](https://doi.org/10.1103/physreva.110.012445) | [Source record](#) | Registered DOI metadata

Q0376 The Role of Initial Entanglement in Adaptive Gibbs State Preparation on Quantum Computers

Sophia E. Economou, Ada Warren, Edwin Barnes.

Conference paper | 2023 | ICASSP 2023 - 2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP) | pp. 1-5.

[DOI 10.1109/icassp49357.2023.10094697](https://doi.org/10.1109/icassp49357.2023.10094697) | [Source record](#) | Registered DOI metadata

Q0377 Variational Quantum Gibbs State Preparation with a Truncated Taylor Series

Youle Wang, Guangxi Li, Xin Wang.

Journal article | 2021 | Physical Review Applied | vol. 16 | no. 5 | article 054035.

[DOI 10.1103/physrevapplied.16.054035](https://doi.org/10.1103/physrevapplied.16.054035) | [Source record](#) | Registered DOI metadata

D07T05 Quantum dynamics and scrambling

Scrambling and transport characterize how information and operators spread through a system. Measured correlators need interpretation that accounts for noise and finite size.

Fine subcategories: OTOCs; operator spreading; transport; many body localization; dynamical phase transitions.

Primary catalog resources: 8. Related topic assignments are searchable in the HTML.

Q0378 Bridging Classical Sensitivity and Quantum Scrambling: A Tutorial on Out-of-Time-Ordered Correlators

Stephen Wiggins.

Journal article | 2026 | International Journal of Bifurcation and Chaos | vol. 36 | no. 12 | article 2650170.

[DOI 10.1142/s0218127426501701](#) | [Source record](#) | Registered DOI metadata

Q0379 Out of Time Order Correlators as Measures of Quantum Information Scrambling with Spectral Diagnostic

Bannishikha Banerjee.

Posted content | 2026 | Springer Science and Business Media LLC.

[DOI 10.21203/rs.3.rs-9329643/v1](#) | [Source record](#) | Registered DOI metadata

Q0380 Analyzing Information Scrambling in Simulated Quantum Systems Using Out-of-Time-Ordered Correlators

Bibiana Firmin R, Sathishkumar M.

Journal article | 2025 | International Journal For Multidisciplinary Research | vol. 7 | no. 6 | article 60321.

[DOI 10.36948/ijfmr.2025.v07i06.60321](#) | [Source record](#) | Registered DOI metadata

Q0381 State estimation with quantum extreme learning machines beyond the scrambling time

Marco Vetrano, Gabriele Lo Monaco, Luca Innocenti, Salvatore Lorenzo, G. Massimo Palma.

Journal article | 2025 | npj Quantum Information | vol. 11 | no. 1 | article 20.

[DOI 10.1038/s41534-024-00927-5](#) | [Source record](#) | Registered DOI metadata

Q0382 Scrambling Dynamics and Out-of-Time-Ordered Correlators in Quantum Many-Body Systems

Shenglong Xu, Brian Swingle.

Journal article | 2024 | PRX Quantum | vol. 5 | no. 1 | article 010201.

[DOI 10.1103/prxquantum.5.010201](#) | [Source record](#) | Registered DOI metadata

Q0383 Imaginary components of out-of-time-order correlator and information scrambling for navigating the learning landscape of a quantum machine learning model

Manas Sajjan, Vinit Singh, Raja Selvarajan, Sabre Kais.

Journal article | 2023 | Physical Review Research | vol. 5 | no. 1 | article 013146.

[DOI 10.1103/physrevresearch.5.013146](#) | [Source record](#) | Registered DOI metadata

Q0384 Author Correction: Unifying scrambling, thermalization and entanglement through measurement of fidelity out-of-time-order correlators in the Dicke model

R. J. Lewis-Swan, A. Safavi-Naini, J. J. Bollinger, A. M. Rey.

Journal article | 2019 | Nature Communications | vol. 10 | no. 1 | article 5007.

[DOI 10.1038/s41467-019-13016-5](#) | [Source record](#) | Registered DOI metadata

Q0385 Unifying scrambling, thermalization and entanglement through measurement of fidelity out-of-time-order correlators in the Dicke model

R. J. Lewis-Swan, A. Safavi-Naini, J. J. Bollinger, A. M. Rey.

Journal article | 2019 | Nature Communications | vol. 10 | no. 1 | article 1581.

[DOI 10.1038/s41467-019-09436-y](#) | [Source record](#) | Registered DOI metadata

D07T06 Lattice gauge theory simulation

Gauge simulation encodes fields subject to local constraints. Truncation and gauge violation must be assessed along with gate noise and observable accuracy.

Fine subcategories: Gauge constraints; truncation; real time dynamics; link encodings; digital gauge simulation.

Primary catalog resources: 8. Related topic assignments are searchable in the HTML.

Q0386 Quantum Simulation of a $\mathbb{Z}_2$ -Lattice Gauge Theory

Sebastian Saner.

Book chapter | 2026 | Springer Theses | pp. 141-168.

[DOI 10.1007/978-3-032-23775-0_8](#) | [Source record](#) | Registered DOI metadata

Q0387 Quantum simulation of baryon scattering in SU(2) lattice gauge theory

João Barata, Juan Hormaza, Zhong-Bo Kang, Wenyang Qian.

Posted content | 2026 | Elsevier BV.

[DOI 10.2139/ssrn.6760550](#) | [Source record](#) | Registered DOI metadata

Q0388 $\mathbb{Z}_2$ lattice gauge theory on non-trivial topology and its quantum simulation

Jiaqi Hu, Shu Tian, Xiaopeng Cui, Rebing Wu, Man-Hong Yung, Yu Shi.

Journal article | 2026 | Journal of High Energy Physics | vol. 2026 | no. 7 | article 67.

[DOI 10.1007/jhep07\(2026\)067](#) | [Source record](#) | Registered DOI metadata

Q0389 Quantum simulation with gauge fixing: From Ising lattice gauge theory to dynamical flux model

Junsen Wang, Xiangxiang Sun, Wei Zheng.

Journal article | 2025 | Physical Review Research | vol. 7 | no. 1 | article 013311.

[DOI 10.1103/physrevresearch.7.013311](#) | [Source record](#) | Registered DOI metadata

Q0390 Quantum simulation of gauge theory via orbifold lattice

Alexander J. Buser, Hrant Gharibyan, Masanori Hanada, Masazumi Honda, Junyu Liu.

Journal article | 2021 | Journal of High Energy Physics | vol. 2021 | no. 9 | article 34.

[DOI 10.1007/jhep09\(2021\)034](#) | [Source record](#) | Registered DOI metadata

Q0391 Circuit-based digital adiabatic quantum simulation and pseudoquantum simulation as new approaches to lattice gauge theory

Xiaopeng Cui, Yu Shi, Ji-Chong Yang.

Journal article | 2020 | Journal of High Energy Physics | vol. 2020 | no. 8 | article 160.

[DOI 10.1007/jhep08\(2020\)160](#) | [Source record](#) | Registered DOI metadata

Fine tags: digital gauge simulation

Q0392 Trotter errors in digital adiabatic quantum simulation of quantum $\mathbb{Z}_2$ lattice gauge theory

Xiaopeng Cui, Yu Shi.

Journal article | 2020 | International Journal of Modern Physics B | vol. 34 | no. 30 | pp. 2050292.

[DOI 10.1142/s0217979220502926](#) | [Source record](#) | Registered DOI metadata

Fine tags: digital gauge simulation

Q0393 Using Euclidean Lattice Field Theory for Efficient Quantum Simulation of a $\mathbb{Z}_2$ Gauge Theory

Erik Gustafson.

Posted content | 2020 | Morressier.

[DOI 10.26226/morressier.5fa409874d4e91fe5c54b96b](#) | [Source record](#) | Registered DOI metadata

D08 Quantum chemistry and materials

Electronic structure is a major application area because quantum states can encode correlated wavefunctions. Practical workflows require molecular mappings, controlled errors, and meaningful classical comparisons.

Prerequisites: Electronic structure, second quantization, and quantum algorithms.

Assessment focus: Basis set, active space, energy accuracy, mapping, and measured observables.

D08T01 Electronic structure algorithms

Electronic structure algorithms estimate properties of interacting electrons in selected basis sets. The physical accuracy target and active space define the application.

Fine subcategories: Ground states; active spaces; basis sets; correlated wavefunctions; energy accuracy.

Primary catalog resources: 12. Related topic assignments are searchable in the HTML.

Q0394 Quantum Computing Beyond Ground-State Electronic Structure: A Review of Progress Toward Quantum Chemistry Out of the Ground State

Alan Bidart, Prateek Vaish, Tilas Kabengele, Yaoqi Pang, Yuan Liu, Brenda M. Rubenstein.

Journal article | 2026 | Annual Review of Physical Chemistry | vol. 77 | no. 1 | pp. 417-441.

[DOI 10.1146/annurev-physchem-082624-084635](https://doi.org/10.1146/annurev-physchem-082624-084635) | [Source record](#) | Registered DOI metadata

Q0395 Analyzing Common Electronic Structure Theory Algorithms for Distributed Quantum Computing

Grier M. Jones, Hans-Arno Jacobsen.

Conference paper | 2025 | 2025 IEEE International Conference on Quantum Computing and Engineering (QCE) | pp. 368-373.

[DOI 10.1109/qce65121.2025.10351](https://doi.org/10.1109/qce65121.2025.10351) | [Source record](#) | Registered DOI metadata

Q0396 NOISY QUANTUM COMPUTING OF ELECTRONIC STRUCTURE OF CRYSTALS

Vojtěch VAŠINA, Ivana MIHÁLIKOVÁ, Martin FRIÁK.

Conference paper | 2025 | METAL Conference Proceedings | vol. 2025 | pp. 538-543.

[DOI 10.37904/metal.2025.5153](https://doi.org/10.37904/metal.2025.5153) | [Source record](#) | Registered DOI metadata

Q0397 Quantum computing of the electronic structure of crystals by the Variational Quantum Deflation algorithm

Michal Ďuriška, Ivana Miháliková, Martin Friák.

Journal article | 2025 | Physica Scripta | vol. 100 | no. 4 | pp. 045105.

[DOI 10.1088/1402-4896/adbb29](https://doi.org/10.1088/1402-4896/adbb29) | [Source record](#) | Registered DOI metadata

Q0398 SUPPRESSING NOISE IN QUANTUM COMPUTING OF ELECTRONIC STRUCTURE OF CRYSTALS

Jan HOJÁČ, Vojtěch VAŠINA, Martin FRIÁK.

Conference paper | 2025 | NANOCON Conference Proceedings | vol. 2025 | pp. 60-65.

[DOI 10.37904/nanocon.2025.5189](https://doi.org/10.37904/nanocon.2025.5189) | [Source record](#) | Registered DOI metadata

Q0399 Electronic structure simulations in the cloud computing environment

Eric J. Bylaska, Ajay Panyala, Nicholas P. Bauman, Bo Peng, Himadri Pathak, Daniel Mejia-Rodriguez, Niranjana Govind, David B. Williams-Young et al..

Journal article | 2024 | The Journal of Chemical Physics | vol. 161 | no. 15 | article 150902.

[DOI 10.1063/5.0226437](https://doi.org/10.1063/5.0226437) | [Source record](#) | Registered DOI metadata

Q0400 Quantum computing for electronic structure analysis: Ground state energy and molecular properties calculations

Nouhaila Innan, Muhammad Al-Zafar Khan, Mohamed Bennai.

Journal article | 2024 | Materials Today Communications | vol. 38 | pp. 107760 | article 107760.

[DOI 10.1016/j.mtcomm.2023.107760](https://doi.org/10.1016/j.mtcomm.2023.107760) | [Source record](#) | Registered DOI metadata

Q0401 Quantum computing quantum Monte Carlo with hybrid tensor network for electronic structure calculations

Shu Kanno, Hajime Nakamura, Takao Kobayashi, Shigeki Gocho, Miho Hatanaka, Naoki Yamamoto, Qi Gao.

Journal article | 2024 | npj Quantum Information | vol. 10 | no. 1 | article 56.

[DOI 10.1038/s41534-024-00851-8](https://doi.org/10.1038/s41534-024-00851-8) | [Source record](#) | Registered DOI metadata

Q0402 Challenges in the Use of Quantum Computing Hardware-Efficient Ansätze in Electronic Structure Theory

Ruhee D'Cunha, T. Daniel Crawford, Mario Motta, Julia E. Rice.

Journal article | 2023 | The Journal of Physical Chemistry A | vol. 127 | no. 15 | pp. 3437-3448.

[DOI 10.1021/acs.jpca.2c08430](https://doi.org/10.1021/acs.jpca.2c08430) | [Source record](#) | Registered DOI metadata

Q0403 Quantum-computing study of the electronic structure of crystals: the case study of Si

Michal ĎURÍŠKA, Ivana MIHÁLIKOVÁ, Martin FRIÁK.

Conference paper | 2023 | NANOCON Conference Proceedings | vol. 2023 | pp. 0-0.

[DOI 10.37904/nanocon.2023.4774](https://doi.org/10.37904/nanocon.2023.4774) | [Source record](#) | Registered DOI metadata

Q0404 GitHub - qiskit-community/qiskit-nature: Qiskit Nature is an open-source, quantum computing, framework for solving quantum mechanical natural science problems. · GitHub

Author metadata not supplied.

Software repository | Undated | Qiskit Nature.

[Official resource](#) | [Source record](#) | Official page retrieved

Q0405 GitHub - quantumlib/OpenFermion: Python package for compiling and analyzing quantum algorithms to simulate electronic structures. · GitHub

Author metadata not supplied.

Software repository | Undated | Google Quantum AI.

[Official resource](#) | [Source record](#) | Official page retrieved

D08T02 Fermion to qubit mappings

Fermion mappings translate anticommutation relations into qubit operators. Operator locality, symmetry reduction, and measurement cost depend on the mapping.

Fine subcategories: Jordan Wigner; Bravyi Kitaev; parity mappings; symmetry tapering; fermionic encodings.

Primary catalog resources: 8. Related topic assignments are searchable in the HTML.

Q0406 Hidden Rotation Symmetry of the Jordan–Wigner Transformation and Its Application to Measurement in Quantum Computation

Grant Davis, James K. Freericks.

Journal article | 2026 | Symmetry | vol. 18 | no. 2 | pp. 251.

[DOI 10.3390/sym18020251](https://doi.org/10.3390/sym18020251) | [Source record](#) | Registered DOI metadata

Fine tags: Jordan Wigner

Q0407 Extension of the Jordan-Wigner mapping to nonorthogonal spin orbitals for quantum computing application to valence bond approaches

Author metadata not supplied.

Book chapter | 2025 | Advances in Quantum Chemistry | pp. 245-266.

[DOI 10.1016/bs.aiq.2025.07.007](#) | [Source record](#) | Registered DOI metadata

Fine tags: Jordan Wigner

Q0408 Jordan-Wigner mapping for nonorthogonal spin orbitals

Alessandro Genoni, Mosè Casalegno, Piero Macchi, Guido Raos.

Posted content | 2025 | American Chemical Society (ACS).

[DOI 10.26434/chemrxiv-2025-9cdvr](#) | [Source record](#) | Registered DOI metadata

Fine tags: Jordan Wigner

Q0409 Simulation of quantum computation with magic states via Jordan-Wigner transformations

Michael Zurel, Lawrence Z. Cohen, Robert Raussendorf.

Journal article | 2025 | Physical Review A | vol. 112 | no. 4 | article 042602.

[DOI 10.1103/ng4l-96kd](#) | [Source record](#) | Registered DOI metadata

Fine tags: Jordan Wigner

Q0410 Geometric Algebra Jordan–Wigner Transformation for Quantum Simulation

Grégoire Veyrac, Zeno Toffano.

Journal article | 2024 | Entropy | vol. 26 | no. 5 | pp. 410.

[DOI 10.3390/e26050410](#) | [Source record](#) | Registered DOI metadata

Fine tags: Jordan Wigner

Q0411 Quaternary Jordan-Wigner mapping and topological extended-kink phase in the interacting Kitaev ring

Zhen-Yu Zheng, Han-Chuan Kou, Peng Li.

Journal article | 2019 | Physical Review B | vol. 100 | no. 23 | article 235127.

[DOI 10.1103/physrevb.100.235127](#) | [Source record](#) | Registered DOI metadata

Fine tags: Jordan Wigner

Q0412 A Comparison of the Bravyi–Kitaev and Jordan–Wigner Transformations for the Quantum Simulation of Quantum Chemistry

Andrew Tranter, Peter J. Love, Florian Mintert, Peter V. Coveney.

Journal article | 2018 | Journal of Chemical Theory and Computation | vol. 14 | no. 11 | pp. 5617-5630.

[DOI 10.1021/acs.jctc.8b00450](#) | [Source record](#) | Registered DOI metadata

Fine tags: Jordan Wigner; Bravyi Kitaev

Q0413 OpenFermion | Google Quantum AI

Author metadata not supplied.

Documentation | Undated | Google Quantum AI.

[Official resource](#) | [Source record](#) | Official page retrieved

D08T03 Excited states and spectroscopy

Excited state and response algorithms target spectra beyond the ground state. State preparation and transition observable estimation are separate resource demands.

Fine subcategories: Subspace methods; equation of motion; response functions; transition amplitudes.

Primary catalog resources: 7. Related topic assignments are searchable in the HTML.

Q0414 Folded Spectrum VQE: A Quantum Computing Method for the Calculation of Molecular Excited States

Lila Cadi Tazi, Alex J. W. Thom.

Journal article | 2024 | Journal of Chemical Theory and Computation | vol. 20 | no. 6 | pp. 2491-2504.

[DOI 10.1021/acs.jctc.3c01378](#) | [Source record](#) | Registered DOI metadata

Q0415 Computing molecular excited states on a D-Wave quantum annealer

Alexander Teplukhin, Brian K. Kendrick, Susan M. Mniszewski, Yu Zhang, Ashutosh Kumar, Christian F. A. Negre, Petr M. Anisimov, Sergei Tretiak et al..

Journal article | 2021 | Scientific Reports | vol. 11 | no. 1 | article 18796.

[DOI 10.1038/s41598-021-98331-y](#) | [Source record](#) | Registered DOI metadata

Q0416 Toward Quantum Computing for High-Energy Excited States in Molecular Systems: Quantum Phase Estimations of Core-Level States

Nicholas P. Bauman, Hongbin Liu, Eric J. Bylaska, Sriram Krishnamoorthy, Guang Hao Low, Christopher E. Granade, Nathan Wiebe, Nathan A. Baker et al..

Journal article | 2020 | Journal of Chemical Theory and Computation | vol. 17 | no. 1 | pp. 201-210.

[DOI 10.1021/acs.jctc.0c00909](#) | [Source record](#) | Registered DOI metadata

Q0417 Quantum Beat Spectroscopy of Excited Molecules.

Soji TSUCHIYA.

Journal article | 1992 | Journal of the Spectroscopical Society of Japan | vol. 41 | no. 1 | pp. 3-20.

[DOI 10.5111/bunkou.41.3](#) | [Source record](#) | Registered DOI metadata

Q0418 Photoelectron spectroscopy of excited molecular states

V. McKoy, M. Braunstein, H. Rudolph, J.A. Stephens, S.N. Dixit, D.L. Lynch.

Journal article | 1990 | Journal of Electron Spectroscopy and Related Phenomena | vol. 52 | pp. 597-612.

[DOI 10.1016/0368-2048\(90\)85051-a](#) | [Source record](#) | Registered DOI metadata

Q0419 Quantum chemistry and spectroscopy of highly excited states of coordination compounds

A. V. Kondratenko, K. M. Neiman, G. L. Gutsev, G. M. Zhidomirov, S. F. Ruzankin.

Journal article | 1989 | Journal of Structural Chemistry | vol. 29 | no. 6 | pp. 899-910.

[DOI 10.1007/bf00748433](#) | [Source record](#) | Registered DOI metadata

Q0420 Excited States in Quantum Chemistry

Author metadata not supplied.

Book | 1978 | Springer Netherlands.

[DOI 10.1007/978-94-009-9902-2](#) | [Source record](#) | Registered DOI metadata

D08T04 Quantum chemistry resource estimates

Chemistry estimates translate an algorithm into logical and physical resources for a chosen accuracy. Molecular basis, code assumptions, and factory architecture should be stated.

Fine subcategories: Logical qubits; T counts; accuracy targets; molecule selection; runtime models.

Primary catalog resources: 4. Related topic assignments are searchable in the HTML.

Q0421 QREChem: quantum resource estimation software for chemistry applications

Matthew Otten, Byeol Kang, Dmitry Fedorov, Joo-Hyoung Lee, Anouar Benali, Salman Habib, Stephen K. Gray, Yuri Alexeev.

Journal article | 2023 | Frontiers in Quantum Science and Technology | vol. 2 | article 1232624.

[DOI 10.3389/frqst.2023.1232624](#) | [Source record](#) | Registered DOI metadata

Q0422 Quantum Resource Estimation for Quantum Chemistry Algorithms

Dmitry Fedorov, Matthew Otten, Byeol Kang, Anouar Benali, Salman Habib, Stephen Gray, Yuri Alexeev.
Conference paper | 2022 | 2022 IEEE International Conference on Quantum Computing and Engineering (QCE) | pp. 859-861.

[DOI 10.1109/qce53715.2022.00144](https://doi.org/10.1109/qce53715.2022.00144) | [Source record](#) | Registered DOI metadata

Q0423 Resource-Optimized Fermionic Local-Hamiltonian Simulation on a Quantum Computer for Quantum Chemistry

Qingfeng Wang, Ming Li, Christopher Monroe, Yunseong Nam.
Journal article | 2021 | Quantum | vol. 5 | pp. 509 | article 509.

[DOI 10.22331/q-2021-07-26-509](https://doi.org/10.22331/q-2021-07-26-509) | [Source record](#) | Registered DOI metadata

Q0424 Accuracy and Resource Estimations for Quantum Chemistry on a Near-Term Quantum Computer

Michael Kühn, Sebastian Zanker, Peter Deglmann, Michael Marthaler, Horst Weiß.
Journal article | 2019 | Journal of Chemical Theory and Computation | vol. 15 | no. 9 | pp. 4764-4780.

[DOI 10.1021/acs.jctc.9b00236](https://doi.org/10.1021/acs.jctc.9b00236) | [Source record](#) | Registered DOI metadata

D08T05 Quantum materials and condensed matter

Materials applications study strongly correlated phases and properties. A proposed workflow should identify observables that inform a materials decision and compare against suitable classical methods.

Fine subcategories: Strong correlation; superconductivity models; defects; electronic phases; materials observables.

Primary catalog resources: 8. Related topic assignments are searchable in the HTML.

Q0425 Condensed Matter Physics of Doped Materials: Theoretical Foundations, Electronic Structure, and Quantum Applications

MANJULA BHARATHI NAGULAPATI.

Journal article | 2025 | International Journal of Scientific Development and Research | vol. 10 | no. 11.

[DOI 10.56975/ijdsr.v10i11.306106](https://doi.org/10.56975/ijdsr.v10i11.306106) | [Source record](#) | Registered DOI metadata

Q0426 Condensed Matter and Materials Theory (NSF)

Author metadata not supplied.

Journal article | 2025 | Federal Grants & Contracts | vol. 49 | no. 7 | pp. 3-3.

[DOI 10.1002/fgc.34235](https://doi.org/10.1002/fgc.34235) | [Source record](#) | Registered DOI metadata

Q0427 Condensed matter chemistry in polymer materials

Wenke Zhang, Yu Song.

Book chapter | 2024 | Introduction to Condensed Matter Chemistry | pp. 105-140.

[DOI 10.1016/b978-0-443-16140-7.00004-3](https://doi.org/10.1016/b978-0-443-16140-7.00004-3) | [Source record](#) | Registered DOI metadata

Q0428 Condensed Matter and Materials Theory (NSF)

Author metadata not supplied.

Journal article | 2023 | Federal Grants & Contracts | vol. 47 | no. 18 | pp. 3-4.

[DOI 10.1002/fgc.33196](https://doi.org/10.1002/fgc.33196) | [Source record](#) | Registered DOI metadata

Q0429 Condensed Matter and Materials Theory (NSF)

Author metadata not supplied.

Journal article | 2022 | Federal Grants & Contracts | vol. 46 | no. 16 | pp. 3-3.

[DOI 10.1002/fgc.32482](https://doi.org/10.1002/fgc.32482) | [Source record](#) | Registered DOI metadata

AQUIRE

Q0430 Condensed Matter and Materials Theory (NSF)

Author metadata not supplied.

Journal article | 2020 | Federal Grants & Contracts | vol. 44 | no. 17 | pp. 2-2.

[DOI 10.1002/fgc.31210](#) | [Source record](#) | Registered DOI metadata

Q0431 Condensed Matter and Materials Physics

Prafulla K. Jha, Arun Pratap, Ashvin R. Jani.

Book | 2013 | Advanced Materials Research.

[DOI 10.4028/b-fa150f](#) | [Source record](#) | Registered DOI metadata

Q0432 Condensed-Matter and Materials Physics

Author metadata not supplied.

Edited book | 2007 | National Academies Press.

[DOI 10.17226/11967](#) | [Source record](#) | Registered DOI metadata

D09 Variational and hybrid quantum methods

Variational algorithms optimize a parameterized circuit with a classical loop. Trainability, sampling, noise, and optimizer costs must be evaluated along with the objective value.

Prerequisites: Optimization, parameterized circuits, and measurement statistics.

Assessment focus: Ansatz bias, trainability, shot cost, noise, and optimizer convergence.

D09T01 Variational quantum eigensolvers

VQE minimizes measured energy with a parameterized circuit and classical optimization. Ansatz bias, shot noise, and optimizer convergence all affect the result.

Fine subcategories: VQE; energy measurement; UCC ansatz; adaptive ansatz; optimizer selection.

Primary catalog resources: 18. Related topic assignments are searchable in the HTML.

Q0433 Variational quantum algorithms

M. Cerezo, Andrew Arrasmith, Ryan Babbush, Simon C. Benjamin, Suguru Endo, Keisuke Fujii, Jarrod R. McClean, Kosuke Mitarai et al..

Journal article | 2021 | Nature Reviews Physics | vol. 3 | no. 9 | pp. 625-644.

[DOI 10.1038/s42254-021-00348-9](https://doi.org/10.1038/s42254-021-00348-9) | [Source record](#) | Registered DOI metadata

Q0434 A variational eigenvalue solver on a photonic quantum processor

Alberto Peruzzo, Jarrod McClean, Peter Shadbolt, Man-Hong Yung, Xiao-Qi Zhou, Peter J. Love, Alán Aspuru-Guzik, Jeremy L. O'Brien.

Journal article | 2014 | Nature Communications | vol. 5 | no. 1 | article 4213.

[DOI 10.1038/ncomms5213](https://doi.org/10.1038/ncomms5213) | [Source record](#) | Registered DOI metadata

Q0435 Efficient variational quantum eigensolver methodologies on quantum processors

Tushar Pandey, Jason Saroni, Abdullah Kazi, Kartik Sharma.

Journal article | 2026 | Physica Scripta | vol. 101 | no. 36 | pp. 365104.

[DOI 10.1088/1402-4896/ae9e06](https://doi.org/10.1088/1402-4896/ae9e06) | [Source record](#) | Registered DOI metadata

Q0436 Geometric analysis of variational quantum eigensolver

Zhen Qin.

Journal article | 2026 | Journal of Physics A: Mathematical and Theoretical | vol. 59 | no. 30 | pp. 305302.

[DOI 10.1088/1751-8121/ae8b60](https://doi.org/10.1088/1751-8121/ae8b60) | [Source record](#) | Registered DOI metadata

Q0437 Quantum Tunneling in Adenine Deamination Using Variational Quantum Eigensolver

Don Roosan, Rubayat Khan, Saif Nirzhor, Brian Provenchar.

Conference paper | 2026 | Proceedings of the 19th International Joint Conference on Biomedical Engineering Systems and Technologies | pp. 591-599.

[DOI 10.5220/0014289200004070](https://doi.org/10.5220/0014289200004070) | [Source record](#) | Registered DOI metadata

Q0438 Weighted approximate quantum natural gradient for variational quantum eigensolver

Chenyu Shi, Vedran Dunjko, Hao Wang.

Journal article | 2026 | Quantum Science and Technology | vol. 11 | no. 1 | pp. 015060.

[DOI 10.1088/2058-9565/ae3fc8](https://doi.org/10.1088/2058-9565/ae3fc8) | [Source record](#) | Registered DOI metadata

Q0439 Drug Discovery Using Variational Quantum EigenSolver

S. Surya Prakash, P. Banu Priya, Sudarshan Someneni, Udendhran, Arnab Banerjee.

Book chapter | 2025 | Lecture Notes in Networks and Systems | pp. 691-703.

[DOI 10.1007/978-981-96-3942-7_51](https://doi.org/10.1007/978-981-96-3942-7_51) | [Source record](#) | Registered DOI metadata

Q0440 Enhancing Drug Design Using Variational Quantum Eigensolver

Rajalakshmi S, Reshmaa SS, Lalith Kannan R, Sree Harish SM, Naveen Kumar S.

Conference paper | 2025 | 2025 IEEE First International Conference on Innovations in Engineering and Next-Generation Technologies for Sustainability (ICINVENTS) | pp. 1-6.

[DOI 10.1109/icinvents64613.2025.11401986](#) | [Source record](#) | Registered DOI metadata

Q0441 Photonic variational quantum eigensolver for NISQ-compatible quantum technology

Kang-Min Hu, Min Namkung, Hyang-Tag Lim.

Journal article | 2025 | Nano Convergence | vol. 12 | no. 1 | article 60.

[DOI 10.1186/s40580-025-00525-x](#) | [Source record](#) | Registered DOI metadata

Q0442 Cascaded variational quantum eigensolver algorithm

Daniel Gunlycke, C. Stephen Hellberg, John P. T. Stenger.

Journal article | 2024 | Physical Review Research | vol. 6 | no. 1 | article 013238.

[DOI 10.1103/physrevresearch.6.013238](#) | [Source record](#) | Registered DOI metadata

Q0443 Hermitian-preserving ansatz and variational open quantum eigensolver

Zhong-Xia Shang.

Journal article | 2024 | Physical Review A | vol. 109 | no. 6 | article 062608.

[DOI 10.1103/physreva.109.062608](#) | [Source record](#) | Registered DOI metadata

Q0444 Photonic variational quantum eigensolver using entanglement measurements

Jinil Lee, Wooyeong Song, Donghwa Lee, Yosep Kim, Seung-Woo Lee, Hyang-Tag Lim, Hojoong Jung, Sang-Wook Han et al..

Journal article | 2024 | Quantum Science and Technology | vol. 9 | no. 4 | pp. 045028.

[DOI 10.1088/2058-9565/ad6d87](#) | [Source record](#) | Registered DOI metadata

Q0445 Variational Quantum Eigensolver Applications in Quantum Machine Learning

M Amshavalli, Shambhu Sharan Srivastava.

Book chapter | 2024 | Hybrid Algorithms for Quantum Computing and Artificial Intelligence | pp. 177-208.

[DOI 10.71443/9788197933646-07](#) | [Source record](#) | Registered DOI metadata

Q0446 Variational Quantum Eigensolver Boosted by Adiabatic Connection

Mikuláš Matoušek, Katarzyna Pernal, Fabijan Pavošević, Libor Veis.

Journal article | 2024 | The Journal of Physical Chemistry A | vol. 128 | no. 3 | pp. 687-698.

[DOI 10.1021/acs.jpca.3c07590](#) | [Source record](#) | Registered DOI metadata

Q0447 Variational denoising for variational quantum eigensolver

Quoc Hoan Tran, Shinji Kikuchi, Hirotaka Oshima.

Journal article | 2024 | Physical Review Research | vol. 6 | no. 2 | article 023181.

[DOI 10.1103/physrevresearch.6.023181](#) | [Source record](#) | Registered DOI metadata

Q0448 Correlated Reference-Assisted Variational Quantum Eigensolver

Nhan Trong Le, Lan Nguyen Tran.

Journal article | 2023 | The Journal of Physical Chemistry A | vol. 127 | no. 24 | pp. 5222-5230.

[DOI 10.1021/acs.jpca.3c00993](#) | [Source record](#) | Registered DOI metadata

Q0449 Orbital expansion variational quantum eigensolver

Yusen Wu, Zigeng Huang, Jinzhao Sun, Xiao Yuan, Jingbo B Wang, Dingshun Lv.

Journal article | 2023 | Quantum Science and Technology | vol. 8 | no. 4 | pp. 045030.

[DOI 10.1088/2058-9565/acf9c7](#) | [Source record](#) | Registered DOI metadata

Q0450 Symmetry enhanced variational quantum spin eigensolver

Chufan Lyu, Xusheng Xu, Man-Hong Yung, Abolfazl Bayat.

Journal article | 2023 | Quantum | vol. 7 | pp. 899 | article 899.

[DOI 10.22331/q-2023-01-19-899](#) | [Source record](#) | Registered DOI metadata

D09T02 Quantum approximate optimization

QAOA alternates cost evolution with a mixing operation to search for good solutions. Depth, constraints, parameter selection, and classical baseline quality determine the assessment.

Fine subcategories: QAOA; mixers; alternating operators; graph problems; approximation ratios.

Primary catalog resources: 16. Related topic assignments are searchable in the HTML.

Q0451 A Quantum Approximate Optimization Algorithm (QAOA) Approach to Portfolio Optimization

Yifei Huang.

Journal article | 2026 | Exploring Science Academic Conference Series | vol. 19 | pp. 628-635.

[DOI 10.70267/icfmb.202619628635](#) | [Source record](#) | Registered DOI metadata

Fine tags: QAOA

Q0452 Enhancing Quantum Approximate Optimization Algorithm Through Manifold Optimization

Qingqing Yu, Yinhui Yu, Rong Jin.

Journal article | 2026 | Quantum Engineering | vol. 2026 | no. 1 | article 3418300.

[DOI 10.1155/que2/3418300](#) | [Source record](#) | Registered DOI metadata

Q0453 Equivariant Quantum Approximate Optimization Algorithm

Boris Tselikhovski, Ilya Safro, Yuri Alexeev.

Journal article | 2026 | IEEE Transactions on Quantum Engineering | vol. 7 | pp. 1-13.

[DOI 10.1109/tqe.2026.3654930](#) | [Source record](#) | Registered DOI metadata

Q0454 Image thresholding using quantum approximate optimization algorithm and Grover's algorithm

Joseph L. Pachau, Nongmeikapam Brajabidhu Singh, Anish Kumar Saha.

Journal article | 2026 | The Journal of Supercomputing | vol. 82 | no. 9 | article 507.

[DOI 10.1007/s11227-026-08618-y](#) | [Source record](#) | Registered DOI metadata

Q0455 Iterative Interpolation Schedules for Quantum Approximate Optimization Algorithm

Anuj Apte, Shree Hari Sureshababu, Ruslan Shaydulin, Sami Boulebnane, Zichang He, Dylan Herman, James Sud, Marco Pistoia.

Journal article | 2026 | ACM Transactions on Quantum Computing | vol. 7 | no. 4 | pp. 1-18.

[DOI 10.1145/3815778](#) | [Source record](#) | Registered DOI metadata

Q0456 Quantum Approximate Optimization Algorithm: Applications, Limitations and Enhancement

P. Herrero-Gómez, M. Alemany-Manzanaro, D. Gil-Méndez, H. Mora-Mora.

Book chapter | 2026 | Communications in Computer and Information Science | pp. 147-157.

[DOI 10.1007/978-3-032-22193-3_11](#) | [Source record](#) | Registered DOI metadata

Q0457 Universal resources for quantum approximate optimization algorithm and quantum annealing

Pablo Díez-Valle, Fernando J. Gómez-Ruiz, Diego Porras, Juan José García-Ripoll.

Journal article | 2026 | Physical Review Research | vol. 8 | no. 1 | article 013211.

[DOI 10.1103/hxv2-sbr7](#) | [Source record](#) | Registered DOI metadata

Q0458 Compressed Space Quantum Approximate Optimization Algorithm for Constrained Combinatorial Optimization

Tatsuhiko Shirai, Nozomu Togawa.

Journal article | 2025 | IEEE Transactions on Quantum Engineering | vol. 6 | pp. 1-14.

[DOI 10.1109/tqe.2025.3602404](#) | [Source record](#) | Registered DOI metadata

Q0459 Exploring graph cuts in quantum approximate optimization algorithm

Nongmeikapam Brajabidhu Singh, Anish Kumar Saha.

Journal article | 2025 | Physica Scripta | vol. 100 | no. 11 | pp. 115104.

[DOI 10.1088/1402-4896/ae1906](#) | [Source record](#) | Registered DOI metadata

Q0460 Higher-Order Portfolio Optimization with Quantum Approximate Optimization Algorithm

Valter Uotila, Julia Ripatti, Bo Zhao.

Conference paper | 2025 | 2025 IEEE International Conference on Quantum Computing and Engineering (QCE) | pp. 01-12.

[DOI 10.1109/qce65121.2025.00244](#) | [Source record](#) | Registered DOI metadata

Q0461 Multiscale quantum approximate optimization algorithm

Ping Zou.

Journal article | 2025 | Physical Review A | vol. 111 | no. 1 | article 012427.

[DOI 10.1103/physreva.111.012427](#) | [Source record](#) | Registered DOI metadata

Q0462 Quantum Approximate Optimization Algorithm on Different Qubit Systems

Seongmin Kim, In-Saeng Suh, Eduardo Antonio Coello Perez, Ryan Landfield, Michael Sandoval, Josh Cunningham, Thomas Beck, Heidi Nelson-Quillin et al..

Conference paper | 2025 | 2025 IEEE International Conference on Quantum Computing and Engineering (QCE) | pp. 542-543.

[DOI 10.1109/qce65121.2025.10436](#) | [Source record](#) | Registered DOI metadata

Q0463 Warm-start adaptive-bias quantum approximate optimization algorithm

Yunlong Yu, Xiang-Bin Wang, Nic Shannon, Robert Joynt.

Journal article | 2025 | Physical Review A | vol. 112 | no. 1 | article 012422.

[DOI 10.1103/nt3w-j4mj](#) | [Source record](#) | Registered DOI metadata

Q0464 Hamiltonian-oriented homotopy quantum approximate optimization algorithm

Akash Kundu, Ludmila Botelho, Adam Glos.

Journal article | 2024 | Physical Review A | vol. 109 | no. 2 | article 022611.

[DOI 10.1103/physreva.109.022611](#) | [Source record](#) | Registered DOI metadata

Q0465 Molecular docking via quantum approximate optimization algorithm

Qi-Ming Ding, Yi-Ming Huang, Xiao Yuan.

Journal article | 2024 | Physical Review Applied | vol. 21 | no. 3 | article 034036.

[DOI 10.1103/physrevapplied.21.034036](#) | [Source record](#) | Registered DOI metadata

Q0466 Quantum Approximate Optimization Algorithm for Test Case Optimization

Xinyi Wang, Shaukat Ali, Tao Yue, Paolo Arcaini.

Journal article | 2024 | IEEE Transactions on Software Engineering | vol. 50 | no. 12 | pp. 3249-3264.

[DOI 10.1109/tse.2024.3479421](#) | [Source record](#) | Registered DOI metadata

D09T03 Ansatz design and expressibility

An ansatz defines the states reachable during optimization. Expressibility alone is not a guarantee of trainability or low resource cost.

Fine subcategories: Hardware efficient circuits; problem inspired circuits; symmetry preservation; entangling structure.

Primary catalog resources: 8. Related topic assignments are searchable in the HTML.

Q0467 Calibration of Variational Quantum Classifiers Under Depolarizing Noise: Expected Calibration Error, Ansatz Expressibility, and Post-Hoc Temperature Scaling

Souvik Ghosh, Amrita Kundu, Mithaguru, Vegi Fernando.

Conference paper | 2026 | 2026 International Conference on Intelligent and Sustainable AI Systems (ICOSAAS) | pp. 297-303.

[DOI 10.1109/icosas68663.2026.11648877](https://doi.org/10.1109/icosas68663.2026.11648877) | [Source record](#) | Registered DOI metadata

Q0468 Expressibility and Trainability Analysis of Hardware-Efficient Ansatz Variants in Variational Quantum Eigensolver with a Linear Mixing Model

Rasyid Ustman Ramadhan, Luthfiya Kurnia Permatahati, Teguh Budi Prayitno, Yanoar P. Sarwono.

Journal article | 2026 | The Journal of Physical Chemistry A | vol. 130 | no. 15 | pp. 3101-3112.

[DOI 10.1021/acs.jpca.5c08292](https://doi.org/10.1021/acs.jpca.5c08292) | [Source record](#) | Registered DOI metadata

Q0469 Genetic optimization of ansatz expressibility for enhanced variational-quantum-algorithm performance

Manish Mallapur, Ronit Raj, Ankur Raina.

Journal article | 2026 | Physical Review A | vol. 113 | no. 3 | article 032437.

[DOI 10.1103/812w-ytnb](https://doi.org/10.1103/812w-ytnb) | [Source record](#) | Registered DOI metadata

Q0470 Hamiltonian expressibility for ansatz selection in variational quantum algorithms

Filippo Brozzi, Gloria Turati, Maurizio Ferrari Dacrema, Filippo Caruso, Paolo Cremonesi.

Journal article | 2026 | Quantum Machine Intelligence | vol. 8 | no. 2 | article 76.

[DOI 10.1007/s42484-026-00407-3](https://doi.org/10.1007/s42484-026-00407-3) | [Source record](#) | Registered DOI metadata

Q0471 Radar Signal Classification with Quantum Machine Learning: Ansatz Depth Impact on Expressibility

Gabriel F. Martinez, Alberto Croci, Francesco Drago, Alessandro Niccolai, Marco Mussetta, Riccardo E. Zich.

Journal article | 2026 | Electronics | vol. 15 | no. 2 | pp. 370.

[DOI 10.3390/electronics15020370](https://doi.org/10.3390/electronics15020370) | [Source record](#) | Registered DOI metadata

Q0472 Learning the Expressibility of Quantum Circuit Ansatz Using Transformer

Fei Zhang, Jie Li, Zhimin He, Haozhen Situ.

Journal article | 2025 | Advanced Quantum Technologies | vol. 8 | no. 6 | article 2400366.

[DOI 10.1002/qute.202400366](https://doi.org/10.1002/qute.202400366) | [Source record](#) | Registered DOI metadata

Q0473 The unified effect of data encoding, ansatz expressibility and entanglement on the trainability of HQNNs

Muhammad Kashif, Saif Al-Kuwari.

Journal article | 2023 | International Journal of Parallel, Emergent and Distributed Systems | vol. 38 | no. 5 | pp. 362-400.

[DOI 10.1080/17445760.2023.2231163](https://doi.org/10.1080/17445760.2023.2231163) | [Source record](#) | Registered DOI metadata

Q0474 Connecting Ansatz Expressibility to Gradient Magnitudes and Barren Plateaus

Zoë Holmes, Kunal Sharma, M. Cerezo, Patrick J. Coles.

Journal article | 2022 | PRX Quantum | vol. 3 | no. 1 | article 010313.

[DOI 10.1103/prxquantum.3.010313](https://doi.org/10.1103/prxquantum.3.010313) | [Source record](#) | Registered DOI metadata

D09T04 Barren plateaus and trainability

Barren plateau analyses study exponentially small or concentrated gradients. The conclusion depends on circuit ensembles, cost locality, initialization, and noise.

Fine subcategories: Gradient concentration; depth dependence; local costs; initialization; noise induced plateaus.

Primary catalog resources: 8. Related topic assignments are searchable in the HTML.

Q0475 BRIDG-Q: Barren-Plateau-Resilient Initialisation with Data-Aware LLM-Generated Quantum Circuits

Ngoc Nhi Nguyen, Thai T. Vu, John Le, Hoa Khanh Dam, Dung Hoang Duong, Dinh Thai Hoang.

Book chapter | 2026 | IFIP Advances in Information and Communication Technology | pp. 434-448.

[DOI 10.1007/978-3-032-27993-4_30](#) | [Source record](#) | Registered DOI metadata

Q0476 Benchmarking Barren Plateau Mitigation Strategies in Quantum Neural Networks on Standard and Medical Image Datasets

Maqsdur Rahman, Rui Liu, Anup Majumder, Pintu Chandra Paul, Kangtong Mo, Amena Begum, Kashmi Sultana, Nahida Akter et al..

Journal article | 2026 | Journal of Imaging | vol. 12 | no. 7 | pp. 275.

[DOI 10.3390/jimaging12070275](#) | [Source record](#) | Registered DOI metadata

Q0477 Deep Variational Quantum Circuits with Barren-Plateau-Free Architectures

Kaining Zhang, Min-Hsiu Hsieh, Dacheng Tao.

Journal article | 2026 | Artificial Intelligence Science and Engineering | vol. 2 | no. 1 | pp. 66-84.

[DOI 10.23919/aise.2026.000005](#) | [Source record](#) | Registered DOI metadata

Q0478 Fourier Analysis of Parameterized Quantum Circuits and the Barren Plateau Problem

Shun Okumura, Masayuki Ohzeki.

Journal article | 2026 | Journal of the Physical Society of Japan | vol. 95 | no. 3 | article 033002.

[DOI 10.7566/jpsj.95.033002](#) | [Source record](#) | Registered DOI metadata

Q0479 Illustration of Barren Plateaus in Quantum Computing

Gerhard Stenzel, Tobias Rohe, Michael Kölle, Leo Sünkel, Jonas Stein, Claudia Linnhoff-Popien.

Conference paper | 2026 | Proceedings of the 18th International Conference on Agents and Artificial Intelligence | pp. 731-742.

[DOI 10.5220/0014310300004052](#) | [Source record](#) | Registered DOI metadata

Q0480 Lie-geometric trainability of quantum dynamical systems: avoiding barren plateaus via low-dimensional Lie subalgebras

Haijian Shao, Yujie Wu, Xing Deng, Yingtao Jiang.

Journal article | 2026 | Physica Scripta | vol. 101 | no. 25 | pp. 255107.

[DOI 10.1088/1402-4896/ae7d6d](#) | [Source record](#) | Registered DOI metadata

Q0481 Mitigating the barren plateau problem in linear optics

Anonymous.

Journal article | 2026 | Physical Review A.

[DOI 10.1103/xndq-tdtj](#) | [Source record](#) | Registered DOI metadata

Q0482 Quantum Optimization for Reconfigurable Intelligent Surface Design: Trainability and Barren Plateau Analysis

Emanuel Colella, Luca Bastianelli, Valter Mariani Primiani, Franco Moglie, Zhen Peng, Gabriele Gradoni.

Journal article | 2026 | IEEE Open Journal of Antennas and Propagation | pp. 1-1.

[DOI 10.1109/ojap.2026.3705248](#) | [Source record](#) | Registered DOI metadata

D09T05 Quantum gradients and optimization

Gradient rules and optimization methods control the classical loop around a quantum circuit. Measurement complexity and gradient variance must accompany iteration counts.

Fine subcategories: Parameter shift rules; finite differences; stochastic gradients; quantum natural gradients.

Primary catalog resources: 8. Related topic assignments are searchable in the HTML.

Q0483 Efficient Optimization of Variational Quantum Algorithms via Gradient-Free Parameter Prediction

Satwik Kundu, Debarshi Kundu, Swaroop Ghosh.

Journal article | 2026 | IEEE Access | vol. 14 | pp. 42485-42499.

[DOI 10.1109/access.2026.3674830](#) | [Source record](#) | Registered DOI metadata

Q0484 Quantum Federated Gradient Aggregation Using the Parameter-Shift Rule

Jaehyun Chung, Chaemoon Im, Soohyun Park, Joongheon Kim, Wonjun Lee.

Journal article | 2026 | IEEE Internet of Things Journal | vol. 13 | no. 11 | pp. 25367-25379.

[DOI 10.1109/jiot.2026.3677003](#) | [Source record](#) | Registered DOI metadata

Q0485 Quantum Optimization via Gradient-Based Hamiltonian Descent

Jiaqi Leng, Bin Shi.

Book chapter | 2026 | Communications in Computer and Information Science | pp. 170-177.

[DOI 10.1007/978-981-95-7829-0_14](#) | [Source record](#) | Registered DOI metadata

Q0486 Gradient-Based Optimization for Linear Quantum Systems

Yunyan Lee, Ian R. Petersen, Daoyi Dong.

Conference paper | 2025 | 2025 IEEE 64th Conference on Decision and Control (CDC) | pp. 6679-6684.

[DOI 10.1109/cdc57313.2025.11312520](#) | [Source record](#) | Registered DOI metadata

Q0487 Natural gradient and parameter estimation for quantum Boltzmann machines

Dhruvil Patel, Mark M. Wilde.

Journal article | 2025 | Physical Review A | vol. 112 | no. 5 | article 052421.

[DOI 10.1103/j8nb-by4l](#) | [Source record](#) | Registered DOI metadata

Q0488 Photonic parameter-shift rule: Enabling gradient computation for photonic quantum computers

Axel Pappalardo, Pierre-Emmanuel Emeriau, Giovanni de Felice, Brian Ventura, Hugo Jaunin, Richie Yeung, Bob Coecke, Shane Mansfield.

Journal article | 2025 | Physical Review A | vol. 111 | no. 3 | article 032429.

[DOI 10.1103/physreva.111.032429](#) | [Source record](#) | Registered DOI metadata

Q0489 Efficient Parameter-Shift Rule Implementation for Computing Gradient on Quantum Simulators

Vu Tuan Hai, Le Vu Trung Duong, Pham Hoai Luan, Yasuhiko Nakashima.

Conference paper | 2024 | 2024 International Conference on Advanced Technologies for Communications (ATC) | pp. 449-454.

[DOI 10.1109/atc63255.2024.10908246](#) | [Source record](#) | Registered DOI metadata

Q0490 General parameter-shift rules for quantum gradients

David Wierichs, Josh Izaac, Cody Wang, Cedric Yen-Yu Lin.

Journal article | 2022 | Quantum | vol. 6 | pp. 677 | article 677.

[DOI 10.22331/q-2022-03-30-677](#) | [Source record](#) | Registered DOI metadata

Fine tags: Parameter shift rules

D09T06 Quantum imaginary time evolution

Imaginary time methods project toward low energy states through nonunitary evolution or approximations. Normalization and locality assumptions determine what is implementable.

Fine subcategories: QITE; variational imaginary time; imaginary time filtering; normalization; thermal preparation.

Primary catalog resources: 16. Related topic assignments are searchable in the HTML.

Q0491 Deterministic Quantum Trajectory via Imaginary Time Evolution

Shivan Mittal, Bin Yan.

Journal article | 2026 | Physical Review Letters | vol. 136 | no. 1 | article 010401.

[DOI 10.1103/fgkg-n2b9](#) | [Source record](#) | Registered DOI metadata

Q0492 Double-Bracket Quantum Algorithms for Quantum Imaginary-Time Evolution

Marek Gluza, Jeongrak Son, Bi Hong Tiang, René Zander, Raphael Seidel, Yudai Suzuki, Zoë Holmes, Nelly H. Y. Ng.

Journal article | 2026 | Physical Review Letters | vol. 136 | no. 2 | article 020601.

[DOI 10.1103/rw81-k8vk](#) | [Source record](#) | Registered DOI metadata

Q0493 Equating quantum imaginary time evolution, Riemannian gradient flows, and stochastic implementations

Nathan A. McMahon, Mahum Pervez, Christian Arenz.

Journal article | 2026 | Physical Review Research | vol. 8 | no. 2 | article 023024.

[DOI 10.1103/ht2m-1j91](#) | [Source record](#) | Registered DOI metadata

Q0494 Operator-projected variational quantum imaginary time evolution

Aeishah Ameerah Anuar, François Jamet, Fabio Gironella, Fedor Šimkovic IV, Riccardo Rossi.

Journal article | 2026 | Journal of Physics A: Mathematical and Theoretical | vol. 59 | no. 24 | pp. 245301.

[DOI 10.1088/1751-8121/ae767d](#) | [Source record](#) | Registered DOI metadata

Fine tags: variational imaginary time

Q0495 Optimizing QUBO on a quantum computer by mimicking imaginary time evolution

Yahui Chai, Alice Di Tucci.

Journal article | 2026 | New Journal of Physics | vol. 28 | no. 6 | pp. 064508.

[DOI 10.1088/1367-2630/ae6ded](#) | [Source record](#) | Registered DOI metadata

Q0496 Physics-inspired transformer quantum states via latent imaginary-time evolution

Kimihiro Yamazaki, Itsushi Sakata, Takuya Konishi, Yoshinobu Kawahara.

Journal article | 2026 | Physical Review Research | vol. 8 | no. 3 | article 033032.

[DOI 10.1103/bjxb-8tsk](#) | [Source record](#) | Registered DOI metadata

Q0497 Quantum imaginary-time evolution with polynomial resources in evolution time

Lei Zhang, Jizhe Lai, Xian Wu, Xin Wang.

Journal article | 2026 | Quantum Science and Technology | vol. 11 | no. 3 | pp. 035041.

[DOI 10.1088/2058-9565/ae7b7f](#) | [Source record](#) | Registered DOI metadata

Q0498 Quasiprobabilistic Imaginary-Time Evolution on Quantum Computers

Annie Ray, Esha Swaroop, Ningping Cao, Michael Vasmer, Anirban Chowdhury.

Journal article | 2026 | Quantum Information & Computation | vol. 26 | no. 1 | pp. 89-113.

[DOI 10.2478/qic-2026-0005](#) | [Source record](#) | Registered DOI metadata

Q0499 Saturable quantum speed limits for imaginary-time evolution

Kohei Kobayashi.

Journal article | 2026 | Physical Review A | vol. 113 | no. 3 | article 032618.

[DOI 10.1103/bln8-8xzy](#) | [Source record](#) | Registered DOI metadata

Q0500 Understanding Quantum Imaginary Time Evolution and Its Variational Form

Andreu Anglés-Castillo, Luca Ion, Tanmoy Pandit, Rafael Gomez-Lurbe, Rodrigo Martínez, Miguel Angel Garcia-March.

Book chapter | 2026 | Lecture Notes in Networks and Systems | pp. 269-279.

[DOI 10.1007/978-3-032-05748-8_22](#) | [Source record](#) | Registered DOI metadata

Fine tags: variational imaginary time

Q0501 Accelerating Large-Scale Linear Algebra Using Variational Quantum Imaginary Time Evolution

Willie Aboumrad, Daiwei Zhu, Claudio Girotto, François-Henry Rouet, Jezer Jojo, Robert Lucas, Jay Pathak, Ananth Kaushik et al..

Conference paper | 2025 | 2025 IEEE International Conference on Quantum Computing and Engineering (QCE) | pp. 1965-1970.

[DOI 10.1109/qce65121.2025.00214](https://doi.org/10.1109/qce65121.2025.00214) | [Source record](#) | Registered DOI metadata

Fine tags: variational imaginary time

Q0502 Accelerating quantum imaginary-time evolution with random measurements

Ioannis Kolotouros, David Joseph, Anand Kumar Narayanan.

Journal article | 2025 | Physical Review A | vol. 111 | no. 1 | article 012424.

[DOI 10.1103/physreva.111.012424](https://doi.org/10.1103/physreva.111.012424) | [Source record](#) | Registered DOI metadata

Q0503 Measurement-Based Variational Quantum Imaginary Time Evolution

Kübra Yeter-Aydeniz, Nora Bauer, George Siopsis.

Conference paper | 2025 | 2025 IEEE International Conference on Quantum Computing and Engineering (QCE) | pp. 408-409.

[DOI 10.1109/qce65121.2025.10369](https://doi.org/10.1109/qce65121.2025.10369) | [Source record](#) | Registered DOI metadata

Fine tags: variational imaginary time

Q0504 Role of Riemannian Geometry in Double-Bracket Quantum Imaginary-Time Evolution

René Zander, Raphael Seidel, Li Xiaoyue, Marek Gluza.

Book chapter | 2025 | Lecture Notes in Computer Science | pp. 105-114.

[DOI 10.1007/978-3-032-03924-8_11](https://doi.org/10.1007/978-3-032-03924-8_11) | [Source record](#) | Registered DOI metadata

Q0505 Adiabatic quantum imaginary time evolution

Kasra Hejazi, Mario Motta, Garnet Kin-Lic Chan.

Journal article | 2024 | Physical Review Research | vol. 6 | no. 3 | article 033084.

[DOI 10.1103/physrevresearch.6.033084](https://doi.org/10.1103/physrevresearch.6.033084) | [Source record](#) | Registered DOI metadata

Q0506 Combinatorial optimization with quantum imaginary time evolution

Nora M. Bauer, Rizwanul Alam, George Siopsis, James Ostrowski.

Journal article | 2024 | Physical Review A | vol. 109 | no. 5 | article 052430.

[DOI 10.1103/physreva.109.052430](https://doi.org/10.1103/physreva.109.052430) | [Source record](#) | Registered DOI metadata

D10 Quantum machine learning

Quantum machine learning includes learning with quantum circuits and learning about quantum systems. Input encoding, sample complexity, classical comparisons, and trainability determine whether a method is useful.

Prerequisites: Machine learning, statistics, and quantum circuits.

Assessment focus: Input encoding, sample complexity, generalization, and classical comparison.

D10T01 Quantum data encoding

Data encodings specify how classical or quantum inputs enter a circuit. State preparation cost and repeated access must be included in learning complexity.

Fine subcategories: Amplitude encoding; angle encoding; basis encoding; data reuploading; state preparation costs.

Primary catalog resources: 8. Related topic assignments are searchable in the HTML.

Q0507 Benchmarking Data Encoding Methods in Quantum Machine Learning

Orlane Zang, Grégoire Barraué, Tony Quertier.

Book chapter | 2026 | Communications in Computer and Information Science | pp. 270-279.

[DOI 10.1007/978-3-032-13852-1_27](https://doi.org/10.1007/978-3-032-13852-1_27) | [Source record](#) | Registered DOI metadata

Q0508 Encoding Numerical Data for Generative Quantum Machine Learning

Michael Krebsbach, Florentin Reiter, Thomas Wellens, Hagen-Henrik Kowalski, Ali Abedi.

Journal article | 2026 | Quantum Science and Technology.

[DOI 10.1088/2058-9565/aeaf7e](https://doi.org/10.1088/2058-9565/aeaf7e) | [Source record](#) | Registered DOI metadata

Q0509 Fast quantum amplitude encoding of typical classical data

Vittorio Pagni, Sigurd Huber, Michael Epping, Michael Felderer.

Journal article | 2026 | EPJ Quantum Technology | vol. 13 | no. 1 | article 24.

[DOI 10.1140/epjqt/s40507-026-00473-3](https://doi.org/10.1140/epjqt/s40507-026-00473-3) | [Source record](#) | Registered DOI metadata

Fine tags: Amplitude encoding

Q0510 Feature Space Dimensional Reduction Based on Quantum Spin-Correlation Encoding

Chee Kian Yap.

Posted content | 2026 | MDPI AG.

[DOI 10.20944/preprints202601.0671.v1](https://doi.org/10.20944/preprints202601.0671.v1) | [Source record](#) | Registered DOI metadata

Q0511 Quantum State Preparation for Classical Data Encoding

Miguel A. A. Lisboa, Victor H. F. Brasil, João V. H. Duarte, Leandro C. Souza.

Conference paper | 2026 | Anais do I Simpósio Brasileiro de Computação e Comunicação Quânticas (SBCCQ 2026) | pp. 119-130.

[DOI 10.5753/sbccq.2026.20984](https://doi.org/10.5753/sbccq.2026.20984) | [Source record](#) | Registered DOI metadata

Q0512 Analisis Performa Variational Quantum Classifier (VQC) dengan ZZ Feature Map dan Angle Encoding Untuk Mengidentifikasi Serangan Jantung

Hilmia Rahma, Dahlan Abdullah, Desvina Yulisda.

Journal article | 2025 | Bulletin of Computer Science Research | vol. 5 | no. 5 | pp. 898-907.

[DOI 10.47065/bulletincsr.v5i5.712](https://doi.org/10.47065/bulletincsr.v5i5.712) | [Source record](#) | Registered DOI metadata

Fine tags: angle encoding

Q0513 Improving Data Encoding for Enhanced AI Performance in Complex Datasets via Quantum Feature Space Mapping: Harnessing Quantum Algorithms

Gurpreet Singh Walia, Sathiya Priya S, KV Karthikeya, D. Suresh, P Sudheer, V. Syambabu.

Conference paper | 2025 | 2025 International Conference on Pervasive Computational Technologies (ICPCT) | pp. 985-990.

[DOI 10.1109/icpct64145.2025.10940638](https://doi.org/10.1109/icpct64145.2025.10940638) | [Source record](#) | Registered DOI metadata

Q0514 A Parallel Quantum Feature Encoding Scheme for Effective Classical Data Classification in Quantum Convolutional Neural Networks

Raisa Mashtura, Jishnu Mahmud, Shaikh Anowarul Fattah, Mohammad Saquib.

Conference paper | 2023 | TENCON 2023 - 2023 IEEE Region 10 Conference (TENCON) | pp. 1-5.

[DOI 10.1109/tencon58879.2023.10322543](https://doi.org/10.1109/tencon58879.2023.10322543) | [Source record](#) | Registered DOI metadata

D10T02 Quantum kernels

Quantum kernels estimate similarities using quantum feature maps. Expressive features can still yield concentrated kernels or costly training measurements.

Fine subcategories: Fidelity kernels; projected kernels; kernel concentration; support vector machines; feature maps.

Primary catalog resources: 16. Related topic assignments are searchable in the HTML.

Q0515 Double Descent in Quantum Kernel Methods

Marie Kempkes, Aroosa Ijaz, Elies Gil-Fuster, Carlos Bravo-Prieto, Jakob Spiegelberg, Evert van Nieuwenburg, Vedran Dunjko.

Journal article | 2026 | PRX Quantum | vol. 7 | no. 1 | article 010312.

[DOI 10.1103/cn64-gs6b](https://doi.org/10.1103/cn64-gs6b) | [Source record](#) | Registered DOI metadata

Q0516 Leveraging Quantum Kernel Methods for Enhanced Intrusion Detection

Anuj Tiwari, Sameer G. Kulkarni.

Conference paper | 2026 | 2026 18th International Conference on COMmunication Systems and NETworks (COMSNETS) | pp. 567-572.

[DOI 10.1109/comsnets67989.2026.11418222](https://doi.org/10.1109/comsnets67989.2026.11418222) | [Source record](#) | Registered DOI metadata

Q0517 Novel Algorithm for Adaptive Circuit Construction for Quantum Kernel Methods

Emanouil Atanassov, Aneta Karaivanova, Svetlozar Yordanov, Stanislav Spasov, Mariya Durchova, Alexandar Kirilov.

Book chapter | 2026 | Lecture Notes in Computer Science | pp. 379-387.

[DOI 10.1007/978-3-032-22221-3_42](https://doi.org/10.1007/978-3-032-22221-3_42) | [Source record](#) | Registered DOI metadata

Q0518 Qmes: Quantum Meta-Learning for Encoding Selection in Quantum Kernel Methods

Duy Tung Dao, Quoc Chuong Nguyen, Tuan Hai Vu, Le Bin Ho, Nguyen Lan Tran.

Posted content | 2026 | Elsevier BV.

[DOI 10.2139/ssrn.7434491](https://doi.org/10.2139/ssrn.7434491) | [Source record](#) | Registered DOI metadata

Q0519 Quantum Kernel Methods for Brain Aneurysm Risk Classification

Sangeeta Yadav, Harshit Yadav, Anusheel Munshi, Roshan M Dsouza.

Book chapter | 2026 | Communications in Computer and Information Science | pp. 66-82.

[DOI 10.1007/978-3-032-17625-7_5](https://doi.org/10.1007/978-3-032-17625-7_5) | [Source record](#) | Registered DOI metadata

Q0520 Quantum Kernel Methods for High Dimensional Data Classification

Maharaj Manohar, Prakash Shanmugam, Chella Pandi.

Journal article | 2026 | International Journal of Quantum Computing and Artificial Intelligence (IJQCAI) | vol. 1 | no. 2 | pp. 31-44.

[DOI 10.67714/3139-5201.vol.1.issue.2.3](https://doi.org/10.67714/3139-5201.vol.1.issue.2.3) | [Source record](#) | Registered DOI metadata

Q0521 Quantum Kernel Methods for Industrial Anomaly Detection

Takao Tomono.

Book chapter | 2026 | Communications in Computer and Information Science | pp. 64-85.

[DOI 10.1007/978-981-95-7829-0_8](#) | [Source record](#) | Registered DOI metadata

Q0522 Quantum Support Vector Machines and Quantum Kernel Methods

Li Xu, Jing Wang, Jiang Chen, Jiamin Xu, Ming Li, Weishan Zhang.

Journal article | 2026 | Software: Practice and Experience | vol. 56 | no. 6 | pp. 786-801.

[DOI 10.1002/spe.70070](#) | [Source record](#) | Registered DOI metadata

Fine tags: support vector machines

Q0523 Quo Vadis, Quantum Machine Learning?: Quantum kernel methods meet Earth observation

Artur Miroszewski, Jakub Nalepa, Agata M. Wijata, Jakub Mielczarek, Bertrand Le Saux, Alessandro Sebastianelli.

Journal article | 2026 | IEEE Geoscience and Remote Sensing Magazine | vol. 14 | no. 1 | pp. 306-334.

[DOI 10.1109/mgrs.2025.3596291](#) | [Source record](#) | Registered DOI metadata

Q0524 Spectral phase encoding for quantum kernel methods

Pablo Herrero Gómez, Antonio Jimeno Morenilla, David Muñoz-Hernández, Higinio Mora Mora.

Journal article | 2026 | EPJ Quantum Technology | vol. 13 | no. 1 | article 66.

[DOI 10.1140/epjqt/s40507-026-00509-8](#) | [Source record](#) | Registered DOI metadata

Q0525 The Cross-Kernel Margin: A Robustness Measure for Quantum Kernel Methods

S. Govender, I. Sinayskiy.

Journal article | 2026 | Advanced Quantum Technologies | vol. 9 | no. 9 | article e70459.

[DOI 10.1002/qute.70459](#) | [Source record](#) | Registered DOI metadata

Q0526 Expressivity vs. Generalization in Quantum Kernel Methods

Markus Gross, Markus Lange, Bogusz Bujnowski, Hans-Martin Rieser.

Conference paper | 2025 | ESANN 2025 proceedings | pp. 543-548.

[DOI 10.14428/esann/2025.es2025-152](#) | [Source record](#) | Registered DOI metadata

Q0527 Quantum Kernel Methods

Yuxuan Du, Xinbiao Wang, Naixu Guo, Zhan Yu, Yang Qian, Kaining Zhang, Min-Hsiu Hsieh, Patrick Rebentrost et al..

Book chapter | 2025 | A Gentle Introduction to Quantum Machine Learning | pp. 69-110.

[DOI 10.1007/978-981-95-1284-3_3](#) | [Source record](#) | Registered DOI metadata

Q0528 Quantum Support Vector Machines and Quantum Kernel Methods

Li Xu, Jing Wang, Chen Jiang, Jiamin Xu, Ming Li, Weishan Zhang.

Posted content | 2025 | Wiley.

[DOI 10.22541/au.175615204.43126822/v1](#) | [Source record](#) | Registered DOI metadata

Fine tags: support vector machines

Q0529 Quantum adversarial learning for kernel methods

Giuseppe Montalbano, Leonardo Banchi.

Journal article | 2025 | Quantum Machine Intelligence | vol. 7 | no. 1 | article 15.

[DOI 10.1007/s42484-025-00238-8](#) | [Source record](#) | Registered DOI metadata

Q0530 Quantum kernel methods under scrutiny: a benchmarking study

Jan Schnabel, Marco Roth.

Journal article | 2025 | Quantum Machine Intelligence | vol. 7 | no. 1 | article 58.

[DOI 10.1007/s42484-025-00273-5](#) | [Source record](#) | Registered DOI metadata

D10T03 Quantum neural networks and classifiers

Quantum neural networks use trainable quantum circuits as learning models. Their evaluation requires held out data, noise analysis, and comparable classical baselines.

Fine subcategories: Variational classifiers; quantum convolution; circuit architectures; supervised learning.

Primary catalog resources: 16. Related topic assignments are searchable in the HTML.

Q0531 Quantum Machine Learning

Andrea Delgado, Kathleen E Hamilton.

Book | 2025 | IOP Publishing.

DOI [10.1088/978-0-7503-4952-9](https://doi.org/10.1088/978-0-7503-4952-9) | [Source record](#) | Registered DOI metadata

Q0532 Quantum machine learning

Jacob Biamonte, Peter Wittek, Nicola Pancotti, Patrick Rebentrost, Nathan Wiebe, Seth Lloyd.

Journal article | 2017 | Nature | vol. 549 | no. 7671 | pp. 195-202.

DOI [10.1038/nature23474](https://doi.org/10.1038/nature23474) | [Source record](#) | Registered DOI metadata

Q0533 Quantum Neural Network Classifier for Cancer Registry System Testing: A Feasibility Study

Xinyi Wang, Shaukat Ali, Paolo Arcaini, Narasimha Raghavan Veeraragavan, Jan F. Nygård.

Journal article | 2026 | ACM Transactions on Software Engineering and Methodology | vol. 35 | no. 5 | pp. 1-24.

DOI [10.1145/3769302](https://doi.org/10.1145/3769302) | [Source record](#) | Registered DOI metadata

Q0534 Quantum Machine Learning Classifier and Neural Network Transfer Learning

Pauline Mosley, Avery Leider.

Book chapter | 2025 | Artificial Intelligence.

DOI [10.5772/intechopen.115051](https://doi.org/10.5772/intechopen.115051) | [Source record](#) | Registered DOI metadata

Q0535 Quantum neural network classifier with differential privacy

Guodong Li, Fangce Yu, Qingle Wang, Lin Liu, Ying Mao, Long Cheng.

Journal article | 2025 | Physica Scripta | vol. 100 | no. 3 | pp. 035109.

DOI [10.1088/1402-4896/adb228](https://doi.org/10.1088/1402-4896/adb228) | [Source record](#) | Registered DOI metadata

Q0536 Hybrid Quantum Machine Learning Classifier with Classical Neural Network Transfer Learning

Avery Leider, Gio Giorgio Abou Jaoude, Pauline Mosley.

Book chapter | 2023 | Lecture Notes in Networks and Systems | pp. 102-116.

DOI [10.1007/978-3-031-28073-3_8](https://doi.org/10.1007/978-3-031-28073-3_8) | [Source record](#) | Registered DOI metadata

Q0537 Infected Plant Leaves Detection Using Multilayered Convolutional Neural Network and Quantum Classifier

Damandeep Kaur, Shamandeep Singh, Simarjeet Kaur, Gurpreet Singh, Rani Kumari.

Book chapter | 2023 | Advances in Bioinformatics and Biomedical Engineering | pp. 110-126.

DOI [10.4018/979-8-3693-1479-1.ch007](https://doi.org/10.4018/979-8-3693-1479-1.ch007) | [Source record](#) | Registered DOI metadata

Fine tags: quantum convolution

Q0538 On Hybrid Artificial Neural Networks and Variational Quantum Classifier for Network Intrusion Detection

Praveen Venkatachalam, David Q. Liu.

Conference paper | 2023 | 2023 International Conference on Cyber-Enabled Distributed Computing and Knowledge Discovery (CyberC) | pp. 410-416.

DOI [10.1109/cyberc58899.2023.00070](https://doi.org/10.1109/cyberc58899.2023.00070) | [Source record](#) | Registered DOI metadata

Q0539 Quantum neural network autoencoder and classifier applied to an industrial case study

Stefano Mangini, Alessia Marruzzo, Marco Piantanida, Dario Gerace, Daniele Bajoni, Chiara Macchiavello.

Journal article | 2022 | Quantum Machine Intelligence | vol. 4 | no. 2 | article 13.

[DOI 10.1007/s42484-022-00070-4](https://doi.org/10.1007/s42484-022-00070-4) | [Source record](#) | Registered DOI metadata

Q0540 Adversarial attacks and robustness for quantum machine learning | PennyLane Demos

Author metadata not supplied.

Tutorial | Undated | PennyLane.

[Official resource](#) | [Source record](#) | Official page retrieved

Q0541 Demos — PennyLane

Author metadata not supplied.

Tutorial | Undated | PennyLane.

[Official resource](#) | [Source record](#) | Official page retrieved

Q0542 GitHub - PennyLaneAI/pennylane: PennyLane is an open-source quantum software platform for quantum computing, quantum machine learning, and quantum chemistry. Create meaningful quantum algorithms, from inspiration to implementation. · GitHub

Author metadata not supplied.

Software repository | Undated | PennyLane.

[Official resource](#) | [Source record](#) | Official page retrieved

Q0543 GitHub - qiskit-community/qiskit-machine-learning: An open-source library built on Qiskit for quantum machine learning tasks at scale on quantum hardware and classical simulators · GitHub

Author metadata not supplied.

Software repository | Undated | Qiskit Machine Learning.

[Official resource](#) | [Source record](#) | Official page retrieved

Q0544 Introduction to Geometric Quantum Machine Learning | PennyLane Demos

Author metadata not supplied.

Tutorial | Undated | PennyLane.

[Official resource](#) | [Source record](#) | Official page retrieved

Q0545 PennyLane Documentation — PennyLane 0.45.1 documentation

Author metadata not supplied.

Documentation | Undated | PennyLane.

[Official resource](#) | [Source record](#) | Official page retrieved

Q0546 Post Variational Quantum Neural Networks | PennyLane Demos

Author metadata not supplied.

Tutorial | Undated | PennyLane.

[Official resource](#) | [Source record](#) | Official page retrieved

D10T04 Quantum generative models

Generative quantum models represent or sample data distributions. Training objectives, sample quality, likelihood access, and the data source define the task.

Fine subcategories: Born machines; quantum GANs; sampling models; generative benchmarks; likelihood estimation.

Primary catalog resources: 16. Related topic assignments are searchable in the HTML.

Q0547 Conditioning in generative quantum denoising diffusion models

Daniel Quinn, Lorenzo Buffoni, Stefano Gherardini, Gabriele De Chiara.

Journal article | 2026 | Quantum Science and Technology | vol. 11 | no. 3 | pp. 035047.

[DOI 10.1088/2058-9565/ae8881](#) | [Source record](#) | Registered DOI metadata

Q0548 Diabatic quantum annealing for training energy-based generative models

Gilhan Kim, Ju-Yeon Gychm, Daniel K. Park.

Journal article | 2026 | Physical Review E | vol. 113 | no. 3 | article 035302.

[DOI 10.1103/2g6m-whm2](#) | [Source record](#) | Registered DOI metadata

Q0549 Enhancing Breast Cancer Diagnosis with Quantum-Driven Generative Adversarial Models

V. Shankar Ganesh, A. Purushothaman, A. Abdul Hayum, K. Karuppusamy.

Journal article | 2026 | International Journal of Drug Delivery Technology | vol. 16 | no. 7.

[DOI 10.25258/ijddt.16.7.110](#) | [Source record](#) | Registered DOI metadata

Q0550 Q-Tag: watermarking quantum circuit generative models

Yang Yang, Yuzhu Long, Han Fang, Zhaoyun Chen, Zhonghui Li, Weiming Zhang, Guoping Guo.

Journal article | 2026 | Science China Information Sciences | vol. 69 | no. 8 | article 180504.

[DOI 10.1007/s11432-025-5016-2](#) | [Source record](#) | Registered DOI metadata

Q0551 Quantum-classical generative models for drug design

Prateek Jain, Param Pathak, Krishna Bhatia, Shalini Devendrababu, Srinjoy Ganguly.

Journal article | 2026 | Quantum Machine Intelligence | vol. 8 | no. 1 | article 30.

[DOI 10.1007/s42484-026-00356-x](#) | [Source record](#) | Registered DOI metadata

Q0552 Towards Stable and Diverse Generative Models with Dual Hybrid Quantum Generative Adversarial Network

S. Kushal Varma, Maski Yashasv, Lanka Dhanush, Kotari Harshith, Srinath Devale.

Book chapter | 2026 | Lecture Notes in Computer Science | pp. 388-399.

[DOI 10.1007/978-3-032-26352-0_32](#) | [Source record](#) | Registered DOI metadata

Q0553 A Review of Quantum-Based Diffusion Models in Generative AI

Glauco Lima, Ernestas Filatovas, Marco Marcozzi, Remigijus Paulavičius.

Journal article | 2025 | Vilnius University Open Series | pp. 109-120.

[DOI 10.15388/lmitt.2025.14](#) | [Source record](#) | Registered DOI metadata

Q0554 Adaptive Quantum Channels as Long-Memory Generative Models

Charlee Stefanski, Vanio Markov, David Novak, Vladimir Rastunkov.

Conference paper | 2025 | Proceedings of the 6th ACM International Conference on AI in Finance | pp. 492-500.

[DOI 10.1145/3768292.3770440](#) | [Source record](#) | Registered DOI metadata

Q0555 Back Cover: Quantum-Noise-Driven Generative Diffusion Models (Adv. Quantum Technol. 12/2025)

Marco Parigi, Stefano Martina, Filippo Caruso.

Journal article | 2025 | Advanced Quantum Technologies | vol. 8 | no. 12 | article e70122.

[DOI 10.1002/qute.70122](#) | [Source record](#) | Registered DOI metadata

Q0556 DisQu: Investigating the Impact of Disorder in Quantum Generative Models

Yannick Werner, Jasmin Frkatovic, Vitor Fortes Rey, Matthias Tschöpe, Sungho Suh, Paul Lukowicz, Nikolaos Palaodimopoulos, Maximilian Kiefer-Emmanouilidis.

Conference paper | 2025 | 2025 International Joint Conference on Neural Networks (IJCNN) | pp. 1-8.

[DOI 10.1109/ijcnn64981.2025.11227841](#) | [Source record](#) | Registered DOI metadata

Q0557 Experimental demonstration of reconstructing quantum states with generative models

Xuegang Li, Wenjie Jiang, Ziyue Hua, Weiting Wang, Xiaoxuan Pan, Weizhou Cai, Zhide Lu, Jiaxiu Han et al..

Journal article | 2025 | Science Bulletin | vol. 70 | no. 10 | pp. 1572-1575.

[DOI 10.1016/j.scib.2025.03.007](#) | [Source record](#) | Registered DOI metadata

Q0558 Exploring biological neuronal correlations with quantum generative models

Vinicius Hernandez, Eliska Greplova.

Journal article | 2025 | Cell Reports Physical Science | vol. 6 | no. 8 | pp. 102682 | article 102682.

[DOI 10.1016/j.xcrp.2025.102682](#) | [Source record](#) | Registered DOI metadata

Q0559 Monitoring and Evaluating Quantum Generative Models Using Spark and MLflow

Saman Siadati.

Journal article | 2025 | Proceedings of the AAAI Symposium Series | vol. 7 | no. 1 | pp. 398-403.

[DOI 10.1609/aaais.v7i1.36911](#) | [Source record](#) | Registered DOI metadata

Q0560 Quantum generative models

Andrea Delgado, Kathleen E Hamilton.

Book chapter | 2025 | Quantum Machine Learning | pp. 7-1-7-18.

[DOI 10.1088/978-0-7503-4952-9ch7](#) | [Source record](#) | Registered DOI metadata

Q0561 Quantum-Enhanced Generative Models for Rare Event Prediction

M.Z. Haider, M.U. Ghouri, Tayyaba Noreen, M. Salman.

Conference paper | 2025 | 2025 Computing, Communications and IoT Applications (ComComAp) | pp. 103-110.

[DOI 10.1109/comcomap68359.2025.11353178](#) | [Source record](#) | Registered DOI metadata

Q0562 A Characterization of Quantum Generative Models

Carlos A. Riofrio, Oliver Mitevski, Caitlin Jones, Florian Krellner, Aleksandar Vuckovic, Joseph Doetsch, Johannes Klepsch, Thomas Ehmer et al..

Journal article | 2024 | ACM Transactions on Quantum Computing | vol. 5 | no. 2 | pp. 1-34.

[DOI 10.1145/3655027](#) | [Source record](#) | Registered DOI metadata

D10T05 Quantum reinforcement learning

Quantum reinforcement learning studies quantum resources in sequential decision tasks. Coherent access to an environment is a substantial assumption and is not always physically available.

Fine subcategories: Agents; policies; value functions; environment access; exploration; speedup assumptions.

Primary catalog resources: 16. Related topic assignments are searchable in the HTML.

Q0563 Noise-resilient quantum reinforcement learning

Jing-Ci Yue, Jun-Hong An.

Journal article | 2026 | Physical Review Applied | vol. 25 | no. 4 | article 044060.

[DOI 10.1103/8gfc-2nsc](#) | [Source record](#) | Registered DOI metadata

Q0564 Quantum reinforcement learning

Samuel Yen-Chi Chen.

Book chapter | 2026 | Quantum Computational AI | pp. 3-23.

[DOI 10.1016/b978-0-44-330259-6.00009-8](#) | [Source record](#) | Registered DOI metadata

Q0565 Quantum reinforcement learning in dynamic environments

Oliver Sefrin, Manuel Radons, Lars Simon, Sabine Wölk.

Journal article | 2026 | Quantum Machine Intelligence | vol. 8 | no. 1 | article 58.

[DOI 10.1007/s42484-026-00383-8](#) | [Source record](#) | Registered DOI metadata

Q0566 Variational Quantum Reinforcement Learning for Quantum Control

Haixu Yu, Lei Ye, Peng Wang.

Conference paper | 2026 | 2026 6th IEEE International Conference on Cybernetics (CYBCONF) | pp. 161-166.

[DOI 10.1109/cybconf69298.2026.11670283](https://doi.org/10.1109/cybconf69298.2026.11670283) | [Source record](#) | Registered DOI metadata

Q0567 Benchmarking Quantum Reinforcement Learning

Georg Kruse, Rodrigo Coelho, Andreas Roskopf, Robert Wille, Jeanette-Miriam Lorenz.

Conference paper | 2025 | Proceedings of the 17th International Conference on Agents and Artificial Intelligence | pp. 773-782.

[DOI 10.5220/0013393200003890](https://doi.org/10.5220/0013393200003890) | [Source record](#) | Registered DOI metadata

Q0568 Quantum Circuit Structure Optimization for Quantum Reinforcement Learning

Seok Bin Son, Joongheon Kim.

Conference paper | 2025 | 2025 IEEE International Conference on Quantum Computing and Engineering (QCE) | pp. 247-251.

[DOI 10.1109/qce65121.2025.10327](https://doi.org/10.1109/qce65121.2025.10327) | [Source record](#) | Registered DOI metadata

Q0569 Quantum Reinforcement Learning with Adaptive Variational Quantum Circuits

Sarvesh Bendre, Samiksha Apake, Ashish Mishra, Atharva Bali, Anikita Bhagade.

Posted content | 2025 | Elsevier BV.

[DOI 10.2139/ssrn.5220513](https://doi.org/10.2139/ssrn.5220513) | [Source record](#) | Registered DOI metadata

Q0570 Quantum noise modeling through reinforcement learning

Simone Bordoni, Andrea Papaluca, Piergiorgio Buttarini, Alejandro Sopena, Stefano Giagu, Stefano Carrazza.

Journal article | 2025 | Quantum Science and Technology | vol. 11 | no. 1 | pp. 015005.

[DOI 10.1088/2058-9565/ae1e98](https://doi.org/10.1088/2058-9565/ae1e98) | [Source record](#) | Registered DOI metadata

Q0571 Quantum-Inspired Reinforcement Learning for Quantum Control

Haixu Yu, Xudong Zhao, Chunlin Chen.

Journal article | 2025 | IEEE Transactions on Control Systems Technology | vol. 33 | no. 1 | pp. 61-76.

[DOI 10.1109/tcst.2024.3437142](https://doi.org/10.1109/tcst.2024.3437142) | [Source record](#) | Registered DOI metadata

Q0572 Model-Based Offline Quantum Reinforcement Learning

Simon Eisenmann, Daniel Hein, Steffen Udluft, Thomas A. Runkler.

Conference paper | 2024 | 2024 IEEE International Conference on Quantum Computing and Engineering (QCE) | pp. 1490-1496.

[DOI 10.1109/qce60285.2024.00175](https://doi.org/10.1109/qce60285.2024.00175) | [Source record](#) | Registered DOI metadata

Q0573 Parametrized Quantum Circuits for Reinforcement Learning

K.T. Mpofu, P. Mthunzi-Kufa.

Conference paper | 2024 | 2024 4th International Multidisciplinary Information Technology and Engineering Conference (IMITEC) | pp. 240-251.

[DOI 10.1109/imitec60221.2024.10850927](https://doi.org/10.1109/imitec60221.2024.10850927) | [Source record](#) | Registered DOI metadata

Q0574 QTRL: Toward Practical Quantum Reinforcement Learning via Quantum-Train

Chen-Yu Liu, Chu-Hsuan Abraham Lin, Chao-Han Huck Yang, Kuan-Cheng Chen, Min-Hsiu Hsieh.

Conference paper | 2024 | 2024 IEEE International Conference on Quantum Computing and Engineering (QCE) | pp. 317-322.

[DOI 10.1109/qce60285.2024.10299](https://doi.org/10.1109/qce60285.2024.10299) | [Source record](#) | Registered DOI metadata

Q0575 Quantum Reinforcement Learning for Cognitive SAR

Jan Meyer, Kay Glatting, Sigurd Huber, Gerhard Krieger.

Conference paper | 2024 | IGARSS 2024 - 2024 IEEE International Geoscience and Remote Sensing Symposium | pp. 794-798.

[DOI 10.1109/igarss53475.2024.10642330](https://doi.org/10.1109/igarss53475.2024.10642330) | [Source record](#) | Registered DOI metadata

Q0576 Quantum Reinforcement Learning: An Overview

Gyu Seon Kim, Soohyun Park, Joongheon Kim.

Conference paper | 2024 | 2024 15th International Conference on Information and Communication Technology Convergence (ICTC) | pp. 1145-1150.

[DOI 10.1109/ictc62082.2024.10826656](#) | [Source record](#) | Registered DOI metadata

Q0577 Quantum Task Impact Learning (QTIL)Algorithm on Quantum Reinforcement Learning

R. Palanivel, P. Muthulakshmi.

Posted content | 2024 | Springer Science and Business Media LLC.

[DOI 10.21203/rs.3.rs-4766272/v1](#) | [Source record](#) | Registered DOI metadata

Q0578 Quantum reinforcement learning

Niels M. P. Neumann, Paolo B. U. L. de Heer, Frank Phillipson.

Journal article | 2023 | Quantum Information Processing | vol. 22 | no. 2 | article 125.

[DOI 10.1007/s11128-023-03867-9](#) | [Source record](#) | Registered DOI metadata

D10T06 Quantum learning theory

Quantum learning theory analyzes sample complexity and generalization under explicit learning models. Results differ for classical data, quantum examples, and unknown quantum states.

Fine subcategories: PAC models; sample complexity; generalization; learnability; quantum examples.

Primary catalog resources: 8. Related topic assignments are searchable in the HTML.

Q0579 Learning complexity gradually in quantum machine learning models

Erik Recio-Armengol, Franz J. Schreiber, Jens Eisert, Carlos Bravo-Prieto.

Journal article | 2026 | Physical Review Research | vol. 8 | no. 3 | article 033006.

[DOI 10.1103/wnlr-q9xd](#) | [Source record](#) | Registered DOI metadata

Q0580 Run-length certificates in quantum learning: Sample complexity and noise thresholds

Jeongho Bang.

Journal article | 2026 | Physical Review A | vol. 113 | no. 6 | article 062601.

[DOI 10.1103/cn9t-dynd](#) | [Source record](#) | Registered DOI metadata

Fine tags: sample complexity

Q0581 Universal Sample Complexity Bounds in Quantum Learning Theory via Fisher Information Matrix

Hyukgun Kwon, Seok Hyung Lie, Liang Jiang.

Journal article | 2026 | PRX Quantum | vol. 7 | no. 3 | article 033063.

[DOI 10.1103/8k5v-ddtw](#) | [Source record](#) | Registered DOI metadata

Fine tags: sample complexity

Q0582 A quadratic sample complexity reduction for agnostic learning via quantum algorithms

Daniel Z. Zanger.

Journal article | 2025 | International Journal of Quantum Information | vol. 23 | no. 06 | article 2550023.

[DOI 10.1142/s0219749925500236](#) | [Source record](#) | Registered DOI metadata

Fine tags: sample complexity

Q0583 On the sample complexity of quantum Boltzmann machine learning

Luuk Coopmans, Marcello Benedetti.

Journal article | 2024 | Communications Physics | vol. 7 | no. 1 | article 274.

[DOI 10.1038/s42005-024-01763-x](#) | [Source record](#) | Registered DOI metadata

Fine tags: sample complexity

Q0584 Sample complexity of learning parametric quantum circuits

Haoyuan Cai, Qi Ye, Dong-Ling Deng.

Journal article | 2022 | Quantum Science and Technology | vol. 7 | no. 2 | pp. 025014.

[DOI 10.1088/2058-9565/ac4f30](#) | [Source record](#) | Registered DOI metadata

Fine tags: sample complexity

Q0585 Tangible reduction in learning sample complexity with large classical samples and small quantum system

Wooyeong Song, Marcin Wieśniak, Nana Liu, Marcin Pawłowski, Jinhyoung Lee, Jaewan Kim, Jeongho Bang.

Journal article | 2021 | Quantum Information Processing | vol. 20 | no. 8 | article 275.

[DOI 10.1007/s11128-021-03217-7](#) | [Source record](#) | Registered DOI metadata

Fine tags: sample complexity

Q0586 Generalization in QML from few training data | PennyLane Demos

Author metadata not supplied.

Tutorial | Undated | PennyLane.

[Official resource](#) | [Source record](#) | Official page retrieved

D10T07 Machine learning for quantum systems

Machine learning can assist calibration, control, characterization, and decoding of quantum hardware. This direction can be useful without claiming a speedup from a quantum learning model.

Fine subcategories: Control optimization; calibration models; learned decoders; state classification; scientific discovery.

Primary catalog resources: 8. Related topic assignments are searchable in the HTML.

Q0587 Adaptive Quantum Error Mitigation Using Machine Learning in Noisy Intermediate-Scale Quantum Systems

Jefin Rojar J, Haribaskar R, Nithya Jayakumar, Ramya Jayakumar.

Conference paper | 2026 | 2026 9th International Conference on Computational Intelligence in Data Science (ICCIDS) | pp. 1-6.

[DOI 10.1109/iccids69108.2026.11407599](#) | [Source record](#) | Registered DOI metadata

Q0588 Error-Corrected Quantum Machine Learning in Financial Forecasting

Mustafa Nazar, Hussein Aljan Hasan, Kahtan Mohammed Adnan, Mahmood Jawad Abu-Alshaeer, Dmytro Hnatchenko.

Book chapter | 2026 | Communications in Computer and Information Science | pp. 458-482.

[DOI 10.1007/978-3-032-39708-9_23](#) | [Source record](#) | Registered DOI metadata

Q0589 Quantum Error Correction and Detection for Quantum Machine Learning

Eromanga Adermann, Haiyue Kang, Martin Sevier, Muhammad Usman.

Book chapter | 2026 | Quantum Science and Technology | pp. 413-436.

[DOI 10.1007/978-3-032-11153-1_17](#) | [Source record](#) | Registered DOI metadata

Q0590 Qyn: FPGA-Based Quantum Error Correction with Integrated Quantum Machine Learning

Claritta AlSaneh, Claudia Mattar, Soraia Oueida.

Conference paper | 2026 | 2026 7th International Conference on Bio-engineering for Smart Technologies (BioSMART) | pp. 1-7.

[DOI 10.1109/biosmart71257.2026.11598218](#) | [Source record](#) | Registered DOI metadata

Q0591 Variational quantum machine learning with quantum error detection

Eromanga Adermann, Hajime Suzuki, Muhammad Usman.

Journal article | 2026 | Quantum Machine Intelligence | vol. 8 | no. 1 | article 1.

[DOI 10.1007/s42484-026-00347-y](#) | [Source record](#) | Registered DOI metadata

Q0592 Efficiently decoding quantum errors with machine learning

Xiu-Hao Deng, Yuan Xu.

Journal article | 2025 | Nature Computational Science | vol. 5 | no. 12 | pp. 1100-1101.

[DOI 10.1038/s43588-025-00907-5](#) | [Source record](#) | Registered DOI metadata

Q0593 Machine learning approach toward quantum error mitigation for accurate molecular energetics

Srushti Patil, Dibyendu Mondal, Rahul Maitra.

Journal article | 2025 | The Journal of Chemical Physics | vol. 163 | no. 2 | article 024129.

[DOI 10.1063/5.0274910](#) | [Source record](#) | Registered DOI metadata

Q0594 Machine-Learning based Decoding of Surface Code Syndromes in Quantum Error Correction

Debasmita Bhounik, Pinaki Sen, Ritajit Majumdar, Susmita Sur-Kolay, Latesh Kumar KJ, Sundaraja Sitharama Iyengar.

Journal article | 2022 | Journal of Engineering Research and Sciences | vol. 1 | no. 6 | pp. 21-35.

[DOI 10.55708/js0106004](#) | [Source record](#) | Registered DOI metadata

D11 Optimization and operations research

Quantum optimization includes annealing, variational methods, and algorithms with provable guarantees. Application results should report feasible solutions and comparisons to strong classical solvers.

Prerequisites: Optimization, constraint modeling, and algorithm analysis.

Assessment focus: Feasibility, approximation quality, embedding overhead, and solver baselines.

D11T01 Combinatorial optimization

Combinatorial methods encode discrete decisions and constraints into quantum workflows. Feasibility, approximation quality, and problem size must be reported.

Fine subcategories: MaxCut; satisfiability; graph coloring; constraint satisfaction; approximation guarantees.

Primary catalog resources: 15. Related topic assignments are searchable in the HTML.

Q0595 Benchmarking variational quantum algorithms for combinatorial optimization in practice

Tim Schwägerl, Yahui Chai, Tobias Hartung, Karl Jansen, Stefan Kühn.

Journal article | 2026 | Quantum Machine Intelligence | vol. 8 | no. 1 | article 26.

[DOI 10.1007/s42484-026-00355-y](https://doi.org/10.1007/s42484-026-00355-y) | [Source record](#) | Registered DOI metadata

Q0596 Enhanced multiscale quantum approximate optimization algorithm in multibody combinatorial optimization problems

Lei-Lei Chen, Ping Zou, Ya-Fei Yu.

Journal article | 2026 | Chinese Physics B | vol. 35 | no. 7 | pp. 070305.

[DOI 10.1088/1674-1056/ae29f7](https://doi.org/10.1088/1674-1056/ae29f7) | [Source record](#) | Registered DOI metadata

Q0597 Quantum-Assisted Hybrid Optimization for Graph-Based Combinatorial Optimization

Srinivasarao Thota, N. Shilpa, Nasr Al Din Ide.

Journal article | 2026 | IEEE Access | vol. 14 | pp. 142245-142261.

[DOI 10.1109/access.2026.3733232](https://doi.org/10.1109/access.2026.3733232) | [Source record](#) | Registered DOI metadata

Q0598 Quantum-Inspired Butterfly Optimization Algorithm for Combinatorial Optimization on Binary Problems

Freddy Alejandro Chaurra Gutierrez, Claudia Feregrino-Urbe, Guohua Sun.

Posted content | 2026 | Elsevier BV.

[DOI 10.2139/ssrn.6968611](https://doi.org/10.2139/ssrn.6968611) | [Source record](#) | Registered DOI metadata

Q0599 Qubit-efficient quantum combinatorial-optimization solver

Bhuvanesh Sundar, Maxime Dupont.

Journal article | 2026 | Physical Review Applied | vol. 25 | no. 3 | article 034071.

[DOI 10.1103/s5jv-jh24](https://doi.org/10.1103/s5jv-jh24) | [Source record](#) | Registered DOI metadata

Q0600 Qubit-Efficient Quantum Local Search for Combinatorial Optimization

Mikhail Podobrii, Viacheslav Kuzmin, Vladimir Voloshinov, Margarita Veshchezerova, Michael R. Perelshtein.

Journal article | 2026 | Advanced Quantum Technologies | vol. 9 | no. 3 | article e00438.

[DOI 10.1002/qute.202500438](https://doi.org/10.1002/qute.202500438) | [Source record](#) | Registered DOI metadata

Q0601 Toward Quantum Combinatorial Optimization

Yunpyo Hong, Jaejoon Gill, Daeyeun Kim, Seungcheol Oh, Byung-Soo Kim, Joongheon Kim.

Journal article | 2026 | IEEE Nanotechnology Magazine | pp. 1-11.

[DOI 10.1109/mnano.2026.3728349](https://doi.org/10.1109/mnano.2026.3728349) | [Source record](#) | Registered DOI metadata

Q0602 Variational quantum algorithm for constrained combinatorial optimization problems

Hui-Min Li, Yuan-Liang Han, Zhi-Xi Wang, Shao-Ming Fei.

Journal article | 2026 | Physical Review A | vol. 113 | no. 3 | article 032406.

[DOI 10.1103/ykxd-h19w](#) | [Source record](#) | Registered DOI metadata

Q0603 Benchmarking Quantum Computing for Combinatorial Optimization

Nathan Kittichaikoonkij, Nutthapat Pongtanyavichai, Poopha Suwananek, Prabhas Chongstitvatana, Kamonluk Suksen.

Conference paper | 2025 | 2025 22nd International Conference on Electrical Engineering/Electronics, Computer, Telecommunications and Information Technology (ECTI-CON) | pp. 1-6.

[DOI 10.1109/ecti-con64996.2025.11100821](#) | [Source record](#) | Registered DOI metadata

Q0604 Combinatorial optimization with quantum computers

Francisco Chicano, Gabiel Luque, Zakaria Abdelmoiz Dahi, Rodrigo Gil-Merino.

Journal article | 2025 | Engineering Optimization | vol. 57 | no. 1 | pp. 208-233.

[DOI 10.1080/0305215x.2024.2435538](#) | [Source record](#) | Registered DOI metadata

Q0605 Quantum Optimization Realized: Quantum Annealing for Combinatorial Spatial Problems in Practice

Amr Magdy.

Conference paper | 2025 | Proceedings of the 1st ACM SIGSPATIAL International Workshop on Quantum Computing for Spatial Data Systems and Applications | pp. 11-14.

[DOI 10.1145/3764923.3777407](#) | [Source record](#) | Registered DOI metadata

Q0606 Quantum Optimization for Enhanced Combinatorial Algorithms

Pradeep Kumar Pandey, Ruchi Chaturvedi, Suraj Bhan Dangi.

Journal article | 2025 | The Nucleus | vol. 62 | no. 1 | pp. 54-58.

[DOI 10.71330/thenucleus.2025.1453](#) | [Source record](#) | Registered DOI metadata

Q0607 Quantum annealing for combinatorial optimization: a benchmarking study

Seongmin Kim, Sang-Woo Ahn, In-Saeng Suh, Alexander W. Dowling, Eungkyu Lee, Tengfei Luo.

Journal article | 2025 | npj Quantum Information | vol. 11 | no. 1 | article 77.

[DOI 10.1038/s41534-025-01020-1](#) | [Source record](#) | Registered DOI metadata

Q0608 Quantum-Guided Cluster Algorithms for Combinatorial Optimization

Peter J. Eder, Aron Kerschbaumer, Jernej Rudi Finžgar, Raimel A. Medina, Martin J. A. Schuetz, Helmut G. Katzgraber, Sarah Braun, Christian B. Mendl.

Conference paper | 2025 | 2025 IEEE International Conference on Quantum Computing and Engineering (QCE) | pp. 219-229.

[DOI 10.1109/qce65121.2025.00033](#) | [Source record](#) | Registered DOI metadata

Q0609 Recursive Quantum Relaxation for Combinatorial Optimization Problems

Ruho Kondo, Yuki Sato, Rudy Raymond, Naoki Yamamoto.

Journal article | 2025 | Quantum | vol. 9 | pp. 1594 | article 1594.

[DOI 10.22331/q-2025-01-15-1594](#) | [Source record](#) | Registered DOI metadata

D11T02 Quantum semidefinite programming

Quantum SDP solvers use structured access to matrices and constraints. Their advantage depends on sparsity, rank, input models, and the required output.

Fine subcategories: Matrix access; feasibility; trace constraints; solver accuracy; oracle costs.

Primary catalog resources: 11. Related topic assignments are searchable in the HTML.

Q0610 Quantum Alternating Direction Method of Multipliers for Semidefinite Programming

Hantao Nie, Dong An, Zaiwen Wen.

Journal article | 2026 | Quantum | vol. 10 | pp. 2154 | article 2154.

[DOI 10.22331/q-2026-07-08-2154](#) | [Source record](#) | Registered DOI metadata

Q0611 Semidefinite programming for quantum channel learning

Mikhail Gennadievich Belov, Victor Victorovich Dubov, Vadim Konstantinovich Ivanov, Alexander Yurievich Maslov, Olga Vladimirovna Proshina, Vladislav Gennadievich Malyskin.

Journal article | 2026 | Physical Review E | vol. 114 | no. 3 | article 035302.

[DOI 10.1103/kdl5-smrh](#) | [Source record](#) | Registered DOI metadata

Q0612 Slack-variable approach for variational quantum semidefinite programming

Jingxuan Chen, Hanna Westerheim, Zoë Holmes, Ivy Luo, Theshani Nuradha, Dhruvil Patel, Soorya Rethinasamy, Kathie Wang et al..

Journal article | 2025 | Physical Review A | vol. 112 | no. 2 | article 022607.

[DOI 10.1103/lwxq-4myj](#) | [Source record](#) | Registered DOI metadata

Q0613 Semidefinite programming in quantum information science

Stefano Scali.

Journal article | 2024 | Contemporary Physics | vol. 65 | no. 3 | pp. 202-203.

[DOI 10.1080/00107514.2024.2415998](#) | [Source record](#) | Registered DOI metadata

Q0614 Semidefinite programming relaxations for quantum correlations

Armin Tavakoli, Alejandro Pozas-Kerstjens, Peter Brown, Mateus Araújo.

Journal article | 2024 | Reviews of Modern Physics | vol. 96 | no. 4 | article 045006.

[DOI 10.1103/revmodphys.96.045006](#) | [Source record](#) | Registered DOI metadata

Q0615 Variational Quantum Algorithms for Semidefinite Programming

Dhruvil Patel, Patrick J. Coles, Mark M. Wilde.

Journal article | 2024 | Quantum | vol. 8 | pp. 1374 | article 1374.

[DOI 10.22331/q-2024-06-17-1374](#) | [Source record](#) | Registered DOI metadata

Q0616 Quantum key distribution rates from semidefinite programming

Mateus Araújo, Marcus Huber, Miguel Navascués, Matej Pivoluska, Armin Tavakoli.

Journal article | 2023 | Quantum | vol. 7 | pp. 1019 | article 1019.

[DOI 10.22331/q-2023-05-24-1019](#) | [Source record](#) | Registered DOI metadata

Q0617 Self-consistent quantum measurement tomography based on semidefinite programming

Marco Cattaneo, Matteo A. C. Rossi, Keijo Korhonen, Elsi-Mari Borrelli, Guillermo García-Pérez, Zoltán Zimborás, Daniel Cavalcanti.

Journal article | 2023 | Physical Review Research | vol. 5 | no. 3 | article 033154.

[DOI 10.1103/physrevresearch.5.033154](#) | [Source record](#) | Registered DOI metadata

Q0618 Semidefinite Programming in Quantum Information Science

Paul Skrzypczyk, Daniel Cavalcanti.

Book | 2023 | IOP Publishing.

[DOI 10.1088/978-0-7503-3343-6](#) | [Source record](#) | Registered DOI metadata

Q0619 Semidefinite programming algorithm for the quantum mechanical bootstrap

David Berenstein, George Hulsey.

Journal article | 2023 | Physical Review E | vol. 107 | no. 5 | article L053301.

[DOI 10.1103/physreve.107.L053301](#) | [Source record](#) | Registered DOI metadata

Q0620 Noisy intermediate-scale quantum algorithm for semidefinite programming

Kishor Bharti, Tobias Haug, Vlatko Vedral, Leong-Chuan Kwek.

Journal article | 2022 | Physical Review A | vol. 105 | no. 5 | article 052445.

[DOI 10.1103/physreva.105.052445](https://doi.org/10.1103/physreva.105.052445) | [Source record](#) | Registered DOI metadata

D11T03 Quantum scheduling and routing

Routing and scheduling applications combine difficult constraints with discrete optimization. Penalties and embedding can create substantial overhead or infeasible solutions.

Fine subcategories: Vehicle routing; job shop scheduling; logistics; constraints; feasibility repair.

Primary catalog resources: 8. Related topic assignments are searchable in the HTML.

Q0621 Hybrid Classical–Quantum Optimization of Wireless Routing Using QAOA and Quantum Walks

Author metadata not supplied.

Journal article | 2026 | International Journal of Intelligent Systems and Applications in Engineering.

[DOI 10.17762/ijisae.v14i1s.8149](https://doi.org/10.17762/ijisae.v14i1s.8149) | [Source record](#) | Registered DOI metadata

Q0622 Purification Strategy Optimization for Entanglement Routing in Quantum Networks

Javier Vecino Peñas, Ana Fernández-Vilas, Rebeca P. Díaz-Redondo, Sergio Gándara Gándara, Manuel Fernández-Veiga.

Conference paper | 2026 | 2026 International Conference on Quantum Control, Computing and Learning (qCCL) | pp. 1-8.

[DOI 10.1109/qcccl70331.2026.11668568](https://doi.org/10.1109/qcccl70331.2026.11668568) | [Source record](#) | Registered DOI metadata

Q0623 Quantum-Inspired Optimization for Robust MANET Routing Protocols

K Subhashini, Kamini Sharma, Aashima Bansal, Jatinder Kaur, Ankit Chamoli, Vipin Kumar Chaudhary.

Conference paper | 2026 | 2026 Second International Conference on Emerging Computational Intelligence (ICECI) | pp. 1-6.

[DOI 10.1109/iceci69159.2026.11519468](https://doi.org/10.1109/iceci69159.2026.11519468) | [Source record](#) | Registered DOI metadata

Q0624 Cyber Risk-Aware Quantum-inspired Optimization of Home Healthcare Routing and Scheduling with Electric Autonomous Vehicles

Hooman Razavi, Shadan Ghaffary, Mostafa Hajiaghaei-Keshteli, Amirhossein Mostofi.

Conference paper | 2025 | 2025 IEEE International Conference on Industrial Engineering and Engineering Management (IEEM) | pp. 1247-1254.

[DOI 10.1109/ieem63636.2025.11357715](https://doi.org/10.1109/ieem63636.2025.11357715) | [Source record](#) | Registered DOI metadata

Fine tags: Vehicle routing

Q0625 Quantum optimization in autonomous underwater vehicle routing navigation

Author metadata not supplied.

Journal article | 2025 | Journal of Transportation Science and Technology | vol. 14 | no. 2.

[DOI 10.55228/jtst.14\(2\).76-83](https://doi.org/10.55228/jtst.14(2).76-83) | [Source record](#) | Registered DOI metadata

Fine tags: Vehicle routing

Q0626 Routing and scheduling optimization for urban air mobility fleet management using quantum annealing

Renichiro Haba, Takuya Mano, Ryosuke Ueda, Genichiro Ebe, Kohei Takeda, Masayoshi Terabe, Masayuki Ohzeki.

Journal article | 2025 | Scientific Reports | vol. 15 | no. 1 | article 4326.

[DOI 10.1038/s41598-025-86843-w](https://doi.org/10.1038/s41598-025-86843-w) | [Source record](#) | Registered DOI metadata

Q0627 An optimization method for maintenance routing and scheduling in offshore wind farms based on chaotic quantum Harris hawks optimization

Ming-Wei Li, Yi-Zhang Lei, Zhong-Yi Yang, Hsin-Pou Huang, Wei-Chiang Hong.

Journal article | 2024 | Ocean Engineering | vol. 308 | pp. 118306 | article 118306.

[DOI 10.1016/j.oceaneng.2024.118306](https://doi.org/10.1016/j.oceaneng.2024.118306) | [Source record](#) | Registered DOI metadata

Q0628 Encoding trade-offs and design toolkits in quantum algorithms for discrete optimization: coloring, routing, scheduling, and other problems

Nicolas PD Sawaya, Albert T Schmitz, Stuart Hadfield.

Journal article | 2023 | Quantum | vol. 7 | pp. 1111 | article 1111.

[DOI 10.22331/q-2023-09-14-1111](https://doi.org/10.22331/q-2023-09-14-1111) | [Source record](#) | Registered DOI metadata

D11T04 Quantum portfolio and financial estimation

Financial algorithms include estimation, pricing, and portfolio models. Model calibration, input access, error tolerances, and real computational costs determine relevance.

Fine subcategories: Portfolio optimization; option pricing; risk estimation; Monte Carlo; data costs.

Primary catalog resources: 8. Related topic assignments are searchable in the HTML.

Q0629 Quantum advantage for multi-option portfolio pricing and valuation adjustments

Jeong Yu Han, Bin Cheng, Dinh-Long Vu, Patrick Rebentrost.

Journal article | 2026 | Quantitative Finance | vol. 26 | no. 3 | pp. 467-489.

[DOI 10.1080/14697688.2026.2614573](https://doi.org/10.1080/14697688.2026.2614573) | [Source record](#) | Registered DOI metadata

Fine tags: option pricing

Q0630 Quantum computing for loan portfolio pricing optimization

Yanbo J. Wang, Xuan Yang, Chao Ju, Zhihang Liu, Jiawei Lu, Yue Zhang, Yiduo Wang, Xinkai Gao et al..

Journal article | 2026 | Frontiers in Physics | vol. 14 | article 1744791.

[DOI 10.3389/fphy.2026.1744791](https://doi.org/10.3389/fphy.2026.1744791) | [Source record](#) | Registered DOI metadata

Fine tags: Portfolio optimization

Q0631 Quantum-Inspired Quantitative Finance for Portfolio Optimization, Derivatives Pricing and Large-Scale Risk Management

Murali Krishna Pasupuleti.

Reference book | 2026 | National Education Services.

[DOI 10.62311/nesx/rb-66ag-978-81-68314-34-4](https://doi.org/10.62311/nesx/rb-66ag-978-81-68314-34-4) | [Source record](#) | Registered DOI metadata

Fine tags: Portfolio optimization

Q0632 Bridging Quantum Algorithms and Classical Finance: Portfolio Optimization Using QAOA and QUBO Framework

Arnav Aggarwal, Pranav Shrivastava, Rajneesh Kler, Purna Agarwal.

Conference paper | 2025 | 2025 1st IEEE Uttar Pradesh Section Women in Engineering International Conference on Electrical Electronics and Computer Engineering (UPWIECON) | pp. 682-687.

[DOI 10.1109/upwiecon67212.2025.11390104](https://doi.org/10.1109/upwiecon67212.2025.11390104) | [Source record](#) | Registered DOI metadata

Fine tags: Portfolio optimization

Q0633 Comprehensive Review of Quantum Computing Applications in Finance: Derivative Pricing, Risk Management, and Portfolio Optimization

Cem Fotocan.

Journal article | 2025 | Next Generation Journal for The Young Researchers | vol. 9 | no. 1 | pp. 65-67.

[DOI 10.62802/dnf08z38](https://doi.org/10.62802/dnf08z38) | [Source record](#) | Registered DOI metadata

Fine tags: Portfolio optimization

Q0634 From portfolio optimization to quantum blockchain and security: a systematic review of quantum computing in finance

Abha Satyavan Naik, Esra Yeniaras, Gerhard Hellstern, Grishma Prasad, Sanjay Kumar Lalta Prasad Vishwakarma.

Journal article | 2025 | Financial Innovation | vol. 11 | no. 1 | article 88.

[DOI 10.1186/s40854-025-00751-6](https://doi.org/10.1186/s40854-025-00751-6) | [Source record](#) | Registered DOI metadata

Fine tags: Portfolio optimization

Q0635 Quantum Algorithms for Portfolio Optimization in Finance

Efe BÜke.

Journal article | 2025 | Next Generation Journal for The Young Researchers | vol. 9 | no. 1 | pp. 57-59.

[DOI 10.62802/krgj1w70](https://doi.org/10.62802/krgj1w70) | [Source record](#) | Registered DOI metadata

Fine tags: Portfolio optimization

Q0636 Quantum Computing and Portfolio Optimization in Finance Services

Alex Khang, Kali Charan Rath, Karteek Madapana, Jagannadha Rao, Lakshmi Prasad Panda, Subhasish Das.

Book chapter | 2025 | Shaping Cutting-Edge Technologies and Applications for Digital Banking and Financial Services | pp. 27-45.

[DOI 10.4324/9781003501947-2](https://doi.org/10.4324/9781003501947-2) | [Source record](#) | Registered DOI metadata

Fine tags: Portfolio optimization

D11T05 Quantum optimization benchmarks

Optimization benchmarking compares solution quality, runtime, and scaling against strong classical solvers. A quantum label or a favorable small instance is insufficient evidence of advantage.

Fine subcategories: Classical baselines; scaling; solution quality; wall time; statistical uncertainty.

Primary catalog resources: 13. Related topic assignments are searchable in the HTML.

Q0637 A Quantum Enhanced Portfolio Optimization Pipeline: An Extensive Backtesting and Benchmarking Approach

Amar G. Nadh, Dushyant K. Sharma, Raghavan Solium.

Conference paper | 2026 | 2026 International Conference on Trends in Quantum Computing and Emerging Business Technologies (TQCEBT) | pp. 1-5.

[DOI 10.1109/tqcebt67648.2026.11681042](https://doi.org/10.1109/tqcebt67648.2026.11681042) | [Source record](#) | Registered DOI metadata

Q0638 Benchmarking Blockchain, Federated Learning, and Quantum Evolutionary Algorithm for Combinatorial Optimization

Hamza Baniata, Attila Kertesz.

Journal article | 2026 | Blockchain: Research and Applications | pp. 100446 | article 100446.

[DOI 10.1016/j.bcra.2026.100446](https://doi.org/10.1016/j.bcra.2026.100446) | [Source record](#) | Registered DOI metadata

Q0639 Benchmarking Classical and Quantum Optimization Approaches for Rider-Order Assignment

Tharmashastha SAPV, Surya Prakash Palanivel, Jasjyot Singh Gulati, M Maruthu Pandi.

Conference paper | 2026 | 2026 International Conference on Quantum Communications, Networking, and Computing (QCNC) | pp. 930-936.

[DOI 10.1109/qcnc69040.2026.00153](https://doi.org/10.1109/qcnc69040.2026.00153) | [Source record](#) | Registered DOI metadata

Q0640 Benchmarking Quantum Optimization Algorithms for the 0–1 Knapsack Problem Using Qiskit

Kampa Lavanya, T. Surekha, J. David Sukeerthi Kumar, K. S. R. K. Sarma, Yaramachu Srikanth.

Book chapter | 2026 | Lecture Notes in Networks and Systems | pp. 532-540.

[DOI 10.1007/978-3-032-30960-0_42](https://doi.org/10.1007/978-3-032-30960-0_42) | [Source record](#) | Registered DOI metadata

Q0641 Benchmarking Quantum Solvers in Noisy Digital Simulations for Financial Portfolio Optimization

Ruizhe Shen, Zichang Hao, Ching Hua Lee.

Journal article | 2026 | Entropy | vol. 28 | no. 8 | pp. 916.

[DOI 10.3390/e28080916](https://doi.org/10.3390/e28080916) | [Source record](#) | Registered DOI metadata

Q0642 Benchmarking optimization algorithms for automated calibration of quantum devices

Kevin Pack, Shai Machnes, Frank K Wilhelm.

Journal article | 2026 | New Journal of Physics | vol. 28 | no. 5 | pp. 054510.

[DOI 10.1088/1367-2630/ae6c54](https://doi.org/10.1088/1367-2630/ae6c54) | [Source record](#) | Registered DOI metadata

Q0643 Benchmarking quantum heuristics: Nonvariational quantum-walk-based optimization algorithm for the weighted MaxCut problem

Tavis Bennett, Aidan Smith, Edric Matwiejew, Jingbo B. Wang.

Journal article | 2026 | Physical Review A | vol. 113 | no. 3 | article 032603.

[DOI 10.1103/sdlc-wl67](https://doi.org/10.1103/sdlc-wl67) | [Source record](#) | Registered DOI metadata

Q0644 Parameter Analysis and Optimization of Layer Fidelity for Quantum Processor Benchmarking at Scale

Maria Jose Lozano Palacio, Hasan Nayfeh, Matthew Ware, David C. McKay.

Journal article | 2026 | IEEE Transactions on Quantum Engineering | vol. 7 | pp. 1-10.

[DOI 10.1109/tqe.2026.3668098](https://doi.org/10.1109/tqe.2026.3668098) | [Source record](#) | Registered DOI metadata

Q0645 QOPTec: a modular platform for benchmarking quantum algorithms through combinatorial optimization problems

Pablo Miranda-Rodríguez, Eneko Osaba.

Journal article | 2026 | SoftwareX | vol. 33 | pp. 102507 | article 102507.

[DOI 10.1016/j.softx.2026.102507](https://doi.org/10.1016/j.softx.2026.102507) | [Source record](#) | Registered DOI metadata

Q0646 Quantum Algorithms for Financial Optimization: Benchmarking QAOA, VQE, and Classical Portfolio Techniques

Subodh Nath Pushpak, Sarika Jain, Siddharth Kalra.

Conference paper | 2026 | 2026 5th International Conference on Innovative Practices in Technology and Management (ICIPTM) | pp. 1-5.

[DOI 10.1109/iciptm69057.2026.11465578](https://doi.org/10.1109/iciptm69057.2026.11465578) | [Source record](#) | Registered DOI metadata

Q0647 The Quantum Optimization Benchmarking Library

Thorsten Koch, David E. Bernal Neira, Ying Chen, Giorgio Cortiana, Daniel J. Egger, Raoul Heese, Narendra N. Hegade, Alejandro Gomez Cadavid et al..

Journal article | 2026 | Nature Computational Science | vol. 6 | no. 6 | pp. 653-671.

[DOI 10.1038/s43588-026-00991-1](https://doi.org/10.1038/s43588-026-00991-1) | [Source record](#) | Registered DOI metadata

Q0648 Benchmarking Adaptive Genetic Pulse-Level Optimization on Standard Quantum Algorithms

William Aguilar-Calvo, Santiago Núñez-Corrales.

Conference paper | 2025 | 2025 IEEE International Conference on Quantum Computing and Engineering (QCE) | pp. 260-263.

[DOI 10.1109/qce65121.2025.10330](https://doi.org/10.1109/qce65121.2025.10330) | [Source record](#) | Registered DOI metadata

Q0649 Benchmarking quantum optimization for the maximum-cut problem on a superconducting quantum computer

Maxime Dupont, Bhuvanesh Sundar, Bram Evert, David E. Bernal Neira, Zedong Peng, Stephen Jeffrey, Mark J. Hodson.

Journal article | 2025 | Physical Review Applied | vol. 23 | no. 1 | article 014045.

[DOI 10.1103/physrevapplied.23.014045](https://doi.org/10.1103/physrevapplied.23.014045) | [Source record](#) | Registered DOI metadata

D12 Superconducting qubit hardware

Superconducting circuits combine microwave control, cryogenics, fabrication, and readout. Device coherence and gate fidelity must be distinguished from logical error performance.

Prerequisites: Microwave circuits, cryogenics, and solid state physics.

Assessment focus: Coherence, leakage, crosstalk, fabrication variation, and system wiring.

D12T01 Transmon and circuit QED

Transmons reduce charge sensitivity through circuit design and couple to microwave resonators. Coherence, anharmonicity, coupling, and control tradeoffs shape performance.

Fine subcategories: Josephson junctions; anharmonicity; dispersive coupling; charge noise; resonators.

Primary catalog resources: 7. Related topic assignments are searchable in the HTML.

Q0650 Circuit quantum electrodynamics

Alexandre Blais, Arne L. Grimsmo, S. M. Girvin, Andreas Wallraff.

Journal article | 2021 | Reviews of Modern Physics | vol. 93 | no. 2 | article 025005.

[DOI 10.1103/revmodphys.93.025005](https://doi.org/10.1103/revmodphys.93.025005) | [Source record](#) | Registered DOI metadata

Q0651 From Macroscopic Quantum Tunneling to Circuit QED: Coherence and Dissipation in 3D Transmon Qubits

Gustavo Moreto, Diego Molina, Saulo Gabriel Alberton, Denys Derlian Carvalho Brito, Ednilson dos Santos Lopes da Cunha, Arthur Rebello, Daniel GONÇALVES BENVENUTTI, Agatha Sandes Dutra et al..

Posted content | 2026 | Springer Science and Business Media LLC.

[DOI 10.21203/rs.3.rs-9657238/v1](https://doi.org/10.21203/rs.3.rs-9657238/v1) | [Source record](#) | Registered DOI metadata

Q0652 Electromagnetically induced acoustic transparency using a superconducting transmon circuit

Abdul Wahab, Muqaddar Abbas, Xiaosen Yang, Yuanping Chen.

Journal article | 2024 | The European Physical Journal Plus | vol. 139 | no. 4 | article 318.

[DOI 10.1140/epjp/s13360-024-05069-3](https://doi.org/10.1140/epjp/s13360-024-05069-3) | [Source record](#) | Registered DOI metadata

Q0653 Electromagnetically induced acoustic transparency amplifier using a superconducting transmon circuit

Syeda Aliya Batool, Rahmatullah, Sajid Qamar.

Journal article | 2023 | Physica Scripta | vol. 98 | no. 2 | pp. 025105.

[DOI 10.1088/1402-4896/acb32f](https://doi.org/10.1088/1402-4896/acb32f) | [Source record](#) | Registered DOI metadata

Q0654 Design and Performance Analysis of Hexagonal Transmon Qubit in a Superconducting Circuit

Seong Hyeon Park, Jeseok Bang, Soobin An, Seungyong Hahn.

Journal article | 2021 | IEEE Transactions on Applied Superconductivity | vol. 31 | no. 5 | pp. 1-5.

[DOI 10.1109/tasc.2021.3070469](https://doi.org/10.1109/tasc.2021.3070469) | [Source record](#) | Registered DOI metadata

Q0655 N Two-Transmon-Qubit Quantum Logic Gates Realized in a Circuit QED System

T., A., M. Said, Chouikh, Benna.

Journal article | 2019 | Applied Mathematics & Information Sciences | vol. 13 | no. 5 | pp. 839-846.

[DOI 10.18576/amis/130518](https://doi.org/10.18576/amis/130518) | [Source record](#) | Registered DOI metadata

Q0656 Transferring arbitrary d-dimensional quantum states of a superconducting transmon qubit in circuit QED

Tong Liu, Qi-Ping Su, Jin-Hu Yang, Yu Zhang, Shao-Jie Xiong, Jin-Ming Liu, Chui-Ping Yang.

Journal article | 2017 | Scientific Reports | vol. 7 | no. 1 | article 7039.

[DOI 10.1038/s41598-017-07225-5](https://doi.org/10.1038/s41598-017-07225-5) | [Source record](#) | Registered DOI metadata

D12T02 Flux and fluxonium qubits

Flux based circuits use magnetic flux and specialized inductive elements to define qubit transitions. Sweet spots and device parameters affect both noise and control.

Fine subcategories: Flux bias; superinductors; sweet spots; tunneling; circuit design.

Primary catalog resources: 16. Related topic assignments are searchable in the HTML.

Q0657 Exploration of Quantum Control Landscapes for Single-qubit Gate Generation in Few-Level Fluxonium Approximations

I. M. Korolev, A. N. Pechen.

Journal article | 2026 | Lobachevskii Journal of Mathematics | vol. 47 | no. 6 | pp. 2541-2549.

[DOI 10.1134/s1995080226619612](#) | [Source record](#) | Registered DOI metadata

Q0658 Fluxonium as a Control Qubit for Bosonic Quantum Information

Ke Nie, J. Nofear Bradford, Supriya Mandal, Aayam Bista, Wolfgang Pfaff, Angela Kou.

Journal article | 2026 | PRX Quantum | vol. 7 | no. 1 | article 010357.

[DOI 10.1103/dgwk-w3jj](#) | [Source record](#) | Registered DOI metadata

Q0659 Fluxonium qubit-based hybrid electromechanical system

Roson Nongthombam, Anshika Ranjan, Amarendra K. Sarma, Vibhor Singh.

Journal article | 2026 | Physical Review A | vol. 113 | no. 6 | article 063707.

[DOI 10.1103/mhnl-b2wp](#) | [Source record](#) | Registered DOI metadata

Q0660 Measurement-induced state transitions across the fluxonium qubit landscape

Anonymous.

Journal article | 2026 | Physical Review Applied.

[DOI 10.1103/n2qr-9p7z](#) | [Source record](#) | Registered DOI metadata

Q0661 Optimization of high-fidelity single-qubit gates for fluxonium qubits using single-flux quantum control

Maxime Lapointe-Major, Boyan Torosov, Bohdan Kulchytsky, Pooya Ronagh.

Journal article | 2026 | Physical Review Applied | vol. 26 | no. 1 | article 014048.

[DOI 10.1103/dz26-n1jc](#) | [Source record](#) | Registered DOI metadata

Q0662 Readout-induced leakage of the fluxonium qubit

Aayam Bista, Matthew Thibodeau, Ke Nie, Kaicheung Chow, Bryan K. Clark, Angela Kou.

Journal article | 2026 | Physical Review Applied | vol. 25 | no. 3 | article 034058.

[DOI 10.1103/wjdb-4814](#) | [Source record](#) | Registered DOI metadata

Q0663 Scalable fluxonium-qubit architecture with tunable interactions between noncomputational levels

Peng Zhao, Guming Zhao, Shaowei Li, Chen Zha, Ming Gong.

Journal article | 2026 | Physical Review Applied | vol. 25 | no. 4 | article 044072.

[DOI 10.1103/mfjl-5rk6](#) | [Source record](#) | Registered DOI metadata

Q0664 Cross-resonance control of an oscillator with an auxiliary fluxonium qubit

Guo Zheng, Simon Lieu, Emma L. Rosenfeld, Kyungjoo Noh, Connor T. Hann.

Journal article | 2025 | Physical Review Applied | vol. 23 | no. 2 | article 024067.

[DOI 10.1103/physrevapplied.23.024067](#) | [Source record](#) | Registered DOI metadata

Q0665 Flux-trapping fluxonium qubit

Kotaro Hida, Kohei Matsuura, Shu Watanabe, Yasunobu Nakamura.

Journal article | 2025 | Physical Review Applied | vol. 24 | no. 6 | article 064069.

[DOI 10.1103/b99c-y22b](#) | [Source record](#) | Registered DOI metadata

Q0666 Niobium-buffered tantalum for a superconducting fluxonium qubit

Zi shuo Li, Ting ting Guo, Wen qu Xu, Li li Shi, Kai xuan Zhang, Quan Zuo, Tian shi Zhou, Guo zhu Sun et al..
Journal article | 2025 | Materials Research Express | vol. 12 | no. 2 | pp. 026001.
[DOI 10.1088/2053-1591/adb091](#) | [Source record](#) | Registered DOI metadata

Q0667 Numerical Analysis of Quantum Control Landscapes for Single-Qubit Gate Generation in Three-Level Fluxonium Systems

I. M. Korolev, A. N. Pechen.
Journal article | 2025 | Lobachevskii Journal of Mathematics | vol. 46 | no. 6 | pp. 2581-2590.
[DOI 10.1134/s1995080225607830](#) | [Source record](#) | Registered DOI metadata

Q0668 Subharmonic Control of a Fluxonium Qubit via a Purcell-Protected Flux Line

J. Schirk, F. Wallner, L. Huang, I. Tsitsilin, N. Bruckmoser, L. Koch, D. Bunch, N.J. Glaser et al..
Journal article | 2025 | PRX Quantum | vol. 6 | no. 3 | article 030315.
[DOI 10.1103/pyx15-jyl7](#) | [Source record](#) | Registered DOI metadata

Q0669 Flux-pulse-assisted readout of a fluxonium qubit

Taryn V. Stefanski, Christian Kraglund Andersen.
Journal article | 2024 | Physical Review Applied | vol. 22 | no. 1 | article 014079.
[DOI 10.1103/physrevapplied.22.014079](#) | [Source record](#) | Registered DOI metadata

Q0670 Fluxonium-based superconducting qubit magnetometer: Optimization of phase estimation algorithms

Vladimir Slepnev, Azat Gubaydullin, Valerii Vinokur.
Journal article | 2024 | Physical Review B | vol. 110 | no. 21 | article 214423.
[DOI 10.1103/physrevb.110.214423](#) | [Source record](#) | Registered DOI metadata

Q0671 Integer Fluxonium Qubit

Raymond A. Mencia, Wei-Ju Lin, Hyunheung Cho, Maxim G. Vavilov, Vladimir E. Manucharyan.
Journal article | 2024 | PRX Quantum | vol. 5 | no. 4 | article 040318.
[DOI 10.1103/prxquantum.5.040318](#) | [Source record](#) | Registered DOI metadata

Q0672 Using Bifluxon Tunneling to Protect the Fluxonium Qubit

Waël Ardati, Sébastien Léger, Shelender Kumar, Vishnu Narayanan Suresh, Dorian Nicolas, Cyril Mori, Francesca D'Esposito, Tereza Vakhtel et al..
Journal article | 2024 | Physical Review X | vol. 14 | no. 4 | article 041014.
[DOI 10.1103/physrevx.14.041014](#) | [Source record](#) | Registered DOI metadata
Fine tags: tunneling

D12T03 Superconducting gates and couplers

Couplers and entangling gates control interactions between superconducting qubits. Gate error, residual coupling, leakage, and spectator effects require joint calibration.

Fine subcategories: Cross resonance; controlled phase; parametric gates; tunable coupling; leakage.

Primary catalog resources: 9. Related topic assignments are searchable in the HTML.

Q0673 Crosstalk-robust superconducting two-qubit geometric gates using tunable couplers

Bo-Xun Deng, Jia-Qi Hu, Cheng-Yun Ding, Zheng-Yuan Xue, Tao Chen.
Journal article | 2026 | Frontiers of Physics | vol. 21 | no. 10 | pp. 103205.
[DOI 10.15302/frontphys.2026.103205](#) | [Source record](#) | Registered DOI metadata

Q0674 Inherent flux crosstalk and coupler-driven single-qubit gates in superconducting circuits

Anonymous.

Journal article | 2026 | Physical Review Applied.

[DOI 10.1103/q2j3-mfff](#) | [Source record](#) | Registered DOI metadata

Q0675 Direct Implementation of High-Fidelity Three-Qubit Gates for Superconducting Processor with Tunable Couplers

Hao-Tian Liu, Bing-Jie Chen, Jia-Chi Zhang, Yong-Xi Xiao, Tian-Ming Li, Kaixuan Huang, Ziting Wang, Hao Li et al..

Journal article | 2025 | Physical Review Letters | vol. 135 | no. 5 | article 050602.

[DOI 10.1103/lvb9-pfr3](#) | [Source record](#) | Registered DOI metadata

Q0676 High-precision pulse calibration of tunable couplers for high-fidelity two-qubit gates in superconducting quantum processors

Tian-Ming Li, Jia-Chi Zhang, Bing-Jie Chen, Kaixuan Huang, Hao-Tian Liu, Yong-Xi Xiao, Cheng-Lin Deng, Gui-Han Liang et al..

Journal article | 2025 | Physical Review Applied | vol. 23 | no. 2 | article 024059.

[DOI 10.1103/physrevapplied.23.024059](#) | [Source record](#) | Registered DOI metadata

Q0677 Simulation Framework and Optimization of Superconducting Transmon-Tunable Coupler-Transmon System for Qudit Gates

Ferris Prima Nugraha, Yuhan Huang, Jiacheng Liu, Qiming Shao.

Conference paper | 2025 | 2025 IEEE/ACM International Conference On Computer Aided Design (ICCAD) | pp. 1-8.

[DOI 10.1109/iccad66269.2025.11240859](#) | [Source record](#) | Registered DOI metadata

Q0678 Three-mode tunable coupler for superconducting two-qubit gates

Elena Yu. Egorova, Alena S. Kazmina, Ilya A. Simakov, Ilya N. Moskalenko, Nikolay N. Abramov, Daria A. Kalacheva, Viktor B. Lubsanov, Alexey N. Bolgar et al..

Journal article | 2025 | Physical Review Applied | vol. 23 | no. 6 | article 064056.

[DOI 10.1103/2h4m-mg2p](#) | [Source record](#) | Registered DOI metadata

Q0679 Controlled-Controlled-Phase Gates for Superconducting Qubits Mediated by a Shared Tunable Coupler

Niklas J. Glaser, Federico Roy, Stefan Filipp.

Journal article | 2023 | Physical Review Applied | vol. 19 | no. 4 | article 044001.

[DOI 10.1103/physrevapplied.19.044001](#) | [Source record](#) | Registered DOI metadata

Fine tags: controlled phase

Q0680 Realization of High-Fidelity Controlled-Phase Gates in Extensible Superconducting Qubits Design with a Tunable Coupler

Yangsen Ye, Sirui Cao, Yulin Wu, Xiawei Chen, Qingling Zhu, Shaowei Li, Fusheng Chen, Ming Gong et al..

Journal article | 2021 | Chinese Physics Letters | vol. 38 | no. 10 | pp. 100301.

[DOI 10.1088/0256-307x/38/10/100301](#) | [Source record](#) | Registered DOI metadata

Fine tags: controlled phase

Q0681 Realization of adiabatic and diabatic CZ gates in superconducting qubits coupled with a tunable coupler*

Huikai Xu, Weiyang Liu, Zhiyuan Li, Jiaxiu Han, Jingning Zhang, Kehuan Linghu, Yongchao Li, Mo Chen et al..

Journal article | 2021 | Chinese Physics B | vol. 30 | no. 4 | pp. 044212.

[DOI 10.1088/1674-1056/abf03a](#) | [Source record](#) | Registered DOI metadata

D12T04 Superconducting readout

Dispersive readout infers a qubit state from a coupled resonator response. Measurement speed, fidelity, backaction, and multiplexing compete.

Fine subcategories: Dispersive measurement; Purcell filters; parametric amplifiers; multiplexing; assignment errors.

Primary catalog resources: 14. Related topic assignments are searchable in the HTML.

Q0682 A Survey of Microwave-Implemented Superconducting Qubit Control and Readout Circuits

Naheel Raza Rizvi, Meraj Ahmad, Arslan Shafique, Hadi Heidari, Martin Weides, Muhammad Ali Imran, Atif Raza Jafri.

Journal article | 2026 | IEEE Transactions on Quantum Engineering | vol. 7 | pp. 1-52.

[DOI 10.1109/tqe.2026.3659400](https://doi.org/10.1109/tqe.2026.3659400) | [Source record](#) | Registered DOI metadata

Q0683 Millimeter wave readout of a superconducting qubit

Anonymous.

Journal article | 2026 | Physical Review Letters.

[DOI 10.1103/c84y-ggqs](https://doi.org/10.1103/c84y-ggqs) | [Source record](#) | Registered DOI metadata

Q0684 Pound-Drever-Hall Method for Superconducting-Qubit Readout

Ibukunoluwa Adisa, Won Chan Lee, Kevin C. Cox, Alicia J. Kollár.

Journal article | 2026 | Physical Review Letters | vol. 136 | no. 23 | article 233601.

[DOI 10.1103/ng1d-6gb8](https://doi.org/10.1103/ng1d-6gb8) | [Source record](#) | Registered DOI metadata

Q0685 Superconducting-qubit readout using next-generation reservoir computing

Robert Kent, Benjamin Lienhard, Gregory Lafyatis, Daniel J. Gauthier.

Journal article | 2026 | Physical Review Applied | vol. 25 | no. 4 | article 044009.

[DOI 10.1103/bnwn-d2p4](https://doi.org/10.1103/bnwn-d2p4) | [Source record](#) | Registered DOI metadata

Q0686 A Survey of Microwave-Implemented Superconducting Qubit Control and Readout Circuits

Naheel Raza Rizvi, Meraj Ahmad, Arslan Shafique, Hadi Heidari, Martin Weides, Muhammad Ali Imran, Atif Raza Jafri.

Posted content | 2025 | Institute of Electrical and Electronics Engineers (IEEE).

[DOI 10.36227/techrxiv.175624606.62662608/v1](https://doi.org/10.36227/techrxiv.175624606.62662608/v1) | [Source record](#) | Registered DOI metadata

Q0687 All-optical superconducting qubit readout

Georg Arnold, Thomas Werner, Rishabh Sahu, Lucky N. Kapoor, Liu Qiu, Johannes M. Fink.

Journal article | 2025 | Nature Physics | vol. 21 | no. 3 | pp. 393-400.

[DOI 10.1038/s41567-024-02741-4](https://doi.org/10.1038/s41567-024-02741-4) | [Source record](#) | Registered DOI metadata

Q0688 Balanced Cross-Kerr Coupling for Superconducting Qubit Readout

Alex A. Chapple, Othmane Benhayoune-Khadraoui, Simon Richer, Alexandre Blais.

Journal article | 2025 | Physical Review Letters | vol. 135 | no. 25 | article 256002.

[DOI 10.1103/r4v5-wyyt](https://doi.org/10.1103/r4v5-wyyt) | [Source record](#) | Registered DOI metadata

Q0689 Benchmarking the Readout of a Superconducting Qubit for Repeated Measurements

S. Hazra, W. Dai, T. Connolly, P. D. Kurilovich, Z. Wang, L. Frunzio, M. H. Devoret.

Journal article | 2025 | Physical Review Letters | vol. 134 | no. 10 | article 100601.

[DOI 10.1103/physrevlett.134.100601](https://doi.org/10.1103/physrevlett.134.100601) | [Source record](#) | Registered DOI metadata

Q0690 Efficient and Scalable Architectures for Multi-level Superconducting Qubit Readout

Chaithanya Naik Mude, Satvik Maurya, Benjamin Lienhard, Swamit Tannu.

Conference paper | 2025 | 2025 62nd ACM/IEEE Design Automation Conference (DAC) | pp. 1-7.

[DOI 10.1109/dac63849.2025.11133314](https://doi.org/10.1109/dac63849.2025.11133314) | [Source record](#) | Registered DOI metadata

Q0691 Josephson Junction-Based Compact Notch Purcell Filters for Superconducting Qubit Readout

Simona Zaccaria, Antonio Gnudi.

Journal article | 2025 | IEEE Transactions on Applied Superconductivity | vol. 35 | no. 8 | pp. 1-7.

[DOI 10.1109/tasc.2025.3610252](https://doi.org/10.1109/tasc.2025.3610252) | [Source record](#) | Registered DOI metadata

Fine tags: Purcell filters

Q0692 Multiphysics Numerical Methods for Designing Multiplexed Superconducting Qubit Readout Chains

Samuel T. Elkin, Michael Haider, Thomas E. Roth.

Conference paper | 2025 | 2025 International Applied Computational Electromagnetics Society Symposium (ACES) | pp. 1-2.

[DOI 10.23919/aces66556.2025.11052488](https://doi.org/10.23919/aces66556.2025.11052488) | [Source record](#) | Registered DOI metadata

Q0693 Optical readout of a superconducting qubit using a piezo-optomechanical transducer

T. C. van Thiel, M. J. Weaver, F. Berto, P. Duivesteyn, M. Lemang, K. L. Schuurman, M. Žemlička, F. Hijazi et al..

Journal article | 2025 | Nature Physics | vol. 21 | no. 3 | pp. 401-405.

[DOI 10.1038/s41567-024-02742-3](https://doi.org/10.1038/s41567-024-02742-3) | [Source record](#) | Registered DOI metadata

Q0694 Superconducting Parametric Amplifiers: Resonator Design and Role in Qubit Readout

Babak Mohammadian.

Book chapter | 2025 | Perspectives on Quantum Technologies - Modeling, Simulation, and Implementation.

[DOI 10.5772/intechopen.1013108](https://doi.org/10.5772/intechopen.1013108) | [Source record](#) | Registered DOI metadata

Fine tags: parametric amplifiers

Q0695 Superconducting Qubit: Readout and Initialization

Hiu Yung Wong.

Book chapter | 2025 | Quantum Computing Architecture and Hardware for Engineers | pp. 309-324.

[DOI 10.1007/978-3-031-78219-0_22](https://doi.org/10.1007/978-3-031-78219-0_22) | [Source record](#) | Registered DOI metadata

D12T05 Cryogenic circuits and packaging

Cryogenic packaging delivers signals while removing heat and controlling electromagnetic environments. System scaling includes wiring, thermal budgets, and electronics placement.

Fine subcategories: Dilution refrigeration; microwave wiring; interposers; thermal budgets; cryogenic electronics.

Primary catalog resources: 8. Related topic assignments are searchable in the HTML.

Q0696 Cryogenic microwave frequency combs based on quantum paraelectric superconducting resonators

Prasad Muragesh, Harikrishnan Sundaresan, Madhu Thalakulam.

Journal article | 2026 | Applied Physics Letters | vol. 129 | no. 7 | article 072601.

[DOI 10.1063/5.0349286](https://doi.org/10.1063/5.0349286) | [Source record](#) | Registered DOI metadata

Q0697 2.5D Cryogenic Packaging for Advanced Quantum Processors

Norhanani Binte Jaafar, Li Hongyu, Ya-Ching Tseng, Yong Chyn Ng, Daniel Lau.

Conference paper | 2025 | 2025 IEEE 27th Electronics Packaging Technology Conference (EPTC) | pp. 1-6.

[DOI 10.1109/eptc67330.2025.11392196](https://doi.org/10.1109/eptc67330.2025.11392196) | [Source record](#) | Registered DOI metadata

Q0698 Cryogenic microwave link for quantum local area networks

W. K. Yam, M. Renger, S. Gandorfer, F. Fesquet, M. Handschuh, K. E. Honasoge, F. Kronowetter, Y. Nojiri et al..

Journal article | 2025 | npj Quantum Information | vol. 11 | no. 1 | article 87.

[DOI 10.1038/s41534-025-01046-5](https://doi.org/10.1038/s41534-025-01046-5) | [Source record](#) | Registered DOI metadata

Q0699 A cryogenic on-chip microwave pulse generator for large-scale superconducting quantum computing

Zenghui Bao, Yan Li, Zhiling Wang, Jiahui Wang, Jize Yang, Haonan Xiong, Yipu Song, Yukai Wu et al..

Journal article | 2024 | Nature Communications | vol. 15 | no. 1 | article 5958.

[DOI 10.1038/s41467-024-50333-w](https://doi.org/10.1038/s41467-024-50333-w) | [Source record](#) | Registered DOI metadata

Q0700 Cryogenic DC and Microwave Characterization and Modeling of 40-nm CMOS for Quantum Computing Applications

Mahesh Kumar Chaubey, Yeke Liu, Po-Chang Wu, Hann-Huei Tsai, Shawn S.H. Hsu.

Conference paper | 2024 | 2024 IEEE Asia-Pacific Microwave Conference (APMC) | pp. 1311-1313.

[DOI 10.1109/apmc60911.2024.10867529](https://doi.org/10.1109/apmc60911.2024.10867529) | [Source record](#) | Registered DOI metadata

Q0701 Cryogenic bridging via propagating microwave quantum teleportation

Vahid Salari, Nasser Gohari Kamel, Farhad Rasekh, Roohollah Ghobadi, Jordan Smith, Daniel Oblak.

Journal article | 2024 | AVS Quantum Science | vol. 6 | no. 4 | article 042001.

[DOI 10.1116/5.0208451](https://doi.org/10.1116/5.0208451) | [Source record](#) | Registered DOI metadata

Q0702 Cryogenic packaging for scalable hybrid quantum PICs

Robert Bernson, Alexander Witte, Genevieve Clark, Kamil Gradkowski, Matthew Saha, Andrew Leenheer, Kevin Chen, Gerald Gilbert et al..

Conference paper | 2024 | Frontiers in Optics + Laser Science 2024 (FiO, LS) | pp. FTh1D.2.

[DOI 10.1364/fio.2024.fth1d.2](https://doi.org/10.1364/fio.2024.fth1d.2) | [Source record](#) | Registered DOI metadata

Q0703 High performance cryogenic packaging for microwave applications of high temperature superconductors

R.B. Greed, R.F. Jeffries, D.C. Voyce, A.J. Barkway, R.G. Humphreys, S.W. Goodyear.

Journal article | 2001 | IEEE Transactions on Applied Superconductivity | vol. 11 | no. 1 | pp. 493-496.

[DOI 10.1109/77.919390](https://doi.org/10.1109/77.919390) | [Source record](#) | Registered DOI metadata

D12T06 Superconducting fabrication and loss

Fabrication and materials affect loss, disorder, yield, and device uniformity. A scalable process must control interfaces as well as individual junction properties.

Fine subcategories: Surface loss; materials interfaces; junction yield; quasiparticles; process control.

Primary catalog resources: 5. Related topic assignments are searchable in the HTML.

Q0704 Design, Fabrication and Characterization of a Superconducting Transmon Qubit Fabricated in Brazil

Diego Molina, Gustavo Moreto, Denys Derlian, Saulo Alberton, Ednilson Cunha, Francisco Rouxinol.

Conference paper | 2026 | 2026 40th Symposium on Microelectronics Technology and Devices (SBMicro) | pp. 1-4.

[DOI 10.1109/sbmicro70495.2026.11684417](https://doi.org/10.1109/sbmicro70495.2026.11684417) | [Source record](#) | Registered DOI metadata

Q0705 Scaffold-assisted window junctions for superconducting qubit fabrication

Chung-Ting Ke, Jun-Yi Tsai, Yen-Chun Chen, Zhen-Wei Xu, Elam Blackwell, Matthew A. Snyder, Spencer Weeden, Peng-Sheng Chen et al..

Journal article | 2026 | npj Quantum Information | vol. 12 | no. 1 | article 98.

[DOI 10.1038/s41534-026-01249-4](https://doi.org/10.1038/s41534-026-01249-4) | [Source record](#) | Registered DOI metadata

Q0706 Materials Science Problems in the Fabrication of Superconducting Qubit Devices

Mingzhao Liu.

Conference paper | 2023 | 2023 IEEE Nanotechnology Materials and Devices Conference (NMDC) | pp. 832-832.

[DOI 10.1109/nmdc57951.2023.10343535](https://doi.org/10.1109/nmdc57951.2023.10343535) | [Source record](#) | Registered DOI metadata

Q0707 Fabrication of Low AC Loss Cu_{0.5}Mn-Based NbTi Superconducting Strands

Qiang Guo, Ruilong Wang, Kailin Zhang, Yanmin Zhu, Jianfeng Li, Xianghong Liu, Yong Feng.

Journal article | 2021 | IEEE Transactions on Applied Superconductivity | vol. 31 | no. 8 | pp. 1-4.

[DOI 10.1109/tasc.2021.3107813](https://doi.org/10.1109/tasc.2021.3107813) | [Source record](#) | Registered DOI metadata

AQUIRE

Q0708 Robust surface code topology against sparse fabrication defects in a superconducting-qubit array

Yong-Chao Tang, Guo-Xing Miao.

Journal article | 2016 | Physical Review A | vol. 93 | no. 3 | article 032322.

[DOI 10.1103/physreva.93.032322](https://doi.org/10.1103/physreva.93.032322) | [Source record](#) | Registered DOI metadata

D13 Trapped ion quantum computing

Ion processors use internal states coupled through collective motion. Scaling requires high quality gates, optical control, shuttling, and modular interconnects.

Prerequisites: Atomic physics, optics, and motional dynamics.

Assessment focus: Gate fidelity, mode management, optical stability, and transport cost.

D13T01 Ion traps and architectures

Ion trap architectures confine and address charged atoms used as qubits. Trap geometry, motional modes, and modular organization affect scaling.

Fine subcategories: Paul traps; Penning traps; surface traps; ion chains; modular architectures.

Primary catalog resources: 9. Related topic assignments are searchable in the HTML.

Q0709 Programmable quantum simulations of spin systems with trapped ions

C. Monroe, W. C. Campbell, L.-M. Duan, Z.-X. Gong, A. V. Gorshkov, P. W. Hess, R. Islam, K. Kim et al..

Journal article | 2021 | Reviews of Modern Physics | vol. 93 | no. 2 | article 025001.

[DOI 10.1103/revmodphys.93.025001](https://doi.org/10.1103/revmodphys.93.025001) | [Source record](#) | Registered DOI metadata

Q0710 Quantum computing architecture with trapped ion crystals and fast Rydberg gates

Han Bao, Jonas Vogel, Ulrich Poschinger, Ferdinand Schmidt-Kaler.

Journal article | 2025 | Physical Review Research | vol. 7 | no. 2 | article 023035.

[DOI 10.1103/physrevresearch.7.023035](https://doi.org/10.1103/physrevresearch.7.023035) | [Source record](#) | Registered DOI metadata

Q0711 Visible Photonic Component Development for Trapped-Ion Quantum Computing

Elliot Lehman, Molly Krogstad, Molly P. Andersen, Sara Campbell, Kirk Cook, Bryan DeBono, Christopher Ertsgaard, Azure Hansen et al..

Conference paper | 2025 | 2025 IEEE International Conference on Quantum Computing and Engineering (QCE) | pp. 518-519.

[DOI 10.1109/qce65121.2025.10424](https://doi.org/10.1109/qce65121.2025.10424) | [Source record](#) | Registered DOI metadata

Q0712 Scalable Architecture for Trapped-Ion Quantum Computing Using rf Traps and Dynamic Optical Potentials

David Schwerdt, Lee Peleg, Yotam Shapira, Nadav Priel, Yanay Florshaim, Avram Gross, Ayelet Zalic, Gadi Afek et al..

Journal article | 2024 | Physical Review X | vol. 14 | no. 4 | article 041017.

[DOI 10.1103/physrevx.14.041017](https://doi.org/10.1103/physrevx.14.041017) | [Source record](#) | Registered DOI metadata

Q0713 Multispecies Segmented Trapped Ion Architecture for Scalable Quantum Computing

M. Popov, N. Sterligov, O. Lakhmanskaya, K. Lakhmanskiy.

Conference paper | 2022 | Quantum 2.0 Conference and Exhibition | pp. QW3A.7.

[DOI 10.1364/quantum.2022.qw3a.7](https://doi.org/10.1364/quantum.2022.qw3a.7) | [Source record](#) | Registered DOI metadata

Q0714 TILT: Achieving Higher Fidelity on a Trapped-Ion Linear-Tape Quantum Computing Architecture

Xin-Chuan Wu, Dripto M. Debroy, Yongshan Ding, Jonathan M. Baker, Yuri Alexeev, Kenneth R. Brown, Frederic T. Chong.

Conference paper | 2021 | 2021 IEEE International Symposium on High-Performance Computer Architecture (HPCA) | pp. 153-166.

[DOI 10.1109/hpca51647.2021.00023](https://doi.org/10.1109/hpca51647.2021.00023) | [Source record](#) | Registered DOI metadata

Q0715 A Two-Dimensional Architecture for Fast Large-Scale Trapped-Ion Quantum Computing

Y.-K. Wu, L.-M. Duan.

Journal article | 2020 | Chinese Physics Letters | vol. 37 | no. 7 | pp. 070302.

[DOI 10.1088/0256-307x/37/7/070302](#) | [Source record](#) | Registered DOI metadata

Q0716 Trapped Ion Architecture for Multi-Dimensional Quantum Simulations

Ulrich Warring, Frederick Hakeberg, Philip Kiefer, Matthias Wittemer, Tobias Schaetz.

Journal article | 2020 | Advanced Quantum Technologies | vol. 3 | no. 11 | article 1900137.

[DOI 10.1002/qute.201900137](#) | [Source record](#) | Registered DOI metadata

Q0717 Integrated optics architecture for trapped-ion quantum information processing

D. Kielpinski, C. Volin, E. W. Streed, F. Lenzini, M. Lobino.

Journal article | 2015 | Quantum Information Processing | vol. 15 | no. 12 | pp. 5315-5338.

[DOI 10.1007/s11128-015-1162-2](#) | [Source record](#) | Registered DOI metadata

D13T02 Ion entangling gates

Entangling ion gates typically use shared motion or geometric phases. Mode crowding, heating, optical noise, and pulse design determine achievable fidelity.

Fine subcategories: Molmer Sorensen gates; geometric phase gates; motional modes; pulse shaping.

Primary catalog resources: 15. Related topic assignments are searchable in the HTML.

Q0718 Analysis of the action of conventional trapped-ion entangling gates in qudit space

Pavel A. Kamenskikh, Nikita V. Semenin, Ilia V. Zalivako, Vasiliy N. Smirnov, Ilya A. Semerikov, Ksenia Yu. Khabarova, Nikolay N. Kolachevsky.

Journal article | 2026 | Physical Review A | vol. 114 | no. 1 | article 012458.

[DOI 10.1103/sxrf-1h7v](#) | [Source record](#) | Registered DOI metadata

Q0719 Computational optimization of two-qubit entangling gates in trapped-ion systems under system frequency drift

Woojun Lee, Chaewon Kim, Taeyoung Choi, Taehyun Kim.

Journal article | 2026 | Current Applied Physics | vol. 83 | pp. 28-37.

[DOI 10.1016/j.cap.2025.11.009](#) | [Source record](#) | Registered DOI metadata

Q0720 Efficient optical configurations for trapped-ion entangling gates

Aditya Milind Kolhatkar, Karan K. Mehta.

Journal article | 2026 | Physical Review A | vol. 113 | no. 4 | article 042424.

[DOI 10.1103/ly5s-tjk2](#) | [Source record](#) | Registered DOI metadata

Q0721 Radial fast entangling gates under micromotion in trapped-ion quantum computers

Phoebe Grosser, Monica Gutierrez Galan, Isabelle Savill-Brown, Alexander K. Ratcliffe, Haonan Liu, Varun D. Vaidya, Simon A. Haine, C. Ricardo Viteri et al..

Journal article | 2026 | Physical Review A | vol. 114 | no. 2 | article 022617.

[DOI 10.1103/zg69-17xc](#) | [Source record](#) | Registered DOI metadata

Q0722 Amplitude-noise-resilient entangling gates for trapped ions

Nguyen H. Le, Modesto Orozco-Ruiz, Sahra A. Kulmiya, James G. Urquhart, Samuel J. Hile, Winfried K. Hensinger, Florian Mintert.

Journal article | 2025 | Physical Review Applied | vol. 24 | no. 5 | article 054060.

[DOI 10.1103/yj7-tvw8](#) | [Source record](#) | Registered DOI metadata

Q0723 Fast, Robust, and Laser-Free Universal Entangling Gates for Trapped-Ion Quantum Computing

Markus Nünnerich, Daniel Cohen, Patrick Barthel, Patrick H. Huber, Dorna Niroomand, Alex Retzker, Christof Wunderlich.

Journal article | 2025 | Physical Review X | vol. 15 | no. 2 | article 021079.

[DOI 10.1103/physrevx.15.021079](https://doi.org/10.1103/physrevx.15.021079) | [Source record](#) | Registered DOI metadata

Q0724 Performance analysis for crosstalk errors between parallel entangling gates in trapped-ion quantum error correction

Fangxuan Liu, Gaoxiang Tang, Luming Duan, Yukai Wu.

Journal article | 2025 | Physical Review Applied | vol. 24 | no. 1 | article 014032.

[DOI 10.1103/c7w3-pxls](https://doi.org/10.1103/c7w3-pxls) | [Source record](#) | Registered DOI metadata

Q0725 Power-optimized amplitude modulation for robust trapped-ion entangling gates: A study of gate-timing errors

Luke Ellert-Beck, Wenchao Ge.

Journal article | 2025 | Physical Review A | vol. 111 | no. 6 | article 062422.

[DOI 10.1103/physreva.111.062422](https://doi.org/10.1103/physreva.111.062422) | [Source record](#) | Registered DOI metadata

Q0726 Transverse Polarization Gradient Entangling Gates for Trapped-Ion Quantum Computation

Jin-Ming Cui, Yan Chen, Yi-Fan Zhou, Quan Long, En-Teng An, Ran He, Yun-Feng Huang, Chuan-Feng Li et al..

Journal article | 2025 | Physical Review Letters | vol. 135 | no. 26 | article 260604.

[DOI 10.1103/w5l6-wmrl](https://doi.org/10.1103/w5l6-wmrl) | [Source record](#) | Registered DOI metadata

Q0727 Laser-free trapped ion entangling gates with AESE: adiabatic elimination of spin-motion entanglement

R Tyler Sutherland, Michael Foss-Feig.

Journal article | 2024 | New Journal of Physics | vol. 26 | no. 1 | pp. 013013.

[DOI 10.1088/1367-2630/ad19f9](https://doi.org/10.1088/1367-2630/ad19f9) | [Source record](#) | Registered DOI metadata

Q0728 Entangling gates for trapped-ion quantum computation and quantum simulation

Zhengyang Cai, Chun -Yang Luan, Lingfeng Ou, Hengchao Tu, Zihan Yin, Jing -Ning Zhang, Kihwan Kim.

Journal article | 2023 | Journal of the Korean Physical Society | vol. 82 | no. 9 | pp. 882-900.

[DOI 10.1007/s40042-023-00772-3](https://doi.org/10.1007/s40042-023-00772-3) | [Source record](#) | Registered DOI metadata

Q0729 Error mitigation on global entangling gates with trapped ions*

Zhengyang Cai, Pengfei Wang, Wentao Chen, Jialiang Zhang, Jing-Ning Zhang, Kihwan Kim.

Posted content | 2023 | Morressier.

[DOI 10.26226/m.646635e94d8a9d0012931f3e](https://doi.org/10.26226/m.646635e94d8a9d0012931f3e) | [Source record](#) | Registered DOI metadata

Q0730 Pairwise-Parallel Entangling Gates on Orthogonal Modes in a Trapped-Ion Chain

Yingyue Zhu, Alaina M. Green, Nhung H. Nguyen, C. Huerta Alderete, Elijah Mossman, Norbert M. Linke.

Journal article | 2023 | Advanced Quantum Technologies | vol. 6 | no. 11 | article 2300056.

[DOI 10.1002/qute.202300056](https://doi.org/10.1002/qute.202300056) | [Source record](#) | Registered DOI metadata

Q0731 Fast multi-qubit global-entangling gates without individual addressing of trapped ions

Kaizhao Wang, Jing-Fan Yu, Pengfei Wang, Chunyang Luan, Jing-Ning Zhang, Kihwan Kim.

Journal article | 2022 | Quantum Science and Technology | vol. 7 | no. 4 | pp. 044005.

[DOI 10.1088/2058-9565/ac84a3](https://doi.org/10.1088/2058-9565/ac84a3) | [Source record](#) | Registered DOI metadata

Q0732 Trapped-Ion Quantum Computer with Robust Entangling Gates and Quantum Coherent Feedback

Tom Manovitz, Yotam Shapira, Lior Gazit, Nitzan Akerman, Roee Ozeri.

Journal article | 2022 | PRX Quantum | vol. 3 | no. 1 | article 010347.

[DOI 10.1103/prxquantum.3.010347](https://doi.org/10.1103/prxquantum.3.010347) | [Source record](#) | Registered DOI metadata

D13T03 Ion transport and QCCD

QCCD architectures move ions between operation regions. Transport must preserve quantum information and manage motional excitation and cooling.

Fine subcategories: Junction transport; sympathetic cooling; motional excitation; trap scheduling.

Primary catalog resources: 5. Related topic assignments are searchable in the HTML.

Q0733 Reinforcement learning for ion shuttling on trapped-ion quantum computers

Maximilian Schier, Lea Richtmann, Christian Staufenbiel, Tobias Schmale, Daniel Borchering, Michèle Heurs, Bodo Rosenhahn.

Journal article | 2026 | Physical Review Research | vol. 8 | no. 3 | article 033360.

[DOI 10.1103/b7ck-8wh4](#) | [Source record](#) | Registered DOI metadata

Q0734 Robust Shuttling Compilation for Trapped-Ion Quantum Computers

Daniel Schoenberger, Robert Wille.

Conference paper | 2026 | 2026 IEEE International Conference on Quantum Software (QSW) | pp. 240-246.

[DOI 10.1109/qsw72780.2026.00036](#) | [Source record](#) | Registered DOI metadata

Q0735 Shuttling for Scalable Trapped-Ion Quantum Computers

Daniel Schoenberger, Stefan Hillmich, Matthias Brandl, Robert Wille.

Journal article | 2025 | IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems | vol. 44 | no. 6 | pp. 2144-2155.

[DOI 10.1109/tcad.2024.3513262](#) | [Source record](#) | Registered DOI metadata

Q0736 CHARGE SHUTTLING: CELL BALANCING FOR PARALLEL LI-ION PACKS

Manish Ramaswamy.

Journal article | 2024 | International Education and Research Journal | vol. 10 | no. 1.

[DOI 10.21276/ierj24376418103884](#) | [Source record](#) | Registered DOI metadata

Q0737 Shuttling-based trapped-ion quantum information processing

V. Kaushal, B. Lekitsch, A. Stahl, J. Hilder, D. Pijn, C. Schmiegelow, A. Bermudez, M. Müller et al..

Journal article | 2020 | AVS Quantum Science | vol. 2 | no. 1 | article 014101.

[DOI 10.1116/1.5126186](#) | [Source record](#) | Registered DOI metadata

D13T04 Ion optical control and readout

Laser control and fluorescence readout link ion states to optical hardware. Addressing errors, stability, collection efficiency, and crosstalk matter.

Fine subcategories: Laser addressing; fluorescence; optical stability; crosstalk; state preparation.

Primary catalog resources: 8. Related topic assignments are searchable in the HTML.

Q0738 Monolithic ion trap integration of superconducting nanowire single-photon detectors and photonics for trapped-ion qubit state readout

Benedikt Hampel, Daniel H. Slichter, Dietrich Leibfried, Colin D. Bruzewicz, John Chiaverini, Ashton Hattori, Dave Kharas, Thomas Mahony et al..

Conference paper | 2026 | Quantum Computing, Communication, and Simulation VI | pp. 63.

[DOI 10.1117/12.3080210](#) | [Source record](#) | Registered DOI metadata

Q0739 Towards site-resolved readout of ions using multimode fibre for scalable trapped ion quantum computing

Thomas Hinde, Emre Pasaogullari, Lorenzo Versini, Catherine E. J. Challoner, Tim F. Wohlers-Reichel, Arjun D. Rao, Mika A. Zalewski, Peter Drmota et al..

Conference paper | 2026 | Quantum Technologies 2026 | pp. 34.

[DOI 10.1117/12.3107916](#) | [Source record](#) | Registered DOI metadata

Q0740 Control and readout of a 13-level trapped ion qudit

Pei Jiang Low, Brendan White, Crystal Senko.

Journal article | 2025 | npj Quantum Information | vol. 11 | no. 1 | article 85.

[DOI 10.1038/s41534-025-01031-y](#) | [Source record](#) | Registered DOI metadata

Q0741 Trap-integrated superconducting nanowire single-photon detectors for trapped-ion qubit state readout

Benedikt Hampel, Daniel H. Slichter, Dietrich G. Leibfried, Richard P. Mirin, Sae Woo Nam, Varun B. Verma.

Conference paper | 2024 | Advanced Photon Counting Techniques XVIII | pp. 8.

[DOI 10.1117/12.3014455](#) | [Source record](#) | Registered DOI metadata

Q0742 Fault-Tolerant Parity Readout on a Shuttling-Based Trapped-Ion Quantum Computer

J. Hilder, D. Pijn, O. Onishchenko, A. Stahl, M. Orth, B. Lekitsch, A. Rodriguez-Blanco, M. Müller et al..

Journal article | 2022 | Physical Review X | vol. 12 | no. 1 | article 011032.

[DOI 10.1103/physrevx.12.011032](#) | [Source record](#) | Registered DOI metadata

Q0743 High-Fidelity Indirect Readout of Trapped-Ion Hyperfine Qubits

Stephen D. Erickson, Jenny J. Wu, Pan-Yu Hou, Daniel C. Cole, Shawn Geller, Alex Kwiatkowski, Scott Glancy, Emanuel Knill et al..

Journal article | 2022 | Physical Review Letters | vol. 128 | no. 16 | article 160503.

[DOI 10.1103/physrevlett.128.160503](#) | [Source record](#) | Registered DOI metadata

Q0744 State Readout of a Trapped Ion Qubit Using a Trap-Integrated Superconducting Photon Detector

S. L. Todaro, V. B. Verma, K. C. McCormick, D. T. C. Allcock, R. P. Mirin, D. J. Wineland, S. W. Nam, A. C. Wilson et al..

Journal article | 2021 | Physical Review Letters | vol. 126 | no. 1 | article 010501.

[DOI 10.1103/physrevlett.126.010501](#) | [Source record](#) | Registered DOI metadata

Q0745 Fast High-Fidelity Readout of a Single Trapped-Ion Qubit via Machine-Learning Methods

Zi-Han Ding, Jin-Ming Cui, Yun-Feng Huang, Chuan-Feng Li, Tao Tu, Guang-Can Guo.

Journal article | 2019 | Physical Review Applied | vol. 12 | no. 1 | article 014038.

[DOI 10.1103/physrevapplied.12.014038](#) | [Source record](#) | Registered DOI metadata

D13T05 Photonic links between ions

Photonic links generate entanglement between distant ion modules. Heralding rates, photon collection, indistinguishability, and memory lifetime constrain execution.

Fine subcategories: Remote entanglement; collection optics; heralding; frequency conversion; modular links.

Primary catalog resources: 7. Related topic assignments are searchable in the HTML.

Q0746 Individual trapped-ion addressing with adjoint-optimized multimode photonic circuits

Melika Momenzadeh, Ke Sun, Qiming Wu, Bingran You, Yu-Lung Tang, Hartmut Häffner, Maxim R. Shcherbakov.

Journal article | 2026 | npj Nanophotonics | vol. 3 | no. 1 | article 3.

[DOI 10.1038/s44310-025-00102-4](#) | [Source record](#) | Registered DOI metadata

Q0747 Integrated photonic structures for photon-mediated entanglement of trapped ions

F. W. Knollmann, E. Clements, P. T. Callahan, M. Gehl, J. D. Hunker, T. Mahony, R. McConnell, R. Swint et al..

Journal article | 2024 | Optica Quantum | vol. 2 | no. 4 | pp. 230.

[DOI 10.1364/opticaq.522128](#) | [Source record](#) | Registered DOI metadata

Q0748 Utilizing integrated photonic technology for trapped-ion quantum computing

May Eun Yeon Kim.

Conference paper | 2024 | Quantum Computing, Communication, and Simulation IV | pp. 73.

[DOI 10.1117/12.3012126](#) | [Source record](#) | Registered DOI metadata

Q0749 High-fidelity trapped-ion qubit operations with scalable photonic modulators

C. W. Hogle, D. Dominguez, M. Dong, A. Leenheer, H. J. McGuinness, B. P. Ruzic, M. Eichenfield, D. Stick.

Journal article | 2023 | npj Quantum Information | vol. 9 | no. 1 | article 74.

[DOI 10.1038/s41534-023-00737-1](https://doi.org/10.1038/s41534-023-00737-1) | [Source record](#) | Registered DOI metadata

Q0750 High-fidelity trapped-ion qubit operations with scalable photonic modulators

Craig Hogle, Daniel Dominguez, Mark Dong, Andrew Leenheer, Hayden McGuinness, Brandon Ruzic, Matt Eichenfield, Daniel Stick.

Conference paper | 2023 | 54th Annual Meeting of the APS Division of Atomic, Molecular and Optical Physics (DAMOP) - Spokane, Washington, United States of America - June - 2023.

[DOI 10.2172/2431086](https://doi.org/10.2172/2431086) | [Source record](#) | Registered DOI metadata

Q0751 Routing single photons from a trapped ion with photonic integrated circuits

Uday Saha, James D. Sivers, John Hannegan, Mihika Prabhu, Eric Bersin, Saumil Bandyopadhyay, Jacques Carolan, Qudsia Quraishi et al..

Conference paper | 2022 | Conference on Lasers and Electro-Optics | pp. FTh5O.1.

[DOI 10.1364/cleo_qels.2022.fth5o.1](https://doi.org/10.1364/cleo_qels.2022.fth5o.1) | [Source record](#) | Registered DOI metadata

Q0752 CMOS Photonic Circuits for Trapped Ion Quantum Computing and Molecular Sensing

Rajeev J. Ram.

Conference paper | 2018 | Conference on Lasers and Electro-Optics | pp. STu3A.5.

[DOI 10.1364/cleo_si.2018.stu3a.5](https://doi.org/10.1364/cleo_si.2018.stu3a.5) | [Source record](#) | Registered DOI metadata

D14 Neutral atom and Rydberg processors

Neutral atom arrays offer reconfigurable geometries and interactions controlled by optical fields. Coherent gates, atom loss, rearrangement, and measurement determine computational reliability.

Prerequisites: Atomic physics, optical trapping, and interacting spin models.

Assessment focus: Atom loading, loss, gate error, geometry, and measurement backaction.

D14T01 Optical tweezer arrays

Optical tweezers create reconfigurable arrays of individual atoms. Loading and rearrangement establish the geometry used by later simulation or gate operations.

Fine subcategories: Loading; rearrangement; trapping; addressing; geometry control.

Primary catalog resources: 8. Related topic assignments are searchable in the HTML.

Q0753 Hybrid Atom Tweezer Array of Nuclear Spin and Optical Clock Qubits

Yuma Nakamura.

Book chapter | 2026 | Springer Theses | pp. 95-119.

[DOI 10.1007/978-981-95-2836-3_6](https://doi.org/10.1007/978-981-95-2836-3_6) | [Source record](#) | Registered DOI metadata

Q0754 Scalable Neutral-Atom Tweezer Arrays Based on TiO₂ Holographic Metasurfaces

Ze Zheng Zhu, Yuan Xu, Aaron Holman, Jiahao Wu, Ximo Sun, Mingxuan Wang, Bojeong Seo, Sebastian Will et al..

Conference paper | 2026 | CLEO 2026 | pp. SM1K.2.

[DOI 10.1364/cleo_si.2026.sm1k.2](https://doi.org/10.1364/cleo_si.2026.sm1k.2) | [Source record](#) | Registered DOI metadata

Q0755 Narrowline cooling of dysprosium atoms in an optical tweezer array

Giulio Biagioni, Britton Hofer, Nathan Bonvalet, Damien Bloch, Antoine Browaeys, Igor Ferrier-Barbut.

Journal article | 2025 | Physical Review A | vol. 112 | no. 1 | article 013316.

[DOI 10.1103/k4vn-y7mb](https://doi.org/10.1103/k4vn-y7mb) | [Source record](#) | Registered DOI metadata

Q0756 An optical tweezer array of ultracold polyatomic molecules

Nathaniel B. Vilas, Paige Robichaud, Christian Hallas, Grace K. Li, Loïc Anderegg, John M. Doyle.

Journal article | 2024 | Nature | vol. 628 | no. 8007 | pp. 282-286.

[DOI 10.1038/s41586-024-07199-1](https://doi.org/10.1038/s41586-024-07199-1) | [Source record](#) | Registered DOI metadata

Q0757 Apparatus for producing single strontium atoms in an optical tweezer array

Kai Wen , Huijin Chen , Xu Yan , Zejian Ren , Chengdong He , Elnur Hajiyev, Preston Tsz Fung Wong , Gyu-Boong Jo.

Journal article | 2024 | Chinese Physics B | vol. 33 | no. 12 | pp. 120703.

[DOI 10.1088/1674-1056/ad84d0](https://doi.org/10.1088/1674-1056/ad84d0) | [Source record](#) | Registered DOI metadata

Q0758 Erbium Atoms in an Optical Tweezer Array

David Ehrenstein.

Journal article | 2024 | Physics | vol. 17 | article s151.

[DOI 10.1103/physics.17.s151](https://doi.org/10.1103/physics.17.s151) | [Source record](#) | Registered DOI metadata

Q0759 Hybrid Atom Tweezer Array of Nuclear Spin and Optical Clock Qubits

Yuma Nakamura, Toshi Kusano, Rei Yokoyama, Keito Saito, Koichiro Higashi, Naoya Ozawa, Tetsushi Takano, Yosuke Takasu et al..

Journal article | 2024 | Physical Review X | vol. 14 | no. 4 | article 041062.

[DOI 10.1103/physrevx.14.041062](https://doi.org/10.1103/physrevx.14.041062) | [Source record](#) | Registered DOI metadata

Q0760 Optical setup for a two-dimensional tweezer array with independently adjustable columns for neutral atom quantum computing

Martin Adams, Martin Traub, Hans-Dieter Hoffmann, Florian Meinert, Philipp Ilzhöfer, Thomas Westphalen, Kersten Ludwig, Jiachen Zhao et al..

Conference paper | 2024 | Quantum Computing, Communication, and Simulation IV | pp. 45.

[DOI 10.1117/12.2692620](#) | [Source record](#) | Registered DOI metadata

D14T02 Rydberg blockade gates

Rydberg excitation enables strong interactions and blockade gates. Decay, motion, pulse control, and blockade imperfections determine gate error.

Fine subcategories: Blockade interactions; excitation errors; entangling gates; pulse design; decay.

Primary catalog resources: 8. Related topic assignments are searchable in the HTML.

Q0761 Floquet quantum gates with ion-enhanced Rydberg blockade

Anonymous.

Journal article | 2026 | Physical Review A.

[DOI 10.1103/247k-6zy1](#) | [Source record](#) | Registered DOI metadata

Q0762 Multitarget Rydberg gates via spatial blockade engineering

Samuel Stein, Chenxu Liu, Shuwen Kan, Eleanor Crane, Yufei Ding, Ying Mao, Alexander Schuckert, Ang Li.

Journal article | 2026 | Physical Review Research | vol. 8 | no. 1 | article 013254.

[DOI 10.1103/k72m-9tn8](#) | [Source record](#) | Registered DOI metadata

Q0763 Design of fast Rydberg blockade SWAP gates with synthetic modulated driving

Xin Wang, Tianze Sheng, Yuan Sun.

Journal article | 2025 | Photonics Research | vol. 13 | no. 4 | pp. 1074.

[DOI 10.1364/prj.550203](#) | [Source record](#) | Registered DOI metadata

Q0764 Geometric gates in atomic arrays without Rydberg blockade

Yue Ming, Zhao-Xin Fu, Yan-Xiong Du.

Journal article | 2025 | Physical Review A | vol. 112 | no. 4 | article 042609.

[DOI 10.1103/tmr4-gtnl](#) | [Source record](#) | Registered DOI metadata

Q0765 Optimization of Entangling Logic Gates Based on the Rydberg Blockade Effect

L. V. Gerasimov, D. V. Kupriyanov, S. S. Straupe.

Journal article | 2023 | Journal of Experimental and Theoretical Physics | vol. 137 | no. 2 | pp. 157-162.

[DOI 10.1134/s1063776123080113](#) | [Source record](#) | Registered DOI metadata

Fine tags: entangling gates

Q0766 Rabi- and Blockade-Error-Resilient All-Geometric Rydberg Quantum Gates

S.-L. Su, Li-Na Sun, B.-J. Liu, L.-L. Yan, M.-H. Yung, W. Li, M. Feng.

Journal article | 2023 | Physical Review Applied | vol. 19 | no. 4 | article 044007.

[DOI 10.1103/physrevapplied.19.044007](#) | [Source record](#) | Registered DOI metadata

Q0767 Two-Qubit Geometric Gates Based on Ground-State Blockade of Rydberg Atoms

Ji-Ze Xu, Li-Na Sun, J.-F. Wei, Y.-L. Du, Ronghui Luo, Lei-Lei Yan, M. Feng, Shi-Lei Su.

Journal article | 2022 | Chinese Physics Letters | vol. 39 | no. 9 | pp. 090301.

[DOI 10.1088/0256-307x/39/9/090301](#) | [Source record](#) | Registered DOI metadata

Q0768 Unselective ground-state blockade of Rydberg atoms for implementing quantum gates

Jin-Lei Wu, Yan Wang, Jin-Xuan Han, Shi-Lei Su, Yan Xia, Yongyuan Jiang, Jie Song.

Journal article | 2021 | Frontiers of Physics | vol. 17 | no. 2 | article 22501.

[DOI 10.1007/s11467-021-1104-7](#) | [Source record](#) | Registered DOI metadata

D14T03 Neutral atom analog simulation

Neutral atom simulators realize programmable interacting spin systems. Interpretation depends on the realized Hamiltonian and measured observable rather than array size alone.

Fine subcategories: Rydberg Hamiltonians; spin systems; many body dynamics; programmable geometry.

Primary catalog resources: 8. Related topic assignments are searchable in the HTML.

Q0769 Analog quantum feature selection with neutral-atom quantum processors

José J. Orquín-Marqués, Carlos Flores-Garrigós, Alejandro Gomez Cadavid, Anton Simen, Enrique Solano, Narendra N. Hegade, José D. Martín-Guerrero, Yolanda Vives-Gilabert.

Journal article | 2026 | Quantum Machine Intelligence | vol. 8 | no. 2 | article 108.

[DOI 10.1007/s42484-026-00456-8](https://doi.org/10.1007/s42484-026-00456-8) | [Source record](#) | Registered DOI metadata

Q0770 Neutral atom quantum computing

J. D. Pritchard.

Journal article | 2026 | Contemporary Physics | pp. 1-19.

[DOI 10.1080/00107514.2026.2675142](https://doi.org/10.1080/00107514.2026.2675142) | [Source record](#) | Registered DOI metadata

Q0771 Qubit 'recycling' gives neutral-atom quantum computing a boost

Anna Demming.

Journal article | 2026 | Physics World | vol. 39 | no. 2 | pp. 10ii-10ii.

[DOI 10.1088/2058-7058/39/02/10](https://doi.org/10.1088/2058-7058/39/02/10) | [Source record](#) | Registered DOI metadata

Q0772 Quantum Simulation Using Ultracold Neutral Atoms

Jongchul MUN, Jeewoo PARK, Yong-il SHIN, Jae-yoon CHOI.

Journal article | 2025 | Physics and High Technology | vol. 34 | no. 6 | pp. 10-18.

[DOI 10.3938/phit.34.017](https://doi.org/10.3938/phit.34.017) | [Source record](#) | Registered DOI metadata

Q0773 Variational Simulation of the Lipkin-Meshkov-Glick Model on a Neutral Atom Quantum Computer

R. Chinnarasu, C. Poole, L. Phuttitarn, A. Noori, T. M. Graham, S. N. Coppersmith, A. B. Balantekin, M. Saffman.

Journal article | 2025 | PRX Quantum | vol. 6 | no. 2 | article 020350.

[DOI 10.1103/prxquantum.6.020350](https://doi.org/10.1103/prxquantum.6.020350) | [Source record](#) | Registered DOI metadata

Q0774 Realistic Neutral Atom Image Simulation

Jonas Winklmann, Dimitrios Tsevas, Martin Schulz.

Conference paper | 2023 | 2023 IEEE International Conference on Quantum Computing and Engineering (QCE) | pp. 1349-1359.

[DOI 10.1109/qce57702.2023.00153](https://doi.org/10.1109/qce57702.2023.00153) | [Source record](#) | Registered DOI metadata

Q0775 Quantum computation and simulation with programmable neutral atom arrays

Alex Keesling.

Conference paper | 2022 | Quantum Computing, Communication, and Simulation II | pp. 13.

[DOI 10.1117/12.2613790](https://doi.org/10.1117/12.2613790) | [Source record](#) | Registered DOI metadata

Q0776 Computer simulation of neutral atom traps

A. Antillón, A. Góngora-T., T. H. Seligman.

Journal article | 1992 | Zeitschrift für Physik D Atoms, Molecules and Clusters | vol. 24 | no. 4 | pp. 347-350.

[DOI 10.1007/bf01426683](https://doi.org/10.1007/bf01426683) | [Source record](#) | Registered DOI metadata

D14T04 Neutral atom error correction

Neutral atom error correction combines logical encodings with atom transport and syndrome measurement. Atom loss and measurement backaction require platform aware analysis.

AQUIRE

Fine subcategories: Logical arrays; atom loss; syndrome extraction; transport; transversal gates.

Primary catalog resources: 6. Related topic assignments are searchable in the HTML.

Q0777 Building Error-Corrected Quantum Computers With Neutral-Atom Qubits

Krish Kotru.

Conference paper | 2025 | CLEO 2025 | pp. AA107_1.

[DOI 10.1364/cleo_at.2025.aa107_1](https://doi.org/10.1364/cleo_at.2025.aa107_1) | [Source record](#) | Registered DOI metadata

Q0778 Coniq: Enabling Concatenated Quantum Error Correction on Neutral Atom Arrays

Pengyu Liu, Mingkuan Xu, Hengyun Zhou, Hanrui Wang, Umut A. Acar, Yunong Shi.

Conference paper | 2025 | 2025 IEEE International Conference on Quantum Computing and Engineering (QCE) | pp. 615-626.

[DOI 10.1109/qce65121.2025.00073](https://doi.org/10.1109/qce65121.2025.00073) | [Source record](#) | Registered DOI metadata

Q0779 State-selective electromagnetically induced transparency for quantum error correction in neutral atom quantum computers

Felipe Giraldo, Aishwarya Kumar, Tsung-Yao Wu, Peng Du, David S. Weiss.

Journal article | 2022 | Physical Review A | vol. 106 | no. 3 | article 032425.

[DOI 10.1103/physreva.106.032425](https://doi.org/10.1103/physreva.106.032425) | [Source record](#) | Registered DOI metadata

Q0780 Quantum Error Correction with a Globally-Coupled Array of Neutral Atom Qubits

Mark Saffman.

Report | 2013 | Defense Technical Information Center.

[DOI 10.21236/ada579666](https://doi.org/10.21236/ada579666) | [Source record](#) | Registered DOI metadata

Q0781 Neutral Atom Quantum Computing

Carl J. Williams.

Conference paper | 2004 | Frontiers in Optics 2004/Laser Science XXII/Diffractive Optics and Micro-Optics/Optical Fabrication and Testing | pp. FMI4.

[DOI 10.1364/fio.2004.fmi4](https://doi.org/10.1364/fio.2004.fmi4) | [Source record](#) | Registered DOI metadata

Q0782 Neutral Atom Quantum Register

D. Schrader, I. Dotsenko, M. Khudaverdyan, Y. Miroshnychenko, A. Rauschenbeutel, D. Meschede.

Journal article | 2004 | Physical Review Letters | vol. 93 | no. 15 | article 150501.

[DOI 10.1103/physrevlett.93.150501](https://doi.org/10.1103/physrevlett.93.150501) | [Source record](#) | Registered DOI metadata

D14T05 Alkaline earth atomic qubits

Alkaline earth and related atoms offer nuclear spin and optical transition resources. Storage, control, readout, and metastable state lifetimes must be considered together.

Fine subcategories: Nuclear spins; optical clocks; metastable states; qubit storage; control transitions.

Primary catalog resources: 8. Related topic assignments are searchable in the HTML.

Q0783 Quantum metrology with alkaline-earth atoms

Nelson Darkwah Oppong.

Conference paper | 2026 | Quantum Sensing, Imaging, and Precision Metrology IV | pp. 158.

[DOI 10.1117/12.3089700](https://doi.org/10.1117/12.3089700) | [Source record](#) | Registered DOI metadata

Q0784 Coherent Control Over the High-Dimensional Space of the Nuclear Spin of Alkaline-Earth Atoms

H. Ahmed, A. Litvinov, P. Guesdon, E. Maréchal, J.H. Huckans, B. Pasquiou, B. Laburthe-Tolra, M. Robert-de-Saint-Vincent.

Journal article | 2025 | PRX Quantum | vol. 6 | no. 2 | article 020352.

[DOI 10.1103/prxquantum.6.020352](https://doi.org/10.1103/prxquantum.6.020352) | [Source record](#) | Registered DOI metadata

Q0785 Quantum metrology with alkaline-earth atom arrays

Nelson Darkwah Oppong.

Conference paper | 2025 | Quantum Sensing, Imaging, and Precision Metrology III | pp. 85.

[DOI 10.1117/12.3053683](#) | [Source record](#) | Registered DOI metadata

Q0786 Erasure conversion for fault-tolerant quantum computing in alkaline earth Rydberg atom arrays

Yue Wu, Shimon Kolkowitz, Shruti Puri, Jeff D. Thompson.

Journal article | 2022 | Nature Communications | vol. 13 | no. 1 | article 4657.

[DOI 10.1038/s41467-022-32094-6](#) | [Source record](#) | Registered DOI metadata

Q0787 Spin-orbit effects in the heavy alkaline-earth atoms

Chris H. Greene, Mireille Aymar.

Book chapter | 2019 | Molecular Applications of Quantum Defect Theory | pp. 421-438.

[DOI 10.1201/9780203746608-25](#) | [Source record](#) | Registered DOI metadata

Q0788 State-dependent lattices for quantum computing with alkaline-earth-metal atoms

A. J. Daley, J. Ye, P. Zoller.

Journal article | 2011 | The European Physical Journal D | vol. 65 | no. 1-2 | pp. 207-217.

[DOI 10.1140/epjd/e2011-20095-2](#) | [Source record](#) | Registered DOI metadata

Q0789 Quantum Computing with Alkaline-Earth-Metal Atoms

Andrew J. Daley, Martin M. Boyd, Jun Ye, Peter Zoller.

Journal article | 2008 | Physical Review Letters | vol. 101 | no. 17 | article 170504.

[DOI 10.1103/physrevlett.101.170504](#) | [Source record](#) | Registered DOI metadata

Q0790 Spin-orbit effects in the heavy alkaline-earth atoms

Chris H. Greene, Mireille Aymar.

Journal article | 1991 | Physical Review A | vol. 44 | no. 3 | pp. 1773-1790.

[DOI 10.1103/physreva.44.1773](#) | [Source record](#) | Registered DOI metadata

D15 Spin qubits and semiconductor platforms

Spin qubits connect quantum control to semiconductor materials and nanofabrication. Scaling involves disorder, charge noise, wiring, exchange coupling, and readout.

Prerequisites: Semiconductor physics, spin dynamics, and nanofabrication.

Assessment focus: Disorder, charge noise, valley structure, readout, and array wiring.

D15T01 Silicon spin qubits

Silicon spin qubits combine quantum dot confinement with magnetic or electrical control. Valley structure, isotopic purity, and disorder are key device variables.

Fine subcategories: Quantum dots; valley splitting; spin orbit coupling; enriched silicon; gate fidelity.

Primary catalog resources: 16. Related topic assignments are searchable in the HTML.

Q0791 A Cryogenic 2-3 GHz LC-VCO for Silicon Spin Qubit Readout

Jiaxuan Wang, Huazi Huang, Lijun Xiao, Hua Chen.

Conference paper | 2026 | 2026 International Conference on Microwave and Millimeter Wave Technology (ICMMT) | pp. 1-3.

[DOI 10.1109/icmmt69626.2026.11678947](https://doi.org/10.1109/icmmt69626.2026.11678947) | [Source record](#) | Registered DOI metadata

Q0792 Control and Readout of a Spin Qubit Register in Silicon Photonics

Hanbin Song, Xueyue Zhang, Lukasz Komza, Niccolo Fiaschi, Magnus L. Madsen, Zi-Huai Zhang, Alp Sipahigil.

Conference paper | 2026 | CLEO 2026 | pp. SM2F.2.

[DOI 10.1364/cleo_si.2026.sm2f.2](https://doi.org/10.1364/cleo_si.2026.sm2f.2) | [Source record](#) | Registered DOI metadata

Q0793 Improved readout accuracy of a silicon spin qubit via log-likelihood ratio

R. Mizokuchi, R. Wada, R. Matsuoka, S. Ota, I. Yanagi, T. Mine, R. Tsuchiya, D. Hisamoto et al..

Journal article | 2026 | Japanese Journal of Applied Physics | vol. 65 | no. 2 | pp. 02SP03.

[DOI 10.35848/1347-4065/ae2d60](https://doi.org/10.35848/1347-4065/ae2d60) | [Source record](#) | Registered DOI metadata

Q0794 Predicting charge stability in donor spin-qubit arrays in silicon

Songqi Jia, Pericles Philippopoulos, Félix Beaudoin, Hong Guo.

Journal article | 2026 | Physical Review Applied | vol. 26 | no. 2 | article 024013.

[DOI 10.1103/vrkm-4x3p](https://doi.org/10.1103/vrkm-4x3p) | [Source record](#) | Registered DOI metadata

Q0795 Robust composite two-qubit gates for silicon-based spin qubits

Yang-Yang Yu, Guang-Hui Zhang, Yan-Jie He, Jun Wu, Xue-Ke Song, Dong Wang.

Journal article | 2026 | Physical Review Applied | vol. 25 | no. 2 | article 024076.

[DOI 10.1103/t8xj-6c5m](https://doi.org/10.1103/t8xj-6c5m) | [Source record](#) | Registered DOI metadata

Q0796 Running a Six-Qubit Quantum Circuit on a Silicon Spin-Qubit Array

I. Fernández de Fuentes, E. Raymenants, B. Undseth, O. Pietx-Casas, S. Philips, M. Mądzik, S.L. de Snoo, S.V. Amitonov et al..

Journal article | 2026 | PRX Quantum | vol. 7 | no. 1 | article 010308.

[DOI 10.1103/f285-l2v5](https://doi.org/10.1103/f285-l2v5) | [Source record](#) | Registered DOI metadata

Q0797 Two-qubit logic and teleportation with mobile spin qubits in silicon

Y. Matsumoto, M. De Smet, L. Tryputen, S. L. de Snoo, S. V. Amitonov, A. Sammak, M. Rimbach-Russ, G. Scappucci et al..

Journal article | 2026 | Nature | vol. 653 | no. 8114 | pp. 391-397.

[DOI 10.1038/s41586-026-10423-9](https://doi.org/10.1038/s41586-026-10423-9) | [Source record](#) | Registered DOI metadata

Q0798 300mm All Silicon Gate-Stack for Si/SiGe Spin Qubit

Clément Godfrin, Arne Loenders, Yosuke Shimura, Roger Loo, Bart Raes, Gulzat Juliel, Sugandha Sharma, Stefan Kubicek et al..

Conference paper | 2025 | Extended Abstracts of the 2025 International Conference on Solid State Devices and Materials.

[DOI 10.7567/ssdm.2025.e-5-01](#) | [Source record](#) | Registered DOI metadata

Q0799 An Efficient Routing Optimization Framework for Silicon-Based Spin-Qubit Devices

Ching-Yao Huang, Wai-Kei Mak.

Conference paper | 2025 | 2025 IEEE/ACM International Conference On Computer Aided Design (ICCAD) | pp. 1-9.

[DOI 10.1109/iccad66269.2025.11240960](#) | [Source record](#) | Registered DOI metadata

Q0800 Entanglement of a nuclear spin qubit register in silicon photonics

Hanbin Song, Xueyue Zhang, Lukasz Komza, Niccolo Fiaschi, Yihuang Xiong, Yiyang Zhi, Scott Dhuey, Adam Schwartzberg et al..

Journal article | 2025 | Nature Nanotechnology | vol. 21 | no. 1 | pp. 53-57.

[DOI 10.1038/s41565-025-02066-0](#) | [Source record](#) | Registered DOI metadata

Q0801 Improved Readout Accuracy of a Silicon Spin Qubit via Maximum Likelihood Estimation

Raisei Mizokuchi, Riku Wada, Ryutaro Matsuoka, Shunsuke Ota, Itaru Yanagi, Toshiyuki Mine, Ryuta Tsuchiya, Digh Hisamoto et al..

Conference paper | 2025 | Extended Abstracts of the 2025 International Conference on Solid State Devices and Materials.

[DOI 10.7567/ssdm.2025.e-6-04](#) | [Source record](#) | Registered DOI metadata

Q0802 Industry-compatible silicon spin-qubit unit cells exceeding 99% fidelity

Paul Steinacker, Nard Dumoulin Stuyck, Wee Han Lim, Tuomo Tanttu, MengKe Feng, Santiago Serrano, Andreas Nickl, Marco Candido et al..

Journal article | 2025 | Nature | vol. 646 | no. 8083 | pp. 81-87.

[DOI 10.1038/s41586-025-09531-9](#) | [Source record](#) | Registered DOI metadata

Q0803 Modeling correlated-noise in silicon spin qubit device

Guoting Cheng, Jing Guo.

Journal article | 2025 | APL Quantum | vol. 2 | no. 1 | article 016101.

[DOI 10.1063/5.0216833](#) | [Source record](#) | Registered DOI metadata

Q0804 A singlet-triplet hole-spin qubit in MOS silicon

S. D. Liles, D. J. Halverson, Z. Wang, A. Shamim, R. S. Eggli, I. K. Jin, J. Hillier, K. Kumar et al..

Journal article | 2024 | Nature Communications | vol. 15 | no. 1 | article 7690.

[DOI 10.1038/s41467-024-51902-9](#) | [Source record](#) | Registered DOI metadata

Q0805 A singlet-triplet hole-spin qubit in MOS silicon

Scott Liles, Daniel Halverson, Zhanning Wang, Aaquib Shamim, Rafael Eggli, Ik Kyeong Jin, Joe Hillier, Kritika Kumar et al..

Posted content | 2024 | Springer Science and Business Media LLC.

[DOI 10.21203/rs.3.rs-3603337/v1](#) | [Source record](#) | Registered DOI metadata

Q0806 Tin as a nuclear spin qubit in silicon

Wayne Witzel.

Conference paper | 2024 | APS March Meeting 2024 - Minneapolis, Minnesota, United States of America - March - 2024.

[DOI 10.2172/2540383](#) | [Source record](#) | Registered DOI metadata

D15T02 Germanium hole qubits

Germanium hole systems provide strong electric control through spin orbit coupling. Strain, confinement, charge noise, and interactions determine operation quality.

Fine subcategories: Heavy holes; strong spin orbit coupling; planar dots; strain; electric control.

Primary catalog resources: 10. Related topic assignments are searchable in the HTML.

Q0807 Comparative assessment of germanium-based spin-qubit modalities: donor, acceptor, gate-defined hole, and gate-defined electron platforms

D-M Mei, K-M Dong, S A Panamaldeniya, A Prem, S Chhetri, N Budhathoki, S Bhattarai.

Journal article | 2026 | Quantum Science and Technology | vol. 11 | no. 4 | pp. 043001.

[DOI 10.1088/2058-9565/ae9cda](https://doi.org/10.1088/2058-9565/ae9cda) | [Source record](#) | Registered DOI metadata

Q0808 Optimising germanium hole spin qubits with a room-temperature magnet

Cécile X. Yu, Barnaby van Straaten, Alexander S. Ivlev, Valentin John, Damien R. Crielgaard, Stefan D. Oosterhout, Lucas E. A. Stehouwer, Francesco Borsoi et al..

Journal article | 2026 | Communications Physics | vol. 9 | no. 1 | article 272.

[DOI 10.1038/s42005-026-02634-3](https://doi.org/10.1038/s42005-026-02634-3) | [Source record](#) | Registered DOI metadata

Q0809 Variability of hole-spin qubits in planar germanium

Biel Martinez, Yann-Michel Niquet.

Journal article | 2026 | Physical Review Applied | vol. 25 | no. 1 | article 014018.

[DOI 10.1103/mtky-93p1](https://doi.org/10.1103/mtky-93p1) | [Source record](#) | Registered DOI metadata

Q0810 Hole spin qubits in unstrained Germanium layers

Lorenzo Mauro, Mauricio J. Rodríguez, Esteban A. Rodríguez-Mena, Yann-Michel Niquet.

Journal article | 2025 | npj Quantum Information | vol. 11 | no. 1 | article 167.

[DOI 10.1038/s41534-025-01108-8](https://doi.org/10.1038/s41534-025-01108-8) | [Source record](#) | Registered DOI metadata

Fine tags: strain

Q0811 Low-noise hole spin qubits in germanium

Danielle Holmes.

Journal article | 2025 | Nature Materials | vol. 24 | no. 12 | pp. 1869-1870.

[DOI 10.1038/s41563-025-02307-6](https://doi.org/10.1038/s41563-025-02307-6) | [Source record](#) | Registered DOI metadata

Q0812 Sweet-spot operation of a germanium hole spin qubit with highly anisotropic noise sensitivity

N. W. Hendrickx, L. Massai, M. Mergenthaler, F. J. Schupp, S. Paredes, S. W. Bedell, G. Salis, A. Fuhrer.

Journal article | 2024 | Nature Materials | vol. 23 | no. 7 | pp. 920-927.

[DOI 10.1038/s41563-024-01857-5](https://doi.org/10.1038/s41563-024-01857-5) | [Source record](#) | Registered DOI metadata

Q0813 Ultrafast and Electrically Tunable Rabi Frequency in a Germanium Hut Wire Hole Spin Qubit

He Liu, Ke Wang, Fei Gao, Jin Leng, Yang Liu, Yu-Chen Zhou, Gang Cao, Ting Wang et al..

Journal article | 2023 | Nano Letters | vol. 23 | no. 9 | pp. 3810-3817.

[DOI 10.1021/acs.nanolett.3c00213](https://doi.org/10.1021/acs.nanolett.3c00213) | [Source record](#) | Registered DOI metadata

Q0814 DFT Analysis of Hole Qubits Spin State in Germanium Thin Layer

Andrey Chibisov, Maxim Aleshin, Mary Chibisova.

Journal article | 2022 | Nanomaterials | vol. 12 | no. 13 | pp. 2244.

[DOI 10.3390/nano12132244](https://doi.org/10.3390/nano12132244) | [Source record](#) | Registered DOI metadata

Q0815 Emergent linear Rashba spin-orbit coupling offers fast manipulation of hole-spin qubits in germanium

Yang Liu, Jia-Xin Xiong, Zhi Wang, Wen-Long Ma, Shan Guan, Jun-Wei Luo, Shu-Shen Li.

Journal article | 2022 | Physical Review B | vol. 105 | no. 7 | article 075313.

[DOI 10.1103/physrevb.105.075313](https://doi.org/10.1103/physrevb.105.075313) | [Source record](#) | Registered DOI metadata

Q0816 Ultrafast coherent control of a hole spin qubit in a germanium quantum dot

Ke Wang, Gang Xu, Fei Gao, He Liu, Rong-Long Ma, Xin Zhang, Zhanning Wang, Gang Cao et al..

Journal article | 2022 | Nature Communications | vol. 13 | no. 1 | article 206.

[DOI 10.1038/s41467-021-27880-7](https://doi.org/10.1038/s41467-021-27880-7) | [Source record](#) | Registered DOI metadata

D15T03 Donor qubits

Donor processors use localized electron and nuclear spins in semiconductors. Placement, hyperfine control, and readout interfaces affect scalability.

Fine subcategories: Phosphorus donors; nuclear spins; hyperfine coupling; atom placement; donor readout.

Primary catalog resources: 7. Related topic assignments are searchable in the HTML.

Q0817 Atomistic first-principles modeling of single donor spin-qubit

Songqi Jia, Félix Beaudoin, Pericles Philippopoulos, Hong Guo.

Journal article | 2024 | Applied Physics Letters | vol. 125 | no. 18 | article 184001.

[DOI 10.1063/5.0221229](https://doi.org/10.1063/5.0221229) | [Source record](#) | Registered DOI metadata

Q0818 Precision tomography of a three-qubit donor quantum processor in silicon

Mateusz T. Mądzik, Serwan Asaad, Akram Youssry, Benjamin Joecker, Kenneth M. Rudinger, Erik Nielsen, Kevin C. Young, Timothy J. Proctor et al..

Journal article | 2022 | Nature | vol. 601 | no. 7893 | pp. 348-353.

[DOI 10.1038/s41586-021-04292-7](https://doi.org/10.1038/s41586-021-04292-7) | [Source record](#) | Registered DOI metadata

Q0819 Coherent control of a donor-molecule electron spin qubit in silicon

Lukas Fricke, Samuel J. Hile, Ludwik Kranz, Yousun Chung, Yu He, Prasanna Pakkiam, Matthew G. House, Joris G. Keizer et al..

Journal article | 2021 | Nature Communications | vol. 12 | no. 1 | article 3323.

[DOI 10.1038/s41467-021-23662-3](https://doi.org/10.1038/s41467-021-23662-3) | [Source record](#) | Registered DOI metadata

Q0820 A two-qubit gate between phosphorus donor electrons in silicon

Y. He, S. K. Gorman, D. Keith, L. Kranz, J. G. Keizer, M. Y. Simmons.

Journal article | 2019 | Nature | vol. 571 | no. 7765 | pp. 371-375.

[DOI 10.1038/s41586-019-1381-2](https://doi.org/10.1038/s41586-019-1381-2) | [Source record](#) | Registered DOI metadata

Q0821 Characterization of a Scalable Donor-Based Singlet-Triplet Qubit Architecture in Silicon

Prasanna Pakkiam, Matthew G. House, Matthias Koch, Michelle Y. Simmons.

Journal article | 2018 | Nano Letters | vol. 18 | no. 7 | pp. 4081-4085.

[DOI 10.1021/acs.nanolett.8b00006](https://doi.org/10.1021/acs.nanolett.8b00006) | [Source record](#) | Registered DOI metadata

Q0822 Optical control of donor spin qubits in silicon

M. J. Gullans, J. M. Taylor.

Journal article | 2015 | Physical Review B | vol. 92 | no. 19 | article 195411.

[DOI 10.1103/physrevb.92.195411](https://doi.org/10.1103/physrevb.92.195411) | [Source record](#) | Registered DOI metadata

Q0823 Spin delocalization in phosphorus donor pairs in silicon

J. H. Pifer.

Journal article | 1985 | Physical Review B | vol. 32 | no. 11 | pp. 7091-7097.

[DOI 10.1103/physrevb.32.7091](https://doi.org/10.1103/physrevb.32.7091) | [Source record](#) | Registered DOI metadata

D15T04 Spin readout and exchange gates

Exchange operations couple nearby spins, while spin to charge conversion enables readout. Control uniformity and charge noise connect gate and measurement performance.

Fine subcategories: Pauli blockade; charge sensing; exchange coupling; resonator readout; spin to charge conversion.

Primary catalog resources: 7. Related topic assignments are searchable in the HTML.

Q0824 Electron readout contrast enhancement in the parallel nuclear regime of an exchange-coupled donor spin qubit system

Holly G. Stemp, Mark R. van Blankestein, Benjamin Wilhelm, Serwan Asaad, Mateusz T. Mądzik, Arne Laucht, Fay E. Hudson, Andrew S. Dzurak et al..

Journal article | 2026 | Physical Review B | vol. 113 | no. 24 | article 245409.

[DOI 10.1103/1jc4-v2zt](#) | [Source record](#) | Registered DOI metadata

Q0825 Exchange-only qubit stabilized by a single-spin qubit

Irina Heinz, Mira Sharma, Joris Kattemölle.

Posted content | 2026 | Springer Science and Business Media LLC.

[DOI 10.21203/rs.3.rs-10700701/v1](#) | [Source record](#) | Registered DOI metadata

Q0826 Singlet-only always-on gapless exchange spin qubits: Charge noise effects and two-qubit gates

Anonymous.

Journal article | 2026 | Physical Review B.

[DOI 10.1103/8bzz-rw2j](#) | [Source record](#) | Registered DOI metadata

Q0827 Integrated silicon qubit platform with single-spin addressability, exchange control and single-shot singlet-triplet readout

M. A. Fogarty, K. W. Chan, B. Hensen, W. Huang, T. Tanttu, C. H. Yang, A. Laucht, M. Veldhorst et al..

Journal article | 2018 | Nature Communications | vol. 9 | no. 1 | article 4370.

[DOI 10.1038/s41467-018-06039-x](#) | [Source record](#) | Registered DOI metadata

Q0828 Quadrupolar Exchange-Only Spin Qubit

Maximilian Russ, J. R. Petta, Guido Burkard.

Journal article | 2018 | Physical Review Letters | vol. 121 | no. 17 | article 177701.

[DOI 10.1103/physrevlett.121.177701](#) | [Source record](#) | Registered DOI metadata

Q0829 Exchange-only singlet-only spin qubit

Arnau Sala, Jeroen Danon.

Journal article | 2017 | Physical Review B | vol. 95 | no. 24 | article 241303.

[DOI 10.1103/physrevb.95.241303](#) | [Source record](#) | Registered DOI metadata

Q0830 Anisotropic Spin Exchange in Pulsed Quantum Gates

N. E. Bonesteel, D. Stepanenko, D. P. DiVincenzo.

Journal article | 2001 | Physical Review Letters | vol. 87 | no. 20 | article 207901.

[DOI 10.1103/physrevlett.87.207901](#) | [Source record](#) | Registered DOI metadata

D15T05 Spin shuttling and scalable arrays

Shuttling and array control seek to connect spin qubits beyond immediate neighbors. Transport coherence and wiring complexity become system level constraints.

Fine subcategories: Electron transport; crossbar control; long range coupling; arrays; cryogenic interfaces.

Primary catalog resources: 8. Related topic assignments are searchable in the HTML.

Q0831 Interplay of Zeeman splitting and tunnel coupling in coherent spin-qubit shuttling

Ssu-Chih Lin, Paul Steinacker, MengKe Feng, Ajit Dash, Santiago Serrano, Wee Han Lim, Kohei M. Itoh, Fay E. Hudson et al..

Journal article | 2026 | Physical Review Applied | vol. 25 | no. 3 | article 034016.

[DOI 10.1103/3d1t-pr7m](#) | [Source record](#) | Registered DOI metadata

Q0832 Numerical simulation of charged-defect-induced decoherence in conveyor-mode spin qubit shuttling in Si/SiGe

Nils Ciroth, Arnau Sala, Ran Xue, Lasse Ermoneit, Thomas Koprucki, Markus Kantner, Lars R. Schreiber.

Journal article | 2026 | Physical Review B | vol. 114 | no. 10 | article 105305.

[DOI 10.1103/styv-ypg9](#) | [Source record](#) | Registered DOI metadata

Q0833 Performance of the spin qubit shuttling architecture for a surface code implementation

Berat Yenilen, Arnau Sala, Hendrik Bluhm, Markus Müller, Manuel Rispler.

Journal article | 2026 | Quantum | vol. 10 | pp. 2219 | article 2219.

[DOI 10.22331/q-2026-09-30-2219](#) | [Source record](#) | Registered DOI metadata

Q0834 Spin Qubit Leapfrogging: Dynamics of shuttling electrons on top of another

Anonymous.

Journal article | 2026 | Physical Review Research.

[DOI 10.1103/11vq-9zhr](#) | [Source record](#) | Registered DOI metadata

Q0835 Synthesizing an optimal spin-qubit shuttling-bus architecture for the surface code

Pau Escofet, Eduard Alarcón, Sergi Abadal, Andrii Semenov, Niall Murphy, Elena Blokhina, Carmen G. Almudéver.

Journal article | 2026 | Physical Review A | vol. 113 | no. 3 | article 032404.

[DOI 10.1103/4p4q-fm3y](#) | [Source record](#) | Registered DOI metadata

Q0836 Dephasing and error dynamics affecting a singlet-triplet qubit during coherent spin shuttling

Natalie D. Foster, Jacob D. Henshaw, Martin Rudolph, Dwight R. Luhman, Ryan M. Jock.

Journal article | 2025 | npj Quantum Information | vol. 11 | no. 1 | article 63.

[DOI 10.1038/s41534-025-00996-0](#) | [Source record](#) | Registered DOI metadata

Q0837 High-fidelity single-electron shuttling in industrially fabricated spin qubit devices

P. Muster, W. Langheinrich, T. Huckemann, S. Pregl, V. Brackmann, M. Friedrich, F. Reichmann, N. D. Komerički et al..

Conference paper | 2025 | 2025 IEEE International Electron Devices Meeting (IEDM) | pp. 1-4.

[DOI 10.1109/iedm50572.2025.11353490](#) | [Source record](#) | Registered DOI metadata

Q0838 Large spin-shuttling oscillations enabling high-fidelity single-qubit gates

Akshay Menon Pazhedath, Alessandro David, Max Oberländer, Matthias M. Müller, Tommaso Calarco, Hendrik Bluhm, Felix Motzoi.

Journal article | 2025 | Physical Review Applied | vol. 24 | no. 3 | article 034029.

[DOI 10.1103/4lky-413f](#) | [Source record](#) | Registered DOI metadata

D16 Photonic quantum computing

Photonic approaches encode information in optical modes and use interference, measurements, and sometimes nonlinear interactions. Loss and resource generation are central engineering constraints.

Prerequisites: Quantum optics, interference, and photodetection.

Assessment focus: Loss, indistinguishability, source rate, detector efficiency, and resource overhead.

D16T01 Linear optical computation

Linear optics uses interference, measurement, and ancillary photons for computation. Probabilistic operations require resource accounting and feed forward.

Fine subcategories: KLM; interferometers; photon counting; postselection; feed forward.

Primary catalog resources: 10. Related topic assignments are searchable in the HTML.

Q0839 A dataflow programming framework for linear optical distributed quantum computing

Giovanni de Felice, Boldizsár Poór, Cole Comfort, Lia Yeh, Mateusz Kupper, William Cashman, Bob Coecke.

Journal article | 2026 | Quantum | vol. 10 | pp. 1972 | article 1972.

[DOI 10.22331/q-2026-01-19-1972](#) | [Source record](#) | Registered DOI metadata

Q0840 Fusion for high-dimensional linear-optical quantum computing with improved success probability

Gözde Üstün, Eleanor G. Rieffel, Simon J. Devitt, Jason Saied.

Journal article | 2025 | Physical Review Applied | vol. 24 | no. 4 | article 044024.

[DOI 10.1103/7bg-hc8c](#) | [Source record](#) | Registered DOI metadata

Q0841 Linear optical quantum computing with a hybrid squeezed-cat code

Shohei Kiryu, Kosuke Fukui, Atsushi Okamoto, Akihisa Tomita.

Journal article | 2025 | Physical Review A | vol. 112 | no. 4 | article 042615.

[DOI 10.1103/jxbx-75m9](#) | [Source record](#) | Registered DOI metadata

Q0842 4-Bit Linear Optical Quantum Computing with Liquid Crystal Devices

Satoshi Yokotsuka, Hiroyuki Okada.

Posted content | 2024 | Elsevier BV.

[DOI 10.2139/ssrn.4792710](#) | [Source record](#) | Registered DOI metadata

Q0843 Four-bit input linear optical quantum computing with liquid crystal devices

Satoshi Yokotsuka, Hiroyuki Okada.

Journal article | 2024 | APL Quantum | vol. 1 | no. 4 | article 046105.

[DOI 10.1063/5.0224043](#) | [Source record](#) | Registered DOI metadata

Q0844 Quantum photonic integrated circuits simulation tool for linear optical quantum computing gates

Argiris Ntanos, Giannis Giannoulis, Aris Stathis, Dimitris Zavitsanos, Hercules Avramopoulos.

Conference paper | 2024 | Quantum Technologies 2024 | pp. 50.

[DOI 10.1117/12.3022597](#) | [Source record](#) | Registered DOI metadata

Q0845 Linear optical quantum computation

Varna University of Management, Andreas Cristoforides, Aaron Miller.

Report | 2020 | Web of Open Science.

[DOI 10.37686/qrl.v1i2.58](#) | [Source record](#) | Registered DOI metadata

Q0846 Scalable controlled-not gate for linear optical quantum computing using microring resonators

Ryan E. Scott, Paul M. Alsing, A. Matthew Smith, Michael L. Fanto, Christopher C. Tison, James Schneeloch, Edwin E. Hach.

Journal article | 2019 | Physical Review A | vol. 100 | no. 2 | article 022322.

[DOI 10.1103/physreva.100.022322](https://doi.org/10.1103/physreva.100.022322) | [Source record](#) | Registered DOI metadata

Q0847 Resource Costs for Fault-Tolerant Linear Optical Quantum Computing

Ying Li, Peter C. Humphreys, Gabriel J. Mendoza, Simon C. Benjamin.

Journal article | 2015 | Physical Review X | vol. 5 | no. 4 | article 041007.

[DOI 10.1103/physrevx.5.041007](https://doi.org/10.1103/physrevx.5.041007) | [Source record](#) | Registered DOI metadata

Q0848 Redirecting to main branch docs

Author metadata not supplied.

Documentation | Undated | Quandela Perceval.

[Official resource](#) | [Source record](#) | Official page retrieved

D16T02 Boson sampling

Boson sampling samples distributions produced by indistinguishable particles passing through an optical network. Loss, distinguishability, and classical simulation affect hardness claims.

Fine subcategories: Permanent estimation; Gaussian boson sampling; distinguishability; loss; classical simulation.

Primary catalog resources: 15. Related topic assignments are searchable in the HTML.

Q0849 Experimental loopback boson sampling

Yu. A. Biriukov, R. D. Morozov, K. I. Okhlopkov, I. V. Dyakonov, N. N. Skryabin, S. A. Zhuravitskii, M. A. Dryazgov, K. V. Taratorin et al..

Journal article | 2026 | Optica Quantum | vol. 4 | no. 5 | pp. 460.

[DOI 10.1364/opticaq.599335](https://doi.org/10.1364/opticaq.599335) | [Source record](#) | Registered DOI metadata

Q0850 Boson Sampling

A.P. Lund, T.C. Ralph.

Book chapter | 2025 | Encyclopedia of Mathematical Physics | pp. 42-56.

[DOI 10.1016/b978-0-323-95703-8.00111-7](https://doi.org/10.1016/b978-0-323-95703-8.00111-7) | [Source record](#) | Registered DOI metadata

Q0851 Boson Sampling Enhanced Quantum Chemistry

Zhong-Xia Shang, Yukun Zhang, Han-Sen Zhong, Cheng-Cheng Yu, Xiao Yuan, Chao-Yang Lu, Jian-Wei Pan, Ming-Cheng Chen.

Journal article | 2025 | PRX Quantum | vol. 6 | no. 4 | article 040357.

[DOI 10.1103/gw1c-5b58](https://doi.org/10.1103/gw1c-5b58) | [Source record](#) | Registered DOI metadata

Q0852 Boson sampling powered image recognition

William J. Munro, Akitada Sakurai, Aoi Hayashi, Kae Nemoto.

Conference paper | 2025 | Quantum Computing, Communication, and Simulation V | pp. 45.

[DOI 10.1117/12.3038939](https://doi.org/10.1117/12.3038939) | [Source record](#) | Registered DOI metadata

Q0853 Unified boson sampling

Luca Bianchi, Carlo Marconi, Laura Ares, Davide Bacco, Jan Sperling.

Journal article | 2025 | Physical Review Research | vol. 7 | no. 4 | article L042068.

[DOI 10.1103/8hy1-m5gg](https://doi.org/10.1103/8hy1-m5gg) | [Source record](#) | Registered DOI metadata

Q0854 Classical sampling from noisy boson sampling and negative probabilities

V. S. Shchesnovich.

Journal article | 2024 | Physical Review A | vol. 109 | no. 3 | article 032610.

[DOI 10.1103/physreva.109.032610](https://doi.org/10.1103/physreva.109.032610) | [Source record](#) | Registered DOI metadata

Q0855 Extending classical boson sampling techniques

S. N. van den Hoven, E. Kanis, J. J. Renema.

Conference paper | 2024 | Quantum 2.0 Conference and Exhibition | pp. QW4A.10.

[DOI 10.1364/quantum.2024.qw4a.10](https://doi.org/10.1364/quantum.2024.qw4a.10) | [Source record](#) | Registered DOI metadata

Q0856 Faster classical boson sampling

Peter Clifford, Raphaël Clifford.

Journal article | 2024 | Physica Scripta | vol. 99 | no. 6 | pp. 065121.

[DOI 10.1088/1402-4896/ad4688](https://doi.org/10.1088/1402-4896/ad4688) | [Source record](#) | Registered DOI metadata

Q0857 Gaussian boson sampling at finite temperature

Gabriele Bressanini, Hyukjoon Kwon, M. S. Kim.

Journal article | 2024 | Physical Review A | vol. 109 | no. 1 | article 013707.

[DOI 10.1103/physreva.109.013707](https://doi.org/10.1103/physreva.109.013707) | [Source record](#) | Registered DOI metadata

Fine tags: Gaussian boson sampling

Q0858 Hybrid Boson Sampling

Vitaly Kocharovsky.

Journal article | 2024 | Entropy | vol. 26 | no. 11 | pp. 926.

[DOI 10.3390/e26110926](https://doi.org/10.3390/e26110926) | [Source record](#) | Registered DOI metadata

Q0859 Gaussian Boson Sampling

Claudio Conti.

Book chapter | 2023 | Quantum Science and Technology | pp. 301-346.

[DOI 10.1007/978-3-031-44226-1_12](https://doi.org/10.1007/978-3-031-44226-1_12) | [Source record](#) | Registered DOI metadata

Fine tags: Gaussian boson sampling

Q0860 Non-linear Boson Sampling

Nicolò Spagnolo, Daniel J. Brod, Ernesto F. Galvão, Fabio Sciarrino.

Journal article | 2023 | npj Quantum Information | vol. 9 | no. 1 | article 3.

[DOI 10.1038/s41534-023-00676-x](https://doi.org/10.1038/s41534-023-00676-x) | [Source record](#) | Registered DOI metadata

Q0861 Programmability empowering quantum boson sampling

Zhaorong Fu, Jueming Bao, Jianwei Wang.

Journal article | 2023 | Nature Computational Science | vol. 3 | no. 10 | pp. 819-820.

[DOI 10.1038/s43588-023-00534-y](https://doi.org/10.1038/s43588-023-00534-y) | [Source record](#) | Registered DOI metadata

Q0862 Boson sampling for generalized bosons

En-Jui Kuo, Yijia Xu, Dominik Hangleiter, Andrey Grankin, Mohammad Hafezi.

Journal article | 2022 | Physical Review Research | vol. 4 | no. 4 | article 043096.

[DOI 10.1103/physrevresearch.4.043096](https://doi.org/10.1103/physrevresearch.4.043096) | [Source record](#) | Registered DOI metadata

Q0863 Timestamp boson sampling

Wen-Hao Zhou, Jun Gao, Zhi-Qiang Jiao, Xiao-Wei Wang, Ruo-Jing Ren, Xiao-Ling Pang, Lu-Feng Qiao, Chao-Ni Zhang et al..

Journal article | 2022 | Applied Physics Reviews | vol. 9 | no. 3 | article 031408.

[DOI 10.1063/5.0066103](https://doi.org/10.1063/5.0066103) | [Source record](#) | Registered DOI metadata

D16T03 Integrated quantum photonics

Integrated photonics places optical components on chips. Loss, phase stability, fabrication variation, and coupling to sources and detectors determine usable performance.

Fine subcategories: Waveguides; interferometer meshes; phase shifters; material platforms; packaging.

Primary catalog resources: 13. Related topic assignments are searchable in the HTML.

Q0864 Programmable integrated quantum photonics

Igor Aharonovich, Kenneth B. Crozier, Dragomir Neshev.

Journal article | 2026 | Nature Photonics | vol. 20 | no. 3 | pp. 254-265.

[DOI 10.1038/s41566-025-01830-x](#) | [Source record](#) | Registered DOI metadata

Q0865 Diamond integrated quantum photonics

Lin Jin, Wolfram H.P. Pernice.

Book chapter | 2025 | Nanophotonics with Diamond and Silicon Carbide for Quantum Technologies | pp. 143-170.

[DOI 10.1016/b978-0-443-13717-4.00008-6](#) | [Source record](#) | Registered DOI metadata

Q0866 Distributed quantum sensing with integrated quantum photonics

Jonathan C. F. Matthews, Beth Puzio.

Conference paper | 2025 | Quantum Sensing, Imaging, and Precision Metrology III | pp. 33.

[DOI 10.1117/12.3053407](#) | [Source record](#) | Registered DOI metadata

Q0867 Enabling quantum sensors with integrated photonics

Neal E. Solmeyer, Argyrios Dellis, Javad Dowran, Chad Fertig, Luke Horstman, Chad Hoyt, Wei Jiang, Karl Nelson et al..

Conference paper | 2025 | Quantum Sensing and Nano Electronics and Photonics XXI | pp. 22.

[DOI 10.1117/12.3042528](#) | [Source record](#) | Registered DOI metadata

Q0868 Integrated Quantum Photonics

Krishna Thyagarajan.

Book | 2025 | Graduate Texts in Physics.

[DOI 10.1007/978-3-031-85728-7](#) | [Source record](#) | Registered DOI metadata

Q0869 Quantum integrated photonics chips and packaging

Stefan F. Preble.

Conference paper | 2024 | Quantum Information Science, Sensing, and Computation XVI | pp. 34.

[DOI 10.1117/12.3021843](#) | [Source record](#) | Registered DOI metadata

Fine tags: packaging

Q0870 Integrated Quantum Photonics

Author metadata not supplied.

Edited book | 2023 | Frontiers Research Topics.

[DOI 10.3389/978-2-88974-074-1](#) | [Source record](#) | Registered DOI metadata

Q0871 Integrated photonics in quantum technologies

Taira Giordani, Francesco Hoch, Gonzalo Carvacho, Nicolò Spagnolo, Fabio Sciarrino.

Journal article | 2023 | La Rivista del Nuovo Cimento | vol. 46 | no. 2 | pp. 71-103.

[DOI 10.1007/s40766-023-00040-x](#) | [Source record](#) | Registered DOI metadata

Q0872 QUANTUM INTEGRATED PHOTONICS

Ryszard Romaniuk.

Journal article | 2023 | ELEKTRONIKA - KONSTRUKCJE, TECHNOLOGIE, ZASTOSOWANIA | vol. 1 | no. 8 | pp. 19-26.

[DOI 10.15199/13.2023.8.4](#) | [Source record](#) | Registered DOI metadata

Q0873 Quantum Information with Integrated Photonics

Paolo Piergentili, Francesco Amanti, Greta Andrini, Fabrizio Armani, Vittorio Bellani, Vincenzo Bonaiuto, Simone Cammarata, Matteo Campostrini et al..

Journal article | 2023 | Applied Sciences | vol. 14 | no. 1 | pp. 387.

[DOI 10.3390/app14010387](https://doi.org/10.3390/app14010387) | [Source record](#) | Registered DOI metadata

Q0874 ScAIN integrated quantum photonics

Zetian Mi.

Conference paper | 2023 | Quantum Sensing and Nano Electronics and Photonics XIX | pp. 1.

[DOI 10.1117/12.2646611](https://doi.org/10.1117/12.2646611) | [Source record](#) | Registered DOI metadata

Q0875 Silicon Nitride Integrated Quantum Photonics

Khaled Mnaymneh, Edith Yeung, David B. Northeast, Jeongwan Jin, Patrick Laferrière, Sofiane Haffouz, Philip J. Poole, Dan Dalacu et al..

Conference paper | 2023 | 2023 23rd International Conference on Transparent Optical Networks (ICTON) | pp. 1-3.

[DOI 10.1109/icton59386.2023.10207494](https://doi.org/10.1109/icton59386.2023.10207494) | [Source record](#) | Registered DOI metadata

Q0876 2022 Roadmap on integrated quantum photonics

Galan Moody, Volker J Sorger, Daniel J Blumenthal, Paul W Juodawlkis, William Loh, Cheryl Sorace-Agaskar, Alex E Jones, Krishna C Balram et al..

Journal article | 2022 | Journal of Physics: Photonics | vol. 4 | no. 1 | pp. 012501.

[DOI 10.1088/2515-7647/ac1ef4](https://doi.org/10.1088/2515-7647/ac1ef4) | [Source record](#) | Registered DOI metadata

D16T04 Single photon sources and detectors

Photon sources and detectors supply and measure optical quantum resources. Efficiency, purity, timing, and indistinguishability directly influence system scale.

Fine subcategories: Heralded sources; quantum dots; SNSPDs; number resolution; efficiency.

Primary catalog resources: 8. Related topic assignments are searchable in the HTML.

Q0877 Improved Heralded Single-Photon Source with a Photon-Number-Resolving Superconducting Nanowire Detector

Samantha I. Davis, Andrew Mueller, Raju Valivarathi, Nikolai Lauk, Lautaro Narvaez, Boris Korzh, Andrew D. Beyer, Olmo Cerri et al..

Journal article | 2022 | Physical Review Applied | vol. 18 | no. 6 | article 064007.

[DOI 10.1103/physrevapplied.18.064007](https://doi.org/10.1103/physrevapplied.18.064007) | [Source record](#) | Registered DOI metadata

Q0878 Absolute calibration of a single-photon avalanche detector using a bright triggered single-photon source based on an InGaAs quantum dot

Hristina Georgieva, Marco López, Helmuth Hofer, Niklas Kanold, Arseny Kaganskiy, Sven Rodt, Stephan Reitzenstein, Stefan Kück.

Journal article | 2021 | Optics Express | vol. 29 | no. 15 | pp. 23500.

[DOI 10.1364/oe.430680](https://doi.org/10.1364/oe.430680) | [Source record](#) | Registered DOI metadata

Q0879 Light Source Monitoring in Quantum Key Distribution With Single-Photon Detector at Room Temperature

Gan Wang, Zhengyu Li, Yucheng Qiao, Ziyang Chen, Xiang Peng, Hong Guo.

Journal article | 2018 | IEEE Journal of Quantum Electronics | vol. 54 | no. 3 | pp. 1-10.

[DOI 10.1109/jqe.2018.2827569](https://doi.org/10.1109/jqe.2018.2827569) | [Source record](#) | Registered DOI metadata

Q0880 A flat spectral photon flux source for single photon detector quantum efficiency calibration

Haiyong Gan, Ruoduan Sun, Nan Xu, Jianwei Li, Yanfei Wang, Guojin Feng, Chundi Zheng, Chong Ma et al..
Conference paper | 2015 | 2015 11th Conference on Lasers and Electro-Optics Pacific Rim (CLEO-PR) | pp. 1-2.
[DOI 10.1109/cleopr.2015.7375863](https://doi.org/10.1109/cleopr.2015.7375863) | [Source record](#) | Registered DOI metadata
Fine tags: efficiency

Q0881 A superconducting nanowire single-photon detector system for single-photon source characterization

C. R. Fitzpatrick, C. M. Natarajan, R. E. Warburton, G. S. Buller, B. Baek, S. Nam, S. Miki, Z. Wang et al..
Conference paper | 2010 | SPIE Proceedings | vol. 7681 | pp. 76810H.
[DOI 10.1117/12.851892](https://doi.org/10.1117/12.851892) | [Source record](#) | Registered DOI metadata

Q0882 Quantum key distribution with a heralded single photon source and a photon number resolving detector

Tomoyuki Horikiri, Yuishi Takeno, Atsushi Yabushita, Haibo Wang, Takayoshi Kobayashi.
Conference paper | 2006 | 2006 Conference on Lasers and Electro-Optics and 2006 Quantum Electronics and Laser Science Conference | pp. 1-2.
[DOI 10.1109/cleo.2006.4628716](https://doi.org/10.1109/cleo.2006.4628716) | [Source record](#) | Registered DOI metadata

Q0883 Single photon source characterization with a superconducting single photon detector

Robert H. Hadfield, Martin J. Stevens, Steven S. Gruber, Aaron J. Miller, Robert E. Schwall, Richard P. Mirin, Sae Woo Nam.
Conference paper | 2006 | 2006 Conference on Lasers and Electro-Optics and 2006 Quantum Electronics and Laser Science Conference | pp. 1-2.
[DOI 10.1109/cleo.2006.4628701](https://doi.org/10.1109/cleo.2006.4628701) | [Source record](#) | Registered DOI metadata

Q0884 Single photon source characterization with a superconducting single photon detector

Robert H. Hadfield, Martin J. Stevens, Steven S. Gruber, Aaron J. Miller, Robert E. Schwall, Richard P. Mirin, Sae Woo Nam.
Journal article | 2005 | Optics Express | vol. 13 | no. 26 | pp. 10846.
[DOI 10.1364/opex.13.010846](https://doi.org/10.1364/opex.13.010846) | [Source record](#) | Registered DOI metadata

D16T05 Photonic cluster and fusion computing

Fusion based schemes combine smaller entangled states into computational resources. Loss tolerance and resource factory overhead are central design questions.

Fine subcategories: Fusion gates; graph states; resource factories; loss tolerance; multiplexing.

Primary catalog resources: 3. Related topic assignments are searchable in the HTML.

Q0885 Next-Gen Quantum Computing—The Fusion of Atoms and Photonic Innovation

Ashok Kumar Patel, K. Saradhi, Gudditi Chetan, Ganesuni Aasish Chowdary, Beebi Naseeba, Nagendra Panini Challa.
Book chapter | 2025 | Quantum Computing | pp. 330-351.
[DOI 10.1201/9781003538950-13](https://doi.org/10.1201/9781003538950-13) | [Source record](#) | Registered DOI metadata

Q0886 Tailoring Fusion-Based Photonic Quantum Computing Schemes to Quantum Emitters

Ming Lai Chan, Thomas J. Bell, Love A. Pettersson, Susan X. Chen, Patrick Yard, Anders S. Sørensen, Stefano Paesani.
Journal article | 2025 | PRX Quantum | vol. 6 | no. 2 | article 020304.
[DOI 10.1103/prxquantum.6.020304](https://doi.org/10.1103/prxquantum.6.020304) | [Source record](#) | Registered DOI metadata

Q0887 Path Qubit Fusion for Photonic Cluster State Generation

Hee Su Park, Sang Min Lee, Jaeyoon Cho, Yoonshik Kang, Sang-Kyung Choi.
Conference paper | 2011 | International Conference on Quantum Information | pp. QMI2.
[DOI 10.1364/icqi.2011.qmi2](https://doi.org/10.1364/icqi.2011.qmi2) | [Source record](#) | Registered DOI metadata

D16T06 Squeezing and Gaussian optics

Gaussian optics controls continuous variable modes with squeezing and linear transformations. Non Gaussian resources determine capabilities beyond Gaussian simulation.

Fine subcategories: Squeezed states; optical quadratures; Gaussian transformations; homodyne detection.

Primary catalog resources: 8. Related topic assignments are searchable in the HTML.

Q0888 Hyperbolic Distance Governs Gaussian Quantum Squeezing

Kenneth A. Menard.

Posted content | 2026 | Elsevier BV.

[DOI 10.2139/ssrn.6985762](https://doi.org/10.2139/ssrn.6985762) | [Source record](#) | Registered DOI metadata

Q0889 Hyperbolic Distance Governs Gaussian Quantum Squeezing

Kenneth Menard.

Posted content | 2026 | Optica Publishing Group.

[DOI 10.1364/opticaopen.32051823.v1](https://doi.org/10.1364/opticaopen.32051823.v1) | [Source record](#) | Registered DOI metadata

Q0890 Quantum squeezing and sensing with anti-parity-time symmetric quantum optics

Shengwang Du, Chuanwei Zhang.

Conference paper | 2023 | Quantum Sensing, Imaging, and Precision Metrology | pp. 73.

[DOI 10.1117/12.2657379](https://doi.org/10.1117/12.2657379) | [Source record](#) | Registered DOI metadata

Q0891 Onset of non-Gaussian quantum physics in pulsed squeezing with mesoscopic fields

Ryotatsu Yanagimoto, Edwin Ng, Atsushi Yamamura, Tatsuhiro Onodera, Logan G. Wright, Marc Jankowski, M. M. Fejer, Peter L. McMahon et al..

Journal article | 2022 | Optica | vol. 9 | no. 4 | pp. 379.

[DOI 10.1364/optica.447782](https://doi.org/10.1364/optica.447782) | [Source record](#) | Registered DOI metadata

Q0892 Non-Gaussian quantum states generation and robust quantum non-Gaussianity via squeezing field

Xu-Bing Tang, Fang Gao, Yao-Xiong Wang, Sen Kuang, Feng Shuang.

Journal article | 2015 | Chinese Physics B | vol. 24 | no. 3 | pp. 034208.

[DOI 10.1088/1674-1056/24/3/034208](https://doi.org/10.1088/1674-1056/24/3/034208) | [Source record](#) | Registered DOI metadata

Q0893 Surmounting intrinsic quantum-measurement uncertainties in Gaussian-state tomography with quadrature squeezing

Jaroslav Řeháček, Yong Siah Teo, Zdeněk Hradil, Sascha Wallentowitz.

Journal article | 2015 | Scientific Reports | vol. 5 | no. 1 | article 12289.

[DOI 10.1038/srep12289](https://doi.org/10.1038/srep12289) | [Source record](#) | Registered DOI metadata

Q0894 Gaussian entanglement for quantum key distribution from a single-mode squeezing source

Tobias Eberle, Vitus Händchen, Jörg Duhme, Torsten Franz, Fabian Furrer, Roman Schnabel, Reinhard F Werner.

Journal article | 2013 | New Journal of Physics | vol. 15 | no. 5 | pp. 053049.

[DOI 10.1088/1367-2630/15/5/053049](https://doi.org/10.1088/1367-2630/15/5/053049) | [Source record](#) | Registered DOI metadata

Q0895 Fisher Information and Quantum Squeezing Properties of Gaussian Pure States

Jia Qiang Zhao, Lian Zhen Cao, Huai Xin Lu.

Journal article | 2012 | Advanced Materials Research | vol. 571 | pp. 283-286.

[DOI 10.4028/www.scientific.net/amr.571.283](https://doi.org/10.4028/www.scientific.net/amr.571.283) | [Source record](#) | Registered DOI metadata

D17 Other qubit platforms and transduction

Additional platforms offer different routes to coherence, control, and connectivity. Quantum transduction connects devices that operate at different physical frequencies.

Prerequisites: Solid state or molecular physics and quantum control.

Assessment focus: Platform specific evidence, scalability, conversion efficiency, and added noise.

D17T01 Diamond NV centers

NV centers combine electronic and nuclear spins with optical interfaces in diamond. Their computing and sensing uses should be distinguished when reading application claims.

Fine subcategories: Electronic spins; nuclear registers; optical interfaces; spin control; fabrication.

Primary catalog resources: 14. Related topic assignments are searchable in the HTML.

Q0896 Low-Frequency Current Detection via Diamond Nitrogen-Vacancy Center Quantum Sensing

Tianfu Huang, Tongrao Lin, Xiaoxu Hu, Xiaofei Li, Han Wang.

Book chapter | 2026 | Lecture Notes in Electrical Engineering | pp. 261-268.

[DOI 10.1007/978-981-95-5304-4_23](#) | [Source record](#) | Registered DOI metadata

Q0897 Nitrogen-vacancy centers in diamond: From fundamental principles to quantum sensing

Yuehui LI, Shaobo CHENG, Chongxin SHAN.

Journal article | 2026 | Acta Physica Sinica | vol. 75 | no. 8 | article 080702.

[DOI 10.7498/aps.75.20251531](#) | [Source record](#) | Registered DOI metadata

Q0898 Probing quantum resources in nitrogen-vacancy centers in diamond

Aicha Chouiba, Samira Elghaayda, Mostafa Mansour.

Journal article | 2026 | Physica Scripta | vol. 101 | no. 21 | pp. 215108.

[DOI 10.1088/1402-4896/ae6e2d](#) | [Source record](#) | Registered DOI metadata

Q0899 Quantum co-magnetometer using diamond nitrogen-vacancy centers and Rubidium cells

Nir Bar-Gill.

Conference paper | 2026 | Optical Sensing and Precision Metrology II | pp. 3.

[DOI 10.1117/12.3092404](#) | [Source record](#) | Registered DOI metadata

Q0900 Unraveling quantum dephasing of nitrogen-vacancy center ensembles in diamond

Jixing Zhang, Cheuk Kit Cheung, Michael Kübler, Magnus Benke, Mathis Brossaud, Yihua Wang, Andrej Denisenko, Ruoming Peng et al..

Journal article | 2026 | npj Quantum Materials | vol. 11 | no. 1 | article 27.

[DOI 10.1038/s41535-026-00869-5](#) | [Source record](#) | Registered DOI metadata

Q0901 Oxygen-terminated hexagonal diamond (001) surfaces for nitrogen-vacancy based quantum sensors

Bo Cui, Zhaolong Sun, Wencui Xiu, You Lv, Nan Gao, Hongdong Li.

Posted content | 2025 | Elsevier BV.

[DOI 10.2139/ssrn.5954749](#) | [Source record](#) | Registered DOI metadata

Q0902 Design of compact integrated diamond nitrogen-vacancy center quantum probe

Sheng-Kai Xia , Wen-Tao Lu , Xu-Tong Zhao , Ya-Wen Xue , Zeng-Bo Xu , Shi-Yu Ge , Yang Wang , Lin-Yan Yu et al..

Journal article | 2024 | Chinese Physics B | vol. 33 | no. 5 | pp. 054202.

[DOI 10.1088/1674-1056/ad2bf6](#) | [Source record](#) | Registered DOI metadata

Q0903 Nitrogen-terminated diamond (111) surface for nitrogen-vacancy based quantum sensors

Li Gaoxian, Cheng Wei, Gao Nan, Cheng Shaoheng, Li Hongdong.

Journal article | 2024 | Diamond and Related Materials | vol. 142 | pp. 110813 | article 110813.

[DOI 10.1016/j.diamond.2024.110813](https://doi.org/10.1016/j.diamond.2024.110813) | [Source record](#) | Registered DOI metadata

Q0904 Quantum sensing of microRNAs with nitrogen-vacancy centers in diamond

Justas Zalieckas, Martin M. Greve, Luca Bellucci, Giuseppe Sacco, Verner Håkonsen, Valentina Tozzini, Riccardo Nifosì.

Journal article | 2024 | Communications Chemistry | vol. 7 | no. 1 | article 101.

[DOI 10.1038/s42004-024-01182-7](https://doi.org/10.1038/s42004-024-01182-7) | [Source record](#) | Registered DOI metadata

Q0905 All-optical nuclear quantum sensing using nitrogen-vacancy centers in diamond

B. Bürgler, T. F. Sjölander, O. Brinza, A. Tallaire, J. Achard, P. Maletinsky.

Journal article | 2023 | npj Quantum Information | vol. 9 | no. 1 | article 56.

[DOI 10.1038/s41534-023-00724-6](https://doi.org/10.1038/s41534-023-00724-6) | [Source record](#) | Registered DOI metadata

Q0906 Fast Quantum State Tomography in the Nitrogen Vacancy Center of Diamond

Jingfu Zhang, Swathi S. Hegde, Dieter Suter.

Journal article | 2023 | Physical Review Letters | vol. 130 | no. 9 | article 090801.

[DOI 10.1103/physrevlett.130.090801](https://doi.org/10.1103/physrevlett.130.090801) | [Source record](#) | Registered DOI metadata

Q0907 Fluorine-terminated diamond (110) surfaces for nitrogen-vacancy quantum sensors

Wei Shen, Gai Wu, Lijie Li, Hui Li, Sheng Liu, Shengnan Shen, Diwei Zou.

Journal article | 2022 | Carbon | vol. 193 | pp. 17-25.

[DOI 10.1016/j.carbon.2022.02.017](https://doi.org/10.1016/j.carbon.2022.02.017) | [Source record](#) | Registered DOI metadata

Q0908 Nitrogen Vacancy-Centered Diamond Qubit: The fabrication, design, and application in quantum computing

Ya-Chi Liu, Yi-Chung Dzung, Chao-Cheng Ting.

Journal article | 2022 | IEEE Nanotechnology Magazine | vol. 16 | no. 4 | pp. 37-43.

[DOI 10.1109/mnano.2022.3175405](https://doi.org/10.1109/mnano.2022.3175405) | [Source record](#) | Registered DOI metadata

Fine tags: fabrication

Q0909 Proximal nitrogen reduces the fluorescence quantum yield of nitrogen-vacancy centres in diamond

Marco Capelli, Lukas Lindner, Tingpeng Luo, Jan Jeske, Hiroshi Abe, Shinobu Onoda, Takeshi Ohshima, Brett Johnson et al..

Journal article | 2022 | New Journal of Physics | vol. 24 | no. 3 | pp. 033053.

[DOI 10.1088/1367-2630/ac5ca9](https://doi.org/10.1088/1367-2630/ac5ca9) | [Source record](#) | Registered DOI metadata

D17T02 Silicon carbide defects

Silicon carbide hosts optically addressable defects compatible with semiconductor fabrication. Coherence and interface properties depend on defect species and material quality.

Fine subcategories: Color centers; divacancies; spin photon interfaces; coherence; integration.

Primary catalog resources: 8. Related topic assignments are searchable in the HTML.

Q0910 In Silico Engineering of Defect Spins in Silicon Carbide for Solid-State Hybrid Quantum Technologies

Dang Huy Ngo, Huyen-Trang T. Le, Tien Lam Pham, Thi Minh Hoa Nghiem, Ngoc Linh Nguyen.

Journal article | 2026 | Journal of Physics: Conference Series | vol. 3284 | no. 1 | pp. 012019.

[DOI 10.1088/1742-6596/3284/1/012019](https://doi.org/10.1088/1742-6596/3284/1/012019) | [Source record](#) | Registered DOI metadata

Q0911 Silicon carbide turns quantum

Jörg Wrachtrup.

Journal article | 2026 | The Innovation Physics | vol. 1 | no. 1 | pp. 100003.

[DOI 10.59717/j.tip.2026.100003](#) | [Source record](#) | Registered DOI metadata

Q0912 A Suspended 4H-Silicon Carbide Membrane Platform for Defect Integration into Quantum Devices

Amberly H. Xie, Aaron M. Day, Jonathan R. Dietz, Chang Jin, Chaoshen Zhang, Eliana Mann, Zhuqing Xu, Marko Loncar et al..

Journal article | 2025 | Nano Letters | vol. 25 | no. 43 | pp. 15637-15642.

[DOI 10.1021/acs.nanolett.5c04169](#) | [Source record](#) | Registered DOI metadata

Fine tags: integration

Q0913 Defects in Silicon Carbide as Quantum Qubits: Recent Advances in Defect Engineering

Ivana Capan.

Journal article | 2025 | Applied Sciences | vol. 15 | no. 10 | pp. 5606.

[DOI 10.3390/app15105606](#) | [Source record](#) | Registered DOI metadata

Q0914 Wafer-scale quantum photonics in silicon carbide

Marina Radulaski.

Conference paper | 2025 | Quantum Computing, Communication, and Simulation V | pp. 23.

[DOI 10.1117/12.3042747](#) | [Source record](#) | Registered DOI metadata

Q0915 Fluorescent Silicon Carbide Quantum Dots

Mahdi Hasanzadeh Azar, Zimo Ji, Jahanbakhsh Jahanzamin, Adrian Kitai.

Book chapter | 2024 | Materials Science.

[DOI 10.5772/intechopen.1007535](#) | [Source record](#) | Registered DOI metadata

Q0916 Scalable Diamond and Silicon Carbide Quantum Systems

Jelena Vuckovic, Souvik Biswas.

Conference paper | 2024 | Frontiers in Optics + Laser Science 2024 (FIO, LS) | pp. FM5A.1.

[DOI 10.1364/ls.2024.fm5a.1](#) | [Source record](#) | Registered DOI metadata

Q0917 Improving Defect-Based Quantum Emitters in Silicon Carbide via Inorganic Passivation

Mark J. Polking, Alan M. Dibos, Nathalie P. de Leon, Hongkun Park.

Journal article | 2017 | Advanced Materials | vol. 30 | no. 4 | article 1704543.

[DOI 10.1002/adma.201704543](#) | [Source record](#) | Registered DOI metadata

D17T03 Topological superconducting devices

Majorana proposals seek protected operations using topological superconductivity. Device observations, zero mode interpretation, and demonstrated logical operations require separate evidence.

Fine subcategories: Zero mode evidence; parity measurement; braiding proposals; poisoning; experimental ambiguity.

Primary catalog resources: 8. Related topic assignments are searchable in the HTML.

Q0918 Complete description of fault-tolerant quantum gate operations for topological Majorana qubit systems

Adrian D. Scheppe, Michael V. Pak.

Journal article | 2022 | Physical Review A | vol. 105 | no. 1 | article 012415.

[DOI 10.1103/physreva.105.012415](#) | [Source record](#) | Registered DOI metadata

Q0919 Manipulation of Majorana-Kramers qubit and its tolerance in time-reversal invariant topological superconductor

Yuki Tanaka, Takumi Sanno, Takeshi Mizushima, Satoshi Fujimoto.

Journal article | 2022 | Physical Review B | vol. 106 | no. 1 | article 014522.

[DOI 10.1103/physrevb.106.014522](https://doi.org/10.1103/physrevb.106.014522) | [Source record](#) | Registered DOI metadata

Q0920 Transport Theory for Topological Josephson Junctions with a Majorana Qubit

Zhi Wang, Jia-Jin Feng, Zhao Huang, Qian Niu.

Journal article | 2022 | Physical Review Letters | vol. 129 | no. 25 | article 257001.

[DOI 10.1103/physrevlett.129.257001](https://doi.org/10.1103/physrevlett.129.257001) | [Source record](#) | Registered DOI metadata

Q0921 Z2 Topological Order and Topological Protection of Majorana Fermion Qubits

Rukhsan Ul Haq, Louis H. Kauffman.

Journal article | 2021 | Condensed Matter | vol. 6 | no. 1 | pp. 11.

[DOI 10.3390/condmat6010011](https://doi.org/10.3390/condmat6010011) | [Source record](#) | Registered DOI metadata

Q0922 Majorana qubits for topological quantum computing

Ramón Aguado, Leo P. Kouwenhoven.

Journal article | 2020 | Physics Today | vol. 73 | no. 6 | pp. 44-50.

[DOI 10.1063/pt.3.4499](https://doi.org/10.1063/pt.3.4499) | [Source record](#) | Registered DOI metadata

Q0923 Phonon-induced Majorana qubit relaxation in tunnel-coupled two-island topological superconductors

Yang Song, S. Das Sarma.

Journal article | 2018 | Physical Review B | vol. 98 | no. 7 | article 075159.

[DOI 10.1103/physrevb.98.075159](https://doi.org/10.1103/physrevb.98.075159) | [Source record](#) | Registered DOI metadata

Q0924 Quantum simulation of topological Majorana bound states and their universal quantum operations using charge-qubit arrays

Ting Mao, Z. D. Wang.

Journal article | 2015 | Physical Review A | vol. 91 | no. 1 | article 012336.

[DOI 10.1103/physreva.91.012336](https://doi.org/10.1103/physreva.91.012336) | [Source record](#) | Registered DOI metadata

Q0925 Topological protection of Majorana qubits

Meng Cheng, Roman M. Lutchyn, S. Das Sarma.

Journal article | 2012 | Physical Review B | vol. 85 | no. 16 | article 165124.

[DOI 10.1103/physrevb.85.165124](https://doi.org/10.1103/physrevb.85.165124) | [Source record](#) | Registered DOI metadata

D17T04 Molecular and nuclear spin processors

Molecular and NMR approaches manipulate coupled spin registers. Ensemble demonstrations can teach control and algorithms while facing distinct scalability limits.

Fine subcategories: NMR; molecular qubits; ensemble control; scalability limitations; spin registers.

Primary catalog resources: 8. Related topic assignments are searchable in the HTML.

Q0926 IMPLEMENTATION OF QUANTUM COMPUTING BASED ON PERSONAL QUANTUM COMPUTERS USING NUCLEAR MAGNETIC RESONANCE

MIREA – Russian Technology University, Moscow, Russia, Vitaly A. Peleshenko.

Journal article | 2024 | SOFT MEASUREMENTS AND COMPUTING | vol. 3 | no. 76 | pp. 14-34.

[DOI 10.36871/2618-9976.2024.03.002](https://doi.org/10.36871/2618-9976.2024.03.002) | [Source record](#) | Registered DOI metadata

Q0927 Nuclear Magnetic Resonance Quantum Computing

Chuck Easttom.

Book chapter | 2024 | Hardware for Quantum Computing | pp. 75-82.

[DOI 10.1007/978-3-031-66477-9_6](#) | [Source record](#) | Registered DOI metadata

Q0928 Advancements in predictive modeling of nuclear magnetic resonance parameters: integrating quantum mechanics, machine learning, and quantum computing

Jonhariono Sihotang, Patrisius Michaud Felix Marsoit.

Journal article | 2022 | Vertex | vol. 12 | no. 1 | pp. 20-29.

[DOI 10.35335/qc6shb61](#) | [Source record](#) | Registered DOI metadata

Q0929 From molecular quantum electrodynamics at finite temperatures to nuclear magnetic resonance

Kolja Them.

Journal article | 2021 | Journal of Physics Communications | vol. 5 | no. 2 | pp. 025011.

[DOI 10.1088/2399-6528/abd8c3](#) | [Source record](#) | Registered DOI metadata

Q0930 Quantum simulations with nuclear magnetic resonance system*

Chudan Qiu, Xinfang Nie, Dawei Lu.

Journal article | 2021 | Chinese Physics B | vol. 30 | no. 4 | pp. 048201.

[DOI 10.1088/1674-1056/abe299](#) | [Source record](#) | Registered DOI metadata

Q0931 Nuclear magnetic resonance for quantum computing: Techniques and recent achievements

Tao Xin, Bi-Xue Wang, Ke-Ren Li, Xiang-Yu Kong, Shi-Jie Wei, Tao Wang, Dong Ruan, Gui-Lu Long.

Journal article | 2018 | Chinese Physics B | vol. 27 | no. 2 | pp. 020308.

[DOI 10.1088/1674-1056/27/2/020308](#) | [Source record](#) | Registered DOI metadata

Q0932 Quantum Information and Nuclear Magnetic Resonance Parameters

Jéssica B. dos R. Lino, Teodorico C. Ramalho.

Journal article | 2018 | Revista Virtual de Química | vol. 10 | no. 4 | pp. 940-962.

[DOI 10.21577/1984-6835.20180067](#) | [Source record](#) | Registered DOI metadata

Q0933 Glycosaminoglycan Monosaccharide Blocks Analysis by Quantum Mechanics, Molecular Dynamics, and Nuclear Magnetic Resonance

Sergey A. Samsonov, Stephan Theisgen, Thomas Riemer, Daniel Huster, M. Teresa Pisabarro.

Journal article | 2014 | BioMed Research International | vol. 2014 | pp. 1-11.

[DOI 10.1155/2014/808071](#) | [Source record](#) | Registered DOI metadata

D17T05 Quantum transduction

Transducers convert quantum information between physical frequencies or platforms. Conversion efficiency and added noise jointly determine whether a link is useful.

Fine subcategories: Microwave to optical conversion; efficiency; added noise; piezoelectricity; optomechanics.

Primary catalog resources: 19. Related topic assignments are searchable in the HTML.

Q0934 Microwave-to-optical quantum transduction of photons for quantum interconnects

Akihiko Sekine, Ryo Murakami, Yoshiyasu Doi.

Journal article | 2026 | npj Nanophotonics | vol. 3 | no. 1 | article 42.

[DOI 10.1038/s44310-026-00130-8](#) | [Source record](#) | Registered DOI metadata

Q0935 Nonlinear quantum transduction of weak topological impurities

Jiahao Duan, Maomao Gong, Yongjun Cheng, Song Bin Zhang.

Journal article | 2026 | APS Open Science | vol. 1 | article 000157.

[DOI 10.1103/8tqj-y91q](#) | [Source record](#) | Registered DOI metadata

Q0936 Quantum Transduction: Enabling Quantum Networking

Marcello Caleffi, Laura d'Avossa, Xu Han, Angela Sara Cacciapuoti.

Journal article | 2026 | IEEE Communications Surveys & Tutorials | vol. 28 | pp. 4195-4214.

[DOI 10.1109/comst.2025.3631150](#) | [Source record](#) | Registered DOI metadata

Q0937 Quantum transduction via generalized continuous-variable teleportation

Quntao Zhuang.

Journal article | 2026 | Physical Review A | vol. 113 | no. 1 | article 012623.

[DOI 10.1103/qwm4-8vxd](#) | [Source record](#) | Registered DOI metadata

Q0938 Quantum-memory-assisted on-demand microwave-optical transduction

Hai-Tao Tu, Kai-Yu Liao, Si-Yuan Qiu, Xiao-Hong Liu, Yi-Qi Guo, Zheng-Qi Du, Yang Xu, Xin-Ding Zhang et al..

Journal article | 2026 | Nature Communications | vol. 17 | no. 1 | article 9397.

[DOI 10.1038/s41467-026-75752-9](#) | [Source record](#) | Registered DOI metadata

Q0939 Antidegradable quantum channel with additional entanglement for quantum transduction

Changchun Zhong.

Journal article | 2025 | Physical Review Applied | vol. 24 | no. 2 | article 024065.

[DOI 10.1103/4g6d-6cmg](#) | [Source record](#) | Registered DOI metadata

Q0940 Microwave-to-optical quantum transduction with antiferromagnets

Akihiko Sekine, Ryo Murakami, Yoshiyasu Doi.

Journal article | 2025 | Physical Review B | vol. 112 | no. 9 | article 094413.

[DOI 10.1103/m52p-sp6d](#) | [Source record](#) | Registered DOI metadata

Q0941 Modeling Quantum Transduction for Multipartite Entanglement Distribution

Laura d'Avossa, Angela Sara Cacciapuoti, Marcello Caleffi.

Journal article | 2025 | IEEE Transactions on Communications | vol. 73 | no. 11 | pp. 11707-11721.

[DOI 10.1109/tcomm.2025.3566987](#) | [Source record](#) | Registered DOI metadata

Q0942 Simulation of Quantum Transduction Strategies for Quantum Networks

Laura d'Avossa, Caitao Zhan, Joaquin Chung, Rajkumar Kettimuthu, Angela Sara Cacciapuoti, Marcello Caleffi.

Conference paper | 2025 | 2025 IEEE International Conference on Quantum Computing and Engineering (QCE) | pp. 1274-1282.

[DOI 10.1109/qce65121.2025.00142](#) | [Source record](#) | Registered DOI metadata

Q0943 Tunable superconducting microwave resonator for quantum transduction

Hana K. Warner, Shima Rajabali, Seunghyun Park, Nayely Rolon-Gomez, Donald Witt, Amir Yacoby, Marko Lončar.

Journal article | 2025 | EPJ Web of Conferences | vol. 335 | pp. 06012.

[DOI 10.1051/epjconf/202533506012](#) | [Source record](#) | Registered DOI metadata

Q0944 A two-dimensional optomechanical crystal for quantum transduction

Felix M. Mayor, Sultan Malik, André G. Primo, Samuel Gyger, Wentao Jiang, Thiago P. M. Alegre, Amir H. Safavi-Naeini.

Conference paper | 2024 | Quantum 2.0 Conference and Exhibition | pp. QTu4A.3.

[DOI 10.1364/quantum.2024.qtu4a.3](#) | [Source record](#) | Registered DOI metadata

Q0945 Efficient quantum transduction using antiferromagnetic topological insulators

Haowei Xu, Changhao Li, Guoqing Wang, Hao Tang, Paola Cappellaro, Ju Li.

Journal article | 2024 | Physical Review B | vol. 110 | no. 8 | article 085136.

[DOI 10.1103/physrevb.110.085136](#) | [Source record](#) | Registered DOI metadata

Q0946 Hybrid Quantum Transduction Systems Based on Magnonic Materials

S. Kazan, N. G. Saribas, S. Ç. Yorulmaz, M. Maksutoglu, E. Avinca, F. Yıldız, S. I. Tarapov, B. Rami.
Book chapter | 2024 | NATO Science for Peace and Security Series B: Physics and Biophysics | pp. 207-219.
[DOI 10.1007/978-94-024-2254-2_10](#) | [Source record](#) | Registered DOI metadata

Q0947 Quantum Transduction Models for Multipartite Entanglement Distribution

Laura d'Avossa, Angela Sara Cacciapuoti, Marcello Caleffi.
Conference paper | 2024 | 2024 International Conference on Quantum Communications, Networking, and Computing (QCNC) | pp. 1-8.
[DOI 10.1109/qcnc62729.2024.00011](#) | [Source record](#) | Registered DOI metadata

Q0948 Time-correlation Transduction in Strong-field Quantum Electrodynamics

zairui li, Wesley Sims, Mirali Shariatdoust, Gabriel Howell, Thomas Searles, Linshan Sun, Sergio Carbajo.
Posted content | 2024 | Springer Science and Business Media LLC.
[DOI 10.21203/rs.3.rs-5124844/v1](#) | [Source record](#) | Registered DOI metadata

Q0949 MW-Magnon Systems for Quantum Transduction Applications

B. Rameev.
Conference paper | 2023 | 2023 Photonics & Electromagnetics Research Symposium (PIERS) | pp. 2086-2092.
[DOI 10.1109/piers59004.2023.10221301](#) | [Source record](#) | Registered DOI metadata

Q0950 Optimized Protocols for Duplex Quantum Transduction

Zhaoyou Wang, Mengzhen Zhang, Yat Wong, Changchun Zhong, Liang Jiang.
Journal article | 2023 | Physical Review Letters | vol. 131 | no. 22 | article 220802.
[DOI 10.1103/physrevlett.131.220802](#) | [Source record](#) | Registered DOI metadata

Q0951 Python in Quantum Transduction

Erica Garcia Badaracco.
Conference paper | 2023 | Python in Quantum Transduction.
[DOI 10.2172/1974707](#) | [Source record](#) | Registered DOI metadata

Q0952 Ultrahigh-precision Mechanical Platform for Quantum Transduction

Julian Delgado.
Conference paper | 2023 | Ultrahigh-precision Mechanical Platform for Quantum Transduction.
[DOI 10.2172/1996938](#) | [Source record](#) | Registered DOI metadata

D18 Control calibration and noise characterization

Control engineering links physical devices to reliable operations. A noise model should be measured and tested rather than assumed to describe all operating regimes.

Prerequisites: Control theory, stochastic processes, and experimental statistics.

Assessment focus: Model identification, robustness, drift, calibration effort, and correlated noise.

D18T01 Optimal quantum control

Optimal control searches for pulses that implement target transformations under constraints. Robustness to parameter errors and hardware bandwidth should be tested.

Fine subcategories: GRAPE; CRAB; bandwidth constraints; robust pulse design; control landscapes.

Primary catalog resources: 8. Related topic assignments are searchable in the HTML.

Q0953 Quantum Measurement and Control

Howard M. Wiseman, Gerard J. Milburn.

Book | 2009 | Cambridge University Press.

[DOI 10.1017/cbo9780511813948](https://doi.org/10.1017/cbo9780511813948) | [Source record](#) | Registered DOI metadata

Q0954 Machine-learning optimal control pulses in an optical quantum memory experiment

Elizabeth Robertson, Luisa Esguerra, Leon Meßner, Guillermo Gallego, Janik Wolters.

Journal article | 2024 | Physical Review Applied | vol. 22 | no. 2 | article 024026.

[DOI 10.1103/physrevapplied.22.024026](https://doi.org/10.1103/physrevapplied.22.024026) | [Source record](#) | Registered DOI metadata

Q0955 Machine Learning Optimal Control Pulses in an Optical Quantum Memory

Elizabeth Robertson, Luisa Esguerra, Leon Meßner, Guillermo Gallego, Janik Wolters.

Conference paper | 2023 | 2023 Conference on Lasers and Electro-Optics Europe & European Quantum Electronics Conference (CLEO/Europe-EQEC) | pp. 1-1.

[DOI 10.1109/cleo/europe-eqec57999.2023.10232106](https://doi.org/10.1109/cleo/europe-eqec57999.2023.10232106) | [Source record](#) | Registered DOI metadata

Q0956 Time-optimal control of two-level quantum systems by piecewise constant pulses

E. Dionis, D. Sugny.

Journal article | 2023 | Physical Review A | vol. 107 | no. 3 | article 032613.

[DOI 10.1103/physreva.107.032613](https://doi.org/10.1103/physreva.107.032613) | [Source record](#) | Registered DOI metadata

Q0957 From Pulses to Circuits and Back Again: A Quantum Optimal Control Perspective on Variational Quantum Algorithms

Alicia B. Magann, Christian Arenz, Matthew D. Grace, Tak-San Ho, Robert L. Kosut, Jarrod R. McClean, Herschel A. Rabitz, Mohan Sarovar.

Journal article | 2021 | PRX Quantum | vol. 2 | no. 1 | article 010101.

[DOI 10.1103/prxquantum.2.010101](https://doi.org/10.1103/prxquantum.2.010101) | [Source record](#) | Registered DOI metadata

Q0958 Optimal control of quantum systems with ultrashort laser pulses and non-adiabatic interactions

José L. Sanz-Vicario, Mariana Ramírez Quiceno.

Conference paper | 2018 | Active Photonic Platforms X | pp. 85.

[DOI 10.1117/12.2320752](https://doi.org/10.1117/12.2320752) | [Source record](#) | Registered DOI metadata

Q0959 Errors in quantum optimal control and strategy for the search of easily implementable control pulses

Antonio Negretti, Rosario Fazio, Tommaso Calarco.

Journal article | 2011 | Journal of Physics B: Atomic, Molecular and Optical Physics | vol. 44 | no. 15 | pp. 154012.

[DOI 10.1088/0953-4075/44/15/154012](https://doi.org/10.1088/0953-4075/44/15/154012) | [Source record](#) | Registered DOI metadata

Q0960 Optimal Control of Quantum Rings by Terahertz Laser Pulses

E. Räsänen, A. Castro, J. Werschnik, A. Rubio, E. K. U. Gross.

Journal article | 2007 | Physical Review Letters | vol. 98 | no. 15 | article 157404.

[DOI 10.1103/physrevlett.98.157404](#) | [Source record](#) | Registered DOI metadata

D18T02 Dynamical decoupling

Dynamical decoupling suppresses selected environmental couplings through pulse sequences. Finite pulse errors and noise spectra limit its effectiveness.

Fine subcategories: Pulse sequences; filter functions; noise spectra; randomized decoupling; storage protection.

Primary catalog resources: 13. Related topic assignments are searchable in the HTML.

Q0961 Crosstalk-robust dynamical decoupling for bipartite-topology quantum processors

Ethan Hickman, Xiaodi Wu, Gregory Quiroz.

Journal article | 2026 | Physical Review Applied | vol. 25 | no. 6 | article 064041.

[DOI 10.1103/r75m-b2d9](#) | [Source record](#) | Registered DOI metadata

Q0962 Dynamical decoupling for quantum metrology with periodically driven Hamiltonians

Qifei Wei, Haorui Chen, Shengshi Pang.

Journal article | 2026 | Physical Review A | vol. 114 | no. 3 | article 032418.

[DOI 10.1103/lb42-hr34](#) | [Source record](#) | Registered DOI metadata

Q0963 Bath dynamical decoupling with a quantum channel

Alexander Hahn, Kazuya Yuasa, Daniel Burgarth.

Journal article | 2025 | Journal of Physics A: Mathematical and Theoretical | vol. 58 | no. 4 | pp. 045305.

[DOI 10.1088/1751-8121/ada219](#) | [Source record](#) | Registered DOI metadata

Q0964 Dynamical-decoupling-protected unconventional nonadiabatic geometric quantum computation

Xuan Wu, Long-Yi Jin, Hong-Fu Wang.

Journal article | 2025 | Physica Scripta | vol. 100 | no. 2 | pp. 025102.

[DOI 10.1088/1402-4896/ada203](#) | [Source record](#) | Registered DOI metadata

Q0965 Empirical Learning of Dynamical Decoupling on Quantum Processors

Christopher Tong, Helena Zhang, Bibek Pokharel.

Journal article | 2025 | PRX Quantum | vol. 6 | no. 3 | article 030319.

[DOI 10.1103/h7pq-s159](#) | [Source record](#) | Registered DOI metadata

Q0966 Multiple quantum many-body clustering probed by dynamical decoupling

Gerónimo Sequeiros, Claudia M. Sánchez, Lisandro Buljubasich, Ana K. Chattah, Horacio M. Pastawski, Rodolfo H. Acosta.

Journal article | 2025 | Physical Review A | vol. 112 | no. 2 | article 022617.

[DOI 10.1103/q39p-yjnm](#) | [Source record](#) | Registered DOI metadata

Q0967 Qudit Dynamical Decoupling on a Superconducting Quantum Processor

Vinay Tripathi, Noah Goss, Arian Vezvaei, Long B. Nguyen, Irfan Siddiqi, Daniel A. Lidar.

Journal article | 2025 | Physical Review Letters | vol. 134 | no. 5 | article 050601.

[DOI 10.1103/physrevlett.134.050601](#) | [Source record](#) | Registered DOI metadata

Q0968 Syncopated dynamical decoupling to suppress crosstalk in quantum circuits

Bram Evert, Zoe Gonzalez Izquierdo, James Sud, Hong-Ye Hu, Shon Grabbe, Eleanor G. Rieffel, Matthew J. Reagor, Zhihui Wang.

Journal article | 2025 | Physical Review Applied | vol. 24 | no. 4 | article 044025.

[DOI 10.1103/8lxc-lvv1](#) | [Source record](#) | Registered DOI metadata

Q0969 Individually Addressed Quantum Gate Interactions Using Dynamical Decoupling

M.C. Smith, A.D. Leu, M.F. Gely, D.M. Lucas.

Journal article | 2024 | PRX Quantum | vol. 5 | no. 3 | article 030321.

[DOI 10.1103/prxquantum.5.030321](#) | [Source record](#) | Registered DOI metadata

Q0970 Dynamical Decoupling (DD) to Improve Fidelity in Quantum Computing

Muhammad Haryo Ramadhani, Maman Abdurrohman, Hilal H. Nuha.

Conference paper | 2023 | 2023 International Conference on Data Science and Its Applications (ICoDSA) | pp. 495-499.

[DOI 10.1109/icodsa58501.2023.10276898](#) | [Source record](#) | Registered DOI metadata

Q0971 Quantum dynamical decoupling by shaking the close environment

Michiel Burgelman, Paolo Forni, Alain Sarlette.

Journal article | 2023 | Journal of the Franklin Institute | vol. 360 | no. 17 | pp. 14022-14074.

[DOI 10.1016/j.jfranklin.2022.08.011](#) | [Source record](#) | Registered DOI metadata

Q0972 Dynamical decoupling and NNN discrete quantum networks

Antonín Hoskovec, Igor Jex.

Journal article | 2022 | International Journal of Quantum Information | vol. 20 | no. 05 | article 2250009.

[DOI 10.1142/s0219749922500095](#) | [Source record](#) | Registered DOI metadata

Q0973 Preserving multilevel quantum coherence by dynamical decoupling

Xinxing Yuan, Yue Li, Mengxiang Zhang, Chang Liu, Mingdong Zhu, Xi Qin, Nikolay V. Vitanov, Yiheng Lin et al..

Journal article | 2022 | Physical Review A | vol. 106 | no. 2 | article 022412.

[DOI 10.1103/physreva.106.022412](#) | [Source record](#) | Registered DOI metadata

D18T03 Quantum noise spectroscopy

Noise spectroscopy infers environmental fluctuations from controlled quantum probes. The reconstruction depends on filter functions and assumptions about noise statistics.

Fine subcategories: Filter functions; spectral reconstruction; correlated noise; non Gaussian noise.

Primary catalog resources: 12. Related topic assignments are searchable in the HTML.

Q0974 Control-centric quantum noise spectroscopy of time-ordered polyspectra

Kaiah Steven, Elliot Coupe, Qi Yu, Gerardo A. Paz-Silva.

Journal article | 2026 | Physical Review A | vol. 114 | no. 3 | article 032606.

[DOI 10.1103/121q-3ddh](#) | [Source record](#) | Registered DOI metadata

Q0975 Fast, accurate, and error-resilient variational quantum noise spectroscopy

Nanako Shitara, Andrés Montoya-Castillo.

Journal article | 2026 | The Journal of Chemical Physics | vol. 164 | no. 6 | article 061101.

[DOI 10.1063/5.0312403](#) | [Source record](#) | Registered DOI metadata

Q0976 Broadband spectroscopy of quantum noise

Yuanlong Wang, Gerardo A. Paz-Silva.

Journal article | 2025 | Physical Review A | vol. 111 | no. 5 | article 052407.

[DOI 10.1103/physreva.111.052407](#) | [Source record](#) | Registered DOI metadata

Q0977 Experimental Application of Variational Quantum Noise Spectroscopy to Nitrogen Vacancy Centers

Connor Desrosiers, Nanako Shitara, Andrés Montoya-Castillo, Shuo Sun.

Conference paper | 2025 | Frontiers in Optics + Laser Science 2025 (FIO, LS) | pp. FW7C.4.

[DOI 10.1364/fio.2025.fw7c.4](https://doi.org/10.1364/fio.2025.fw7c.4) | [Source record](#) | Registered DOI metadata

Q0978 Machine learning non-Markovian two-level quantum noise spectroscopy

Juan M. Scarpetta, John H. Reina, Morten Hjorth-Jensen.

Journal article | 2025 | Physical Review Research | vol. 7 | no. 4 | article 043285.

[DOI 10.1103/2lzl-vpjd](https://doi.org/10.1103/2lzl-vpjd) | [Source record](#) | Registered DOI metadata

Q0979 Quantum noise spectroscopy by qudit spectators

Clara Javaherian, Chris Ferrie.

Journal article | 2025 | Physical Review A | vol. 112 | no. 3 | article 032411.

[DOI 10.1103/drc2-8qbg](https://doi.org/10.1103/drc2-8qbg) | [Source record](#) | Registered DOI metadata

Q0980 Characterization and mitigation of axial motion induced noise on trapped ions using quantum noise spectroscopy

Sandia National Laboratories (SNL-NM), Albuquerque, NM (United States), Matthew Chow, Advanced Scientific Computing Research, Vivian Maloney, Ashlyn Burch, Megan Ivory, Daniel Lobser, Gregory Quiroz et al..

Report | 2024 | Office of Scientific and Technical Information (OSTI).

[DOI 10.2172/2585236](https://doi.org/10.2172/2585236) | [Source record](#) | Registered DOI metadata

Q0981 Digital noise spectroscopy with a quantum sensor

Guoqing Wang (), Yuan Zhu, Boning Li, Changhao Li, Lorenza Viola, Alexandre Cooper, Paola Cappellaro.

Journal article | 2024 | Quantum Science and Technology | vol. 9 | no. 3 | pp. 035006.

[DOI 10.1088/2058-9565/ad3846](https://doi.org/10.1088/2058-9565/ad3846) | [Source record](#) | Registered DOI metadata

Q0982 Electrical noise spectroscopy of magnons in a quantum Hall ferromagnet

Ravi Kumar, Saurabh Kumar Srivastav, Ujjal Roy, Jinhong Park, Christian Spånslätt, K. Watanabe, T. Taniguchi, Yuval Gefen et al..

Journal article | 2024 | Nature Communications | vol. 15 | no. 1 | article 4998.

[DOI 10.1038/s41467-024-49446-z](https://doi.org/10.1038/s41467-024-49446-z) | [Source record](#) | Registered DOI metadata

Q0983 Quantum-Enhanced Dual-Comb Spectroscopy Beyond the Shot Noise Limit

Daniel I. Herman, Mathieu Walsh, Molly Kate Kreider, Noah Lordi, Eugene J. Tsao, Alexander J. Lind, Matthew Heyrich, Joshua Combes et al..

Conference paper | 2024 | Optica Imaging Congress 2024 (3D, AOMS, COSI, ISA, pcAOP) | pp. JTu5A.3.

[DOI 10.1364/3d.2024.jtu5a.3](https://doi.org/10.1364/3d.2024.jtu5a.3) | [Source record](#) | Registered DOI metadata

Q0984 Quantum Noise Spectroscopy of Dynamical Critical Phenomena

Francisco Machado, Eugene A. Demler, Norman Y. Yao, Shubhayu Chatterjee.

Journal article | 2023 | Physical Review Letters | vol. 131 | no. 7 | article 070801.

[DOI 10.1103/physrevlett.131.070801](https://doi.org/10.1103/physrevlett.131.070801) | [Source record](#) | Registered DOI metadata

Q0985 Quantum noise spectroscopy as an incoherent imaging problem

Mankei Tsang.

Journal article | 2023 | Physical Review A | vol. 107 | no. 1 | article 012611.

[DOI 10.1103/physreva.107.012611](https://doi.org/10.1103/physreva.107.012611) | [Source record](#) | Registered DOI metadata

D18T04 Leakage crosstalk and correlated errors

Leakage and correlated errors violate simple independent Pauli models. Detection, reset, and system context are essential for realistic reliability estimates.

AQUIRE

Fine subcategories: Leakage detection; spectator errors; spatial correlation; temporal drift; coherent errors.

Primary catalog resources: 8. Related topic assignments are searchable in the HTML.

Q0986 Quantum sensing in the presence of pulse errors and qubit leakage

Anonymous.

Journal article | 2026 | Physical Review Research.

[DOI 10.1103/lb3n-qply](#) | [Source record](#) | Registered DOI metadata

Q0987 Secure quantum key distribution against correlated leakage source

Jia-Xuan Li, Yang-Guang Shan, Rong Wang, Feng-Yu Lu, Zhen-Qiang Yin, Shuang Wang, Wei Chen, De-Yong He et al..

Journal article | 2026 | Science Advances | vol. 12 | no. 15 | article eaed2420.

[DOI 10.1126/sciadv.aed2420](#) | [Source record](#) | Registered DOI metadata

Q0988 Characterization of leakage errors in a transmon qubit due to resonant digital control

M. A. Castellanos-Beltran, A. J. Sirois, D. I. Olaya, J. Biesecker, S. P. Benz, P. F. Hopkins.

Journal article | 2025 | Applied Physics Letters | vol. 127 | no. 23 | article 232601.

[DOI 10.1063/5.0304764](#) | [Source record](#) | Registered DOI metadata

Q0989 Mitigating qubit leakage errors in quantum circuits with gadgets and post-selection

Karl Mayer.

Conference paper | 2022 | 2022 IEEE International Conference on Quantum Computing and Engineering (QCE) | pp. 809-809.

[DOI 10.1109/qce53715.2022.00126](#) | [Source record](#) | Registered DOI metadata

Q0990 Removing leakage-induced correlated errors in superconducting quantum error correction

M. McEwen, D. Kafri, Z. Chen, J. Atalaya, K. J. Satzinger, C. Quintana, P. V. Klimov, D. Sank et al..

Journal article | 2021 | Nature Communications | vol. 12 | no. 1 | article 1761.

[DOI 10.1038/s41467-021-21982-y](#) | [Source record](#) | Registered DOI metadata

Q0991 Protecting quantum entanglement from leakage and qubit errors via repetitive parity measurements

C. C. Bultink, T. E. O'Brien, R. Vollmer, N. Muthusubramanian, M. W. Beekman, M. A. Rol, X. Fu, B. Tarasinski et al..

Journal article | 2020 | Science Advances | vol. 6 | no. 12 | article eaay3050.

[DOI 10.1126/sciadv.aay3050](#) | [Source record](#) | Registered DOI metadata

Q0992 Leakage mitigation for quantum error correction using a mixed qubit scheme

Natalie C. Brown, Kenneth R. Brown.

Journal article | 2019 | Physical Review A | vol. 100 | no. 3 | article 032325.

[DOI 10.1103/physreva.100.032325](#) | [Source record](#) | Registered DOI metadata

Q0993 Tradeoff between leakage and dephasing errors in the fluxonium qubit

David A. Herrera-Martí, Ahsan Nazir, Sean D. Barrett.

Journal article | 2013 | Physical Review B | vol. 88 | no. 9 | article 094512.

[DOI 10.1103/physrevb.88.094512](#) | [Source record](#) | Registered DOI metadata

D18T05 Automated quantum calibration

Automated calibration estimates and updates control parameters from measurements. Drift, latency, identifiability, and calibration cost affect closed loop performance.

Fine subcategories: Closed loop optimization; drift tracking; parameter estimation; scheduling; calibration transfer.

Primary catalog resources: 5. Related topic assignments are searchable in the HTML.

Q0994 Qubit-efficient quantum intrusion detection using dual-parameter encoding and multi-metric calibration

Lubna Khan, Burhan Ul Islam Khan, Aabid A. Mir, Khang Wen Goh, Dwi Sudarno Putra, Uzair Ishtiaq, Mesith Chaimanee.

Journal article | 2026 | Scientific Reports | vol. 16 | no. 1 | article 23621.

[DOI 10.1038/s41598-026-55614-6](https://doi.org/10.1038/s41598-026-55614-6) | [Source record](#) | Registered DOI metadata

Q0995 CaliQEC: In-situ Qubit Calibration for Surface Code Quantum Error Correction

Xiang Fang, Keyi Yin, Yuchen Zhu, Jixuan Ruan, Dean Tullsen, Zhiding Liang, Andrew Sornborger, Ang Li et al..

Conference paper | 2025 | Proceedings of the 52nd Annual International Symposium on Computer Architecture | pp. 1402-1416.

[DOI 10.1145/3695053.3731042](https://doi.org/10.1145/3695053.3731042) | [Source record](#) | Registered DOI metadata

Q0996 Realization and Calibration of Continuously Parameterized Two-Qubit Gates on a Trapped-Ion Quantum Processor

Christopher G. Yale, Ashlyn D. Burch, Matthew N. H. Chow, Brandon P. Ruzic, Daniel S. Lobser, Brian K. McFarland, Melissa C. Revelle, Susan M. Clark.

Journal article | 2025 | IEEE Transactions on Quantum Engineering | vol. 6 | pp. 1-17.

[DOI 10.1109/tqe.2025.3600216](https://doi.org/10.1109/tqe.2025.3600216) | [Source record](#) | Registered DOI metadata

Q0997 High-Speed Calibration and Characterization of Superconducting Quantum Processors without Qubit Reset

M. Werninghaus, D.J. Egger, S. Filipp.

Journal article | 2021 | PRX Quantum | vol. 2 | no. 2 | article 020324.

[DOI 10.1103/prxquantum.2.020324](https://doi.org/10.1103/prxquantum.2.020324) | [Source record](#) | Registered DOI metadata

Q0998 Autonomous calibration of single spin qubit operations

Florian Frank, Thomas Uden, Jonathan Zoller, Ressa S. Said, Tommaso Calarco, Simone Montangero, Boris Naydenov, Fedor Jelezko.

Journal article | 2017 | npj Quantum Information | vol. 3 | no. 1 | article 48.

[DOI 10.1038/s41534-017-0049-8](https://doi.org/10.1038/s41534-017-0049-8) | [Source record](#) | Registered DOI metadata

D19 Quantum error correction codes

Error correction encodes logical information into larger physical systems and extracts error syndromes without directly measuring the logical state. Code performance depends on the physical error model and decoder.

Prerequisites: Pauli operators, circuits, and coding theory.

Assessment focus: Code distance, check structure, physical noise, syndrome extraction, and decoding.

D19T01 Stabilizer and CSS codes

Stabilizer and CSS constructions specify logical subspaces through commuting checks. Syndrome measurement must reveal error information without revealing the logical state.

Fine subcategories: Pauli stabilizers; CSS construction; syndrome extraction; distance; logical operators.

Primary catalog resources: 7. Related topic assignments are searchable in the HTML.

Q0999 Quantum Error Correction

Giuliano Gadioli La Guardia.

Book | 2020 | Quantum Science and Technology.

[DOI 10.1007/978-3-030-48551-1](https://doi.org/10.1007/978-3-030-48551-1) | [Source record](#) | Registered DOI metadata

Q1000 Error Correcting Codes in Quantum Theory

A. M. Steane.

Journal article | 1996 | Physical Review Letters | vol. 77 | no. 5 | pp. 793-797.

[DOI 10.1103/physrevlett.77.793](https://doi.org/10.1103/physrevlett.77.793) | [Source record](#) | Registered DOI metadata

Q1001 Scheme for reducing decoherence in quantum computer memory

Peter W. Shor.

Journal article | 1995 | Physical Review A | vol. 52 | no. 4 | pp. R2493-R2496.

[DOI 10.1103/physreva.52.r2493](https://doi.org/10.1103/physreva.52.r2493) | [Source record](#) | Registered DOI metadata

Q1002 Exploring the relationship between twisted centralizers and stabilizer codes for quantum error correction

Galih Pradananta, Rendra Erdkhadifa.

Conference paper | 2026 | AIP Conference Proceedings | vol. 3389 | pp. 020025.

[DOI 10.1063/5.0317717](https://doi.org/10.1063/5.0317717) | [Source record](#) | Registered DOI metadata

Q1003 Introduction to quantum error correction with stabilizer codes

Zachary P. Bradshaw, Jeffrey J. Dale, Ethan N. Evans.

Journal article | 2026 | Annals of Physics | vol. 487 | pp. 170353 | article 170353.

[DOI 10.1016/j.aop.2026.170353](https://doi.org/10.1016/j.aop.2026.170353) | [Source record](#) | Registered DOI metadata

Q1004 Advancing Quantum Reliability: A Study on Topological Codes and Stabilizer Techniques for Scalable Error Correction

Krishna Kumar.

Conference paper | 2025 | 2025 International Conference on Computing Technologies (ICOCT) | pp. 1-9.

[DOI 10.1109/icoct64433.2025.11118845](https://doi.org/10.1109/icoct64433.2025.11118845) | [Source record](#) | Registered DOI metadata

Q1005 Quantum Information Science II | Physics | MIT OpenCourseWare

Author metadata not supplied.

Course | 2018 | MIT OpenCourseWare.

[Official resource](#) | [Source record](#) | Official page retrieved

D19T02 Surface and toric codes

Surface and toric codes use structured local checks and topological logical operators. Thresholds depend on the noise and syndrome circuit model.

Fine subcategories: Planar patches; toric boundaries; syndrome circuits; thresholds; defects.

Primary catalog resources: 7. Related topic assignments are searchable in the HTML.

Q1006 Surface codes: Towards practical large-scale quantum computation

Austin G. Fowler, Matteo Mariantoni, John M. Martinis, Andrew N. Cleland.

Journal article | 2012 | Physical Review A | vol. 86 | no. 3 | article 032324.

[DOI 10.1103/physreva.86.032324](https://doi.org/10.1103/physreva.86.032324) | [Source record](#) | Registered DOI metadata

Q1007 Quantum error correction with the surface code

Author metadata not supplied.

Book chapter | 2025 | Quantum Algorithms | pp. 320-324.

[DOI 10.1017/9781009639651.030](https://doi.org/10.1017/9781009639651.030) | [Source record](#) | Registered DOI metadata

Q1008 FPGA-Based Quantum Emulator for Surface Code

Youngchul Kim, Gyuil Cha.

Conference paper | 2024 | 2024 15th International Conference on Information and Communication Technology Convergence (ICTC) | pp. 1320-1324.

[DOI 10.1109/ictc62082.2024.10827261](https://doi.org/10.1109/ictc62082.2024.10827261) | [Source record](#) | Registered DOI metadata

Q1009 Quantum double aspects of surface code models

Alexander Cowtan, Shahn Majid.

Journal article | 2022 | Journal of Mathematical Physics | vol. 63 | no. 4 | pp. 042202.

[DOI 10.1063/5.0063768](https://doi.org/10.1063/5.0063768) | [Source record](#) | Registered DOI metadata

Q1010 A silicon-based surface code quantum computer

Joe O’Gorman, Naomi H Nickerson, Philipp Ross, John JL Morton, Simon C Benjamin.

Journal article | 2016 | npj Quantum Information | vol. 2 | no. 1 | article 15019.

[DOI 10.1038/npjqi.2015.19](https://doi.org/10.1038/npjqi.2015.19) | [Source record](#) | Registered DOI metadata

Q1011 A surface code quantum computer in silicon

Charles D. Hill, Eldad Peretz, Samuel J. Hile, Matthew G. House, Martin Fuechsle, Sven Rogge, Michelle Y. Simmons, Lloyd C. L. Hollenberg.

Journal article | 2015 | Science Advances | vol. 1 | no. 9 | article e1500707.

[DOI 10.1126/sciadv.1500707](https://doi.org/10.1126/sciadv.1500707) | [Source record](#) | Registered DOI metadata

Q1012 INTRODUCTION TO SURFACE CODE QUANTUM COMPUTATION

YIDUN WAN.

Conference paper | 2012 | Quantum Information and Quantum Computing | pp. 63-65.

[DOI 10.1142/9789814425223_0004](https://doi.org/10.1142/9789814425223_0004) | [Source record](#) | Registered DOI metadata

D19T03 Quantum LDPC codes

Quantum LDPC codes aim for sparse checks and improved encoding efficiency. Required connectivity, measurement circuits, and decoding are important implementation constraints.

Fine subcategories: Sparse checks; hypergraph products; lifted products; bivariate bicycle codes; connectivity.

Primary catalog resources: 16. Related topic assignments are searchable in the HTML.

Q1013 ADMM Decoding of Quantum LDPC Codes

Jiaxin Lyu, Guanghui He.

Conference paper | 2026 | 2026 IEEE International Symposium on Information Theory (ISIT) | pp. 1-6.

[DOI 10.1109/isit62367.2026.11653953](#) | [Source record](#) | Registered DOI metadata

Q1014 Absorbing sets in quantum LDPC codes

Kirsten D. Morris, Tefjol Pillaha, Christine A. Kelley.

Journal article | 2026 | Advances in Mathematics of Communications | vol. 23 | no. 0 | pp. 234-254.

[DOI 10.3934/amc.2026023](#) | [Source record](#) | Registered DOI metadata

Q1015 Breaking the Orthogonality Barrier in Quantum LDPC Codes

Kenta Kasai.

Journal article | 2026 | Quantum | vol. 10 | pp. 2205 | article 2205.

[DOI 10.22331/q-2026-09-09-2205](#) | [Source record](#) | Registered DOI metadata

Q1016 Decoding correlated errors in quantum LDPC codes

Arshpreet Singh Maan, Francisco Miguel Garcia Herrero, Alexandru Paler, Valentin Savin.

Journal article | 2026 | Nature Communications | vol. 17 | no. 1 | article 3965.

[DOI 10.1038/s41467-026-70556-3](#) | [Source record](#) | Registered DOI metadata

Q1017 Efficient post-selection for general quantum LDPC Codes

Seok-Hyung Lee, Lucas H. English, Stephen D. Bartlett.

Journal article | 2026 | npj Quantum Information | vol. 12 | no. 1 | article 96.

[DOI 10.1038/s41534-026-01242-x](#) | [Source record](#) | Registered DOI metadata

Q1018 Quantum stabilizer and quantum LDPC codes

Ivan B. Djordjevic.

Book chapter | 2026 | Quantum Communication, Quantum Networks, and Quantum Sensing | pp. 305-365.

[DOI 10.1016/b978-0-443-40568-6.00005-7](#) | [Source record](#) | Registered DOI metadata

Q1019 Quantum CSS LDPC Codes with Quasi-Dyadic Structure

Alessio Baldelli, Massimo Battaglioni, Paolo Santini.

Conference paper | 2025 | 2025 13th International Symposium on Topics in Coding (ISTC) | pp. 1-5.

[DOI 10.1109/istc65386.2025.11154589](#) | [Source record](#) | Registered DOI metadata

Q1020 Decoding Quantum LDPC Codes Using Graph Neural Networks

Vukan Ninkovic, Ognjen Kundacina, Dejan Vukobratovic, Christian Häger, Alexandre Graell i Amat.

Conference paper | 2024 | GLOBECOM 2024 - 2024 IEEE Global Communications Conference | pp. 3479-3484.

[DOI 10.1109/globecom52923.2024.10901425](#) | [Source record](#) | Registered DOI metadata

Q1021 Decoding Quasi-Cyclic Quantum LDPC Codes

Louis Golowich, Venkatesan Guruswami.

Conference paper | 2024 | 2024 IEEE 65th Annual Symposium on Foundations of Computer Science (FOCS) | pp. 344-368.

[DOI 10.1109/focs61266.2024.00029](#) | [Source record](#) | Registered DOI metadata

Q1022 Girth Analysis of Quantum Quasi-Cyclic LDPC Codes

Farzane Amirzade, Daniel Panario, Mohammad-Reza Sadeghi.

Journal article | 2024 | Problems of Information Transmission | vol. 60 | no. 2 | pp. 71-89.

[DOI 10.1134/s0032946024020017](#) | [Source record](#) | Registered DOI metadata

Q1023 New quantum LDPC codes based on Euclidean Geometry

Ya'nan Feng, Chuchen Tang, Chenming Bai.

Journal article | 2024 | Laser Physics | vol. 34 | no. 6 | pp. 065201.

[DOI 10.1088/1555-6611/ad3aed](#) | [Source record](#) | Registered DOI metadata

Q1024 New quantum LDPC codes based on projective geometry

Chuchen Tang, Chenming Bai, Ya'nan Feng.

Journal article | 2024 | Scientific Reports | vol. 14 | no. 1 | article 17014.

[DOI 10.1038/s41598-024-67786-0](#) | [Source record](#) | Registered DOI metadata

Q1025 Quantum LDPC Codes From Intersecting Subsets

Dimitar Ostrev.

Journal article | 2024 | IEEE Transactions on Information Theory | vol. 70 | no. 8 | pp. 5692-5709.

[DOI 10.1109/tit.2024.3402091](#) | [Source record](#) | Registered DOI metadata

Q1026 Rateless Protograph LDPC Codes for Quantum Key Distribution

Alberto Tarable, Rudi Paolo Paganelli, Marco Ferrari.

Journal article | 2024 | IEEE Transactions on Quantum Engineering | vol. 5 | pp. 1-11.

[DOI 10.1109/tqe.2024.3361810](#) | [Source record](#) | Registered DOI metadata

Q1027 Single-Shot Decoding of Good Quantum LDPC Codes

Shouzhen Gu, Eugene Tang, Libor Caha, Shin Ho Choe, Zhiyang He, Aleksander Kubica.

Journal article | 2024 | Communications in Mathematical Physics | vol. 405 | no. 3 | article 85.

[DOI 10.1007/s00220-024-04951-6](#) | [Source record](#) | Registered DOI metadata

Q1028 Viderman's algorithm for quantum LDPC codes

Anirudh Krishna, Inbal Livni Navon, Mary Wootters.

Book chapter | 2024 | Proceedings of the 2024 Annual ACM-SIAM Symposium on Discrete Algorithms (SODA) | pp. 2481-2507.

[DOI 10.1137/1.9781611977912.88](#) | [Source record](#) | Registered DOI metadata

D19T04 Bosonic error correction

Bosonic codes encode a logical system into oscillator states. Photon loss, finite energy, state preparation, and syndrome extraction determine protection.

Fine subcategories: Cat codes; GKP codes; binomial codes; oscillators; photon loss.

Primary catalog resources: 8. Related topic assignments are searchable in the HTML.

Q1029 Bosonic quantum error correction with microwave cavities for quantum repeaters

S. Siddardha Chelluri, Sanchar Sharma, Frank Schmidt, Silvia Viola Kusminskiy, Peter van Loock.

Journal article | 2026 | Physical Review A | vol. 114 | no. 1 | article 012616.

[DOI 10.1103/hkvx-rks6](#) | [Source record](#) | Registered DOI metadata

Q1030 Holonomic quantum gates via continuous measurement in bosonic codes: GKP and cat states

Juan Garcia Nila, Anirudh Lanka, Todd Brun.

Posted content | 2026 | Springer Science and Business Media LLC.

[DOI 10.21203/rs.3.rs-10693919/v1](#) | [Source record](#) | Registered DOI metadata

Fine tags: Cat codes; GKP codes

Q1031 Temporal Susceptibility Feedback for Hardware-efficient Bosonic Quantum Error Correction beyond Markovian Approximation

Isamu Ohnishi.

Posted content | 2026 | Elsevier BV.

[DOI 10.2139/ssrn.7422499](#) | [Source record](#) | Registered DOI metadata

Q1032 Bosonic quantum error correction using squeezed Fock states

E. N. Bashmakova, S. B. Korolev, T. Yu. Golubeva.

Journal article | 2025 | Physical Review A | vol. 112 | no. 3 | article 032434.

[DOI 10.1103/97yt-nzg2](#) | [Source record](#) | Registered DOI metadata

Q1033 Bosonic quantum error correction with neutral atoms in optical dipole traps

Leon H. Bohnmann, David F. Locher, Johannes Zeiher, Markus Müller.

Journal article | 2025 | Physical Review A | vol. 111 | no. 2 | article 022432.

[DOI 10.1103/physreva.111.022432](#) | [Source record](#) | Registered DOI metadata

Q1034 Hardware-efficient quantum error correction via concatenated bosonic qubits

Harald Putterman, Kyungjoo Noh, Connor T. Hann, Gregory S. MacCabe, Shahriar Aghaeimeibodi, Rishi N. Patel, Menyoung Lee, William M. Jones et al..

Journal article | 2025 | Nature | vol. 638 | no. 8052 | pp. 927-934.

[DOI 10.1038/s41586-025-08642-7](#) | [Source record](#) | Registered DOI metadata

Q1035 Quantum logic with bosonic error correction

Zhubing Jia.

Journal article | 2025 | Nature Physics | vol. 21 | no. 10 | pp. 1528-1529.

[DOI 10.1038/s41567-025-03011-7](#) | [Source record](#) | Registered DOI metadata

Q1036 Continuous-variable quantum repeaters based on bosonic error-correction and teleportation: architecture and applications

Bo-Han Wu, Zheshen Zhang, Quntao Zhuang.

Journal article | 2022 | Quantum Science and Technology | vol. 7 | no. 2 | pp. 025018.

[DOI 10.1088/2058-9565/ac4f6b](#) | [Source record](#) | Registered DOI metadata

D19T05 Subsystem and color codes

Subsystem and color codes offer alternative check and logical gate structures. Gauge choices and measurement schedules affect both error correction and computation.

Fine subcategories: Gauge fixing; Bacon Shor; gauge color codes; transversal structure; local checks.

Primary catalog resources: 8. Related topic assignments are searchable in the HTML.

Q1037 Distributed Realization of Color Codes for Quantum Error Correction

Nitish Kumar Chandra, David Tipper, Reza Nejabati, Eneet Kaur, Kaushik P. Seshadreesan.

Conference paper | 2025 | 2025 IEEE International Conference on Quantum Computing and Engineering (QCE) | pp. 2482-2492.

[DOI 10.1109/qce65121.2025.00269](#) | [Source record](#) | Registered DOI metadata

Q1038 Decoding quantum color codes with MaxSAT

Lucas Berent, Lukas Burgholzer, Peter-Jan H.S. Derks, Jens Eisert, Robert Wille.

Journal article | 2024 | Quantum | vol. 8 | pp. 1506 | article 1506.

[DOI 10.22331/q-2024-10-23-1506](#) | [Source record](#) | Registered DOI metadata

Q1039 Facilitating practical fault-tolerant quantum computing based on color codes

Jiaxuan Zhang, Yu-Chun Wu, Guo-Ping Guo.

Journal article | 2024 | Physical Review Research | vol. 6 | no. 3 | article 033086.

[DOI 10.1103/physrevresearch.6.033086](#) | [Source record](#) | Registered DOI metadata

Q1040 Construction of New Quantum Color Codes

Author metadata not supplied.

Journal article | 2022 | Quantum Physics Letters | vol. 11 | no. 3 | pp. 45-51.

[DOI 10.18576/qpl/110302](#) | [Source record](#) | Registered DOI metadata

Q1041 Construction of new quantum color codes

Avaz Naghipour, Duc Manh Nguyen.

Posted content | 2022 | Research Square Platform LLC.

[DOI 10.21203/rs.3.rs-1633740/v1](https://doi.org/10.21203/rs.3.rs-1633740/v1) | [Source record](#) | Registered DOI metadata

Q1042 New Quantum Color Codes Based on Hyperbolic Geometry

Avaz Naghipour, Duc Manh Nguyen.

Journal article | 2022 | Journal of Quantum Computing | vol. 4 | no. 2 | pp. 113-120.

[DOI 10.32604/jqc.2022.033712](https://doi.org/10.32604/jqc.2022.033712) | [Source record](#) | Registered DOI metadata

Q1043 Quantum computation with charge-and-color-permuting twists in qudit color codes

Manoj G. Gowda, Pradeep Kiran Sarvepalli.

Journal article | 2022 | Physical Review A | vol. 105 | no. 2 | article 022621.

[DOI 10.1103/physreva.105.022621](https://doi.org/10.1103/physreva.105.022621) | [Source record](#) | Registered DOI metadata

Q1044 Pseudocodeword-based Decoding of Quantum Color Codes

July X. Li, Joseph M. Renes, Pascal O. Vontobel.

Conference paper | 2021 | 2021 IEEE International Symposium on Information Theory (ISIT) | pp. 1558-1563.

[DOI 10.1109/isit45174.2021.9518077](https://doi.org/10.1109/isit45174.2021.9518077) | [Source record](#) | Registered DOI metadata

D19T06 Erasure and biased noise codes

Tailored codes exploit information about erasures or noise bias. Their benefit can disappear if detection or bias assumptions fail.

Fine subcategories: Loss detection; XZZX codes; biased dephasing; erasure conversion; tailored decoding.

Primary catalog resources: 8. Related topic assignments are searchable in the HTML.

Q1045 Developments in superconducting erasure qubits for hardware-efficient quantum error correction

Maria Violaris, Luciana Henaut, James Wills, Gioele Consani, Jamie Friel, Brian Vlastakis.

Journal article | 2026 | Materials for Quantum Technology | vol. 6 | no. 3 | pp. 033001.

[DOI 10.1088/2633-4356/ae9236](https://doi.org/10.1088/2633-4356/ae9236) | [Source record](#) | Registered DOI metadata

Q1046 Erasure Minesweeper: Exploring Hybrid-Erasure Surface Code Architectures for Efficient Quantum Error Correction

Jason D. Chadwick, Mariesa H. Teo, Joshua Visslai, Willers Yang, Frederic T. Chong.

Conference paper | 2025 | 2025 IEEE International Conference on Quantum Computing and Engineering (QCE) | pp. 658-669.

[DOI 10.1109/qce65121.2025.00077](https://doi.org/10.1109/qce65121.2025.00077) | [Source record](#) | Registered DOI metadata

Q1047 Fault-Tolerant Quantum Architectures Based on Erasure Qubits:Optimizing Code Conversion for Biased Noise Environments

Anand Thipperudra, Randeep Singh, Shambhulinga Aralekallu.

Posted content | 2025 | Elsevier BV.

[DOI 10.2139/ssrn.5750803](https://doi.org/10.2139/ssrn.5750803) | [Source record](#) | Registered DOI metadata

Fine tags: erasure conversion

Q1048 Optimizing Quantum Error-Correction Protocols with Erasure Qubits

Shouzheng Gu, Yotam Vaknin, Alex Retzker, Aleksander Kubica.

Journal article | 2025 | PRX Quantum | vol. 6 | no. 4 | article 040354.

[DOI 10.1103/985g-58gd](https://doi.org/10.1103/985g-58gd) | [Source record](#) | Registered DOI metadata

Q1049 Erasure Qubits for Abridged Error Correction

Matteo Rini.

Journal article | 2024 | Physics | vol. 17 | article s35.

[DOI 10.1103/physics.17.s35](#) | [Source record](#) | Registered DOI metadata

Q1050 Quantum Error Correction with Metastable States of Trapped Ions Using Erasure Conversion

Mingyu Kang, Wesley C. Campbell, Kenneth R. Brown.

Journal article | 2023 | PRX Quantum | vol. 4 | no. 2 | article 020358.

[DOI 10.1103/prxquantum.4.020358](#) | [Source record](#) | Registered DOI metadata

Fine tags: erasure conversion

Q1051 Channel Correction via Quantum Erasure

Francesco Buscemi.

Journal article | 2007 | Physical Review Letters | vol. 99 | no. 18 | article 180501.

[DOI 10.1103/physrevlett.99.180501](#) | [Source record](#) | Registered DOI metadata

Q1052 Landauer's erasure, error correction and entanglement

Vlatko Vedral.

Journal article | 2000 | Proceedings of the Royal Society of London. Series A: Mathematical, Physical and Engineering Sciences | vol. 456 | no. 1996 | pp. 969-984.

[DOI 10.1098/rspa.2000.0545](#) | [Source record](#) | Registered DOI metadata

D20 Fault tolerance and logical operations

Fault tolerance ensures that errors during correction and computation do not spread uncontrollably. Logical gates and routing must be evaluated with the entire correction cycle.

Prerequisites: Quantum error correction and logical circuits.

Assessment focus: Logical error budgets, gate factories, connectivity, decoder latency, and runtime.

D20T01 Thresholds and logical error scaling

Threshold analyses determine when larger encodings can suppress logical error. Finite distance performance and realistic correlations are needed for engineering decisions.

Fine subcategories: Threshold theorems; pseudothresholds; finite size scaling; correlated noise; logical error budgets.

Primary catalog resources: 8. Related topic assignments are searchable in the HTML.

Q1053 The Decade of Fault-Tolerant Quantum Computing: From Threshold Crossing to Scalable Logical Qubits

Volkan Erol.

Posted content | 2025 | MDPI AG.

[DOI 10.20944/preprints202509.2238.v1](https://doi.org/10.20944/preprints202509.2238.v1) | [Source record](#) | Registered DOI metadata

Q1054 Fast universal quantum gate above the fault-tolerance threshold in silicon

Akito Noiri, Kenta Takeda, Takashi Nakajima, Takashi Kobayashi, Amir Sammak, Giordano Scappucci, Seigo Tarucha.

Journal article | 2022 | Nature | vol. 601 | no. 7893 | pp. 338-342.

[DOI 10.1038/s41586-021-04182-y](https://doi.org/10.1038/s41586-021-04182-y) | [Source record](#) | Registered DOI metadata

Q1055 High-Threshold Fault-Tolerant Quantum Computation with Analog Quantum Error Correction

Kosuke Fukui, Akihisa Tomita, Atsushi Okamoto, Keisuke Fujii.

Journal article | 2018 | Physical Review X | vol. 8 | no. 2 | article 021054.

[DOI 10.1103/physrevx.8.021054](https://doi.org/10.1103/physrevx.8.021054) | [Source record](#) | Registered DOI metadata

Q1056 Superconducting quantum circuits at the surface code threshold for fault tolerance

R. Barends, J. Kelly, A. Megrant, A. Veitia, D. Sank, E. Jeffrey, T. C. White, J. Mutus et al..

Journal article | 2014 | Nature | vol. 508 | no. 7497 | pp. 500-503.

[DOI 10.1038/nature13171](https://doi.org/10.1038/nature13171) | [Source record](#) | Registered DOI metadata

Q1057 Error-Detection-Based Quantum Fault-Tolerance Threshold

Ben W. Reichardt.

Journal article | 2007 | Algorithmica | vol. 55 | no. 3 | pp. 517-556.

[DOI 10.1007/s00453-007-9069-7](https://doi.org/10.1007/s00453-007-9069-7) | [Source record](#) | Registered DOI metadata

Q1058 Fault-Tolerance Threshold for a Distance-Three Quantum Code

Ben W. Reichardt.

Book chapter | 2006 | Lecture Notes in Computer Science | pp. 50-61.

[DOI 10.1007/11786986_6](https://doi.org/10.1007/11786986_6) | [Source record](#) | Registered DOI metadata

Q1059 Threshold error penalty for fault-tolerant quantum computation with nearest neighbor communication

T. Szkopek, P.O. Boykin, Heng Fan, V.P. Roychowdhury, E. Yablonovitch, G. Simms, M. Gyure, B. Fong.

Journal article | 2006 | IEEE Transactions On Nanotechnology | vol. 5 | no. 1 | pp. 42-49.

[DOI 10.1109/tnano.2005.861402](https://doi.org/10.1109/tnano.2005.861402) | [Source record](#) | Registered DOI metadata

Q1060 Microscopic quantum dynamics study on the noise threshold of fault-tolerant quantum error correction

Y. C. Cheng, R. J. Silbey.

Journal article | 2005 | Physical Review A | vol. 72 | no. 1 | article 012320.

[DOI 10.1103/physreva.72.012320](https://doi.org/10.1103/physreva.72.012320) | [Source record](#) | Registered DOI metadata

D20T02 Magic state distillation

Magic state factories produce resources for non Clifford logical gates. Distillation yield, scheduling, and physical volume can dominate a large computation.

Fine subcategories: T factories; distillation protocols; acceptance rates; factory scheduling; space time cost.

Primary catalog resources: 16. Related topic assignments are searchable in the HTML.

Q1061 Fragility of Magic State Distillation under Imperfect Measurements

Yunzhe Zheng, Yuanchen Zhao, Dong E. Liu.

Journal article | 2026 | npj Quantum Information.

[DOI 10.1038/s41534-026-01373-1](https://doi.org/10.1038/s41534-026-01373-1) | [Source record](#) | Registered DOI metadata

Q1062 High-threshold magic state distillation with quantum quadratic residue codes

Michael Zurel, Santanil Jana, Nadish de Silva.

Journal article | 2026 | Quantum Science and Technology | vol. 11 | no. 4 | pp. 045052.

[DOI 10.1088/2058-9565/aea362](https://doi.org/10.1088/2058-9565/aea362) | [Source record](#) | Registered DOI metadata

Q1063 Magic state injection on IBM quantum processors above the distillation threshold

Younghun Kim, Martin Sevier, Muhammad Usman.

Journal article | 2026 | Scientific Reports | vol. 16 | no. 1 | article 11189.

[DOI 10.1038/s41598-026-40381-1](https://doi.org/10.1038/s41598-026-40381-1) | [Source record](#) | Registered DOI metadata

Q1064 Noise-canceling quantum feedback: Non-Hermitian dynamics with applications to state preparation and magic state distillation

Tathagata Karmakar, Philippe Lewalle, Yipei Zhang, K. Birgitta Whaley.

Journal article | 2026 | Physical Review A | vol. 113 | no. 4 | article 042403.

[DOI 10.1103/cnxj-m1wn](https://doi.org/10.1103/cnxj-m1wn) | [Source record](#) | Registered DOI metadata

Q1065 Search for high-threshold qutrit magic-state distillation routines

Shiroman Prakash, Rishabh Singhal.

Journal article | 2026 | Physical Review A | vol. 113 | no. 4 | article 042404.

[DOI 10.1103/61q5-f754](https://doi.org/10.1103/61q5-f754) | [Source record](#) | Registered DOI metadata

Q1066 Constant-overhead magic state distillation

Adam Wills, Min-Hsiu Hsieh, Hayata Yamasaki.

Journal article | 2025 | Nature Physics | vol. 21 | no. 11 | pp. 1842-1846.

[DOI 10.1038/s41567-025-03026-0](https://doi.org/10.1038/s41567-025-03026-0) | [Source record](#) | Registered DOI metadata

Q1067 Efficient Magic State Distillation by Zero-Level Distillation

Tomohiro Itogawa, Yugo Takada, Yutaka Hirano, Keisuke Fujii.

Journal article | 2025 | PRX Quantum | vol. 6 | no. 2 | article 020356.

[DOI 10.1103/thxx-njr6](https://doi.org/10.1103/thxx-njr6) | [Source record](#) | Registered DOI metadata

Q1068 Experimental demonstration of logical magic state distillation

Pedro Sales Rodriguez, John M. Robinson, Paul Niklas Jepsen, Zhiyang He, Casey Duckering, Chen Zhao, Kai-Hsin Wu, Joseph Campo et al..

Journal article | 2025 | Nature | vol. 645 | no. 8081 | pp. 620-625.

[DOI 10.1038/s41586-025-09367-3](https://doi.org/10.1038/s41586-025-09367-3) | [Source record](#) | Registered DOI metadata

Q1069 From Magic State Distillation to Dynamical Systems

Yunzhe Zheng, Dong E. Liu.

Journal article | 2025 | Quantum | vol. 9 | pp. 1858 | article 1858.

[DOI 10.22331/q-2025-09-15-1858](#) | [Source record](#) | Registered DOI metadata

Q1070 Low Overhead Qutrit Magic State Distillation

Shiroman Prakash, Tanay Saha.

Journal article | 2025 | Quantum | vol. 9 | pp. 1768 | article 1768.

[DOI 10.22331/q-2025-06-12-1768](#) | [Source record](#) | Registered DOI metadata

Q1071 Low-Overhead Magic State Distillation with Color Codes

Seok-Hyung Lee, Felix Thomsen, Nicholas Fazio, Benjamin J. Brown, Stephen D. Bartlett.

Journal article | 2025 | PRX Quantum | vol. 6 | no. 3 | article 030317.

[DOI 10.1103/ch5r-cnfq](#) | [Source record](#) | Registered DOI metadata

Q1072 Magic State Injection on IBM Quantum Processors Above the Distillation Threshold

Younghun Kim, Martin Sevier, Muhammad Usman.

Posted content | 2025 | Springer Science and Business Media LLC.

[DOI 10.21203/rs.3.rs-8303387/v1](#) | [Source record](#) | Registered DOI metadata

Q1073 Magic state distillation without measurements and post-selection

Sascha Heußen.

Journal article | 2025 | APL Quantum | vol. 2 | no. 4 | article 046113.

[DOI 10.1063/5.0280551](#) | [Source record](#) | Registered DOI metadata

Q1074 Evaluation of Y Magic State Distillation Circuit

Youngchul Kim, Soo-Cheol Oh, Sangmin Lee, Ki-Sung Jin, Gyuil Cha.

Conference paper | 2024 | 2024 26th International Conference on Advanced Communications Technology (ICACT) | pp. 221-225.

[DOI 10.23919/icact60172.2024.10471972](#) | [Source record](#) | Registered DOI metadata

Q1075 Magic State Distillation from Qudit Stabilizer Codes

Abhi Kumar Sharma, Shayan Srinivasa Garani.

Conference paper | 2024 | ICC 2024 - IEEE International Conference on Communications | pp. 1-6.

[DOI 10.1109/icc51166.2024.10623071](#) | [Source record](#) | Registered DOI metadata

Q1076 Magic State Distillation with Reduced Time Cost

Yujin Kang, Youshin Chung, Huidan Zheng, Sungyeon Kook, Jun Heo.

Conference paper | 2024 | 2024 IEEE International Conference on Quantum Computing and Engineering (QCE) | pp. 416-417.

[DOI 10.1109/qce60285.2024.10333](#) | [Source record](#) | Registered DOI metadata

D20T03 Lattice surgery

Lattice surgery implements logical parity operations through changing code boundaries. Routing and time steps must be included in algorithm mapping.

Fine subcategories: Parity measurements; patch merging; patch splitting; logical routing; surgery scheduling.

Primary catalog resources: 9. Related topic assignments are searchable in the HTML.

Q1077 Non-Clifford quantum gate teleportation with generalized lattice surgery

Yifei Wang, Yingfei Gu.

Journal article | 2026 | Physical Review A | vol. 113 | no. 6 | article 062410.

[DOI 10.1103/1xq4-856m](#) | [Source record](#) | Registered DOI metadata

Q1078 Quantum Resource Comparison for Two Leading Surface Code Lattice Surgery Approaches

Tyler LeBlond, Ryan S. Bennink.

Journal article | 2026 | Quantum | vol. 10 | pp. 2187 | article 2187.

[DOI 10.22331/q-2026-08-10-2187](https://doi.org/10.22331/q-2026-08-10-2187) | [Source record](#) | Registered DOI metadata

Q1079 Ultra-Low Logical Depth Fault-Tolerant Quantum Circuit Synthesis via Lattice Surgery

Chien-Tung Cherie Kuo, Cheng-En Tsai, Chung-Yang Ric Huang.

Conference paper | 2026 | 2026 Design, Automation & Test in Europe Conference (DATE) | pp. 1-7.

[DOI 10.23919/date69613.2026.11539371](https://doi.org/10.23919/date69613.2026.11539371) | [Source record](#) | Registered DOI metadata

Q1080 Lattice Surgery for Dummies

Avimita Chatterjee, Subrata Das, Swaroop Ghosh.

Journal article | 2025 | Sensors | vol. 25 | no. 6 | pp. 1854.

[DOI 10.3390/s25061854](https://doi.org/10.3390/s25061854) | [Source record](#) | Registered DOI metadata

Q1081 Network-Integrated Decoding System for Real-Time Quantum Error Correction with Lattice Surgery

Namitha Liyanage, Yue Wu, Emmet Houghton, Lin Zhong.

Conference paper | 2025 | 2025 IEEE International Conference on Quantum Computing and Engineering (QCE) | pp. 1148-1159.

[DOI 10.1109/qce65121.2025.00129](https://doi.org/10.1109/qce65121.2025.00129) | [Source record](#) | Registered DOI metadata

Q1082 Multi-FPGA System for Quantum Error Correction with Lattice Surgery

Namitha Liyanage, Yue Wu, Emmet Houghton, Lin Zhong.

Conference paper | 2024 | 2024 IEEE International Conference on Quantum Computing and Engineering (QCE) | pp. 622-623.

[DOI 10.1109/qce60285.2024.10435](https://doi.org/10.1109/qce60285.2024.10435) | [Source record](#) | Registered DOI metadata

Q1083 Lattice surgery-based logical operations in a fault-tolerant quantum software framework

Youngchul Kim, Soo-Cheol Oh, Sangmin Lee, Ki-Sung Jin, Gyuil Cha.

Conference paper | 2023 | 2023 14th International Conference on Information and Communication Technology Convergence (ICTC) | pp. 173-176.

[DOI 10.1109/ictc58733.2023.10393193](https://doi.org/10.1109/ictc58733.2023.10393193) | [Source record](#) | Registered DOI metadata

Q1084 Universal Quantum Computing with Twist-Free and Temporally Encoded Lattice Surgery

Christopher Chamberland, Earl T. Campbell.

Journal article | 2022 | PRX Quantum | vol. 3 | no. 1 | article 010331.

[DOI 10.1103/prxquantum.3.010331](https://doi.org/10.1103/prxquantum.3.010331) | [Source record](#) | Registered DOI metadata

Q1085 Surface code quantum computing by lattice surgery

Dominic Horsman, Austin G Fowler, Simon Devitt, Rodney Van Meter.

Journal article | 2012 | New Journal of Physics | vol. 14 | no. 12 | pp. 123011.

[DOI 10.1088/1367-2630/14/12/123011](https://doi.org/10.1088/1367-2630/14/12/123011) | [Source record](#) | Registered DOI metadata

D20T04 Transversal gates and code switching

Transversal gates and code switching manage logical operations while limiting error spread. Universality restrictions motivate combining multiple methods.

Fine subcategories: Eastin Knill limits; gauge fixing; code conversion; logical Clifford gates.

Primary catalog resources: 8. Related topic assignments are searchable in the HTML.

Q1086 Co-Designing Quantum Codes with Transversal Diagonal Gates via Multi-Agent Systems

Xi He, Sirui Lu, Bei Zeng.

Conference paper | 2026 | 2026 IEEE 19th Dallas Circuits and Systems Conference (DCAS) | pp. 1-6.

[DOI 10.1109/dcas69364.2026.11544392](https://doi.org/10.1109/dcas69364.2026.11544392) | [Source record](#) | Registered DOI metadata

Q1087 Homological Origin of the Transversal Implementability of Logical Diagonal Gates in Quantum CSS Codes

Junichi Haruna.

Journal article | 2026 | Progress of Theoretical and Experimental Physics | vol. 2026 | no. 8 | article 083A03.

[DOI 10.1093/ptep/ptag133](https://doi.org/10.1093/ptep/ptag133) | [Source record](#) | Registered DOI metadata

Q1088 Asymptotically Good Quantum Codes with Transversal Non-Clifford Gates

Louis Golowich, Venkatesan Guruswami.

Conference paper | 2025 | Proceedings of the 57th Annual ACM Symposium on Theory of Computing | pp. 707-717.

[DOI 10.1145/3717823.3718234](https://doi.org/10.1145/3717823.3718234) | [Source record](#) | Registered DOI metadata

Q1089 Near-Asymptotically-Good Quantum Codes with Transversal CCZ Gates and Sublinear-Weight Parity-Checks

Louis Golowich, Venkatesan Guruswami.

Conference paper | 2025 | 2025 IEEE 66th Annual Symposium on Foundations of Computer Science (FOCS) | pp. 1561-1569.

[DOI 10.1109/focs63196.2025.00082](https://doi.org/10.1109/focs63196.2025.00082) | [Source record](#) | Registered DOI metadata

Q1090 Permutation-Invariant Quantum Codes With Transversal Generalized Phase Gates

Eric Kubischta, Ian Teixeira.

Journal article | 2025 | IEEE Transactions on Information Theory | vol. 71 | no. 1 | pp. 485-498.

[DOI 10.1109/tit.2024.3487964](https://doi.org/10.1109/tit.2024.3487964) | [Source record](#) | Registered DOI metadata

Q1091 Quantum LDPC Codes with Transversal Non-Clifford Gates via Products of Algebraic Codes

Louis Golowich, Ting-Chun Lin.

Conference paper | 2025 | Proceedings of the 57th Annual ACM Symposium on Theory of Computing | pp. 689-696.

[DOI 10.1145/3717823.3718139](https://doi.org/10.1145/3717823.3718139) | [Source record](#) | Registered DOI metadata

Q1092 Fold-Transversal Clifford Gates for Quantum Codes

Nikolas P. Breuckmann, Simon Burton.

Journal article | 2024 | Quantum | vol. 8 | pp. 1372 | article 1372.

[DOI 10.22331/q-2024-06-13-1372](https://doi.org/10.22331/q-2024-06-13-1372) | [Source record](#) | Registered DOI metadata

Q1093 Family of Quantum Codes with Exotic Transversal Gates

Eric Kubischta, Ian Teixeira.

Journal article | 2023 | Physical Review Letters | vol. 131 | no. 24 | article 240601.

[DOI 10.1103/physrevlett.131.240601](https://doi.org/10.1103/physrevlett.131.240601) | [Source record](#) | Registered DOI metadata

D20T05 Fault tolerant resource estimation

Resource estimation turns logical circuits into physical hardware and runtime requirements. Error budgets, code distance, connectivity, and factory assumptions should be explicit.

Fine subcategories: Physical qubits; logical qubits; T depth; runtime; code distance; architectural assumptions.

Primary catalog resources: 7. Related topic assignments are searchable in the HTML.

Q1094 Integration and Resource Estimation of Cryoelectronics for Superconducting Fault-Tolerant Quantum Computers

Shiro KAWABATA.

Journal article | 2026 | IEICE Transactions on Electronics | vol. E109.C | no. 10 | pp. 512-520 | article 2025SEI0001.

[DOI 10.1587/transele.2025sei0001](https://doi.org/10.1587/transele.2025sei0001) | [Source record](#) | Registered DOI metadata

Q1095 Optomechanical Resource for Fault-Tolerant Quantum Computing

Margaret Pavlovich, Peter T. Rakich, Shruti Puri.

Journal article | 2026 | PRX Quantum | vol. 7 | no. 1 | article 010316.

[DOI 10.1103/4k7h-4vwc](https://doi.org/10.1103/4k7h-4vwc) | [Source record](#) | Registered DOI metadata

Q1096 Resource Allocation for Distributed Fault-Tolerant Quantum Computing With Surface Codes

Xu Xu, Yu Liu, Yingling Mao, Yuanyuan Yang.

Conference paper | 2026 | 2026 IEEE 46th International Conference on Distributed Computing Systems (ICDCS) | pp. 1004-1014.

[DOI 10.1109/2575-8411.2026.00100](https://doi.org/10.1109/2575-8411.2026.00100) | [Source record](#) | Registered DOI metadata

Q1097 Photon distillation with reduced resource costs for fault-tolerant quantum computation

Frank Somhorst, Kite Sauër, Stefan van den Hoven, Jelmer Renema.

Conference paper | 2025 | Quantum Computing, Communication, and Simulation V | pp. 79.

[DOI 10.1117/12.3041144](https://doi.org/10.1117/12.3041144) | [Source record](#) | Registered DOI metadata

Q1098 Resource-efficient fault-tolerant one-way quantum repeater with code concatenation

Kah Jen Wo, Guus Avis, Filip Rozpędek, Maria Flors Mor-Ruiz, Gregor Pieplow, Tim Schröder, Liang Jiang, Anders S. Sørensen et al..

Journal article | 2023 | npj Quantum Information | vol. 9 | no. 1 | article 123.

[DOI 10.1038/s41534-023-00792-8](https://doi.org/10.1038/s41534-023-00792-8) | [Source record](#) | Registered DOI metadata

Q1099 Using Azure Quantum Resource Estimator for Assessing Performance of Fault Tolerant Quantum Computation

Wim van Dam, Mariia Mykhailova, Mathias Soeken.

Conference paper | 2023 | Proceedings of the SC '23 Workshops of the International Conference on High Performance Computing, Network, Storage, and Analysis | pp. 1414-1419.

[DOI 10.1145/3624062.3624211](https://doi.org/10.1145/3624062.3624211) | [Source record](#) | Registered DOI metadata

Q1100 Fault-Tolerant Resource Estimation of Quantum Random-Access Memories

Olivia Di Matteo, Vlad Gheorghiu, Michele Mosca.

Journal article | 2020 | IEEE Transactions on Quantum Engineering | vol. 1 | pp. 1-13.

[DOI 10.1109/tqe.2020.2965803](https://doi.org/10.1109/tqe.2020.2965803) | [Source record](#) | Registered DOI metadata

D20T06 Quantum error decoding

Decoders infer likely errors from noisy syndrome data. Accuracy must be evaluated together with throughput, latency, and robustness to model mismatch.

Fine subcategories: Matching; belief propagation; union find; neural decoders; latency; streaming inference.

Primary catalog resources: 12. Related topic assignments are searchable in the HTML.

Q1101 Backlog Metastability in Windowed Quantum Error Correction Decoding

Jean-Jacques Dubray.

Posted content | 2026 | MDPI AG.

[DOI 10.20944/preprints202608.1425.v3](https://doi.org/10.20944/preprints202608.1425.v3) | [Source record](#) | Registered DOI metadata

Q1102 Two-Stage Neural Residual Decoding for FastSurface-Code Quantum Error Correction

Jiajun Chen, Jierui Peng, Jingyi Liu.

Posted content | 2026 | Springer Science and Business Media LLC.

[DOI 10.21203/rs.3.rs-10123128/v1](https://doi.org/10.21203/rs.3.rs-10123128/v1) | [Source record](#) | Registered DOI metadata

Q1103 Vibe Decoding Quantum Error Correction with CUDA

Stergios Koutsoumpas, Joschka Roffe.

Posted content | 2026 | Front Matter.

[DOI 10.59350/qxsf5-r9s02](https://doi.org/10.59350/qxsf5-r9s02) | [Source record](#) | Registered DOI metadata

Q1104 Estimating decoding graphs and hypergraphs of memory quantum error-correction experiments

Evangelia Takou, Kenneth R. Brown.

Journal article | 2025 | Physical Review A | vol. 112 | no. 5 | article 052414.

[DOI 10.1103/qcz4-nx4r](https://doi.org/10.1103/qcz4-nx4r) | [Source record](#) | Registered DOI metadata

Q1105 Parallel Minimum-Weight Parity Factor Decoding for Quantum Error Correction

Liu Yang, Yue Wu, Lin Zhong.

Conference paper | 2024 | 2024 IEEE International Conference on Quantum Computing and Engineering (QCE) | pp. 581-582.

[DOI 10.1109/qce60285.2024.10415](https://doi.org/10.1109/qce60285.2024.10415) | [Source record](#) | Registered DOI metadata

Q1106 Better Than Worst-Case Decoding for Quantum Error Correction

Gokul Subramanian Ravi, Jonathan M. Baker, Arash Fayyazi, Sophia Fuhui Lin, Ali Javadi-Abhari, Massoud Pedram, Frederic T. Chong.

Conference paper | 2023 | Proceedings of the 28th ACM International Conference on Architectural Support for Programming Languages and Operating Systems, Volume 2 | pp. 88-102.

[DOI 10.1145/3575693.3575733](https://doi.org/10.1145/3575693.3575733) | [Source record](#) | Registered DOI metadata

Q1107 Quantum Error Correction Via Noise Guessing Decoding

Diogo Cruz, Francisco A. Monteiro, Bruno C. Coutinho.

Journal article | 2023 | IEEE Access | vol. 11 | pp. 119446-119461.

[DOI 10.1109/access.2023.3327214](https://doi.org/10.1109/access.2023.3327214) | [Source record](#) | Registered DOI metadata

Q1108 Decoding Quantum Error Correction Codes With Local Variation

Michael Hanks, William J. Munro, Kae Nemoto.

Journal article | 2020 | IEEE Transactions on Quantum Engineering | vol. 1 | pp. 1-8.

[DOI 10.1109/tqe.2020.2967890](https://doi.org/10.1109/tqe.2020.2967890) | [Source record](#) | Registered DOI metadata

Q1109 Error correction and decoding for quantum stabilizer codes

Xiao Fang-Ying, Chen Han-Wu, School of Computer Science and Engineering, Southeast University, Nanjing 211189, China; Key Laboratory of Computer Network and Information Integration of Ministry of Education, Southeast University, Nanjing 211189, China.

Journal article | 2011 | Acta Physica Sinica | vol. 60 | no. 8 | pp. 080303.

[DOI 10.7498/aps.60.080303](https://doi.org/10.7498/aps.60.080303) | [Source record](#) | Registered DOI metadata

Q1110 GitHub - oscarhiggott/PyMatching: PyMatching: A Python/C++ library for decoding quantum error correcting codes with minimum-weight perfect matching. · GitHub

Author metadata not supplied.

Software repository | Undated | PyMatching.

[Official resource](#) | [Source record](#) | Official page retrieved

AQUIRE

Q1111 GitHub - quantumlib/Stim: A fast stabilizer circuit library. · GitHub

Author metadata not supplied.

Software repository | Undated | Google Quantum AI.

[Official resource](#) | [Source record](#) | Official page retrieved

Q1112 PyMatching 2 — PyMatching 2.1.0 documentation

Author metadata not supplied.

Documentation | Undated | PyMatching.

[Official resource](#) | [Source record](#) | Official page retrieved

D21 Error mitigation and near term reliability

Error mitigation changes estimators or experiments to reduce bias without providing the full protection of a fault tolerant encoding. Sampling overhead and assumptions must be reported.

Prerequisites: Noise channels, statistical estimation, and circuit execution.

Assessment focus: Estimator bias, variance, model mismatch, calibration drift, and shot overhead.

D21T01 Zero noise extrapolation

Zero noise extrapolation fits results measured at increased noise to estimate a lower noise limit. Extrapolation introduces variance and model dependent bias.

Fine subcategories: Noise scaling; folding; regression; bias; variance; extrapolation failure.

Primary catalog resources: 16. Related topic assignments are searchable in the HTML.

Q1113 A useful metric for the NISQ era: Qubit error probability and its role in zero noise extrapolation

Nahual Sobrino, Unai Aseginolaza, Joaquim Jornet-Somoza, Juan Borge.

Journal article | 2026 | AVS Quantum Science | vol. 8 | no. 1 | article 013803.

[DOI 10.1116/5.0287324](#) | [Source record](#) | Registered DOI metadata

Q1114 Improving Zero-Noise Extrapolation for Quantum-Gate Error Mitigation Using a Noise-Aware Folding Method

Leanghok Hour, Myeongseong Go, Youngsun Han.

Journal article | 2026 | IEEE Access | vol. 14 | pp. 37362-37372.

[DOI 10.1109/access.2026.3672056](#) | [Source record](#) | Registered DOI metadata

Fine tags: folding

Q1115 QNBAD: Quantum Noise-induced Backdoor Attacks against Zero Noise Extrapolation

Cheng Chu, Qian Lou, Fan Chen, Lei Jiang.

Conference paper | 2026 | Proceedings 2026 Network and Distributed System Security Symposium.

[DOI 10.14722/ndss.2026.241665](#) | [Source record](#) | Registered DOI metadata

Q1116 Quantum Noise Mitigation With Adaptive Zero-Noise Extrapolation: A Contextual Multi-Armed Bandits Approach

Ratun Rahman, Dinh C. Nguyen.

Journal article | 2026 | IEEE Journal on Selected Areas in Communications | vol. 44 | pp. 5600-5614.

[DOI 10.1109/jsac.2026.3721807](#) | [Source record](#) | Registered DOI metadata

Q1117 Zero-noise extrapolation via cyclic permutations of quantum circuit layouts

Zahar Sayapin, Daniil Rabinovich, Nikita Korolev, Kirill Lakhmanskiy.

Journal article | 2026 | Physical Review A | vol. 114 | no. 1 | article 012620.

[DOI 10.1103/wb3p-1sm4](#) | [Source record](#) | Registered DOI metadata

Q1118 A Novel Approach for Mitigating Gate- Level Noise in IBM Quantum Hardware via Zero Noise Extrapolation (ZNE) and Probabilistic Error Cancellation (PEC) Techniques

Mohammed Suhaib, V. Karthick, Monica D.

Posted content | 2025 | Springer Science and Business Media LLC.

[DOI 10.21203/rs.3.rs-7883122/v1](#) | [Source record](#) | Registered DOI metadata

Q1119 Application of zero-noise-extrapolation-based quantum error mitigation to a silicon spin qubit

Hanseo Sohn, Jaewon Jung, Jaemin Park, Hyeongyu Jang, Lucas E. A. Stehouwer, Davide Degli Esposti, Giordano Scappucci, Dohun Kim.

Journal article | 2025 | Physical Review A | vol. 112 | no. 1 | article 012408.

[DOI 10.1103/925y-b4s1](#) | [Source record](#) | Registered DOI metadata

Q1120 Digital Zero-Noise Extrapolation with Quantum Circuit Unoptimization

Elijah Pelofske, Vincent Russo.

Conference paper | 2025 | 2025 IEEE International Conference on Quantum Computing and Engineering (QCE) | pp. 01-13.

[DOI 10.1109/qce65121.2025.00020](https://doi.org/10.1109/qce65121.2025.00020) | [Source record](#) | Registered DOI metadata

Q1121 Evaluating the Role of Noise Mitigation in QAOA: A Study of Zero-Noise Extrapolation and CVaR

Adriano Lusso, Marco Venere, Victor Onofre, Alberto Maldonado-Romo, Alejandro Montanez-Barrera, Marco D. Santambrogio.

Conference paper | 2025 | 2025 IEEE International Conference on Quantum Computing and Engineering (QCE) | pp. 602-603.

[DOI 10.1109/qce65121.2025.10466](https://doi.org/10.1109/qce65121.2025.10466) | [Source record](#) | Registered DOI metadata

Q1122 ML-Driven Quantum Portfolio Optimization: Hybrid CNN-LSTM Architecture with Adaptive Zero-Noise Extrapolation on NISQ Devices

Maryam Arif, Soban Saeed.

Posted content | 2025 | Springer Science and Business Media LLC.

[DOI 10.21203/rs.3.rs-8423464/v1](https://doi.org/10.21203/rs.3.rs-8423464/v1) | [Source record](#) | Registered DOI metadata

Q1123 Noise-Mitigated Variational Quantum Eigensolver with Pre-training and Zero-Noise Extrapolation

Wanqi Sun, Jungang Xu, Chenghua Duan.

Conference paper | 2025 | ICASSP 2025 - 2025 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP) | pp. 1-5.

[DOI 10.1109/icassp49660.2025.10887977](https://doi.org/10.1109/icassp49660.2025.10887977) | [Source record](#) | Registered DOI metadata

Q1124 Back Cover: Purity-Assisted Zero-Noise Extrapolation for Quantum Error Mitigation (Adv. Quantum Technol. 12/2024)

Tian-Ren Jin, Yun-Hao Shi, Zheng-An Wang, Tian-Ming Li, Kai Xu, Heng Fan.

Journal article | 2024 | Advanced Quantum Technologies | vol. 7 | no. 12 | article 2470037.

[DOI 10.1002/qute.202470037](https://doi.org/10.1002/qute.202470037) | [Source record](#) | Registered DOI metadata

Q1125 Increasing the Measured Effective Quantum Volume with Zero Noise Extrapolation

Elijah Pelofske, Vincent Russo, Ryan Larose, Andrea Mari, Dan Strano, Andreas Bärttschi, Stephan Eidenbenz, William Zeng.

Journal article | 2024 | ACM Transactions on Quantum Computing | vol. 5 | no. 3 | pp. 1-18.

[DOI 10.1145/3680290](https://doi.org/10.1145/3680290) | [Source record](#) | Registered DOI metadata

Q1126 Inverted-circuit zero-noise extrapolation for quantum-gate error mitigation

Kathrin F. Koenig, Finn Reinecke, Walter Hahn, Thomas Wellens.

Journal article | 2024 | Physical Review A | vol. 110 | no. 4 | article 042625.

[DOI 10.1103/physreva.110.042625](https://doi.org/10.1103/physreva.110.042625) | [Source record](#) | Registered DOI metadata

Q1127 Mitigating quantum gate errors for variational eigensolvers using hardware-inspired zero-noise extrapolation

Alexey Uvarov, Daniil Rabinovich, Olga Lakhmanskaya, Kirill Lakhmanskiy, Jacob Biamonte, Soumik Adhikary.

Journal article | 2024 | Physical Review A | vol. 110 | no. 1 | article 012404.

[DOI 10.1103/physreva.110.012404](https://doi.org/10.1103/physreva.110.012404) | [Source record](#) | Registered DOI metadata

Q1128 Purity-Assisted Zero-Noise Extrapolation for Quantum Error Mitigation

Tian-Ren Jin, Yun-Hao Shi, Zheng-An Wang, Tian-Ming Li, Kai Xu, Heng Fan.

Journal article | 2024 | Advanced Quantum Technologies | vol. 7 | no. 12 | article 2400150.

[DOI 10.1002/qute.202400150](https://doi.org/10.1002/qute.202400150) | [Source record](#) | Registered DOI metadata

D21T02 Probabilistic error cancellation

Probabilistic cancellation represents an inverse noise process through weighted sampling. Reliable channel knowledge and sampling overhead limit applicability.

Fine subcategories: Quasiprobabilities; channel learning; sampling overhead; model mismatch.

Primary catalog resources: 7. Related topic assignments are searchable in the HTML.

Q1129 CV4Quantum: Reducing the Sampling Overhead in Probabilistic Error Cancellation

US Department of Energy, Prasanth Shyamsundar, Fermi National Accelerator Laboratory (FNAL), Batavia, IL (United States).

Report | 2025 | Office of Scientific and Technical Information (OSTI).

[DOI 10.2172/2568558](#) | [Source record](#) | Registered DOI metadata

Fine tags: sampling overhead

Q1130 Variational quantum algorithms with invariant probabilistic error cancellation on noisy quantum processors

Yulin Chi, Hongyi Shi, Wen Zheng, Haoyang Cai, Yu Zhang, Xinsheng Tan, Shaoxiong Li, Jianwei Wang et al..

Journal article | 2025 | Science China Physics, Mechanics & Astronomy | vol. 69 | no. 1 | article 210312.

[DOI 10.1007/s11433-025-2779-x](#) | [Source record](#) | Registered DOI metadata

Q1131 Probabilistic error cancellation for dynamic quantum circuits

Riddhi S. Gupta, Ewout van den Berg, Maika Takita, Diego Ristè, Kristan Temme, Abhinav Kandala.

Journal article | 2024 | Physical Review A | vol. 109 | no. 6 | article 062617.

[DOI 10.1103/physreva.109.062617](#) | [Source record](#) | Registered DOI metadata

Q1132 Noise-Assisted Digital Quantum Simulation of Open Systems Using Partial Probabilistic Error Cancellation

José D. Guimarães, James Lim, Mikhail I. Vasilevskiy, Susana F. Huelga, Martin B. Plenio.

Journal article | 2023 | PRX Quantum | vol. 4 | no. 4 | article 040329.

[DOI 10.1103/prxquantum.4.040329](#) | [Source record](#) | Registered DOI metadata

Q1133 Probabilistic error cancellation with sparse Pauli–Lindblad models on noisy quantum processors

Ewout van den Berg, Zlatko K. Mineev, Abhinav Kandala, Kristan Temme.

Journal article | 2023 | Nature Physics | vol. 19 | no. 8 | pp. 1116-1121.

[DOI 10.1038/s41567-023-02042-2](#) | [Source record](#) | Registered DOI metadata

Q1134 Automated quantum error mitigation based on probabilistic error reduction

Benjamin McDonough, Andrea Mari, Nathan Shammah, Nathaniel T. Stemen, Misty Wahl, William J. Zeng, Peter P. Orth. Conference paper | 2022 | 2022 IEEE/ACM Third International Workshop on Quantum Computing Software (QCS) | pp. 83-93.

[DOI 10.1109/qcs56647.2022.00015](#) | [Source record](#) | Registered DOI metadata

Q1135 Extending quantum probabilistic error cancellation by noise scaling

Andrea Mari, Nathan Shammah, William J. Zeng.

Journal article | 2021 | Physical Review A | vol. 104 | no. 5 | article 052607.

[DOI 10.1103/physreva.104.052607](#) | [Source record](#) | Registered DOI metadata

D21T03 Readout error mitigation

Readout mitigation adjusts observed outcome statistics using a measurement error model. Correlated errors and calibration drift can invalidate simple assignment matrices.

Fine subcategories: Assignment matrices; correlated readout; unfolding; scalable calibration.

Primary catalog resources: 8. Related topic assignments are searchable in the HTML.

Q1136 Improving VQE parameter quality on noisy quantum processors with cost-effective readout error mitigation

Nacer Eddine Belaloui, Abdellah Tounsi, Abdelmouheymen Rabah Khamadja, Hamza Benkadour, Mohamed Messaoud Louamri, Achour Benslama, Mohamed Taha Rouabah.

Journal article | 2026 | Physica Scripta | vol. 101 | no. 3 | pp. 035101.

[DOI 10.1088/1402-4896/ae2f3e](#) | [Source record](#) | Registered DOI metadata

Q1137 Adaptive Quantum Readout Error Mitigation with Transfer Learning

Usama Inam Paracha, Syed Muhammad Abuzar Rizvi, Muhammad Mustafa Umar Gondel, Wook Park, Hyundong Shin.

Conference paper | 2024 | 2024 15th International Conference on Information and Communication Technology Convergence (ICTC) | pp. 506-511.

[DOI 10.1109/ictc62082.2024.10826964](#) | [Source record](#) | Registered DOI metadata

Q1138 Quantum generative adversarial networks based on a readout error mitigation method with fault tolerant mechanism

Run-Sheng Zhao, Hong-Yang Ma, Tao Cheng, Shuang Wang, Xing-Kui Fan.

Journal article | 2024 | Chinese Physics B | vol. 33 | no. 4 | pp. 040304.

[DOI 10.1088/1674-1056/ad02e7](#) | [Source record](#) | Registered DOI metadata

Q1139 SpREM: Exploiting Hamming Sparsity for Fast Quantum Readout Error Mitigation

Hanyu Zhang, Liqiang Lu, Siwei Tan, Size Zheng, Jia Yu, Jianwei Yin.

Conference paper | 2024 | Proceedings of the 61st ACM/IEEE Design Automation Conference | pp. 1-6.

[DOI 10.1145/3649329.3655675](#) | [Source record](#) | Registered DOI metadata

Q1140 Information-theoretic approach to readout-error mitigation for quantum computers

Hai-Chau Nguyen.

Journal article | 2023 | Physical Review A | vol. 108 | no. 5 | article 052419.

[DOI 10.1103/physreva.108.052419](#) | [Source record](#) | Registered DOI metadata

Q1141 Perturbative readout-error mitigation for near-term quantum computers

Evan Peters, Andy C. Y. Li, Gabriel N. Perdue.

Journal article | 2023 | Physical Review A | vol. 107 | no. 6 | article 062426.

[DOI 10.1103/physreva.107.062426](#) | [Source record](#) | Registered DOI metadata

Q1142 Configurable Readout Error Mitigation in Quantum Workflows

Martin Beisel, Johanna Barzen, Frank Leymann, Felix Truger, Benjamin Weder, Vladimir Yussupov.

Journal article | 2022 | Electronics | vol. 11 | no. 19 | pp. 2983.

[DOI 10.3390/electronics11192983](#) | [Source record](#) | Registered DOI metadata

Q1143 Quantum readout error mitigation via deep learning

Jihye Kim, Byungdu Oh, Yonuk Chong, Euyheon Hwang, Daniel K Park.

Journal article | 2022 | New Journal of Physics | vol. 24 | no. 7 | pp. 073009.

[DOI 10.1088/1367-2630/ac7b3d](#) | [Source record](#) | Registered DOI metadata

D21T04 Symmetry verification and postselection

Symmetry checks reject or reweight outcomes that violate a known constraint. Acceptance probability and postselection bias must be reported.

Fine subcategories: Conserved quantities; acceptance rates; bias; verification checks.

Primary catalog resources: 4. Related topic assignments are searchable in the HTML.

Q1144 Symmetry verification for noisy quantum simulations of non-Abelian lattice gauge theories

Edoardo Ballini, Julius Mildenerberger, Matteo M. Wauters, Philipp Hauke.

Journal article | 2025 | Quantum | vol. 9 | pp. 1802 | article 1802.

[DOI 10.22331/q-2025-07-22-1802](#) | [Source record](#) | Registered DOI metadata

Q1145 Circuit Symmetry Verification Mitigates Quantum-Domain Impairments

Yifeng Xiong, Daryus Chandra, Soon Xin Ng, Lajos Hanzo.

Journal article | 2023 | IEEE Transactions on Signal Processing | vol. 71 | pp. 477-493.

[DOI 10.1109/tsp.2023.3244666](#) | [Source record](#) | Registered DOI metadata

Q1146 Experimental error mitigation via symmetry verification in a variational quantum eigensolver

R. Sagastizabal, X. Bonet-Monroig, M. Singh, M. A. Rol, C. C. Bultink, X. Fu, C. H. Price, V. P. Ostroukh et al..

Journal article | 2019 | Physical Review A | vol. 100 | no. 1 | article 010302.

[DOI 10.1103/physreva.100.010302](#) | [Source record](#) | Registered DOI metadata

Q1147 Quantum symmetry

Murray Gerstenhaber, Anthony Giaquinto, Samuel D. Schack.

Book chapter | 1992 | Lecture Notes in Mathematics | pp. 9-46.

[DOI 10.1007/bfb0101176](#) | [Source record](#) | Registered DOI metadata

D21T05 Virtual distillation and purification

Virtual distillation uses multiple copies or related estimators to suppress some mixed state errors. Coherent errors and measurement costs remain relevant.

Fine subcategories: Multiple copies; purity estimation; coherent errors; measurement cost.

Primary catalog resources: 4. Related topic assignments are searchable in the HTML.

Q1148 Low depth virtual distillation of quantum circuits by deterministic circuit decomposition

Akib Karim, Shaobo Zhang, Muhammad Usman.

Journal article | 2024 | Physical Review Research | vol. 6 | no. 3 | article 033223.

[DOI 10.1103/physrevresearch.6.033223](#) | [Source record](#) | Registered DOI metadata

Q1149 Study of noise in virtual distillation circuits for quantum error mitigation

Pontus Vikstål, Giulia Ferrini, Shruti Puri.

Journal article | 2024 | Quantum | vol. 8 | pp. 1441 | article 1441.

[DOI 10.22331/q-2024-08-14-1441](#) | [Source record](#) | Registered DOI metadata

Q1150 Enhancing Virtual Distillation with Circuit Cutting for Quantum Error Mitigation

Peiyi Li, Ji Liu, Hrushikesh Pramod Patil, Paul Hovland, Huiyang Zhou.

Conference paper | 2023 | 2023 IEEE 41st International Conference on Computer Design (ICCD) | pp. 94-101.

[DOI 10.1109/iccd58817.2023.00024](#) | [Source record](#) | Registered DOI metadata

Q1151 Virtual Distillation for Quantum Error Mitigation

William J. Huggins, Sam McArdle, Thomas E. O'Brien, Joonho Lee, Nicholas C. Rubin, Sergio Boixo, K. Birgitta Whaley, Ryan Babbush et al..

Journal article | 2021 | Physical Review X | vol. 11 | no. 4 | article 041036.

[DOI 10.1103/physrevx.11.041036](#) | [Source record](#) | Registered DOI metadata

D22 Characterization verification and benchmarks

Characterization measures device behavior; verification tests specified claims; benchmarking compares workloads. Each answers a different question and needs an appropriate uncertainty model.

Prerequisites: Quantum measurements, estimation, and statistics.

Assessment focus: Identifiability, confidence intervals, trust assumptions, and workload definition.

D22T01 Quantum state and process tomography

Tomography reconstructs a state or channel from informationally complete data. Reconstruction method, confidence, and physical constraints affect the estimate.

Fine subcategories: Informational completeness; reconstruction; compressed sensing; confidence regions.

Primary catalog resources: 9. Related topic assignments are searchable in the HTML.

Q1152 Efficient noisy quantum state and process tomography

Chenyang Li, Shengxin Zhuang, Yukun Zhang, Jingbo B Wang, Xiao Yuan, Yusen Wu, Chuan Wang.

Journal article | 2026 | Quantum Science and Technology.

[DOI 10.1088/2058-9565/aeae6e](https://doi.org/10.1088/2058-9565/aeae6e) | [Source record](#) | Registered DOI metadata

Q1153 Wigner state and process tomography on near-term quantum devices

Amit Devra, Niklas J. Glaser, Dennis Huber, Steffen J. Glaser.

Journal article | 2024 | Quantum Information Processing | vol. 23 | no. 10 | article 359.

[DOI 10.1007/s11128-024-04550-3](https://doi.org/10.1007/s11128-024-04550-3) | [Source record](#) | Registered DOI metadata

Q1154 Scheme for coherent-state quantum process tomography via normally-ordered moments

M. Ghalaii, A. T. Rezakhani.

Journal article | 2017 | Physical Review A | vol. 95 | no. 3 | article 032336.

[DOI 10.1103/physreva.95.032336](https://doi.org/10.1103/physreva.95.032336) | [Source record](#) | Registered DOI metadata

Q1155 Quantum state and process tomography via adaptive measurements

HengYan Wang, WenQiang Zheng, NengKun Yu, KeRen Li, DaWei Lu, Tao Xin, Carson Li, ZhengFeng Ji et al..

Journal article | 2016 | Science China Physics, Mechanics & Astronomy | vol. 59 | no. 10 | article 100313.

[DOI 10.1007/s11433-016-0287-y](https://doi.org/10.1007/s11433-016-0287-y) | [Source record](#) | Registered DOI metadata

Q1156 Characterization of conditional state-engineering quantum processes by coherent state quantum process tomography

Merlin Cooper, Eirion Slade, Michał Karpiński, Brian J Smith.

Journal article | 2015 | New Journal of Physics | vol. 17 | no. 3 | pp. 033041.

[DOI 10.1088/1367-2630/17/3/033041](https://doi.org/10.1088/1367-2630/17/3/033041) | [Source record](#) | Registered DOI metadata

Q1157 Continuous-variable quantum process tomography with squeezed-state probes

Jaromír Fiurášek.

Journal article | 2015 | Physical Review A | vol. 92 | no. 2 | article 022101.

[DOI 10.1103/physreva.92.022101](https://doi.org/10.1103/physreva.92.022101) | [Source record](#) | Registered DOI metadata

Q1158 Holistic Quantum State and Process Tomography

Nicolás Quesada, Agata M. Brańczyk, Daniel F.V. James.

Conference paper | 2013 | Frontiers in Optics 2013 | pp. FW1C.6.

[DOI 10.1364/fio.2013.fw1c.6](https://doi.org/10.1364/fio.2013.fw1c.6) | [Source record](#) | Registered DOI metadata

Q1159 Selective and efficient quantum state tomography and its application to quantum process tomography

Ariel Bendersky, Juan Pablo Paz.

Journal article | 2013 | Physical Review A | vol. 87 | no. 1 | article 012122.

[DOI 10.1103/physreva.87.012122](#) | [Source record](#) | Registered DOI metadata

Q1160 GitHub - qiskit-community/qiskit-experiments: Qiskit Experiments · GitHub

Author metadata not supplied.

Software repository | Undated | Qiskit Experiments.

[Official resource](#) | [Source record](#) | Official page retrieved

D22T02 Randomized benchmarking

Randomized benchmarking estimates average error behavior through randomized sequences. Interpreting the fitted decay requires checking gate dependence and noise assumptions.

Fine subcategories: Clifford sequences; interleaved benchmarking; cycle benchmarking; gate dependence.

Primary catalog resources: 8. Related topic assignments are searchable in the HTML.

Q1161 Characterization of Quantum Gate Noise Using Randomized Benchmarking

Goutam Paul.

Book chapter | 2026 | Advanced Tutorials on Quantum Circuits | pp. 13-32.

[DOI 10.1201/9788743807087-3](#) | [Source record](#) | Registered DOI metadata

Q1162 Quantum non-Markovian noise in randomized benchmarking of spin-boson models

Srilekha Gandhari, Michael J. Gullans.

Journal article | 2026 | Physical Review Research | vol. 8 | no. 2 | article 023075.

[DOI 10.1103/z8zh-n1hl](#) | [Source record](#) | Registered DOI metadata

Q1163 Randomized benchmarking with synthetic quantum circuits

Yale Fan, Riley Murray, Thaddeus D Ladd, Kevin Young, Robin Blume-Kohout.

Journal article | 2026 | Quantum Science and Technology | vol. 11 | no. 3 | pp. 035052.

[DOI 10.1088/2058-9565/ae8b56](#) | [Source record](#) | Registered DOI metadata

Q1164 Fault-Detecting Randomized Benchmarking for Testing Quantum Processors

Cynthia Kuan, Cheng-Yun Hsieh, Shan-Chi Shih, James Chien-Mo Li.

Conference paper | 2025 | 2025 IEEE International Test Conference in Asia (ITC-Asia) | pp. 54-59.

[DOI 10.1109/itc-asia67627.2025.00018](#) | [Source record](#) | Registered DOI metadata

Q1165 Modeling quantum volume using randomized benchmarking of Room-Temperature NV center quantum registers

Tom Jäger, MinSik Kwon, Max Keller, Rouven Maier, Nicholas Bronn, Regina Finsterhoelzl, Guido Burkard, Leon Büttner et al..

Journal article | 2025 | npj Quantum Information | vol. 12 | no. 1 | article 6.

[DOI 10.1038/s41534-025-01164-0](#) | [Source record](#) | Registered DOI metadata

Q1166 Randomized Benchmarking of Local Zeroth-Order Optimizers for Variational Quantum Systems

Lucas Tecot, Cho-Jui Hsieh.

Conference paper | 2024 | 2024 IEEE 6th International Conference on Trust, Privacy and Security in Intelligent Systems, and Applications (TPS-ISA) | pp. 461-470.

[DOI 10.1109/tps-isa62245.2024.00062](#) | [Source record](#) | Registered DOI metadata

Q1167 Randomized benchmarking into the quantum advantage regime

Sandia National Laboratories (SNL-NM), Albuquerque, NM (United States), Jordan Hines, Advanced Scientific Computing Research, Daniel Hothem, Marie Lu, Ravi Naik, Akel Hashim, Jean-Loup Ville et al..

Report | 2023 | Office of Scientific and Technical Information (OSTI).

[DOI 10.2172/3023952](#) | [Source record](#) | Registered DOI metadata

Q1168 Scalable Randomized Benchmarking of Quantum Computers Using Mirror Circuits

Timothy Proctor, Stefan Seritan, Kenneth Rudinger, Erik Nielsen, Robin Blume-Kohout, Kevin Young.

Journal article | 2022 | Physical Review Letters | vol. 129 | no. 15 | article 150502.

[DOI 10.1103/physrevlett.129.150502](#) | [Source record](#) | Registered DOI metadata

D22T03 Gate set tomography

Gate set tomography jointly estimates operations and preparation and measurement errors. Gauge freedom means some fitted quantities are not uniquely identifiable.

Fine subcategories: Self consistency; gauge freedom; SPAM; long sequence experiments.

Primary catalog resources: 8. Related topic assignments are searchable in the HTML.

Q1169 A Diary of a Faulty Logical Qubit: Logical Gate Set Tomography

Sandia National Laboratories (SNL-NM), Albuquerque, NM (United States), Aliza Siddiqui, Intelligence Advanced Research Projects Activity, USDOE National Nuclear Security Administration (NNSA), Stefan Seritan, Kenneth Rudinger.

Report | 2025 | Office of Scientific and Technical Information (OSTI).

[DOI 10.2172/3023921](#) | [Source record](#) | Registered DOI metadata

Q1170 Characterizing a trapped ion logical qubit with gate set tomography

Sandia National Laboratories (SNL-NM), Albuquerque, NM (United States), Kenneth Rudinger, Intelligence Advanced Research Projects Activity, USDOE National Nuclear Security Administration (NNSA), Piper Wysocki, Daniel Hothem, Jalan Ziyad, Aliza Siddiqui et al..

Report | 2025 | Office of Scientific and Technical Information (OSTI).

[DOI 10.2172/3453489](#) | [Source record](#) | Registered DOI metadata

Q1171 Transformer models for quantum gate set tomography

King Yiu Yu, Aritra Sarkar, Maximilian Rimbach-Russ, Ryoichi Ishihara, Sebastian Feld.

Journal article | 2025 | Quantum Machine Intelligence | vol. 7 | no. 1 | article 10.

[DOI 10.1007/s42484-025-00237-9](#) | [Source record](#) | Registered DOI metadata

Q1172 A DIARY OF A FAULTY LOGICAL QUBIT: LOGICAL GATE SET TOMOGRAPHY

Sandia National Laboratories (SNL-NM), Albuquerque, NM (United States), Aliza Siddiqui, Intelligence Advanced Research Projects Activity, USDOE National Nuclear Security Administration (NNSA), Stefan Seritan, Kenneth Rudinger.

Report | 2024 | Office of Scientific and Technical Information (OSTI).

[DOI 10.2172/3454429](#) | [Source record](#) | Registered DOI metadata

Q1173 Non-Markovian quantum gate set tomography

Ze-Tong Li, Cong-Cong Zheng, Fan-Xu Meng, Han Zeng, Tian Luan, Zai-Chen Zhang, Xu-Tao Yu.

Journal article | 2024 | Quantum Science and Technology | vol. 9 | no. 3 | pp. 035027.

[DOI 10.1088/2058-9565/ad3d80](#) | [Source record](#) | Registered DOI metadata

Q1174 Streaming Quantum Gate Set Tomography Using the Extended Kalman Filter

J. P. Marceaux, Kevin Young.

Conference paper | 2023 | 2023 IEEE International Conference on Quantum Computing and Engineering (QCE) | pp. 1401-1411.

[DOI 10.1109/qce57702.2023.00159](#) | [Source record](#) | Registered DOI metadata

Q1175 Characterizing Midcircuit Measurements on a Superconducting Qubit Using Gate Set Tomography

Kenneth Rudinger, Guilhem J. Ribeill, Luke C.G. Govia, Matthew Ware, Erik Nielsen, Kevin Young, Thomas A. Ohki, Robin Blume-Kohout et al..

Journal article | 2022 | Physical Review Applied | vol. 17 | no. 1 | article 014014.

[DOI 10.1103/physrevapplied.17.014014](https://doi.org/10.1103/physrevapplied.17.014014) | [Source record](#) | Registered DOI metadata

Q1176 Gate Set Tomography of a Logical Qubit.

Kenneth Rudinger, Julie Campos, Mario Morford-Oberst, Erik Nielsen, Stefan Seritan, Tzvetan Metodi, Robin Blume-Kohout.

Conference paper | 2022 | Proposed for presentation at the American Physical Society March Meeting held March 14-18, 2022 in Chicago, IL..

[DOI 10.2172/2001890](https://doi.org/10.2172/2001890) | [Source record](#) | Registered DOI metadata

D22T04 Classical shadows

Classical shadows use randomized measurements to predict many observables. Sample requirements depend on the measurement ensemble and target observable family.

Fine subcategories: Random measurements; observable prediction; sample complexity; derandomization.

Primary catalog resources: 16. Related topic assignments are searchable in the HTML.

Q1177 A computationally efficient approach to quantum state reconstruction using robust classical shadows

Sanjay Sharma, Shyam Akashe, Govind Murari Upadhyay, Amanjot Kaur Lamba.

Journal article | 2026 | Scientific Reports | vol. 16 | no. 1 | article 6927.

[DOI 10.1038/s41598-026-35442-4](https://doi.org/10.1038/s41598-026-35442-4) | [Source record](#) | Registered DOI metadata

Q1178 Large-scale implementation of quantum subspace expansion with classical shadows

Anonymous.

Journal article | 2026 | Physical Review Letters.

[DOI 10.1103/2w21-qs38](https://doi.org/10.1103/2w21-qs38) | [Source record](#) | Registered DOI metadata

Q1179 Quantum-classical auxiliary field quantum Monte Carlo with matchgate shadows on trapped ion quantum computers

Luning Zhao, Joshua J. Goings, Willie Aboumradi, Andrew Arrasmith, Lazaro Calderin, Spencer Churchill, Dor Gabay, Thea Harvey-Brown et al..

Journal article | 2026 | Physical Review Research | vol. 8 | no. 3 | article 033061.

[DOI 10.1103/n1tf-8kr7](https://doi.org/10.1103/n1tf-8kr7) | [Source record](#) | Registered DOI metadata

Q1180 The efficiency frontier: Classical shadows versus direct quantum measurement

Shuowei Ma, Junyu Liu.

Journal article | 2026 | Quantum Machine Intelligence | vol. 8 | no. 2 | article 67.

[DOI 10.1007/s42484-026-00413-5](https://doi.org/10.1007/s42484-026-00413-5) | [Source record](#) | Registered DOI metadata

Q1181 Enhancing quantum state reconstruction with structured classical shadows

Zhen Qin, Joseph M. Lukens, Brian T. Kirby, Zhihui Zhu.

Journal article | 2025 | npj Quantum Information | vol. 11 | no. 1 | article 147.

[DOI 10.1038/s41534-025-01101-1](https://doi.org/10.1038/s41534-025-01101-1) | [Source record](#) | Registered DOI metadata

Q1182 Non-semisimple quantum invariants and abelian classical shadows

Renaud Detcherry.

Journal article | 2025 | Annales de la Faculté des sciences de Toulouse : Mathématiques | vol. 34 | no. 2 | pp. 395-412.

[DOI 10.5802/afst.1816](https://doi.org/10.5802/afst.1816) | [Source record](#) | Registered DOI metadata

Q1183 Quantum Computing Approach to Fixed-Node Monte Carlo Using Classical Shadows

Nick S. Blunt, Laura Caune, Javiera Quiroz-Fernandez.

Journal article | 2025 | Journal of Chemical Theory and Computation | vol. 21 | no. 4 | pp. 1652-1666.

[DOI 10.1021/acs.jctc.4c01468](https://doi.org/10.1021/acs.jctc.4c01468) | [Source record](#) | Registered DOI metadata

Q1184 Quantum Phases Beyond Classical Shadows: A New Framework for Gravitational Wave Effects

Partha Nandi.

Posted content | 2025 | Cassyni.

[DOI 10.52843/cassyni.q6svlz](https://doi.org/10.52843/cassyni.q6svlz) | [Source record](#) | Registered DOI metadata

Q1185 Regularizing least squares quantum state tomography with classical shadows

Zhihui Zhu, Joseph M. Lukens, Brian T. Kirby.

Conference paper | 2025 | CLEO 2025 | pp. FF120_6.

[DOI 10.1364/cleo_fs.2025.ff120_6](https://doi.org/10.1364/cleo_fs.2025.ff120_6) | [Source record](#) | Registered DOI metadata

Q1186 Classical shadows for quantum process tomography on near-term quantum computers

Ryan Levy, Di Luo, Bryan K. Clark.

Journal article | 2024 | Physical Review Research | vol. 6 | no. 1 | article 013029.

[DOI 10.1103/physrevresearch.6.013029](https://doi.org/10.1103/physrevresearch.6.013029) | [Source record](#) | Registered DOI metadata

Q1187 Classical shadows meet quantum optimal mass transport

Giacomo De Palma, Tristan Klein, Davide Pastorello.

Journal article | 2024 | Journal of Mathematical Physics | vol. 65 | no. 9 | article 092201.

[DOI 10.1063/5.0178897](https://doi.org/10.1063/5.0178897) | [Source record](#) | Registered DOI metadata

Q1188 Error-mitigated fermionic classical shadows on noisy quantum devices

Bujiao Wu, Dax Enshan Koh.

Journal article | 2024 | npj Quantum Information | vol. 10 | no. 1 | article 39.

[DOI 10.1038/s41534-024-00836-7](https://doi.org/10.1038/s41534-024-00836-7) | [Source record](#) | Registered DOI metadata

Q1189 Evaluating a quantum-classical quantum Monte Carlo algorithm with Matchgate shadows

Benchen Huang, Yi-Ting Chen, Brajesh Gupta, Martin Suchara, Anh Tran, Sam McArdle, Giulia Galli.

Journal article | 2024 | Physical Review Research | vol. 6 | no. 4 | article 043063.

[DOI 10.1103/physrevresearch.6.043063](https://doi.org/10.1103/physrevresearch.6.043063) | [Source record](#) | Registered DOI metadata

Q1190 Quantum Circuit Cutting for Classical Shadows

Daniel Tzu Shiuan Chen, Zain Hamid Saleem, Michael Alexandrovich Perlin.

Journal article | 2024 | ACM Transactions on Quantum Computing | vol. 5 | no. 2 | pp. 1-21.

[DOI 10.1145/3665335](https://doi.org/10.1145/3665335) | [Source record](#) | Registered DOI metadata

Q1191 Quantum Error Mitigated Classical Shadows

Hamza Jnane, Jonathan Steinberg, Zhenyu Cai, H. Chau Nguyen, Bálint Koczor.

Journal article | 2024 | PRX Quantum | vol. 5 | no. 1 | article 010324.

[DOI 10.1103/prxquantum.5.010324](https://doi.org/10.1103/prxquantum.5.010324) | [Source record](#) | Registered DOI metadata

Q1192 Restoring symmetries in quantum computing using Classical Shadows

Edgar Andres Ruiz Guzman, Denis Lacroix.

Journal article | 2024 | The European Physical Journal A | vol. 60 | no. 5 | article 112.

[DOI 10.1140/epja/s10050-024-01314-6](https://doi.org/10.1140/epja/s10050-024-01314-6) | [Source record](#) | Registered DOI metadata

D22T05 Quantum volume and application benchmarks

System benchmarks summarize performance on specified circuit or application families. Different metrics are complementary and do not supply a universal ranking.

AQUIRE

Fine subcategories: Quantum volume; mirror circuits; CLOPS; application performance; statistical comparison.

Primary catalog resources: 1. Related topic assignments are searchable in the HTML.

Q1193 Volumetric Benchmarking of IQM-Spark: Parity-Preserving Quantum Volume and Compilation Overheads in Odra 5

Andrzej Gnatowski, Jarosław Rudy, Marcin Rudziński, Rafał Bistróń, Krzysztof Święcicki, Teodor Niżyński, Wojciech Bożejko, Karol Życzkowski.

Book chapter | 2026 | Lecture Notes in Computer Science | pp. 171-185.

[DOI 10.1007/978-3-032-36868-3_12](#) | [Source record](#) | Registered DOI metadata

Fine tags: Quantum volume

D22T06 Verification and quantum certification

Verification and certification test a precise state or computation claim under a trust model. Protocol guarantees need to be read together with device assumptions.

Fine subcategories: Interactive verification; state certification; self testing; trusted measurements.

Primary catalog resources: 8. Related topic assignments are searchable in the HTML.

Q1194 Enhancing quantum audio watermarking security through joint verification and certification

Zheng Xing, Chan-Tong Lam, Xiaochen Yuan.

Journal article | 2026 | Scientific Reports | vol. 16 | no. 1 | article 5616.

[DOI 10.1038/s41598-026-36535-w](#) | [Source record](#) | Registered DOI metadata

Q1195 AutoQ 2.0: From Verification of Quantum Circuits to Verification of Quantum Programs

Yu-Fang Chen, Kai-Min Chung, Min-Hsiu Hsieh, Wei-Jia Huang, Ondřej Lengál, Jyun-Ao Lin, Wei-Lun Tsai.

Book chapter | 2025 | Lecture Notes in Computer Science | pp. 87-108.

[DOI 10.1007/978-3-031-90660-2_5](#) | [Source record](#) | Registered DOI metadata

Q1196 Verification of Quantum Circuits

Robert Wille, Lukas Burgholzer.

Book chapter | 2024 | Handbook of Computer Architecture | pp. 1413-1440.

[DOI 10.1007/978-981-97-9314-3_43](#) | [Source record](#) | Registered DOI metadata

Q1197 Verification of quantum programs

Mingsheng Ying.

Book chapter | 2024 | Foundations of Quantum Programming | pp. 137-167.

[DOI 10.1016/b978-0-44-315942-8.00018-6](#) | [Source record](#) | Registered DOI metadata

Q1198 Quantum Robustness Verification: A Hybrid Quantum-Classical Neural Network Certification Algorithm

Nicola Franco, Tom Wollschläger, Nicholas Gao, Jeanette Miriam Lorenz, Stephan Gunnemann.

Conference paper | 2022 | 2022 IEEE International Conference on Quantum Computing and Engineering (QCE) | pp. 142-153.

[DOI 10.1109/qce53715.2022.00033](#) | [Source record](#) | Registered DOI metadata

Q1199 Sample-Efficient Device-Independent Quantum State Verification and Certification

Aleksandra Gočanin, Ivan Šupić, Borivoje Dakić.

Journal article | 2022 | PRX Quantum | vol. 3 | no. 1 | article 010317.

[DOI 10.1103/prxquantum.3.010317](#) | [Source record](#) | Registered DOI metadata

Fine tags: state certification

AQUIRE

Q1200 Verification of Quantum Circuits

Robert Wille, Lukas Burgholzer.

Book chapter | 2022 | Handbook of Computer Architecture | pp. 1-28.

[DOI 10.1007/978-981-15-6401-7_43-1](https://doi.org/10.1007/978-981-15-6401-7_43-1) | [Source record](#) | Registered DOI metadata

Q1201 Verification of Quantum Computation

Elham Kashefi.

Conference paper | 2014 | Research in Optical Sciences | pp. QTh2A.2.

[DOI 10.1364/qim.2014.qth2a.2](https://doi.org/10.1364/qim.2014.qth2a.2) | [Source record](#) | Registered DOI metadata

D23 Software programming and compilation

Quantum software translates a problem into executable circuits and measured results. Compiler decisions should preserve semantics while adapting operations to hardware constraints.

Prerequisites: Programming, compiler concepts, and circuit semantics.

Assessment focus: Semantic preservation, target constraints, verification, and reproducibility.

D23T01 Quantum programming languages

Programming languages express quantum operations alongside classical computation. Control flow, type safety, and provider execution models determine practical capabilities.

Fine subcategories: Q sharp; Qiskit; Cirq; functional languages; type systems; quantum control flow.

Primary catalog resources: 13. Related topic assignments are searchable in the HTML.

Q1202 Fundamentals of Quantum Programming in IBM's Quantum Computers

Weng-Long Chang, Athanasios V. Vasilakos.

Book | 2021 | Studies in Big Data.

[DOI 10.1007/978-3-030-63583-1](https://doi.org/10.1007/978-3-030-63583-1) | [Source record](#) | Registered DOI metadata

Q1203 Quantum Computer Programming Languages for Aerospace Applications

Fred C. Briggs.

Conference paper | 2026 | AIAA SCITECH 2026 Forum.

[DOI 10.2514/6.2026-0441](https://doi.org/10.2514/6.2026-0441) | [Source record](#) | Registered DOI metadata

Q1204 Quantum Programming: Languages and Frameworks

Ashwin Prakash Nalwade, Khan Shariya Hasan Upoma.

Book chapter | 2026 | Quantum Ops | pp. 43-62.

[DOI 10.1007/978-3-032-10775-6_3](https://doi.org/10.1007/978-3-032-10775-6_3) | [Source record](#) | Registered DOI metadata

Q1205 Quantum Programming Paradigms and Description Languages

Sonia Lopez Alarcón, Elaine Wong, Travis S. Humble, Eugene Dumitrescu.

Journal article | 2023 | Computing in Science & Engineering | vol. 25 | no. 6 | pp. 33-38.

[DOI 10.1109/mcse.2024.3375432](https://doi.org/10.1109/mcse.2024.3375432) | [Source record](#) | Registered DOI metadata

Q1206 Linear Dependent Type Theory for Quantum Programming Languages

Peng Fu, Kohei Kishida, Peter Selinger.

Journal article | 2022 | Logical Methods in Computer Science | vol. Volume 18, Issue 3 | article 6930.

[DOI 10.46298/lmcs-18\(3:28\)2022](https://doi.org/10.46298/lmcs-18(3:28)2022) | [Source record](#) | Registered DOI metadata

Q1207 Semantics of quantum programming languages: Classical control, quantum control

Benoît Valiron.

Journal article | 2022 | Journal of Logical and Algebraic Methods in Programming | vol. 128 | pp. 100790 | article 100790.

[DOI 10.1016/j.jlamp.2022.100790](https://doi.org/10.1016/j.jlamp.2022.100790) | [Source record](#) | Registered DOI metadata

Q1208 Linear Dependent Type Theory for Quantum Programming Languages

Peng Fu, Kohei Kishida, Peter Selinger.

Conference paper | 2020 | Proceedings of the 35th Annual ACM/IEEE Symposium on Logic in Computer Science | pp. 440-453.

[DOI 10.1145/3373718.3394765](https://doi.org/10.1145/3373718.3394765) | [Source record](#) | Registered DOI metadata

Q1209 Quantum programming languages

Bettina Heim, Mathias Soeken, Sarah Marshall, Chris Granade, Martin Roetteler, Alan Geller, Matthias Troyer, Krysta Svore.

Journal article | 2020 | Nature Reviews Physics | vol. 2 | no. 12 | pp. 709-722.

[DOI 10.1038/s42254-020-00245-7](#) | [Source record](#) | Registered DOI metadata

Q1210 Cirq basics | Google Quantum AI

Author metadata not supplied.

Tutorial | Undated | Google Quantum AI.

[Official resource](#) | [Source record](#) | Official page retrieved

Q1211 Cirq | Google Quantum AI

Author metadata not supplied.

Documentation | Undated | Google Quantum AI.

[Official resource](#) | [Source record](#) | Official page retrieved

Q1212 GitHub - quantumlib/Cirq: Python framework for creating, editing, and running Noisy Intermediate-Scale Quantum (NISQ) circuits. · GitHub

Author metadata not supplied.

Software repository | Undated | Google Quantum AI.

[Official resource](#) | [Source record](#) | Official page retrieved

Q1213 Introduction to the Quantum Programming Language Q# - Azure Quantum | Microsoft Learn

Author metadata not supplied.

Documentation | Undated | Microsoft.

[Official resource](#) | [Source record](#) | Official page retrieved

Q1214 What is Azure Quantum? - Azure Quantum | Microsoft Learn

Author metadata not supplied.

Documentation | Undated | Microsoft.

[Official resource](#) | [Source record](#) | Official page retrieved

D23T02 Quantum intermediate representations

Intermediate representations support compilation and interchange across tools. Semantics and classical control must survive translation between representations.

Fine subcategories: OpenQASM; QIR; MLIR; classical control; interoperability.

Primary catalog resources: 8. Related topic assignments are searchable in the HTML.

Q1215 Quantum Circuit Synthesis from C via Multi-Level Intermediate Representation

Giacomo Lancellotti, Francesco Rosnati, Leonardo Ignazio Pagliochini, Giovanni Agosta.

Conference paper | 2026 | Proceedings of the 41st ACM/SIGAPP Symposium on Applied Computing | pp. 1359-1366.

[DOI 10.1145/3748522.3779947](#) | [Source record](#) | Registered DOI metadata

Q1216 Quantum Oracle Synthesis from HDL Designs via Multi Level Intermediate Representation

Giacomo Lancellotti, Filippo Buda, Giacomo Carugati, Daniele Gazzola, Alessandro Barenghi, Giovanni Agosta, Gerardo Pelosi.

Conference paper | 2026 | 2026 31st Asia and South Pacific Design Automation Conference (ASP-DAC) | pp. 191-197.

[DOI 10.1109/asp-dac66049.2026.11420750](#) | [Source record](#) | Registered DOI metadata

Q1217 QIRopt: An Optimization Method for Quantum Intermediate Representation

Junjie Luo, Haoyu Zhang, Jianjun Zhao.

Conference paper | 2025 | 2025 IEEE International Conference on Quantum Computing and Engineering (QCE) | pp. 459-469.

[DOI 10.1109/qce65121.2025.00058](https://doi.org/10.1109/qce65121.2025.00058) | [Source record](#) | Registered DOI metadata

Fine tags: QIR

Q1218 Towards a Pulse-Level Intermediate Representation for Diverse Quantum Control Systems

Jude Alnas, Aniket S. Dalvi, Kenneth R. Brown.

Conference paper | 2025 | 2025 IEEE International Conference on Quantum Computing and Engineering (QCE) | pp. 448-458.

[DOI 10.1109/qce65121.2025.00057](https://doi.org/10.1109/qce65121.2025.00057) | [Source record](#) | Registered DOI metadata

Q1219 Enhancing Code Safety in Quantum Intermediate Representation

Junjie Luo, Jianjun Zhao.

Conference paper | 2023 | 2023 38th IEEE/ACM International Conference on Automated Software Engineering (ASE) | pp. 1771-1775.

[DOI 10.1109/ase56229.2023.00195](https://doi.org/10.1109/ase56229.2023.00195) | [Source record](#) | Registered DOI metadata

Q1220 Retargetable Optimizing Compilers for Quantum Accelerators via a Multilevel Intermediate Representation

Thien Nguyen, Alexander McCaskey.

Journal article | 2022 | IEEE Micro | vol. 42 | no. 5 | pp. 17-33.

[DOI 10.1109/mm.2022.3179654](https://doi.org/10.1109/mm.2022.3179654) | [Source record](#) | Registered DOI metadata

Q1221 GitHub - openqasm/openqasm: Quantum assembly language for extended quantum circuits · GitHub

Author metadata not supplied.

Software repository | Undated | OpenQASM.

[Official resource](#) | [Source record](#) | Official page retrieved

Q1222 GitHub - qir-alliance/qir-spec: QIR specification defining how to represent quantum programs within the LLVM IR · GitHub

Author metadata not supplied.

Software repository | Undated | QIR Alliance.

[Official resource](#) | [Source record](#) | Official page retrieved

D23T03 Quantum circuit synthesis

Circuit synthesis decomposes target transformations into allowed operations. Approximation accuracy and target gate set determine circuit size.

Fine subcategories: Unitary decomposition; Clifford T synthesis; arithmetic circuits; gate minimization.

Primary catalog resources: 17. Related topic assignments are searchable in the HTML.

Q1223 Error-Tolerant Quantum State Discrimination: Optimization and Quantum Circuit Synthesis

Chien-Kai Ma, Bo-Hung Chen, Tian-Fu Chen, Dah-Wei Chiou, Jie-Hong R. Jiang.

Book chapter | 2026 | Lecture Notes in Computer Science | pp. 440-458.

[DOI 10.1007/978-3-032-22749-2_22](https://doi.org/10.1007/978-3-032-22749-2_22) | [Source record](#) | Registered DOI metadata

Q1224 Learning State Separation for Quantum Circuit Synthesis

Christoph Stein, Stefan Klikovits, Manuel Wimmer.

Conference paper | 2026 | Proceedings of the 7th IEEE/ACM International Workshop on Quantum Software Engineering | pp. 17-24.

[DOI 10.1145/3786150.3788615](https://doi.org/10.1145/3786150.3788615) | [Source record](#) | Registered DOI metadata

Q1225 Quantum Circuit Synthesis Based on LimTDD

Xin Hong, Chenjian Li, Aochu Dai, Runhong He, Shenggang Ying.

Conference paper | 2026 | 2026 Design, Automation & Test in Europe Conference (DATE) | pp. 1-7.

[DOI 10.23919/date69613.2026.11539289](#) | [Source record](#) | Registered DOI metadata

Q1226 Revisiting Crossover Effectiveness in Quantum Circuit Synthesis

Christoph Stein, Stefan Klikovits, Manuel Wimmer.

Conference paper | 2026 | Proceedings of the Genetic and Evolutionary Computation Conference Companion | pp. 1577-1584.

[DOI 10.1145/3795101.3814690](#) | [Source record](#) | Registered DOI metadata

Q1227 Synthesis of Compact and Expressive Quantum-Circuit Optimizations

Wei Qiang, Ronghui Gu.

Journal article | 2026 | Proceedings of the ACM on Programming Languages | vol. 10 | no. OOPSLA2 | pp. 91-118.

[DOI 10.1145/3839450](#) | [Source record](#) | Registered DOI metadata

Q1228 Amplitude-Ensemble Quantum-Inspired Tabu Search Algorithm for Quantum Boolean Circuit Synthesis

Kuo-Chun Tseng, Zi-Yi Sun, Pei-Lun Ho, Yu-Chieh Cho, Yu-Syuan Liu, Wei-Chieh Lai, Wei-Chun Huang, Jen-Shin Hong.

Conference paper | 2025 | 2025 IEEE International Conference on Systems, Man, and Cybernetics (SMC) | pp. 4845-4852.

[DOI 10.1109/smc58881.2025.11343273](#) | [Source record](#) | Registered DOI metadata

Q1229 Quantum Circuit Synthesis Using Fuzzy-Logic-Assisted Genetic Algorithms

Ishraq Islam, Vinayak Jha, Sneha Thomas, Kieran F. Egan, Alvir Nobel, Serom Kim, Manu Chaudhary, Sunday Ogundele et al..

Journal article | 2025 | Algorithms | vol. 18 | no. 4 | pp. 178.

[DOI 10.3390/a18040178](#) | [Source record](#) | Registered DOI metadata

Q1230 Quantum circuit synthesis with SQiSW

Jialiang Tang, Jialin Zhang, Xiaoming Sun.

Journal article | 2025 | Quantum | vol. 9 | pp. 1889 | article 1889.

[DOI 10.22331/q-2025-10-20-1889](#) | [Source record](#) | Registered DOI metadata

Q1231 Study on Machine Learning Application in Quantum Circuit Synthesis

Jyoti, Amitabh Wahi, S.B.L Tripathi.

Journal article | 2025 | RESEARCH REVIEW International Journal of Multidisciplinary | vol. 10 | no. 4 | pp. 06-13.

[DOI 10.31305/rrijm.2025.v10.n4.002](#) | [Source record](#) | Registered DOI metadata

Q1232 Discovering quantum circuit components with program synthesis

Leopoldo Sarra, Kevin Ellis, Florian Marquardt.

Journal article | 2024 | Machine Learning: Science and Technology | vol. 5 | no. 2 | pp. 025029.

[DOI 10.1088/2632-2153/ad4252](#) | [Source record](#) | Registered DOI metadata

Q1233 Optimizing Quantum Circuit Synthesis with Dominator Analysis

Giacomo Lancellotti, Giovanni Agosta, Alessandro Barenghi, Gerardo Pelosi.

Conference paper | 2024 | 2024 IEEE 42nd International Conference on Computer Design (ICCD) | pp. 24-27.

[DOI 10.1109/iccd63220.2024.00015](#) | [Source record](#) | Registered DOI metadata

Q1234 Quantum Circuit Partitioning for Scalable Noise-Aware Quantum Circuit Re-Synthesis

Mohammad Walid Charrwi, Georgios Ioannou, Ed Younis, Wibe Albert de Jong, Samah Mohamed Saeed.

Conference paper | 2024 | 2024 IEEE International Conference on Quantum Computing and Engineering (QCE) | pp. 359-364.

[DOI 10.1109/qce60285.2024.10306](#) | [Source record](#) | Registered DOI metadata

Q1235 Quantum circuit synthesis on noisy intermediate-scale quantum devices

Shuai Yang, Guojing Tian, Jialin Zhang, Xiaoming Sun.

Journal article | 2024 | Physical Review A | vol. 109 | no. 1 | article 012602.

[DOI 10.1103/physreva.109.012602](#) | [Source record](#) | Registered DOI metadata

Q1236 Quantum circuit synthesis with diffusion models

Florian Furrutter, Gorka Muñoz-Gil, Hans J. Briegel.

Journal article | 2024 | Nature Machine Intelligence | vol. 6 | no. 5 | pp. 515-524.

[DOI 10.1038/s42256-024-00831-9](#) | [Source record](#) | Registered DOI metadata

Q1237 Review Quantum Circuit Synthesis for Grover's Algorithm Oracle

Miguel A. Naranjo, Luis A. Fletscher.

Journal article | 2024 | Algorithms | vol. 17 | no. 9 | pp. 382.

[DOI 10.3390/a17090382](#) | [Source record](#) | Registered DOI metadata

Q1238 Synthetiq: Fast and Versatile Quantum Circuit Synthesis

Anouk Paradis, Jasper Dekoninck, Benjamin Bichsel, Martin Vechev.

Journal article | 2024 | Proceedings of the ACM on Programming Languages | vol. 8 | no. OOPSLA1 | pp. 55-82.

[DOI 10.1145/3649813](#) | [Source record](#) | Registered DOI metadata

Q1239 GitHub - zxcalc/pyzx: Python library for quantum circuit rewriting and optimisation using the ZX-calculus · GitHub

Author metadata not supplied.

Software repository | Undated | PyZX.

[Official resource](#) | [Source record](#) | Official page retrieved

D23T04 Qubit mapping and routing

Placement and routing adapt logical circuits to hardware connectivity. Inserted operations change depth, noise exposure, and execution cost.

Fine subcategories: Placement; SWAP insertion; topology constraints; scheduling; crosstalk awareness.

Primary catalog resources: 8. Related topic assignments are searchable in the HTML.

Q1240 DMapS: End-to-End Qubit Mapping and Routing for Distributed Quantum Computing Architectures

Tingyu Luo, Yuzhen Zheng, Yuxin Deng, X. Fu.

Journal article | 2026 | IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems | vol. 45 | no. 5 | pp. 2095-2108.

[DOI 10.1109/tcad.2025.3611153](#) | [Source record](#) | Registered DOI metadata

Q1241 Generating Compilers for Qubit Mapping and Routing

Abtin Molavi, Amanda Xu, Ethan Cecchetti, Swamit Tannu, Aws Albarghouthi.

Journal article | 2026 | Proceedings of the ACM on Programming Languages | vol. 10 | no. POPL | pp. 2265-2294.

[DOI 10.1145/3776720](#) | [Source record](#) | Registered DOI metadata

Q1242 Nested Qubit Routing

Harshit Dhankhar, Tristan Cazenave.

Book chapter | 2026 | Communications in Computer and Information Science | pp. 70-81.

[DOI 10.1007/978-3-032-15931-1_6](#) | [Source record](#) | Registered DOI metadata

Q1243 QDP: Worst-case Fidelity-aware Qubit Mapping and Routing using Dynamic Programming

Shui Jiang, Wen Cheng, Yi-Hua Chung, Tsung-Yi Ho, Tsung-Wei Huang.

Journal article | 2026 | IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems | pp. 1-1.

[DOI 10.1109/tcad.2026.3693628](#) | [Source record](#) | Registered DOI metadata

Q1244 Unifying Qubit Routing Across Diverse Quantum ISAs via Canonical Representation

Zhaohui Yang, Kai Zhang, Xinyang Tian, Xiangyu Ren, Yingjian Liu, Yunfeng Li, Dawei Ding, Jianxin Chen et al..

Conference paper | 2026 | 2026 ACM/IEEE 53rd Annual International Symposium on Computer Architecture (ISCA) | pp. 2302-2317.

[DOI 10.1109/isca66397.2026.00162](https://doi.org/10.1109/isca66397.2026.00162) | [Source record](#) | Registered DOI metadata

Q1245 Robust Qubit Mapping Algorithm via Double-Source Optimal Routing on Large Quantum Circuits

Chin-Yi Cheng, Chien-Yi Yang, Yi-Hsiang Kuo, Ren-Chu Wang, Hao-Chung Cheng, Chung-Yang (Ric) Huang.

Journal article | 2024 | ACM Transactions on Quantum Computing | vol. 5 | no. 3 | pp. 1-26.

[DOI 10.1145/3680291](https://doi.org/10.1145/3680291) | [Source record](#) | Registered DOI metadata

Q1246 Dynamic Qubit Routing with CNOT Circuit Synthesis for Quantum Compilation

Arianne Meijer-van de Griend, Sarah Meng Li.

Journal article | 2023 | Electronic Proceedings in Theoretical Computer Science | vol. 394 | pp. 363-399.

[DOI 10.4204/eptcs.394.18](https://doi.org/10.4204/eptcs.394.18) | [Source record](#) | Registered DOI metadata

Q1247 Qubit Mapping and Routing via MaxSAT

Abtin Molavi, Amanda Xu, Martin Diges, Lauren Pick, Swamit Tannu, Aws Albarghouthi.

Conference paper | 2022 | 2022 55th IEEE/ACM International Symposium on Microarchitecture (MICRO) | pp. 1078-1091.

[DOI 10.1109/micro56248.2022.00077](https://doi.org/10.1109/micro56248.2022.00077) | [Source record](#) | Registered DOI metadata

D23T05 Quantum compiler optimization

Compiler optimizations simplify circuits or exploit device characteristics. Semantic correctness and the validity of a noise objective should be checked independently.

Fine subcategories: Rewrite rules; commutation; cancellation; approximate synthesis; noise aware passes.

Primary catalog resources: 12. Related topic assignments are searchable in the HTML.

Q1248 Kernpiler: Compiler Optimization for Quantum Hamiltonian Simulation with Partial Trotterization

Ethan Decker, Lucas Goetz, Evan McKinney, Erik Gustafson, Junyu Zhou, Yuhao Liu, Alex K. Jones, Ang Li et al..

Conference paper | 2026 | 2026 ACM/IEEE 53rd Annual International Symposium on Computer Architecture (ISCA) | pp. 905-920.

[DOI 10.1109/isca66397.2026.00073](https://doi.org/10.1109/isca66397.2026.00073) | [Source record](#) | Registered DOI metadata

Q1249 A Detailed Survey and Analysis of Recent Advancements in Quantum Compiler Optimization Techniques

Pooja Vaikar, Amol Potgantwar.

Book chapter | 2025 | Lecture Notes in Networks and Systems | pp. 681-692.

[DOI 10.1007/978-981-96-8043-6_52](https://doi.org/10.1007/978-981-96-8043-6_52) | [Source record](#) | Registered DOI metadata

Q1250 A Universal Quantum Compiler GPT: Multi-Framework Optimization and Translation Using Large Language Models

Somrak Petchartee.

Conference paper | 2025 | 2025 29th International Computer Science and Engineering Conference (ICSEC) | pp. 58-63.

[DOI 10.1109/icsec67360.2025.11298108](https://doi.org/10.1109/icsec67360.2025.11298108) | [Source record](#) | Registered DOI metadata

Q1251 Inverse-Transpilation: Reverse-Engineering Quantum Compiler Optimization Passes from Circuit Snapshots

Satwik Kundu, Swaroop Ghosh.

Conference paper | 2025 | Proceedings of the Great Lakes Symposium on VLSI 2025 | pp. 273-277.

[DOI 10.1145/3716368.3735298](https://doi.org/10.1145/3716368.3735298) | [Source record](#) | Registered DOI metadata

Q1252 MarQSim: Reconciling Determinism and Randomness in Compiler Optimization for Quantum Simulation

Xiuqi Cao, Junyu Zhou, Yuhao Liu, Yunong Shi, Gushu Li.

Journal article | 2025 | Proceedings of the ACM on Programming Languages | vol. 9 | no. PLDI | pp. 576-600.

[DOI 10.1145/3729269](#) | [Source record](#) | Registered DOI metadata

Q1253 Quantum Software Engineering: Algorithm Design, Error Mitigation, and Compiler Optimization for Fault-Tolerant Quantum Computing

Author metadata not supplied.

Journal article | 2025 | International Journal of Computer Applications Technology and Research.

[DOI 10.7753/ijcatr1404.1003](#) | [Source record](#) | Registered DOI metadata

Q1254 Bosehdral: Compiler Optimization for Bosonic Quantum Computing

Junyu Zhou, Yuhao Liu, Yunong Shi, Ali Javadi-Abhari, Gushu Li.

Conference paper | 2024 | 2024 ACM/IEEE 51st Annual International Symposium on Computer Architecture (ISCA) | pp. 261-276.

[DOI 10.1109/isca59077.2024.00028](#) | [Source record](#) | Registered DOI metadata

Q1255 Compiler Optimization for Quantum Computing Using Reinforcement Learning

Nils Quetschlich, Lukas Burgholzer, Robert Wille.

Conference paper | 2023 | 2023 60th ACM/IEEE Design Automation Conference (DAC) | pp. 1-6.

[DOI 10.1109/dac56929.2023.10248002](#) | [Source record](#) | Registered DOI metadata

Q1256 Parity Quantum Optimization: Compiler

Kilian Ender, Roeland ter Hoeven, Benjamin E. Niehoff, Maike Drieb-Schön, Wolfgang Lechner.

Journal article | 2023 | Quantum | vol. 7 | pp. 950 | article 950.

[DOI 10.22331/q-2023-03-17-950](#) | [Source record](#) | Registered DOI metadata

Q1257 GitHub - Qiskit/qiskit: Qiskit is an open-source SDK for working with quantum computers at the level of extended quantum circuits, operators, and primitives. · GitHub

Author metadata not supplied.

Software repository | Undated | Qiskit.

[Official resource](#) | [Source record](#) | Official page retrieved

Q1258 GitHub - Quantinuum/tket: Source code for the TKET quantum compiler, Python bindings and utilities · GitHub

Author metadata not supplied.

Software repository | Undated | Quantinuum.

[Official resource](#) | [Source record](#) | Official page retrieved

Q1259 TKET — Quantinuum Documentation

Author metadata not supplied.

Documentation | Undated | Quantinuum.

[Official resource](#) | [Source record](#) | Official page retrieved

D23T06 Quantum software testing and formal methods

Testing and formal methods address errors in quantum programs and compilation. Assertions, equivalence, and statistical testing have different guarantees.

Fine subcategories: Assertions; equivalence checking; program logics; property tests; debugging.

Primary catalog resources: 17. Related topic assignments are searchable in the HTML.

Q1260 A Methodological Analysis of Empirical Studies in Quantum Software Testing

Yuechen Li, Minqi Shao, Jianjun Zhao, Qichen Wang.

Journal article | 2026 | ACM Transactions on Software Engineering and Methodology | article 3819590.

[DOI 10.1145/3819590](#) | [Source record](#) | Registered DOI metadata

Q1261 A Practical Specification Language for Automatic Quantum Program Verification

Wei-Lun Tsai, Yu-Fang Chen, Ondřej Lengál.

Book chapter | 2026 | Lecture Notes in Computer Science | pp. 302-325.

[DOI 10.1007/978-3-032-32537-2_15](#) | [Source record](#) | Registered DOI metadata

Q1262 Industry Expectations and Skill Demands in Quantum Software Testing

Ronnie de Souza Santos, Maria Teresa Baldassarre, Cesar França.

Conference paper | 2026 | Proceedings of the 7th IEEE/ACM International Workshop on Quantum Software Engineering | pp. 9-16.

[DOI 10.1145/3786150.3788614](#) | [Source record](#) | Registered DOI metadata

Q1263 Relational Verification for Cost-Aware Quantum Program Optimization

Ziming Zhao, Tingting Li, Zhaoxuan Li, Jianwei Yin.

Journal article | 2026 | Proceedings of the AAAI Conference on Artificial Intelligence | vol. 40 | no. 17 | pp. 14414-14422.

[DOI 10.1609/aaai.v40i17.38457](#) | [Source record](#) | Registered DOI metadata

Q1264 Software Testing in the Quantum World

Rui Abreu, Shaukat Ali, Paolo Arcaini, José Campos, Michael Felderer, Claude Gravel, Fuyuki Ishikawa, Stefan Klikovits et al..

Journal article | 2026 | Computer | vol. 59 | no. 4 | pp. 135-138.

[DOI 10.1109/mc.2026.3655854](#) | [Source record](#) | Registered DOI metadata

Q1265 Embedding Quantum Program Verification into Dafny

Feifei Cheng, Sushen Vangeepuram, Henry Allard, Seyed Mohammad Reza Jafari, Alex Potanin, Liyi Li.

Journal article | 2025 | Proceedings of the ACM on Programming Languages | vol. 9 | no. OOPSLA2 | pp. 2981-3007.

[DOI 10.1145/3763157](#) | [Source record](#) | Registered DOI metadata

Q1266 Exact Inference for Quantum Circuits: A Testing Oracle for Quantum Software Stacks

Kanguk Lee, Jaemin Hong, Sukyoung Ryu.

Conference paper | 2025 | 2025 40th IEEE/ACM International Conference on Automated Software Engineering (ASE) | pp. 2465-2477.

[DOI 10.1109/ase63991.2025.00203](#) | [Source record](#) | Registered DOI metadata

Q1267 Mutation-Based Quantum Software Testing

Macario Polo-Usaola, Manuel Serrano, Ignacio García-Rodríguez de Guzmán.

Conference paper | 2025 | Proceedings of the 1st International Conference on Quantum Software | pp. 138-145.

[DOI 10.5220/0013561000004525](#) | [Source record](#) | Registered DOI metadata

Q1268 SQUAD: software testing for quantum distributed learning software

Soohyun Park, Jae Hyun Cho, Hyun Jun Yook, Ga San Jhun, Youn Kyu Lee, Joongheon Kim.

Journal article | 2025 | The Journal of Supercomputing | vol. 81 | no. 9 | article 1071.

[DOI 10.1007/s11227-025-07556-5](#) | [Source record](#) | Registered DOI metadata

Q1269 The Landscape of Quantum Software Testing Tools

Xinyi Wang, Shaukat Ali, Davide Taibi.

Journal article | 2025 | IEEE Software | vol. 42 | no. 5 | pp. 136-140.

[DOI 10.1109/ms.2025.3578154](#) | [Source record](#) | Registered DOI metadata

Q1270 Automated Quantum Program Verification in Dynamic Quantum Logic

Tsubasa Takagi, Canh Minh Do, Kazuhiro Ogata.

Book chapter | 2024 | Lecture Notes in Computer Science | pp. 68-84.

[DOI 10.1007/978-3-031-51777-8_5](#) | [Source record](#) | Registered DOI metadata

Q1271 Gate Branch Coverage: A Metric for Quantum Software Testing

Daniel Fortunato, José Campos, Rui Abreu.

Conference paper | 2024 | Proceedings of the 1st ACM International Workshop on Quantum Software Engineering: The Next Evolution | pp. 15-18.

[DOI 10.1145/3663531.3664753](#) | [Source record](#) | Registered DOI metadata

Q1272 Mitigating Noise in Quantum Software Testing Using Machine Learning

Asmar Muqeet, Tao Yue, Shaukat Ali, Paolo Arcaini.

Journal article | 2024 | IEEE Transactions on Software Engineering | vol. 50 | no. 11 | pp. 2947-2961.

[DOI 10.1109/tse.2024.3462974](#) | [Source record](#) | Registered DOI metadata

Q1273 Quantum Machine Learning Techniques Integrated in Quantum Software Testing

Bala Gangadhara Gutam, Sunil Kumar Malchi.

Posted content | 2024 | Springer Science and Business Media LLC.

[DOI 10.21203/rs.3.rs-4653367/v1](#) | [Source record](#) | Registered DOI metadata

Q1274 Quantum Software Testing 101

Shaukat Ali.

Conference paper | 2024 | Proceedings of the 2024 IEEE/ACM 46th International Conference on Software Engineering: Companion Proceedings | pp. 426-427.

[DOI 10.1145/3639478.3643059](#) | [Source record](#) | Registered DOI metadata

Q1275 QuCAT: A Combinatorial Testing Tool for Quantum Software

Xinyi Wang, Paolo Arcaini, Tao Yue, Shaukat Ali.

Conference paper | 2023 | 2023 38th IEEE/ACM International Conference on Automated Software Engineering (ASE) | pp. 2066-2069.

[DOI 10.1109/ase56229.2023.00062](#) | [Source record](#) | Registered DOI metadata

Q1276 Quantum Software Testing: A Brief Introduction

Shaukat Ali, Tao Yue.

Conference paper | 2023 | 2023 IEEE/ACM 45th International Conference on Software Engineering: Companion Proceedings (ICSE-Companion) | pp. 332-333.

[DOI 10.1109/icse-companion58688.2023.00093](#) | [Source record](#) | Registered DOI metadata

D24 Classical simulation and hybrid systems

Classical simulators support development, verification, and baselines. Their scalability depends on entanglement, circuit structure, precision, and memory bandwidth.

Prerequisites: Numerical linear algebra, parallel computing, and circuit structure.

Assessment focus: Memory, entanglement, precision, communication, and comparison to hardware.

D24T01 State vector and density matrix simulation

State vector simulation tracks amplitudes, while density matrix simulation tracks mixed states. Memory grows exponentially and differs substantially between these representations.

Fine subcategories: Memory scaling; distributed state vectors; mixed states; precision; GPU execution.

Primary catalog resources: 8. Related topic assignments are searchable in the HTML.

Q1277 DAG-Aware Gate Fusion for Efficient State-Vector Quantum-Circuit Simulation

Shangshu Li, Zheng-An Wang, Heng Fan.

Journal article | 2026 | IEEE Transactions on Quantum Engineering | vol. 7 | pp. 1-10.

[DOI 10.1109/tqe.2026.3705783](#) | [Source record](#) | Registered DOI metadata

Q1278 Memory-Efficient State-Vector Simulation of Quantum Computer in Massively Parallel Supercomputers

Naoki Yoshioka, Nobuyasu Ito.

Book chapter | 2026 | Springer Proceedings in Physics | pp. 177-187.

[DOI 10.1007/978-981-92-0844-9_21](#) | [Source record](#) | Registered DOI metadata

Q1279 phase2: full-state vector simulation of quantum time evolution at scale

Marek Miller, Jakob Günther, Freek Witteveen, Matthew S. Teynor, Mihael Erakovic, Markus Reiher, Gemma C. Solomon, Matthias Christandl.

Journal article | 2026 | Communications AI & Computing | vol. 1 | no. 1 | article 5.

[DOI 10.1038/s44488-026-00002-2](#) | [Source record](#) | Registered DOI metadata

Q1280 Lazy Qubit Reordering for Accelerating Parallel State-Vector-based Quantum Circuit Simulation

Yusuke Teranishi, Shoma Hiraoka, Wataru Mizukami, Masao Okita, Fumihiko Ino.

Journal article | 2025 | ACM Transactions on Quantum Computing | vol. 6 | no. 4 | pp. 1-33.

[DOI 10.1145/3748261](#) | [Source record](#) | Registered DOI metadata

Q1281 Q2SV: A High-Level Synthesis Approach for State Vector Quantum Simulation

Ahmad Bennakhi, Gregory T. Byrd, Paul Franzon.

Journal article | 2025 | Quantum Engineering | vol. 2025 | no. 1 | article 9017796.

[DOI 10.1155/que2/9017796](#) | [Source record](#) | Registered DOI metadata

Q1282 GitHub - Qiskit/qiskit-aer: Aer is a high performance simulator for quantum circuits that includes noise models · GitHub

Author metadata not supplied.

Software repository | Undated | Qiskit Aer.

[Official resource](#) | [Source record](#) | Official page retrieved

Q1283 GitHub - quantumlib/qsim: Fast C++ and Python library for state-vector simulation of quantum circuits. · GitHub

Author metadata not supplied.

Software repository | Undated | Google Quantum AI.

[Official resource](#) | [Source record](#) | Official page retrieved

Q1284 Simulate a circuit | Cirq | Google Quantum AI

Author metadata not supplied.

Documentation | Undated | Google Quantum AI.

[Official resource](#) | [Source record](#) | Official page retrieved

D24T02 Tensor network simulation

Tensor network methods exploit circuit structure and limited entanglement. Bond dimensions and contraction order control practical cost.

Fine subcategories: MPS; PEPS; contraction order; bond dimension; circuit contraction.

Primary catalog resources: 15. Related topic assignments are searchable in the HTML.

Q1285 Scalable Quantum Molecular Generation via GPU-Accelerated Tensor-Network Simulation

Yu-Cheng Xiao, Jen-Yu Chang, Tzu-Ling Kuo, Aninda Astuti, Shu-Chi Wu, Ka-Lok Ng, Yun-Yuan Wang, Yu-Ze Chen et al..

Conference paper | 2026 | 2026 International Conference on Quantum Communications, Networking, and Computing (QCNC) | pp. 852-856.

[DOI 10.1109/qcnc69040.2026.00140](#) | [Source record](#) | Registered DOI metadata

Q1286 Scalable Tensor Network Simulation for Quantum-Classical Dual Kernel

Mei Ian Sam, Tai-Yu Li.

Conference paper | 2026 | 2026 International Conference on Quantum Communications, Networking, and Computing (QCNC) | pp. 900-905.

[DOI 10.1109/qcnc69040.2026.00148](#) | [Source record](#) | Registered DOI metadata

Q1287 Sparse Tensor-Network Quantum Machine Learning for Simulation, Cybersecurity and Scientific Discovery

Murali Krishna Pasupuleti.

Reference book | 2026 | National Education Services.

[DOI 10.62311/nesx/rb8jy-978-81-688729-2-9](#) | [Source record](#) | Registered DOI metadata

Q1288 Tensor-Network Neural Operators for Quantum-AI Scientific Simulation

Murali Krishna Pasupuleti.

Reference book | 2026 | National Education Services.

[DOI 10.62311/nesx/rb8m-978-81-686966-2-4](#) | [Source record](#) | Registered DOI metadata

Q1289 Approximate Quantum Compiling for Quantum Simulation: A Tensor Network Based Approach

Niall Robertson, Albert Akhriev, Jiri Vala, Sergiy Zhuk.

Journal article | 2025 | ACM Transactions on Quantum Computing | vol. 6 | no. 3 | pp. 1-15.

[DOI 10.1145/3731251](#) | [Source record](#) | Registered DOI metadata

Q1290 Efficient simulation of leakage errors in quantum error correcting codes using tensor network methods

Hidetaka Manabe, Yasunari Suzuki, Andrew S Darmawan.

Journal article | 2025 | New Journal of Physics | vol. 27 | no. 11 | pp. 114512.

[DOI 10.1088/1367-2630/ae1529](#) | [Source record](#) | Registered DOI metadata

Q1291 Hybrid Tree Tensor Networks for Quantum Simulation

Julian Schuhmacher, Marco Ballarin, Alberto Baiardi, Giuseppe Magnifico, Francesco Tacchino, Simone Montangero, Ivano Tavernelli.

Journal article | 2025 | PRX Quantum | vol. 6 | no. 1 | article 010320.

[DOI 10.1103/prxquantum.6.010320](#) | [Source record](#) | Registered DOI metadata

Q1292 Numerical Simulation of Two-Dimensional Quantum XYX Model Based on Tensor Network Algorithm

Sheng-Hao Li, Yuan-Cheng Luo, Yuan-Yuan Wu.

Conference paper | 2025 | 2025 2nd International Conference on Intelligent Computing and Robotics (ICICR) | pp. 1208-1212.

[DOI 10.1109/icicr65456.2025.00214](https://doi.org/10.1109/icicr65456.2025.00214) | [Source record](#) | Registered DOI metadata

Q1293 Parallel Tensor Network Contraction for Efficient Quantum Circuit Simulation on Multicore CPUs and GPUs

Alfred Pastor, Maribel Castillo, Jose Badia.

Conference paper | 2025 | Proceedings of the 1st International Conference on Quantum Software | pp. 120-127.

[DOI 10.5220/0013551400004525](https://doi.org/10.5220/0013551400004525) | [Source record](#) | Registered DOI metadata

Fine tags: circuit contraction

Q1294 Coded Computing Meets Quantum Circuit Simulation: Coded Parallel Tensor Network Contraction Algorithm

Jin Lee, Zheng Zhang, Sofía González-García, Haewon Jeong.

Conference paper | 2024 | 2024 IEEE International Symposium on Information Theory (ISIT) | pp. 3095-3100.

[DOI 10.1109/isit57864.2024.10619404](https://doi.org/10.1109/isit57864.2024.10619404) | [Source record](#) | Registered DOI metadata

Fine tags: circuit contraction

Q1295 Efficient Quantum Circuit Simulation by Tensor Network Methods on Modern GPUs

Feng Pan, Hanfeng Gu, Lvlin Kuang, Bing Liu, Pan Zhang.

Journal article | 2024 | ACM Transactions on Quantum Computing | vol. 5 | no. 4 | pp. 1-26.

[DOI 10.1145/3696465](https://doi.org/10.1145/3696465) | [Source record](#) | Registered DOI metadata

Q1296 Efficient tensor network simulation of IBM's largest quantum processors

Siddhartha Patra, Saeed S. Jahromi, Sukhbinder Singh, Román Orús.

Journal article | 2024 | Physical Review Research | vol. 6 | no. 1 | article 013326.

[DOI 10.1103/physrevresearch.6.013326](https://doi.org/10.1103/physrevresearch.6.013326) | [Source record](#) | Registered DOI metadata

Q1297 Optimal tree tensor network operators for tensor network simulations: Applications to open quantum systems

Weitang Li, Jiajun Ren, Hengrui Yang, Haobin Wang, Zhigang Shuai.

Journal article | 2024 | The Journal of Chemical Physics | vol. 161 | no. 5 | article 054116.

[DOI 10.1063/5.0218773](https://doi.org/10.1063/5.0218773) | [Source record](#) | Registered DOI metadata

Q1298 Tensor network simulation of chains of non-Markovian open quantum systems

Gerald E. Fux, Dainius Kilda, Brendon W. Lovett, Jonathan Keeling.

Journal article | 2023 | Physical Review Research | vol. 5 | no. 3 | article 033078.

[DOI 10.1103/physrevresearch.5.033078](https://doi.org/10.1103/physrevresearch.5.033078) | [Source record](#) | Registered DOI metadata

Q1299 GitHub - jcmgray/quimb: A python library for quantum information and many-body calculations including tensor networks. · GitHub

Author metadata not supplied.

Software repository | Undated | Quimb.

[Official resource](#) | [Source record](#) | Official page retrieved

D24T03 Stabilizer circuit simulation

Stabilizer simulation efficiently handles Clifford operations and suitable states. Non Clifford extensions require additional resources or approximation.

Fine subcategories: Gottesman Knill; tableau methods; Clifford circuits; non Clifford extensions.

Primary catalog resources: 8. Related topic assignments are searchable in the HTML.

Q1300 GPU-Accelerated Quantum Simulation of Stabilizer Circuits

Muhammad Osama, Dimitrios Thanos, Alfons Laarman.

Journal article | 2026 | Quantum | vol. 10 | pp. 2225 | article 2225.

[DOI 10.22331/q-2026-10-01-2225](#) | [Source record](#) | Registered DOI metadata

Q1301 Hybrid variational quantum circuit approach for stabilizer-state classifiers

Hamna Aslam, Frédéric Holweck.

Journal article | 2026 | Physical Review A | vol. 113 | no. 5 | article 052429.

[DOI 10.1103/tl4h-vs8y](#) | [Source record](#) | Registered DOI metadata

Q1302 quESTab: Towards Scalable Quantum Circuit Simulation on Multi-GPU using an Extended Stabilizer Formalism

Hyunjoon Shin, Seokhyeon Lee, Myeongjin Kwak, Yongtae Kim.

Conference paper | 2026 | Proceedings of the 40th ACM International Conference on Supercomputing | pp. 1296-1309.

[DOI 10.1145/3797905.3816723](#) | [Source record](#) | Registered DOI metadata

Q1303 Abstraqt: Analysis of Quantum Circuits via Abstract Stabilizer Simulation

Benjamin Bichsel, Anouk Paradis, Maximilian Baader, Martin Vechev.

Journal article | 2023 | Quantum | vol. 7 | pp. 1185 | article 1185.

[DOI 10.22331/q-2023-11-20-1185](#) | [Source record](#) | Registered DOI metadata

Q1304 SIMULATION OF THE STABILIZER BOOST CIRCUIT FROM A CAPACITIVE STORAGE

Kamchatka State Technical University, Sergei Yu. Trudnev, Kamchatka State Technical University, Aleksei A. Marchenko.

Journal article | 2020 | Vestnik Gosudarstvennogo universiteta morskogo i rechnogo flota imeni admirala S. O. Makarova | vol. 12 | no. 6 | pp. 1118-1127.

[DOI 10.21821/2309-5180-2020-12-6-1118-1127](#) | [Source record](#) | Registered DOI metadata

Q1305 Simulation of quantum circuits by low-rank stabilizer decompositions

Sergey Bravyi, Dan Browne, Padraic Calpin, Earl Campbell, David Gosset, Mark Howard.

Journal article | 2019 | Quantum | vol. 3 | pp. 181 | article 181.

[DOI 10.22331/q-2019-09-02-181](#) | [Source record](#) | Registered DOI metadata

Q1306 Simulation of Quantum Circuits via Stabilizer Frames

Hector J. Garcia, Igor L. Markov.

Journal article | 2015 | IEEE Transactions on Computers | vol. 64 | no. 8 | pp. 2323-2336.

[DOI 10.1109/tc.2014.2360532](#) | [Source record](#) | Registered DOI metadata

Q1307 Quipu: High-performance simulation of quantum circuits using stabilizer frames

Héctor J. García, Igor L. Markov.

Conference paper | 2013 | 2013 IEEE 31st International Conference on Computer Design (ICCD) | pp. 404-410.

[DOI 10.1109/iccd.2013.6657072](#) | [Source record](#) | Registered DOI metadata

D24T04 Quantum GPU and HPC integration

GPU and distributed HPC methods accelerate selected classical simulation operations. Memory bandwidth, communication, and precision determine real performance.

Fine subcategories: Accelerators; distributed memory; kernels; HPC scheduling; numerical validation.

Primary catalog resources: 6. Related topic assignments are searchable in the HTML.

Q1308 Performance analysis and modeling for quantum computing simulation on distributed GPU platforms

Armin Ahmadzadeh, Hamid Sarbazi-Azad.

Journal article | 2024 | Quantum Information Processing | vol. 23 | no. 11 | article 373.

[DOI 10.1007/s11128-024-04580-x](#) | [Source record](#) | Registered DOI metadata

Q1309 High performance computing and quantum trajectory method in CPU and GPU systems

Joanna Wiśniewska, Marek Sawerwain, Wiesław Leoński.

Journal article | 2015 | Journal of Physics: Conference Series | vol. 574 | pp. 012127.

[DOI 10.1088/1742-6596/574/1/012127](#) | [Source record](#) | Registered DOI metadata

Q1310 GPU-aware distributed quantum simulation

Anderson Avila, Adriano Maron, Renata Reiser, Mauricio Pilla, Adenauer Yamin.

Conference paper | 2014 | Proceedings of the 29th Annual ACM Symposium on Applied Computing | pp. 860-865.

[DOI 10.1145/2554850.2554892](#) | [Source record](#) | Registered DOI metadata

Q1311 CUDA-Q — NVIDIA CUDA-Q documentation

Author metadata not supplied.

Documentation | Undated | NVIDIA.

[Official resource](#) | [Source record](#) | Official page retrieved

Q1312 Getting Started — NVIDIA cuQuantum

Author metadata not supplied.

Documentation | Undated | NVIDIA.

[Official resource](#) | [Source record](#) | Official page retrieved

Q1313 Quick Start — NVIDIA CUDA-Q documentation

Author metadata not supplied.

Tutorial | Undated | NVIDIA.

[Official resource](#) | [Source record](#) | Official page retrieved

D24T05 Hybrid quantum classical workflows

Hybrid workflows coordinate classical processing with quantum execution. Batching, data movement, scheduling, and asynchronous control affect end to end cost.

Fine subcategories: Batching; asynchronous jobs; optimization loops; data movement; heterogeneous computing.

Primary catalog resources: 8. Related topic assignments are searchable in the HTML.

Q1314 Hybrid Quantum Classical Integration

Aayushee G.

Journal article | 2026 | International Journal of Creative and Open Research in Engineering and Management | vol. 02 | no. 08 | pp. 1-9.

[DOI 10.55041/ijcope.v2i8.063](#) | [Source record](#) | Registered DOI metadata

Q1315 Hybrid Quantum Computing Workflow Optimization

Khalid Abdul Jaleel, Gold Sharon R, Nivi V.

Conference paper | 2026 | 2026 International Conference on Innovative Trends in Information Technology (ICITIIT) | pp. 1-6.

[DOI 10.1109/icitit68860.2026.11499461](#) | [Source record](#) | Registered DOI metadata

Q1316 Hybrid Quantum–Classical Pipelines for Big Data Analytics: Workflow Design with Resource and Throughput Analyses

Laith S. Ismail, Bushra Jabbar Abdul-Kareem, Doaa Thamer Mohammed, Hussain Kassim Ahmad, Alona Desiatko.

Book chapter | 2026 | Communications in Computer and Information Science | pp. 431-457.

[DOI 10.1007/978-3-032-39708-9_22](#) | [Source record](#) | Registered DOI metadata

Q1317 Hybrid quantum–classical machine translation

Mina Abbaszadeh, Mariam Zomorodi, Mehrnoosh Sadrzadeh, Vahid Salari, Philip Kurian.

Journal article | 2026 | Quantum Information Processing | vol. 25 | no. 2 | article 49.

[DOI 10.1007/s11128-025-05032-w](#) | [Source record](#) | Registered DOI metadata

Q1318 QSplit: A Workflow-Oriented Hybrid Quantum–Classical Optimization Framework

Mario Bifulco, Francesco Medina, Dorian Medić, Luca Roversi, Marco Aldinucci.

Book chapter | 2026 | Lecture Notes in Computer Science | pp. 195-209.

[DOI 10.1007/978-3-032-35251-4_14](#) | [Source record](#) | Registered DOI metadata

Q1319 RIGOLETTO: A Workflow Definition Language for Hybrid Quantum-Classical Scientific Applications

Vincenzo De Maio, Dominik Bork, Ivona Brandic.

Conference paper | 2024 | 2024 26th International Conference on Business Informatics (CBI) | pp. 40-49.

[DOI 10.1109/cbi62504.2024.00015](#) | [Source record](#) | Registered DOI metadata

Q1320 A hybrid classical-quantum workflow for natural language processing

Lee J O'Riordan, Myles Doyle, Fabio Baruffa, Venkatesh Kannan.

Journal article | 2020 | Machine Learning: Science and Technology | vol. 2 | no. 1 | pp. 015011.

[DOI 10.1088/2632-2153/abbd2e](#) | [Source record](#) | Registered DOI metadata

Q1321 GitHub - NVIDIA/cuda-quantum: C++ and Python support for the CUDA Quantum programming model for heterogeneous quantum-classical workflows · GitHub

Author metadata not supplied.

Software repository | Undated | NVIDIA.

[Official resource](#) | [Source record](#) | Official page retrieved

D25 Architecture cloud and distributed computing

Architecture combines qubits, classical control, compilation, and communication into a system. Distributed designs must include entanglement costs, synchronization, and failures.

Prerequisites: Quantum hardware, compilation, and distributed systems.

Assessment focus: Connectivity, scheduling, communication, queueing, and complete system cost.

D25T01 Quantum processor architecture

Processor architecture combines connectivity, instruction execution, control, and packaging. The best design depends on the workload and error correction strategy.

Fine subcategories: Connectivity; modularity; instruction sets; control planes; co design.

Primary catalog resources: 9. Related topic assignments are searchable in the HTML.

Q1322 Quantum Computing: An Applied Approach

Jack D. Hidary.

Book | 2021 | Springer International Publishing.

[DOI 10.1007/978-3-030-83274-2](#) | [Source record](#) | Registered DOI metadata

Q1323 Differentiable architecture search for adversarially robust quantum computer vision

Mohamed Afane, Quanjiang Long, Haoting Shen, Ying Mao, Junaid Farooq, Ying Wang, Juntao Chen.

Journal article | 2026 | Quantum Machine Intelligence | vol. 8 | no. 1 | article 2.

[DOI 10.1007/s42484-026-00353-0](#) | [Source record](#) | Registered DOI metadata

Q1324 Quantum Computer Hardware and Architecture—An Overview

Hiu Yung Wong.

Book chapter | 2025 | Quantum Computing Architecture and Hardware for Engineers | pp. 3-11.

[DOI 10.1007/978-3-031-78219-0_1](#) | [Source record](#) | Registered DOI metadata

Q1325 The Tri-Computer Architecture: Unifying Space, Time, and Quantum Dimensions

Ethereum Computer, Jameson Joseph Bednarski, Rafael Henrique do Nascimento Oliveira.

Posted content | 2025 | DeSci Labs AG.

[DOI 10.62891/3b97db99](#) | [Source record](#) | Registered DOI metadata

Q1326 A modular entanglement-based quantum computer architecture

Ferran Riera-Sàbat, Wolfgang Dür.

Journal article | 2024 | New Journal of Physics | vol. 26 | no. 12 | pp. 123015.

[DOI 10.1088/1367-2630/ad9945](#) | [Source record](#) | Registered DOI metadata

Q1327 Quantum Computing Impact on Traditional Computer Architecture Models

Alnoor University, Salih Mahmoud Attia, Suhad Qasim G. Haddad, Al Mansour University College, Hamid Kareem Radam Al-Zaidi, Al Hikma University College, Wafaa Mustafa Hameed, Cihan University Sulaimaniya et al..

Journal article | 2024 | Radioelectronics. Nanosystems. Information Technologies. | vol. 16 | no. 5 | pp. 691-704.

[DOI 10.17725/j.rensit.2024.16.691](#) | [Source record](#) | Registered DOI metadata

Q1328 Quantum Computing Impact on Traditional Computer Architecture Models

Университет Алнур, Салих Махмуд Аттъя, Сухад Касим Дж. Хаддад, Университетский колледж Аль-Мансура, Хамид Карим Радам Аль-Заиди, Университетский колледж Аль-Хикма, Вафаа Мустафа Хамид, Университет Джихан et al..

Journal article | 2024 | Radioelectronics. Nanosystems. Information Technologies. | vol. 16 | no. 5 | pp. 691-704.

[DOI 10.17725/rensit.2024.16.691](#) | [Source record](#) | Registered DOI metadata

Q1329 ION TRAP QUANTUM COMPUTER ARCHITECTURE

Grzegorz Kasprówicz.

Journal article | 2023 | ELEKTRONIKA - KONSTRUKCJE, TECHNOLOGIE, ZASTOSOWANIA | vol. 1 | no. 8 | pp. 14-18.

[DOI 10.15199/13.2023.8.3](#) | [Source record](#) | Registered DOI metadata

Q1330 Quantum Computer Architecture: A Quantum Circuit-Based Approach Towards Quantum Neural Network

Tariq Mahmood, Talab Hussain, Maqsood Ahmed.

Journal article | 2023 | Proceedings of the Pakistan Academy of Sciences: A. Physical and Computational Sciences | vol. 60 | no. 2.

[DOI 10.53560/ppasa\(60-2\)668](#) | [Source record](#) | Registered DOI metadata

D25T02 Distributed quantum computing

Distributed computation uses entanglement or remote operations across processors. Partitioning and communication overhead must be included in scaling claims.

Fine subcategories: Remote gates; partitioning; entanglement consumption; synchronization; latency.

Primary catalog resources: 16. Related topic assignments are searchable in the HTML.

Q1331 A Quantum Circuit Optimization Framework for Distributed Quantum Computing

Fengsheng Liu, Fudong Liu, Yangyang Fei, Hong Wang, Junchao Wang, Haodong Jiang, Zhi Ma.

Journal article | 2026 | IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems | vol. 45 | no. 10 | pp. 4762-4771.

[DOI 10.1109/tcad.2026.3656760](#) | [Source record](#) | Registered DOI metadata

Q1332 Cost-Aware Quantum Bit Mapping for Distributed Quantum Computing

Furong Zhan, Xiaoyu Wang, Yangming Zhao, Hongli Xu.

Conference paper | 2026 | 2026 International Wireless Communications and Mobile Computing (IWCMC) | pp. 857-862.

[DOI 10.1109/iwcmc69287.2026.11580025](#) | [Source record](#) | Registered DOI metadata

Q1333 Distributed Quantum Computing for Scalable Systems

Daniel Casado Faulí, Parfait Atchade-Adelomou, David Pérez de Lara, Rodrigo Gil-Merino.

Book chapter | 2026 | Lecture Notes in Networks and Systems | pp. 243-253.

[DOI 10.1007/978-3-032-05748-8_20](#) | [Source record](#) | Registered DOI metadata

Q1334 On Distributed Quantum Computing with Distributed Fan-Out Operations

Seng W. Loke.

Conference paper | 2026 | 2026 IEEE International Conference on Quantum Software (QSW) | pp. 1-6.

[DOI 10.1109/qsw72780.2026.00034](#) | [Source record](#) | Registered DOI metadata

Q1335 Optimized Compilation for Distributed Quantum Computing

Michele Bandini, Davide Ferrari, Stefano Carretta, Michele Amoretti.

Journal article | 2026 | IEEE Access | vol. 14 | pp. 97220-97231.

[DOI 10.1109/access.2026.3706203](#) | [Source record](#) | Registered DOI metadata

Q1336 Distributed Quantum Computing: Applications and Challenges

Juan C. Boschero, Niels M. P. Neumann, Ward van der Schoot, Thom Sijpesteijn, Robert Wezeman.

Book chapter | 2025 | Lecture Notes in Networks and Systems | pp. 100-116.

[DOI 10.1007/978-3-031-92602-0_6](#) | [Source record](#) | Registered DOI metadata

Q1337 Distributed Quantum Neural Networks on Distributed Photonic Quantum Computing

Kuan-Cheng Chen, Chen-Yu Liu, Yu Shang, Felix Burt, Kin K. Leung.

Conference paper | 2025 | 2025 IEEE International Conference on Quantum Computing and Engineering (QCE) | pp. 1477-1488.

[DOI 10.1109/qce65121.2025.00165](#) | [Source record](#) | Registered DOI metadata

Q1338 Error correction for distributed quantum computing

Daowen Qiu, Ligang Xiao, Le Luo, Paulo Mateus.

Journal article | 2025 | EPJ Quantum Technology | vol. 12 | no. 1 | article 142.

[DOI 10.1140/epjqt/s40507-025-00455-x](#) | [Source record](#) | Registered DOI metadata

Q1339 Evaluating Variational Quantum Circuit Architectures for Distributed Quantum Computing

Leo Sünkel, Jonas Stein, Jonas Nüßlein, Tobias Rohe, Claudia Linnhoff-Popien.

Conference paper | 2025 | 2025 IEEE International Conference on Quantum Artificial Intelligence (QAI) | pp. 343-350.

[DOI 10.1109/qai63978.2025.00060](#) | [Source record](#) | Registered DOI metadata

Q1340 Online Locality Meets Distributed Quantum Computing

Amirreza Akbari, Xavier Coiteux-Roy, Francesco d'Amore, François Le Gall, Henrik Lievonen, Darya Melnyk, Augusto Modanese, Shreyas Pai et al..

Conference paper | 2025 | Proceedings of the 57th Annual ACM Symposium on Theory of Computing | pp. 1295-1306.

[DOI 10.1145/3717823.3718211](#) | [Source record](#) | Registered DOI metadata

Q1341 Optimizing Distributed Systems With Quantum Computing

Manish Patil, Shraddha Sanjay Mhetre.

Journal article | 2025 | INTERNATIONAL JOURNAL OF ADVANCES IN SIGNAL AND IMAGE SCIENCES | vol. 11 | no. 5s | pp. 683-700.

[DOI 10.29284/4rsdh697](#) | [Source record](#) | Registered DOI metadata

Q1342 Review of Distributed Quantum Computing: From single QPU to High Performance Quantum Computing

David Barral, F. Javier Cardama, Guillermo Díaz-Camacho, Daniel Faílde, Iago F. Llovo, Mariamo Mussa-Juane, Jorge Vázquez-Pérez, Juan Villasuso et al..

Journal article | 2025 | Computer Science Review | vol. 57 | pp. 100747 | article 100747.

[DOI 10.1016/j.cosrev.2025.100747](#) | [Source record](#) | Registered DOI metadata

Q1343 Towards Distributed Quantum Error Correction for Distributed Quantum Computing

Shahram Babaie, Chunming Qiao.

Conference paper | 2025 | 2025 International Conference on Quantum Communications, Networking, and Computing (QCNC) | pp. 66-73.

[DOI 10.1109/qcnc64685.2025.00019](#) | [Source record](#) | Registered DOI metadata

Q1344 Transversal Fault Tolerant Distributed Quantum Computing Operations

Frank Mueller, Ming Wang, John Stack.

Posted content | 2025 | Springer Science and Business Media LLC.

[DOI 10.21203/rs.3.rs-7633777/v1](#) | [Source record](#) | Registered DOI metadata

Q1345 Distributed Quantum Computing via Integrating Quantum and Classical Computing

Wei Tang, Margaret Martonosi.

Journal article | 2024 | Computer | vol. 57 | no. 4 | pp. 131-136.

[DOI 10.1109/mc.2024.3360569](#) | [Source record](#) | Registered DOI metadata

Q1346 Distributed quantum computing: A survey

Marcello Caleffi, Michele Amoretti, Davide Ferrari, Jessica Illiano, Antonio Manzalini, Angela Sara Cacciapuoti.

Journal article | 2024 | Computer Networks | vol. 254 | pp. 110672 | article 110672.

[DOI 10.1016/j.comnet.2024.110672](#) | [Source record](#) | Registered DOI metadata

D25T03 Quantum cloud execution

Cloud execution provides remote access and job orchestration for quantum devices. Queue delays, device changes, and preserved experiment metadata affect reproducibility.

Fine subcategories: Queues; job orchestration; access models; reproducibility; provider abstraction.

Primary catalog resources: 16. Related topic assignments are searchable in the HTML.

Q1347 Cloud Quantum Computing

Marcus Stephen Edwards.

Book chapter | 2026 | An Introduction to Quantum Computing for Computer Engineers | pp. 265-295.

[DOI 10.1007/978-3-032-03650-6_10](#) | [Source record](#) | Registered DOI metadata

Q1348 Distributed Scheduling for Modular Quantum Computing in Cloud

Vinoth Rao Kulkarni, Vipin Chaudhary.

Conference paper | 2026 | Proceedings of the 35th International Symposium on High-Performance Parallel and Distributed Computing | pp. 572-574.

[DOI 10.1145/3806645.3820071](#) | [Source record](#) | Registered DOI metadata

Q1349 Quantum Computing and Cloud

Premanand Narasimhan, Aruna Buvaneswari.

Journal article | 2026 | International Journal of Scientific Research in Computer Science, Engineering and Information Technology | vol. 12 | no. 2 | pp. 56-61.

[DOI 10.32628/cseit26121327](#) | [Source record](#) | Registered DOI metadata

Q1350 Cloud Computing Framework based on Quantum-Neuromorphic Technology

Surya Bhushan Kumar, Kuntal Mukherjee, Swapan Dey.

Conference paper | 2025 | 2025 International Conference on Sustainability, Innovation & Technology (ICSIT) | pp. 1-6.

[DOI 10.1109/icsit65336.2025.11295176](#) | [Source record](#) | Registered DOI metadata

Q1351 Enhancing Cloud Security Via Quantum Computing

Praveen Chaitanya Jakku, Prasad Sundaramoorthy, Sharath Chandra Kondaparthi, Tanvi Desai, Rohit Jarubula, Anusha Nerella.

Book chapter | 2025 | Lecture Notes in Networks and Systems | pp. 377-386.

[DOI 10.1007/978-3-032-03558-5_31](#) | [Source record](#) | Registered DOI metadata

Q1352 Qonductor: A Cloud Orchestrator for Quantum Computing

Emmanouil Giortamis, Francisco Romao, Nathaniel Tornow, Dmitry Lugovoy, Pramod Bhatotia.

Conference paper | 2025 | Proceedings of the International Conference for High Performance Computing, Networking, Storage and Analysis | pp. 728-745.

[DOI 10.1145/3712285.3759785](#) | [Source record](#) | Registered DOI metadata

Q1353 Quantum Cloud Computing Breakthrough

MCA, Zeal Institute of Business Administration, Computer Application & Research, Prof. Nitin Ramarao Yadav, Anurag Omprakash Shastri.

Journal article | 2025 | INTERNATIONAL JOURNAL OF SCIENTIFIC RESEARCH IN ENGINEERING AND MANAGEMENT | vol. 09 | no. 11 | pp. 1-9.

[DOI 10.55041/ijssrem54158](#) | [Source record](#) | Registered DOI metadata

Q1354 Quantum Cloud Ecosystems: Building Scalable, Multi-Tenant Quantum Computing Platforms

Murali Krishna Pasupuleti.

Reference book | 2025 | National Education Services.

[DOI 10.62311/nesx/rb68](#) | [Source record](#) | Registered DOI metadata

Q1355 Quantum Computing, AI, ML, and the Cloud

Arthur M. Langer.

Book chapter | 2025 | Analysis and Design of Next-Generation Software Architectures | pp. 167-181.

[DOI 10.1007/978-3-031-76212-3_8](#) | [Source record](#) | Registered DOI metadata

Q1356 Quantum Edge Cloud Computing: Revolutionizing IoT

Aanya Tiwari, Ahsaan Ahmad Ahanger, Anjali Meena, Pawan Singh Mehra.

Conference paper | 2025 | 2025 3rd International Conference on Device Intelligence, Computing and Communication Technologies (DICCT) | pp. 626-631.

[DOI 10.1109/dicct64131.2025.10986380](#) | [Source record](#) | Registered DOI metadata

Q1357 Quantum as a Service in Cloud Computing

Abdelkader Laouid, Mostefa Kara, Khaled Chait.

Book chapter | 2025 | Quantum Technology Applications, Impact, and Future Challenges | pp. 124-140.

[DOI 10.1201/9781003537243-8](#) | [Source record](#) | Registered DOI metadata

Q1358 Quantum cloud computing: Enterprise strategies for hybrid quantum-classical workloads

Clement Praveen Xavier Pakkam Isaac.

Journal article | 2025 | World Journal of Advanced Research and Reviews | vol. 26 | no. 1 | pp. 2245-2262.

[DOI 10.30574/wjarr.2025.26.1.1248](#) | [Source record](#) | Registered DOI metadata

Q1359 Quantum Process Tomography on Cloud-accessible Quantum Computing Platforms

P. E. Vedrukov, A. D. Ivlev, A. V. Liniov, I. B. Meyerov, M. V. Ivanchenko.

Journal article | 2024 | Lobachevskii Journal of Mathematics | vol. 45 | no. 1 | pp. 119-129.

[DOI 10.1134/s1995080224010529](#) | [Source record](#) | Registered DOI metadata

Q1360 Quantum cloud computing: Trends and challenges

Muhammed Golec, Emir Sahin Hatay, Mustafa Golec, Murat Uyar, Merve Golec, Sukhpal Singh Gill.

Journal article | 2024 | Journal of Economy and Technology | vol. 2 | pp. 190-199.

[DOI 10.1016/j.ject.2024.05.001](#) | [Source record](#) | Registered DOI metadata

Q1361 Secure Quantum Cloud Computing

Ming-Xing Luo.

Book chapter | 2024 | Quantum Networks | pp. 249-291.

[DOI 10.1007/978-981-97-6226-2_7](#) | [Source record](#) | Registered DOI metadata

Q1362 Universal terminal for cloud quantum computing

Mohammadsadegh Khazali.

Journal article | 2024 | Scientific Reports | vol. 14 | no. 1 | article 15412.

[DOI 10.1038/s41598-024-65899-0](#) | [Source record](#) | Registered DOI metadata

D25T04 Quantum circuit cutting

Circuit cutting reconstructs a larger circuit from smaller experiments and classical postprocessing. Sampling overhead can grow rapidly with the number and type of cuts.

Fine subcategories: Wire cutting; gate cutting; reconstruction; shot overhead; partitioning.

Primary catalog resources: 16. Related topic assignments are searchable in the HTML.

Q1363 Co-design threading model and circuit cutting for static and adaptive quantum circuits

Waldemir Cambiucci, Regina Melo Silveira, Wilson Vicente Ruggiero.

Journal article | 2026 | Quantum Information Processing | vol. 25 | no. 4 | article 118.

[DOI 10.1007/s11128-026-05136-x](#) | [Source record](#) | Registered DOI metadata

Q1364 DAScut: density-and-structure-aware circuit cutting for scalable quantum simulations

Theodora Adufu, Yoonhee Kim.

Journal article | 2026 | Cluster Computing | vol. 29 | no. 3 | article 163.

[DOI 10.1007/s10586-025-05911-y](#) | [Source record](#) | Registered DOI metadata

Q1365 Gate Teleportation vs. Circuit Cutting in Distributed Quantum Computing

Shobhit Gupta, Daniel J. Dilley, Nikolay Sheshko, Alvin Gonzales, Sean E. Sullivan, Manish K. Singh, Zain H. Saleem.

Journal article | 2026 | Advanced Quantum Technologies | vol. 9 | no. 9 | article e70424.

[DOI 10.1002/qute.70424](#) | [Source record](#) | Registered DOI metadata

Fine tags: gate cutting

Q1366 QCutSim: Accelerating Quantum Circuit Cutting Simulation on Consumer-Grade Classical Systems

Po-Hsuan Huang, Chun-Yen Tai, Chia-Heng Tu, Shih-Hao Hung.

Journal article | 2026 | ACM Transactions on Design Automation of Electronic Systems | article 3833423.

[DOI 10.1145/3833423](#) | [Source record](#) | Registered DOI metadata

Q1367 QuMod: Parallel Quantum Job Scheduling on Modular QPUs Using Circuit Cutting

Vinooth Kulkarni, Aaron Orenstein, Xinpeng Li, Shuai Xu, Daniel Blankenberg, Vipin Chaudhary.

Conference paper | 2026 | 2026 International Conference on Quantum Communications, Networking, and Computing (QCNC) | pp. 75-82.

[DOI 10.1109/qcnc69040.2026.00020](#) | [Source record](#) | Registered DOI metadata

Q1368 Reducing Maximum Subcircuits Depth in Quantum Circuit Cutting

Milad Eslaminia, Sébastien Le Beux.

Journal article | 2026 | IEEE Transactions on Quantum Engineering | vol. 7 | pp. 3103209-3103209.

[DOI 10.1109/tqe.2026.3690593](#) | [Source record](#) | Registered DOI metadata

Q1369 Scalable Quantum Circuit Simulation via Circuit Cutting and FPGA Acceleration

Xiaokun Yang, Xinpeng Li, Jeremy W Turner, Cameron D Disomma, Yunhe Feng, Xuechen Zhang, Vipin Chaudhary, Shuai Xu.

Conference paper | 2026 | Proceedings of the 35th International Symposium on High-Performance Parallel and Distributed Computing | pp. 642-650.

[DOI 10.1145/3806645.3816155](#) | [Source record](#) | Registered DOI metadata

Q1370 A generalized reconstruction model in circuit cutting and nonlocal-gate-based distributed quantum computation

Yi Sun, Changhua Zhu, Yuan Zhao, Guangwu Hou.

Journal article | 2025 | New Journal of Physics | vol. 27 | no. 11 | pp. 114508.

[DOI 10.1088/1367-2630/ae1866](#) | [Source record](#) | Registered DOI metadata

Fine tags: gate cutting; reconstruction

Q1371 DevQCC: Device-Aware Quantum Circuit Cutting framework with applications in quantum machine learning

Himanshu Sahu, Hari Prabhat Gupta, Vishnu Vardhan Puvvada, Rahul Mishra.

Journal article | 2025 | Quantum Machine Intelligence | vol. 7 | no. 2 | article 89.

[DOI 10.1007/s42484-025-00313-0](#) | [Source record](#) | Registered DOI metadata

Q1372 Improved sampling bounds and scalable partitioning for quantum circuit cutting beyond bipartitions

Junya Nakamura, Takahiko Satoh, Shinichiro Sanji.

Journal article | 2025 | Physical Review A | vol. 112 | no. 4 | article 042422.

[DOI 10.1103/xnw7-mtbd](#) | [Source record](#) | Registered DOI metadata

Fine tags: partitioning

Q1373 Is Circuit Cutting Scalable for Practical Quantum Applications?

Songqinghao Yang, Prakash Murali.

Conference paper | 2025 | 2025 IEEE International Conference on Quantum Computing and Engineering (QCE) | pp. 428-438.

[DOI 10.1109/qce65121.2025.00055](https://doi.org/10.1109/qce65121.2025.00055) | [Source record](#) | Registered DOI metadata

Q1374 Orchestrating Quantum-HPC Workflows with Distributed Quantum Circuit Cutting

Mar Tejedor, Berta Casas, Javier Conejero, Alba Cervera-Lierta, Rosa M. Badia.

Conference paper | 2025 | Proceedings of the SC '25 Workshops of the International Conference for High Performance Computing, Networking, Storage and Analysis | pp. 1898-1906.

[DOI 10.1145/3731599.3767547](https://doi.org/10.1145/3731599.3767547) | [Source record](#) | Registered DOI metadata

Q1375 QuFlex: Parallel Quantum Job Scheduling using Adaptive Circuit Cutting

Vinoth Kulkarni, Aaron Orenstein, Xinpeng Li, Shuai Xu, Daniel Blankenberg, Vipin Chaudhary.

Conference paper | 2025 | 2025 Supercomputing India (SCI) | pp. 1-8.

[DOI 10.1109/sci68648.2025.11333863](https://doi.org/10.1109/sci68648.2025.11333863) | [Source record](#) | Registered DOI metadata

Q1376 Quantum Circuit Cutting: A Security Methodology

George Typaldos, Theodoros Trochatos, Jakub Szefer.

Conference paper | 2025 | 2025 IEEE International Conference on Quantum Computing and Engineering (QCE) | pp. 417-427.

[DOI 10.1109/qce65121.2025.00054](https://doi.org/10.1109/qce65121.2025.00054) | [Source record](#) | Registered DOI metadata

Q1377 State Dependent Optimization with Quantum Circuit Cutting

Xinpeng Li, Ji Liu, Jeffrey M. Larson, Shuai Xu, Sundararaja Sitharama Iyengar, Paul Hovland, Vipin Chaudhary.

Conference paper | 2025 | 2025 IEEE Computer Society Annual Symposium on VLSI (ISVLSI) | pp. 1-6.

[DOI 10.1109/isvlsi65124.2025.11130234](https://doi.org/10.1109/isvlsi65124.2025.11130234) | [Source record](#) | Registered DOI metadata

Q1378 Patterns for Quantum Circuit Cutting

Author metadata not supplied.

Conference paper | 2024 | Proceedings of the 30th Conference on Pattern Languages of Programs.

[DOI 10.64346/plop2023p25](https://doi.org/10.64346/plop2023p25) | [Source record](#) | Registered DOI metadata

D25T05 Quantum memory architecture

Quantum memories preserve states for later operations or communication. Lifetime, retrieval efficiency, fidelity, and interface compatibility determine their role.

Fine subcategories: Storage fidelity; memory lifetime; interfaces; multiplexing; logical memory.

Primary catalog resources: 8. Related topic assignments are searchable in the HTML.

Q1379 Photonic Quantum Computing on Spin Memory Architecture with Tree-Encoded Fusion

Xiangyu Ren, Yuexun Huang, Zhemin Zhang, Yuchen Zhu, Tsung-Yi Ho, Antonio Barbalace, Zhiding Liang.

Conference paper | 2026 | 2026 ACM/IEEE 53rd Annual International Symposium on Computer Architecture (ISCA) | pp. 2333-2348.

[DOI 10.1109/isca66397.2026.00164](https://doi.org/10.1109/isca66397.2026.00164) | [Source record](#) | Registered DOI metadata

Q1380 Quantum-enhanced reconfigurable in-memory stochastic computing

Hong-Zhe Yang, Jian-Peng Dou, Feng Lu, Xiao-Wen Shang, Chao-Ni Zhang, Heng Zhou, Hao Tang, Xian-Min Jin.

Journal article | 2026 | Light: Science & Applications | vol. 15 | no. 1 | article 178.

[DOI 10.1038/s41377-025-02181-6](https://doi.org/10.1038/s41377-025-02181-6) | [Source record](#) | Registered DOI metadata

Q1381 Quantum Cognition: A Cognitive Architecture for Human-AI and In-Memory Computing

Fariborz Farahmand.

Journal article | 2023 | Computer | vol. 56 | no. 4 | pp. 135-138.

[DOI 10.1109/mc.2023.3242056](https://doi.org/10.1109/mc.2023.3242056) | [Source record](#) | Registered DOI metadata

Q1382 Quantum memory decoherence-mitigating architecture for quantum repeaters

Siddhartha Santra, Liang Jiang, Vladimir S. Malinovsky.

Conference paper | 2019 | Quantum Information and Measurement (QIM) V: Quantum Technologies | pp. S1D.4.

[DOI 10.1364/qim.2019.s1d.4](https://doi.org/10.1364/qim.2019.s1d.4) | [Source record](#) | Registered DOI metadata

Q1383 Quantum probabilistic associative memory architecture

Fernando M de Paula Neto, Adenilton J da Silva, Wilson R de Oliveira, Teresa B. Ludermit.

Journal article | 2019 | Neurocomputing | vol. 351 | pp. 101-110.

[DOI 10.1016/j.neucom.2019.03.078](https://doi.org/10.1016/j.neucom.2019.03.078) | [Source record](#) | Registered DOI metadata

Q1384 Architectures of Quantum Memory-driven Computing

Vladimir Hahanov, Svetlana Chumachenko, Eugenia Litvinova, Hanna Khakhanova.

Conference paper | 2018 | 2018 IEEE East-West Design & Test Symposium (EWDTS) | pp. 1-7.

[DOI 10.1109/ewdts.2018.8524843](https://doi.org/10.1109/ewdts.2018.8524843) | [Source record](#) | Registered DOI metadata

Q1385 Realization of processing In-memory computing architecture using Quantum Dot Cellular Automata

P.P. Chougule, B. Sen, T.D. Dongale.

Journal article | 2017 | Microprocessors and Microsystems | vol. 52 | pp. 49-58.

[DOI 10.1016/j.micpro.2017.04.022](https://doi.org/10.1016/j.micpro.2017.04.022) | [Source record](#) | Registered DOI metadata

Q1386 Single-flux-quantum cache memory architecture

Koki Ishida, Masamitsu Tanaka, Takatsugu Ono, Koji Inoue.

Conference paper | 2016 | 2016 International SoC Design Conference (ISOC) | pp. 105-106.

[DOI 10.1109/isoc.2016.7799755](https://doi.org/10.1109/isoc.2016.7799755) | [Source record](#) | Registered DOI metadata

D26 Quantum networks and communication

Quantum networking distributes quantum states or entanglement between nodes. Protocols and hardware must account for loss, timing, storage, and classical coordination.

Prerequisites: Quantum channels, entanglement, and networking.

Assessment focus: Loss, memory lifetime, heralding, synchronization, and capacity assumptions.

D26T01 Quantum teleportation

Teleportation transfers a state using shared entanglement, measurements, and classical messages. It does not transmit usable information faster than light.

Fine subcategories: Bell measurements; classical feed forward; fidelity; network teleportation.

Primary catalog resources: 8. Related topic assignments are searchable in the HTML.

Q1387 Quantum Teleportation

F. J. Duarte.

Book chapter | 2026 | Quantum Clear | pp. 55-56.

[DOI 10.1201/9781003649434-13](#) | [Source record](#) | Registered DOI metadata

Q1388 Quantum Teleportation

Meryem El Kirdi, Hanane El Hadfi, Lalla Btissam Drissi, Rachid Ahl Laamara, Abdallah Slaoui.

Book chapter | 2026 | Quantum Computing for Multimedia Processing, Teleportation, and Secure Networks | pp. 173-199.

[DOI 10.1201/9781003621423-11](#) | [Source record](#) | Registered DOI metadata

Q1389 Boosted quantum teleportation

Simone E. D'Aurelio, Matthias J. Bayerbach, Stefanie Barz.

Journal article | 2025 | npj Quantum Information | vol. 11 | no. 1 | article 37.

[DOI 10.1038/s41534-025-00992-4](#) | [Source record](#) | Registered DOI metadata

Q1390 Quantum Cryptography and Quantum Teleportation

F.J. Duarte.

Book chapter | 2024 | Quantum Optics for Engineers | pp. 258-268.

[DOI 10.1201/9781003398707-20](#) | [Source record](#) | Registered DOI metadata

Q1391 Quantum Energy Teleportation versus Information Teleportation

Jinzhao Wang, Shunyu Yao.

Journal article | 2024 | Quantum | vol. 8 | pp. 1564 | article 1564.

[DOI 10.22331/q-2024-12-12-1564](#) | [Source record](#) | Registered DOI metadata

Q1392 Multiphoton quantum teleportation

A.V. Belinsky, A.P. Grigorieva, I.I. Dzhadan.

Journal article | 2023 | Vestnik Moskovskogo Universiteta, Seriya 3: Fizika, Astronomiya | no. №5_2023 | pp. 2350104–1-2350104–6.

[DOI 10.55959/msu0579-9392.78.2350104](#) | [Source record](#) | Registered DOI metadata

Q1393 Quantum Teleportation

Ahmed Banafa.

Book chapter | 2023 | Introduction to Quantum Computing | pp. 13-16.

[DOI 10.1201/9781003440239-4](#) | [Source record](#) | Registered DOI metadata

Q1394 Quantum Teleportation

Reinhold A. Bertlmann, Nicolai Friis.

Book chapter | 2023 | Modern Quantum Theory | pp. 403-433.

[DOI 10.1093/oso/9780199683338.003.0014](https://doi.org/10.1093/oso/9780199683338.003.0014) | [Source record](#) | Registered DOI metadata

D26T02 Quantum repeaters

Repeaters extend entanglement distribution beyond direct transmission limits. Rates depend on memories, losses, purification or coding, and coordination.

Fine subcategories: Heralding; purification; error corrected repeaters; rate loss tradeoffs.

Primary catalog resources: 12. Related topic assignments are searchable in the HTML.

Q1395 Merging-based quantum repeater

Maria Flors Mor-Ruiz, Jorge Miguel-Ramiro, Julius Wallnöfer, Tim Coopmans, Wolfgang Dür.

Journal article | 2026 | npj Quantum Information.

[DOI 10.1038/s41534-026-01340-w](https://doi.org/10.1038/s41534-026-01340-w) | [Source record](#) | Registered DOI metadata

Q1396 Quantum Repeater Protocol Using Quantum Error Correction for Distillation

Ashlesha Patil, Michele Pacenti, Bane Vasić, Saikat Guha, Narayanan Rengaswamy.

Journal article | 2026 | IEEE Internet Computing | vol. 30 | no. 1 | pp. 38-46.

[DOI 10.1109/mic.2026.3656561](https://doi.org/10.1109/mic.2026.3656561) | [Source record](#) | Registered DOI metadata

Q1397 Continuous-variable multiplexed quantum repeater networks

Pei-Zhe Li, William J Munro, Kae Nemoto, Nicolò Lo Piparo.

Journal article | 2025 | Quantum Science and Technology | vol. 10 | no. 2 | pp. 025057.

[DOI 10.1088/2058-9565/adc500](https://doi.org/10.1088/2058-9565/adc500) | [Source record](#) | Registered DOI metadata

Q1398 Finding the perfect quantum dot for a quantum repeater

Katie McDonnell, Sai Sreesh Venuturumilli, Bera Yavuz, Rubayet Al Maruf, Dan Dalacu, Philip J. Poole, Michael E. Reimer, Michal Bajcsy.

Conference paper | 2025 | Photonics for Quantum 2025 | pp. 15.

[DOI 10.1117/12.3063250](https://doi.org/10.1117/12.3063250) | [Source record](#) | Registered DOI metadata

Q1399 Generalized Quantum Repeater Graph States

Bikun Li, Kenneth Goodenough, Filip Rozpędek, Liang Jiang.

Journal article | 2025 | Physical Review Letters | vol. 134 | no. 19 | article 190801.

[DOI 10.1103/physrevlett.134.190801](https://doi.org/10.1103/physrevlett.134.190801) | [Source record](#) | Registered DOI metadata

Q1400 Imbalanced Quantum Channels in Quantum Teleportation and Quantum Repeater Schemes

Lucas Vogeli, Phillip Cornett, Albert B. Dinkins V, Kwang-Cheng Chen.

Conference paper | 2025 | 2025 28th International Symposium on Wireless Personal Multimedia Communications (WPMC) | pp. 1-6.

[DOI 10.1109/wpmc67460.2025.11351266](https://doi.org/10.1109/wpmc67460.2025.11351266) | [Source record](#) | Registered DOI metadata

Q1401 Teleportation fidelity of quantum repeater networks

Ganesh Mylavarapu, Subrata Ghosh, Chittaranjan Hens, Indranil Chakrabarty, Subhadip Mitra.

Journal article | 2025 | Physical Review A | vol. 112 | no. 3 | article 032618.

[DOI 10.1103/72jp-37k6](https://doi.org/10.1103/72jp-37k6) | [Source record](#) | Registered DOI metadata

Q1402 Towards a spectrally multiplexed quantum repeater

Tanmoy Chakraborty, Antariksha Das, Hedser van Brug, Oriol Pietx-Casas, Peng-Cheng Wang, Gustavo Castro do Amaral, Anna L. Tchebotareva, Wolfgang Tittel.

Journal article | 2025 | npj Quantum Information | vol. 11 | no. 1 | article 3.

[DOI 10.1038/s41534-024-00946-2](https://doi.org/10.1038/s41534-024-00946-2) | [Source record](#) | Registered DOI metadata

Q1403 Asynchronous quantum repeater using multiple quantum memory

Chen-Long Li, Hua-Lei Yin, Zeng-Bing Chen.

Journal article | 2024 | Reports on Progress in Physics | vol. 87 | no. 12 | pp. 127901.

[DOI 10.1088/1361-6633/ad91de](https://doi.org/10.1088/1361-6633/ad91de) | [Source record](#) | Registered DOI metadata

Q1404 Towards a Practical Quantum Repeater

Mehdi Namazi.

Conference paper | 2024 | Frontiers in Optics + Laser Science 2024 (FIO, LS) | pp. FM5A.2.

[DOI 10.1364/ls.2024.fm5a.2](https://doi.org/10.1364/ls.2024.fm5a.2) | [Source record](#) | Registered DOI metadata

Q1405 Quantum Repeater Goes the Distance

Michal Hajdušek.

Journal article | 2023 | Physics | vol. 16 | article 84.

[DOI 10.1103/physics.16.84](https://doi.org/10.1103/physics.16.84) | [Source record](#) | Registered DOI metadata

Q1406 Scaling Limits of Quantum Repeater Networks

Mahdi Chehimi, Shahrooz Pouryousef, Nitish K. Panigrahy, Don Towsley, Walid Saad.

Conference paper | 2023 | 2023 IEEE International Conference on Quantum Computing and Engineering (QCE) | pp. 1205-1210.

[DOI 10.1109/qce57702.2023.00136](https://doi.org/10.1109/qce57702.2023.00136) | [Source record](#) | Registered DOI metadata

D26T03 Entanglement distribution and routing

Entanglement routing allocates links and swapping operations to network tasks. Probabilistic success and memory decoherence require different metrics from ordinary packet routing.

Fine subcategories: Entanglement swapping; routing metrics; scheduling; multipath distribution.

Primary catalog resources: 16. Related topic assignments are searchable in the HTML.

Q1407 Asynchronous Routing for Multipartite Entanglement in Quantum Networks

Chenliang Tian, Zebo Yang, Raj Jain, Ramana Kompella, Reza Nejabatati, Eneet Kaur, Aiman Erbad, Mounir Hamdi et al..

Conference paper | 2026 | 2026 IEEE 16th Annual Computing and Communication Workshop and Conference (CCWC) | pp. 0533-0541.

[DOI 10.1109/ccwc67433.2026.11393739](https://doi.org/10.1109/ccwc67433.2026.11393739) | [Source record](#) | Registered DOI metadata

Q1408 Improved Routing of Multiparty Entanglement over Quantum Networks

Nirupam Basak, Goutam Paul.

Journal article | 2026 | ACM Transactions on Quantum Computing | vol. 7 | no. 3 | pp. 1-26.

[DOI 10.1145/3811537](https://doi.org/10.1145/3811537) | [Source record](#) | Registered DOI metadata

Q1409 Maximize Quantum Network Throughput via EPS Placement and Lightweight Entanglement Routing

Yangming Zhao, Qiucheng Zhu, Bingyi Liu, Nai Xia, Chen Tian, Hongli Xu, Liusheng Huang, Kun Yang et al..

Journal article | 2026 | IEEE Transactions on Networking | vol. 34 | pp. 5349-5364.

[DOI 10.1109/ton.2026.3696466](https://doi.org/10.1109/ton.2026.3696466) | [Source record](#) | Registered DOI metadata

Q1410 On Utility-Optimal Entanglement Routing in Quantum Networks

Sounak Kar, Arpan Mukhopadhyay.

Conference paper | 2026 | 2026 International Conference on Quantum Communications, Networking, and Computing (QCNC) | pp. 457-464.

[DOI 10.1109/qcnc69040.2026.00082](https://doi.org/10.1109/qcnc69040.2026.00082) | [Source record](#) | Registered DOI metadata

Q1411 Optimal Multipartite Entanglement Routing Model in Quantum Networks

Yudai Ogata, Imran Ahmed, Eiji Oki.

Conference paper | 2026 | 2026 IEEE 27th International Conference on High Performance Switching and Routing (HPSR) | pp. 1-6.

[DOI 10.1109/hpsr68369.2026.11615194](https://doi.org/10.1109/hpsr68369.2026.11615194) | [Source record](#) | Registered DOI metadata

Q1412 Resource estimation for entanglement routing in quantum networks

Manik Dawar, Ralf Riedinger, Nilesh Vyas, Paulo Mendes.

Journal article | 2026 | Physical Review A | vol. 113 | no. 2 | article 022622.

[DOI 10.1103/vf4x-rjsd](https://doi.org/10.1103/vf4x-rjsd) | [Source record](#) | Registered DOI metadata

Q1413 Routing entanglement through quantum networks

Karl Pelka, Matteo Aquilina, André Xuereb.

Journal article | 2026 | Physical Review Research | vol. 8 | no. 2 | article L022062.

[DOI 10.1103/hsc8-pv2s](https://doi.org/10.1103/hsc8-pv2s) | [Source record](#) | Registered DOI metadata

Q1414 Differentiated service entanglement routing for quantum networks

Hui Han, Bo Liu, Bang-Ying Tang, Si-Yu Xiong, Jin-Quan Huang, Wan-Rong Yu, Shu-Hui Chen.

Journal article | 2025 | Quantum Science and Technology | vol. 10 | no. 3 | pp. 035013.

[DOI 10.1088/2058-9565/adc82b](https://doi.org/10.1088/2058-9565/adc82b) | [Source record](#) | Registered DOI metadata

Q1415 DyQNet: Optimizing Dynamic Entanglement Routing with Online Request in Quantum Network

Tianyao Chu, Liqiang Lu, Shiyu Li, Xinghui Jia, Chenren Xu, Siwei Tan, Jianwei Yin.

Book chapter | 2025 | Lecture Notes in Computer Science | pp. 186-200.

[DOI 10.1007/978-981-95-1021-4_14](https://doi.org/10.1007/978-981-95-1021-4_14) | [Source record](#) | Registered DOI metadata

Q1416 Entanglement Routing in Quantum Networks: A Comprehensive Survey

Amar Abane, Michael Cubeddu, Van Sy Mai, Abdella Battou.

Journal article | 2025 | IEEE Transactions on Quantum Engineering | vol. 6 | pp. 1-39.

[DOI 10.1109/tqe.2025.3541123](https://doi.org/10.1109/tqe.2025.3541123) | [Source record](#) | Registered DOI metadata

Q1417 Entanglement-Reusable Dynamic Routing for Quantum Networks

Junwei Wu, Songshi Dou, Kwan L. Yeung.

Conference paper | 2025 | 2025 IEEE International Conference on Consumer Electronics (ICCE) | pp. 1-4.

[DOI 10.1109/icce63647.2025.10929901](https://doi.org/10.1109/icce63647.2025.10929901) | [Source record](#) | Registered DOI metadata

Q1418 Optimizing qubit transfer in multi-host quantum network using security-oriented entanglement routing algorithm

Saumya Priyadarshini, Chandrashekar Jatoth, Rajesh Doriya, Rajkumar Buyya.

Journal article | 2025 | Optics Communications | vol. 596 | pp. 132549 | article 132549.

[DOI 10.1016/j.optcom.2025.132549](https://doi.org/10.1016/j.optcom.2025.132549) | [Source record](#) | Registered DOI metadata

Q1419 Entanglement Routing Design Over Quantum Networks

Yiming Zeng, Jiarui Zhang, Ji Liu, Zhenhua Liu, Yuanyuan Yang.

Journal article | 2024 | IEEE/ACM Transactions on Networking | vol. 32 | no. 1 | pp. 352-367.

[DOI 10.1109/tnet.2023.3282560](https://doi.org/10.1109/tnet.2023.3282560) | [Source record](#) | Registered DOI metadata

Q1420 High-fidelity entanglement routing in quantum networks

HaoRan Hu, HuaZhi Lun, ZhiFeng Deng, Jie Tang, JiaHao Li, YueXiang Cao, Ya Wang, Ying Liu et al..

Journal article | 2024 | Results in Physics | vol. 60 | pp. 107682 | article 107682.

[DOI 10.1016/j.rinp.2024.107682](https://doi.org/10.1016/j.rinp.2024.107682) | [Source record](#) | Registered DOI metadata

Q1421 Quantum Error Correction Based Entanglement Routing in Socially-Aware Quantum Networks

Shao-Min Huang, Ming-Huang Chien, Ting-Yuan Wen, Qian-Jing Wang, Jian-Jhih Kuo.

Conference paper | 2024 | GLOBECOM 2024 - 2024 IEEE Global Communications Conference | pp. 4503-4508.

[DOI 10.1109/globecom52923.2024.10901515](#) | [Source record](#) | Registered DOI metadata

Q1422 qRL: Reinforcement Learning Routing for Quantum Entanglement Networks

Diego Abreu, Antonio Abelém.

Conference paper | 2024 | 2024 IEEE Symposium on Computers and Communications (ISCC) | pp. 1-6.

[DOI 10.1109/iscc61673.2024.10733623](#) | [Source record](#) | Registered DOI metadata

D26T04 Quantum internet protocols

Quantum network protocols coordinate physical links, entanglement, and applications. Layer interfaces must represent limited and consumable quantum resources.

Fine subcategories: Link layers; network stacks; resource management; control planes; application interfaces.

Primary catalog resources: 8. Related topic assignments are searchable in the HTML.

Q1423 Efficient and Quantum-Safe Internet Key Exchange Protocols for Satellite Communications

Davide De Zuane, Marco Baldi, Paolo Santini, Grégoire Anachelergues, Daniele Romano.

Conference paper | 2026 | 2026 IEEE 32nd International Symposium on Local and Metropolitan Area Networks (LANMAN) | pp. 1-6.

[DOI 10.1109/lanman69841.2026.11623493](#) | [Source record](#) | Registered DOI metadata

Q1424 Quantum Internet: Technologies, Protocols, and Research Challenges

Vinay Kumar, Claudio Cicconetti, Marco Conti, Andrea Passarella.

Journal article | 2025 | International Journal of Networked and Distributed Computing | vol. 13 | no. 2 | article 22.

[DOI 10.1007/s44227-025-00060-5](#) | [Source record](#) | Registered DOI metadata

Q1425 A Survey of Quantum Internet Protocols From a Layered Perspective

Yuan Li, Hao Zhang, Chen Zhang, Tao Huang, F. Richard Yu.

Journal article | 2024 | IEEE Communications Surveys & Tutorials | vol. 26 | no. 3 | pp. 1606-1634.

[DOI 10.1109/comst.2024.3361662](#) | [Source record](#) | Registered DOI metadata

Q1426 Guest Editorial The Quantum Internet: Principles, Protocols and Architectures

Angela Sara Cacciapuoti, Anne Broadbent, Eleni Diamanti, Jacqueline Romero, Stephanie Wehner.

Journal article | 2024 | IEEE Journal on Selected Areas in Communications | vol. 42 | no. 7 | pp. 1719-1722.

[DOI 10.1109/jsac.2024.3379106](#) | [Source record](#) | Registered DOI metadata

Q1427 Quantum Internet: Functionalities, Protocols and Applications

Eftekar Abdulaziz, Khalid Al-Hussaini, Fua'Ad Abdulrazzak.

Conference paper | 2024 | 2024 1st International Conference on Emerging Technologies for Dependable Internet of Things (ICETI) | pp. 1-6.

[DOI 10.1109/iceti63946.2024.10777267](#) | [Source record](#) | Registered DOI metadata

Q1428 Secure and Efficient Entanglement Distribution Protocols for Near-Term Quantum Internet

Nicholas Skjellum, Mohamed Shaban, Muhammad Ismail.

Conference paper | 2024 | 2024 33rd International Conference on Computer Communications and Networks (ICCCN) | pp. 1-9.

[DOI 10.1109/icccn61486.2024.10637640](#) | [Source record](#) | Registered DOI metadata

Q1429 Towards Secure and Efficient Practical Consensus Protocols in Quantum Internet

Woonghee Lee, Donghee Kim, Junbeom Hur.

Conference paper | 2024 | 2024 15th International Conference on Information and Communication Technology Convergence (ICTC) | pp. 759-761.

[DOI 10.1109/ictc62082.2024.10826676](https://doi.org/10.1109/ictc62082.2024.10826676) | [Source record](#) | Registered DOI metadata

Q1430 Use of hybrid post-quantum key exchange in internet protocols

Valery Smyslov.

Journal article | 2024 | Journal of Computer Virology and Hacking Techniques | vol. 20 | no. 3 | pp. 447-454.

[DOI 10.1007/s11416-024-00515-3](https://doi.org/10.1007/s11416-024-00515-3) | [Source record](#) | Registered DOI metadata

D26T05 Quantum communication capacity

Channel capacities quantify asymptotic communication rates under specified assistance. Classical, quantum, and private capacities answer different coding questions.

Fine subcategories: Classical capacity; quantum capacity; private capacity; coding theorems.

Primary catalog resources: 8. Related topic assignments are searchable in the HTML.

Q1431 Channel capacity of relativistic quantum communication with rapid interaction

Erickson Tjoa, Kensuke Gallock-Yoshimura.

Journal article | 2022 | Physical Review D | vol. 105 | no. 8 | article 085011.

[DOI 10.1103/physrevd.105.085011](https://doi.org/10.1103/physrevd.105.085011) | [Source record](#) | Registered DOI metadata

Fine tags: quantum capacity

Q1432 Drastic increase of channel capacity in quantum secure direct communication using masking

Gui-Lu Long, Haoran Zhang.

Journal article | 2021 | Science Bulletin | vol. 66 | no. 13 | pp. 1267-1269.

[DOI 10.1016/j.scib.2021.04.016](https://doi.org/10.1016/j.scib.2021.04.016) | [Source record](#) | Registered DOI metadata

Fine tags: quantum capacity

Q1433 Quantum-channel capacity of distributing orbital-angular-momentum states for underwater optical quantum communication

Shuang Zhai, Jicheng Wang, Yun Zhu, Yixin Zhang, Zheng-Da Hu.

Journal article | 2020 | Journal of the Optical Society of America A | vol. 38 | no. 1 | pp. 36.

[DOI 10.1364/josaa.402794](https://doi.org/10.1364/josaa.402794) | [Source record](#) | Registered DOI metadata

Fine tags: quantum capacity

Q1434 Classical communication with quantum receivers: towards violating classical Shannon channel capacity (Conference Presentation)

Ivan Burenkov, Sergey V. Polyakov.

Conference paper | 2019 | Advances in Photonics of Quantum Computing, Memory, and Communication XII | pp. 17.

[DOI 10.1117/12.2510091](https://doi.org/10.1117/12.2510091) | [Source record](#) | Registered DOI metadata

Fine tags: Classical capacity; quantum capacity

Q1435 Cluster state based controlled quantum secure direct communication protocol with controllable channel capacity

Zheng Xiao-Yi, Long Yin-Xiang, Automation Engineering Department, Guangdong Technical College of Water Resource and Electric Engineering, Guangzhou 510635, China.

Journal article | 2017 | Acta Physica Sinica | vol. 66 | no. 18 | pp. 180303.

[DOI 10.7498/aps.66.180303](https://doi.org/10.7498/aps.66.180303) | [Source record](#) | Registered DOI metadata

Fine tags: quantum capacity

Q1436 Capacity of Quantum Erasure Channel Assisted by Backwards Classical Communication

Debbie Leung, Joungkeun Lim, Peter Shor.

Journal article | 2009 | Physical Review Letters | vol. 103 | no. 24 | article 240505.

[DOI 10.1103/physrevlett.103.240505](https://doi.org/10.1103/physrevlett.103.240505) | [Source record](#) | Registered DOI metadata

Fine tags: Classical capacity; quantum capacity

Q1437 Quantum Communication with Zero-Capacity Channels

Graeme Smith, Jon Yard.

Journal article | 2008 | Science | vol. 321 | no. 5897 | pp. 1812-1815.

[DOI 10.1126/science.1162242](https://doi.org/10.1126/science.1162242) | [Source record](#) | Registered DOI metadata

Fine tags: quantum capacity

Q1438 Information capacity of a quantum communication channel. II.

R. L. Stratonovich.

Journal article | 1966 | Soviet Radiophysics | vol. 8 | no. 1 | pp. 92-101.

[DOI 10.1007/bf01038471](https://doi.org/10.1007/bf01038471) | [Source record](#) | Registered DOI metadata

Fine tags: quantum capacity

D27 Quantum cryptography and security

Quantum cryptography uses quantum systems for security tasks, while post quantum cryptography uses classical algorithms designed to resist quantum attacks. Their mechanisms and deployment requirements are different.

Prerequisites: Cryptographic definitions, probability, and quantum algorithms.

Assessment focus: Threat model, composable security, finite size effects, and implementation attacks.

D27T01 Quantum key distribution

QKD establishes secret keys under a stated device and security model. Finite size analysis, authentication, and side channels are part of practical security.

Fine subcategories: BB84; decoy states; finite keys; composable security; implementation attacks.

Primary catalog resources: 8. Related topic assignments are searchable in the HTML.

Q1439 Quantum cryptography

Nicolas Gisin, Grégoire Ribordy, Wolfgang Tittel, Hugo Zbinden.

Journal article | 2002 | Reviews of Modern Physics | vol. 74 | no. 1 | pp. 145-195.

[DOI 10.1103/revmodphys.74.145](#) | [Source record](#) | Registered DOI metadata

Q1440 Quantum Key Distribution

R. Naveenkumar, N. M. Sivamangai, C. M. Lingesh, G. Saranya, G. Indhumathi, N. Vini Antony Grace.

Book chapter | 2026 | Quantum Computing | pp. 207-234.

[DOI 10.1201/9781003587835-10](#) | [Source record](#) | Registered DOI metadata

Q1441 Quantum key distribution

Ivan B. Djordjevic.

Book chapter | 2026 | Quantum Communication, Quantum Networks, and Quantum Sensing | pp. 257-304.

[DOI 10.1016/b978-0-443-40568-6.00001-x](#) | [Source record](#) | Registered DOI metadata

Q1442 Quantum Key Distribution

Ray LaPierre.

Book chapter | 2025 | The Materials Research Society Series | pp. 97-105.

[DOI 10.1007/978-3-031-90731-9_6](#) | [Source record](#) | Registered DOI metadata

Q1443 Quantum Key Distribution

Ri-Gui Zhou, Xiao-Xue Zhang, Lin-Tao Du.

Book chapter | 2024 | Design and Analysis of Secure Quantum Communication Schemes | pp. 35-46.

[DOI 10.1007/978-3-031-78428-6_3](#) | [Source record](#) | Registered DOI metadata

Q1444 Quantum Key Distribution

Jasper Rödiger.

Book chapter | 2023 | Trends in Data Protection and Encryption Technologies | pp. 41-45.

[DOI 10.1007/978-3-031-33386-6_9](#) | [Source record](#) | Registered DOI metadata

Q1445 Quantum Key Distribution

Filip Wojcieszyn.

Book chapter | 2022 | Quantum Science and Technology | pp. 181-211.

[DOI 10.1007/978-3-030-99379-5_6](#) | [Source record](#) | Registered DOI metadata

Q1446 Quantum key distribution

Ivan B. Djordjevic.

Book chapter | 2022 | Quantum Communication, Quantum Networks, and Quantum Sensing | pp. 215-272.

[DOI 10.1016/b978-0-12-822942-2.00012-1](#) | [Source record](#) | Registered DOI metadata

D27T02 Device independent cryptography

Device independent protocols infer security from observed nonlocal correlations. Loophole closure, randomness, and finite statistics are demanding requirements.

Fine subcategories: Bell violations; loophole closure; entropy accumulation; security assumptions.

Primary catalog resources: 8. Related topic assignments are searchable in the HTML.

Q1447 Partially device-independent quantum cryptography in asymmetric networks

Jun-Hao Wei, Shu-Ming Hu, Nuo-Ya Yang, Shuai Zhao, Li Li, Nai-Le Liu, Kai Chen.

Journal article | 2026 | Physical Review Research | vol. 8 | no. 3 | article 033060.

[DOI 10.1103/53w8-6v5z](#) | [Source record](#) | Registered DOI metadata

Q1448 A Direct Product Theorem for Quantum Communication Complexity with Applications to Device-Independent Cryptography

Rahul Jain, Srijita Kundu.

Journal article | 2025 | SIAM Journal on Computing | vol. 54 | no. 4 | pp. 964-1020.

[DOI 10.1137/23m1549353](#) | [Source record](#) | Registered DOI metadata

Q1449 Seedless extractors for device-independent quantum cryptography

Cameron Foreman, Lluís Masanes.

Journal article | 2025 | Quantum | vol. 9 | pp. 1654 | article 1654.

[DOI 10.22331/q-2025-03-06-1654](#) | [Source record](#) | Registered DOI metadata

Q1450 Boosting device-independent cryptography with tripartite nonlocality

Federico Grasselli, Gláucia Murta, Hermann Kampermann, Dagmar Bruß.

Journal article | 2023 | Quantum | vol. 7 | pp. 980 | article 980.

[DOI 10.22331/q-2023-04-13-980](#) | [Source record](#) | Registered DOI metadata

Q1451 Autocompensating Measurement-Device-Independent Quantum Cryptography in Space Division Multiplexing Optical Fibers

Jesús Liñares, Gabriel M. Carral, Xesús Prieto-Blanco, Daniel Balado.

Posted content | 2021 | Springer Science and Business Media LLC.

[DOI 10.21203/rs.3.rs-141385/v1](#) | [Source record](#) | Registered DOI metadata

Q1452 Autocompensating measurement-device-independent quantum cryptography in space division multiplexing optical fibers

J. Liñares, G. M. Carral, X. Prieto-Blanco, D. Balado.

Journal article | 2021 | Journal of the European Optical Society-Rapid Publications | vol. 17 | no. 1 | article 19.

[DOI 10.1186/s41476-021-00166-7](#) | [Source record](#) | Registered DOI metadata

Q1453 Device-Independent Quantum Cryptography

Federico Grasselli.

Book chapter | 2021 | Quantum Science and Technology | pp. 105-148.

[DOI 10.1007/978-3-030-64360-7_7](#) | [Source record](#) | Registered DOI metadata

Q1454 Autocompensating Measurement-Device-Independent quantum cryptography in few-mode optical fibers

Jesús Liñares-Beiras, Xesús Prieto-Blanco, Daniel Balado, Gabriel M. Carral.

Journal article | 2020 | EPJ Web of Conferences | vol. 238 | pp. 09002.

[DOI 10.1051/epjconf/202023809002](https://doi.org/10.1051/epjconf/202023809002) | [Source record](#) | Registered DOI metadata

D27T03 Blind and delegated computing

Blind computing hides some information about a delegated task, while verifiable protocols also test correctness. Client capability and server trust distinguish protocols.

Fine subcategories: Hidden circuits; verifiable delegation; trap protocols; server trust.

Primary catalog resources: 8. Related topic assignments are searchable in the HTML.

Q1455 Blind quantum computing with different qudit resource state architectures

Alena Romanova, Wolfgang Dür.

Journal article | 2026 | Physical Review A | vol. 113 | no. 3 | article 032606.

[DOI 10.1103/1p5p-zywx](https://doi.org/10.1103/1p5p-zywx) | [Source record](#) | Registered DOI metadata

Q1456 Practical blind quantum computation with parity quantum computing framework

Yuxun Wang, Qin Li, Shao-Ming Fei, Vlatko Vedral.

Journal article | 2026 | Quantum Science and Technology | vol. 11 | no. 2 | pp. 025033.

[DOI 10.1088/2058-9565/ae54c5](https://doi.org/10.1088/2058-9565/ae54c5) | [Source record](#) | Registered DOI metadata

Q1457 Secure Multi-Party Computation Based on Blind Quantum Computing

Yongli Wang, Tianci Cao, Yixin Wang, Rui Zhang, Yinghui Yang, Yongli Tang.

Posted content | 2026 | Springer Science and Business Media LLC.

[DOI 10.21203/rs.3.rs-9186808/v1](https://doi.org/10.21203/rs.3.rs-9186808/v1) | [Source record](#) | Registered DOI metadata

Q1458 Blind quantum computation with fewer quantum and delegated cost of client

Yu-Zhan Yan, Zhen Yang, Guang-Yang Wu, Yuan-Mao Luo, Ming-Qiang Bai, Zhi-Wen Mo.

Journal article | 2025 | International Journal of Quantum Information | vol. 23 | no. 02 | article 2530001.

[DOI 10.1142/s0219749925300013](https://doi.org/10.1142/s0219749925300013) | [Source record](#) | Registered DOI metadata

Q1459 Universal blind quantum computing assisted by quantum teleportation

Xiaoqian Zhang.

Journal article | 2025 | Quantum Information Processing | vol. 24 | no. 6 | article 182.

[DOI 10.1007/s11128-025-04798-3](https://doi.org/10.1007/s11128-025-04798-3) | [Source record](#) | Registered DOI metadata

Q1460 Universal distributed blind quantum computing with solid-state qubits

Y.-C. Wei, P.-J. Stas, A. Suleymanzade, G. Baranes, F. Machado, Y. Q. Huan, C. M. Knaut, S. W. Ding et al..

Journal article | 2025 | Science | vol. 388 | no. 6746 | pp. 509-513.

[DOI 10.1126/science.adu6894](https://doi.org/10.1126/science.adu6894) | [Source record](#) | Registered DOI metadata

Q1461 Private Set Intersection with Delegated Blind Quantum Computing

Michele Amoretti.

Conference paper | 2021 | 2021 IEEE Global Communications Conference (GLOBECOM) | pp. 1-6.

[DOI 10.1109/globecom46510.2021.9685125](https://doi.org/10.1109/globecom46510.2021.9685125) | [Source record](#) | Registered DOI metadata

Q1462 Delegated Preparation of Quantum Error Correction Code for Blind Quantum Computation

Qiang Zhao, Qiong Li.

Book chapter | 2019 | Smart Innovation, Systems and Technologies | pp. 147-154.

[DOI 10.1007/978-981-13-9710-3_15](https://doi.org/10.1007/978-981-13-9710-3_15) | [Source record](#) | Registered DOI metadata

D27T04 Quantum cryptanalysis

Quantum cryptanalysis studies attacks on classical or quantum security systems. Algorithmic complexity and physical fault tolerant cost should both be considered.

Fine subcategories: Factoring attacks; discrete logarithms; Grover search; resource estimates.

Primary catalog resources: 14. Related topic assignments are searchable in the HTML.

Q1463 Cryptanalysis of Post-Quantum Cryptographic Schemes

Veda Chatiyode, Baghavathi Priya S..

Book chapter | 2025 | Post-Quantum Cryptography and IoT Communications for Sustainable Urban Development | pp. 171-204.

[DOI 10.4018/979-8-3373-3166-9.ch007](#) | [Source record](#) | Registered DOI metadata

Q1464 Quantum Cryptanalysis: Breaking Classical Encryption with Shor's and Grover's Algorithms

Godwin Olaoye.

Posted content | 2025 | Wiley.

[DOI 10.22541/au.174431280.02434905/v1](#) | [Source record](#) | Registered DOI metadata

Q1465 Quantum Cryptanalysis: Evaluating the Impact of Shor's and Grover's Algorithms on Modern Encryption Standards

Mithal Hadi Jebur, Sumar Khaleel.

Journal article | 2025 | CyberSystem Journal | vol. 2 | no. 2 | pp. 41-55.

[DOI 10.57238/csj.2025.1012](#) | [Source record](#) | Registered DOI metadata

Q1466 Scalable, Fault-Tolerant Quantum Algorithms for Cryptanalysis-Resistant Communications

Ankith Konda.

Conference paper | 2025 | 2025 13th International Conference on Intelligent Systems and Embedded Design (ISED) | pp. 936-943.

[DOI 10.1109/ised67359.2025.11405294](#) | [Source record](#) | Registered DOI metadata

Q1467 Lattice-Based Cryptosystems and Quantum Cryptanalysis

Bruce Schneier.

Journal article | 2024 | Communications of the ACM | article 3665224.

[DOI 10.1145/3665224](#) | [Source record](#) | Registered DOI metadata

Q1468 Quantum Cryptanalysis of Affine Cipher

Mahima Mary Mathews, Panchami V, Vishnu Ajith.

Journal article | 2024 | IEEE Journal on Emerging and Selected Topics in Circuits and Systems | vol. 14 | no. 3 | pp. 507-519.

[DOI 10.1109/jetcas.2024.3428436](#) | [Source record](#) | Registered DOI metadata

Q1469 Quantum cryptanalysis

Author metadata not supplied.

Book chapter | 2024 | Mathematical Surveys and Monographs | pp. 53-62.

[DOI 10.1090/surv/278/06](#) | [Source record](#) | Registered DOI metadata

Q1470 Quantum related-key differential cryptanalysis

Hongyu Wu, Xiaoning Feng.

Journal article | 2024 | Quantum Information Processing | vol. 23 | no. 7 | article 269.

[DOI 10.1007/s11128-024-04472-0](#) | [Source record](#) | Registered DOI metadata

Q1471 The problem of finding periodicity in quantum cryptanalysis of group cryptography algorithms

Y. Kotukh, G. Khalimov, I. Dzhura.

Journal article | 2024 | Radiotekhnika | no. 218 | pp. 103-109.

[DOI 10.30837/rt.2024.3.218.08](#) | [Source record](#) | Registered DOI metadata

Q1472 Applied Quantum Cryptanalysis

Alexei Petrenko.

Book | 2023 | River Publishers.

[DOI 10.1201/9781003392873](#) | [Source record](#) | Registered DOI metadata

Q1473 Basic Algorithms Quantum Cryptanalysis

Saint Petersburg Electrotechnical University LETI, Alexei Petrenko, Sergei Petrenko, Saint Petersburg Electrotechnical University LETI.

Journal article | 2023 | Voprosy kiberbezopasnosti | no. 1(53) | pp. 100-115.

[DOI 10.21681/2311-3456-2023-1-100-115](#) | [Source record](#) | Registered DOI metadata

Q1474 Cryptanalysis of three quantum money schemes

Andriyan Bilyk, Javad Doliskani, Zhiyong Gong.

Journal article | 2023 | Quantum Information Processing | vol. 22 | no. 4 | article 177.

[DOI 10.1007/s11128-023-03919-0](#) | [Source record](#) | Registered DOI metadata

Q1475 Development of Quantum Cryptanalysis Algorithms

Alexei Petrenko, Sergei Petrenko.

Book chapter | 2023 | Applied Quantum Cryptanalysis | pp. 97-128.

[DOI 10.1201/9781003392873-4](#) | [Source record](#) | Registered DOI metadata

Q1476 The Relevance of Quantum Cryptanalysis

Alexei Petrenko, Sergei Petrenko.

Book chapter | 2023 | Applied Quantum Cryptanalysis | pp. 7-65.

[DOI 10.1201/9781003392873-2](#) | [Source record](#) | Registered DOI metadata

D27T05 Post quantum cryptography interface

Post quantum cryptography uses classical schemes intended to resist quantum attacks. It is included as a security interface to quantum computing, not as a quantum hardware protocol.

Fine subcategories: Lattice schemes; code based schemes; hash signatures; migration; threat models.

Primary catalog resources: 8. Related topic assignments are searchable in the HTML.

Q1477 Post-Quantum Cryptography

Author metadata not supplied.

Book | 2026 | Lecture Notes in Computer Science.

[DOI 10.1007/978-3-032-22698-3](#) | [Source record](#) | Registered DOI metadata

Q1478 Post-Quantum Cryptography: can Classical Algorithms Resist Quantum Attacks

Dinh Doan Xuan Phuong, Pham Truong Son.

Conference paper | 2025 | 2025 RIVF International Conference on Computing and Communication Technologies (RIVF) | pp. 717-722.

[DOI 10.1109/rivf68649.2025.11365084](#) | [Source record](#) | Registered DOI metadata

Q1479 Cryptographic Protocols Resilient to Quantum Attacks: Advancements in Post-Quantum Cryptography

Pallavi Niraj Vithalkar.

Journal article | 2024 | Communications on Applied Nonlinear Analysis | vol. 31 | no. 3s | pp. 520-532.

[DOI 10.52783/cana.v31.805](#) | [Source record](#) | Registered DOI metadata

Q1480 FIPS 203, Module-Lattice-Based Key-Encapsulation Mechanism Standard | CSRC

Author metadata not supplied.

Standard | 2024 | NIST.

[Official resource](#) | [Source record](#) | Official page retrieved

Q1481 FIPS 204, Module-Lattice-Based Digital Signature Standard | CSRC

Author metadata not supplied.

Standard | 2024 | NIST.

[Official resource](#) | [Source record](#) | Official page retrieved

Q1482 FIPS 205, Stateless Hash-Based Digital Signature Standard | CSRC

Author metadata not supplied.

Standard | 2024 | NIST.

[Official resource](#) | [Source record](#) | Official page retrieved

Q1483 Post-Quantum Cryptography: Securing Future Communication Networks Against Quantum Attacks

Dr. N Krishnamoorthy, Dr. S. Subbaiah, J Revathi.

Journal article | 2024 | Nanotechnology Perceptions | pp. 264-278.

[DOI 10.62441/nano-ntp.vi.2774](#) | [Source record](#) | Registered DOI metadata

Q1484 Post-Quantum Cryptography: Securing Future Communication Networks Against Quantum Attacks

Author metadata not supplied.

Journal article | 2024 | Nanotechnology Perceptions | vol. 20 | no. S14.

[DOI 10.62441/nano-ntp.v20is14.16](#) | [Source record](#) | Registered DOI metadata

D27T06 Quantum random number generation

Quantum random number generators estimate and extract entropy from measured quantum processes. Security depends on the source model and the amount of trusted hardware.

Fine subcategories: Entropy sources; extractors; certification; device trust; finite size analysis.

Primary catalog resources: 15. Related topic assignments are searchable in the HTML.

Q1485 Interferometric Quantum Random Number Generation via Michelson Configuration

Ram Soorat, Aakash Chilakamarri, Priyanka, Barath Sai Kumar Jakkuva, Syed Areebuddin.

Journal article | 2026 | International Journal of Theoretical Physics | vol. 65 | no. 6 | article 169.

[DOI 10.1007/s10773-026-06369-3](#) | [Source record](#) | Registered DOI metadata

Q1486 Quantum random number generation using spatial quantum noise of light

Mohamed Armoon Shaliq, Ashok Kumar.

Journal article | 2026 | Applied Physics Letters | vol. 129 | no. 7 | article 074002.

[DOI 10.1063/5.0342762](#) | [Source record](#) | Registered DOI metadata

Q1487 Certified random number generation using quantum computers

Pingal Pratyush Nath, Aninda Sinha, Urbasi Sinha.

Journal article | 2025 | Frontiers in Quantum Science and Technology | vol. 4 | article 1661544.

[DOI 10.3389/frqst.2025.1661544](#) | [Source record](#) | Registered DOI metadata

Q1488 Quantum Random Number Generation and ML-Based Authentication

Liviu Ionut Epure.

Posted content | 2025 | Wiley.

[DOI 10.22541/au.176281462.20133004/v1](#) | [Source record](#) | Registered DOI metadata

Q1489 Quantum Random Number Generation via Von Neumann Projection

Nawres A. Alwan, Suzan J. Obaiys, Nadia M. G. Al-Saidi, Nurul Fazmidar Binti Mohd Noor.

Book chapter | 2025 | Lecture Notes in Computer Science | pp. 176-193.

[DOI 10.1007/978-3-031-97000-9_11](#) | [Source record](#) | Registered DOI metadata

Q1490 Exploring quantum systems for pseudo-random number generation

Luis José Mantilla Santa Cruz, Luis Fernando Faina, João Henrique de Souza Pereira.

Journal article | 2024 | Quantum Studies: Mathematics and Foundations | vol. 12 | no. 1 | article 3.

[DOI 10.1007/s40509-024-00348-1](#) | [Source record](#) | Registered DOI metadata

Q1491 Investigating device-independent quantum random number generation

Vardaan Mongia, Abhishek Kumar, Shashi Prabhakar, Anindya Banerji, R.P. Singh.

Journal article | 2024 | Physics Letters A | vol. 526 | pp. 129954 | article 129954.

[DOI 10.1016/j.physleta.2024.129954](#) | [Source record](#) | Registered DOI metadata

Q1492 Quantum Random Number Generation

Christoph Capellaro, Mohammad Hammoudeh.

Book chapter | 2024 | Quantum Computing | pp. 132-141.

[DOI 10.1201/9781003475286-8](#) | [Source record](#) | Registered DOI metadata

Q1493 Quantum random number generation based on phase reconstruction

Jialiang Li, Zitao Huang, Chunlin Yu, Jiajie Wu, Tongge Zhao, Xiangwei Zhu, Shihai Sun.

Journal article | 2024 | Optics Express | vol. 32 | no. 4 | pp. 5056.

[DOI 10.1364/oe.515390](#) | [Source record](#) | Registered DOI metadata

Q1494 Quantum random number generation using Quandela photonic quantum computer

Muriel A. de Souza, Flavia P. Agostini, Luiz Vicente G. Tarelho.

Journal article | 2024 | Quantum Information Processing | vol. 23 | no. 11 | article 381.

[DOI 10.1007/s11128-024-04593-6](#) | [Source record](#) | Registered DOI metadata

Q1495 Secure quantum random number generation with perovskite photonics

Joakim Argillander, Alvaro Alarcon, Chunxiong Bao, Chaoyang Kuang, Gustavo Lima, Feng Gao, Guilherme B. Xavier.

Conference paper | 2024 | Quantum Computing, Communication, and Simulation IV | pp. 103.

[DOI 10.1117/12.2692061](#) | [Source record](#) | Registered DOI metadata

Q1496 Exploring Quantum Systems for Pseudo-Random Number Generation

Luis José Mantilla Santa Cruz, Luis F. Faina, João Henrique de Souza Pereira.

Posted content | 2023 | Research Square Platform LLC.

[DOI 10.21203/rs.3.rs-3630730/v1](#) | [Source record](#) | Registered DOI metadata

Q1497 Multiplexing quantum tunneling diodes for random number generation

Kanin Aungskunsiri, Ratthasart Amarit, Sakdinan Jantarachote, Kruawan Wongpanya, Pongpun Punpetch, Sarun Sumriddetchkajorn.

Journal article | 2023 | Review of Scientific Instruments | vol. 94 | no. 1 | pp. 014704.

[DOI 10.1063/5.0113995](#) | [Source record](#) | Registered DOI metadata

Q1498 Quantum random number generation with uncharacterised homodyne detection

Chao Wang, Ignatius William Primaatmaja, Hong Jie Ng, Jing Yan Haw, Raymond Ho, Jianran Zhang, Gong Zhang, Charles Lim.

Conference paper | 2023 | CLEO 2023 | pp. JTh2A.20.

[DOI 10.1364/cleo_at.2023.jth2a.20](#) | [Source record](#) | Registered DOI metadata

AQUIRE

Q1499 Semi-Quantum Random Number Generation

Julia Guskind, Walter O. Krawec.

Conference paper | 2023 | 2023 IEEE International Conference on Quantum Computing and Engineering (QCE) | pp. 1211-1219.

[DOI 10.1109/qce57702.2023.00137](https://doi.org/10.1109/qce57702.2023.00137) | [Source record](#) | [Registered DOI metadata](#)

D28 Quantum sensing and metrology

Sensing is an adjacent quantum technology rather than a synonym for computation. It is included because control, estimation, and quantum resources connect it to computing research.

Prerequisites: Quantum mechanics, parameter estimation, and experimental physics.

Assessment focus: Sensitivity, bandwidth, noise, trust, and whether bounds are attainable.

D28T01 Quantum parameter estimation

Quantum estimation studies how states and measurements encode unknown parameters. Bounds do not automatically imply an experimentally achievable estimator.

Fine subcategories: Quantum Fisher information; Cramer Rao bounds; multiparameter estimation; estimators.

Primary catalog resources: 8. Related topic assignments are searchable in the HTML.

Q1500 Quantum sensing

C. L. Degen, F. Reinhard, P. Cappellaro.

Journal article | 2017 | Reviews of Modern Physics | vol. 89 | no. 3 | article 035002.

[DOI 10.1103/revmodphys.89.035002](https://doi.org/10.1103/revmodphys.89.035002) | [Source record](#) | Registered DOI metadata

Q1501 Channel correlation effects on quantum Fisher information in channel parameter estimation

Min Yu, Youneng Guo.

Journal article | 2025 | Physica Scripta | vol. 100 | no. 6 | pp. 065101.

[DOI 10.1088/1402-4896/add05b](https://doi.org/10.1088/1402-4896/add05b) | [Source record](#) | Registered DOI metadata

Fine tags: Quantum Fisher information

Q1502 Simulated Annealing-Assisted GRAPE for Superior Quantum Fisher Information in Quantum Parameter Estimation

Yi Zhang, Zibo Miao, Zigui Zhang.

Conference paper | 2025 | 2025 IEEE 26th China Conference on System Simulation Technology and its Applications (CCSSTA) | pp. 649-654.

[DOI 10.1109/ieeeeconf65522.2025.11137080](https://doi.org/10.1109/ieeeeconf65522.2025.11137080) | [Source record](#) | Registered DOI metadata

Fine tags: Quantum Fisher information

Q1503 Fisher-information susceptibility for multiparameter quantum estimation

Francesco Albarelli, Ilaria Gianani, Marco G. Genoni, Marco Barbieri.

Journal article | 2024 | Physical Review A | vol. 110 | no. 3 | article 032436.

[DOI 10.1103/physreva.110.032436](https://doi.org/10.1103/physreva.110.032436) | [Source record](#) | Registered DOI metadata

Fine tags: Quantum Fisher information; multiparameter estimation

Q1504 Fisher information analysis for quantum-enhanced parameter estimation in an electromagnetically-induced-transparency spectrum with single photons

Pin-Ju Tsai, Lun-Ping Yuan, Ying-Cheng Chen.

Journal article | 2023 | Physical Review A | vol. 108 | no. 3 | article 033711.

[DOI 10.1103/physreva.108.033711](https://doi.org/10.1103/physreva.108.033711) | [Source record](#) | Registered DOI metadata

Fine tags: Quantum Fisher information

Q1505 Quantum Fisher information and parameter estimation in non-Hermitian Hamiltonians

Jing Li, Hai-Tao Ding, Dan-Wei Zhang, Key Laboratory of Atomic and Subatomic Structure and Quantum Control, Ministry of Education, School of Physics, South China Normal University, Guangzhou 510006, China, National Key Laboratory of Solid State Microstructures, School of Physics, Nanjing University, Nanjing 210093, China.

Journal article | 2023 | Acta Physica Sinica | vol. 72 | no. 20 | pp. 200601.

[DOI 10.7498/aps.72.20230862](https://doi.org/10.7498/aps.72.20230862) | [Source record](#) | Registered DOI metadata

Fine tags: Quantum Fisher information

Q1506 Quantum Reservoir Parameter Estimation via Fisher Information

Ufuk KORKMAZ, Deniz TÜRKPENÇE.

Journal article | 2022 | Sakarya University Journal of Science | vol. 26 | no. 2 | pp. 388-396.

[DOI 10.16984/saufenbilder.1018716](https://doi.org/10.16984/saufenbilder.1018716) | [Source record](#) | Registered DOI metadata

Fine tags: Quantum Fisher information

Q1507 Coherence and quantum Fisher information in general single-qubit parameter estimation processes

Jun-Long Zhao, Dong-Xu Chen, Yu Zhang, Yu-Liang Fang, Ming Yang, Qi-Cheng Wu, Chui-Ping Yang.

Journal article | 2021 | Physical Review A | vol. 104 | no. 6 | article 062608.

[DOI 10.1103/physreva.104.062608](https://doi.org/10.1103/physreva.104.062608) | [Source record](#) | Registered DOI metadata

Fine tags: Quantum Fisher information

D28T02 Quantum magnetometry

Magnetometry estimates fields using spin or other quantum probes. Sensitivity must be quoted with bandwidth, integration time, and spatial resolution.

Fine subcategories: NV sensing; spin ensembles; sensitivity; bandwidth; spatial resolution.

Primary catalog resources: 17. Related topic assignments are searchable in the HTML.

Q1508 Operando Quantum Magnetometry for Battery Diagnostics

Avetik R. Harutyunyan.

Journal article | 2026 | ECS Meeting Abstracts | vol. MA2026-01 | no. 17 | pp. 1143-1143.

[DOI 10.1149/ma2026-01171143mtgabs](https://doi.org/10.1149/ma2026-01171143mtgabs) | [Source record](#) | Registered DOI metadata

Q1509 Quantum entanglement for magnetometry

Vladimir Slepnev, Azat Gubaydullin, Valerii Vinokur.

Journal article | 2026 | EPJ Quantum Technology | vol. 13 | no. 1 | article 76.

[DOI 10.1140/epjqt/s40507-026-00524-9](https://doi.org/10.1140/epjqt/s40507-026-00524-9) | [Source record](#) | Registered DOI metadata

Q1510 Quantum magnetometry enhanced by machine learning

Isabell Jauch, Thomas Strohm, Tino Fuchs, Fedor Jelezko.

Journal article | 2026 | Quantum Science and Technology | vol. 11 | no. 1 | pp. 015055.

[DOI 10.1088/2058-9565/ae3acf](https://doi.org/10.1088/2058-9565/ae3acf) | [Source record](#) | Registered DOI metadata

Q1511 Massively Multiplexed Nanoscale Magnetometry with Diamond Quantum Sensors

Kai-Hung Cheng, Zeeshawn Kazi, Jared Rovny, Bichen Zhang, Lila S. Nassar, Jeff D. Thompson, Nathalie P. de Leon.

Journal article | 2025 | Physical Review X | vol. 15 | no. 3 | article 031014.

[DOI 10.1103/t8fz-3tzs](https://doi.org/10.1103/t8fz-3tzs) | [Source record](#) | Registered DOI metadata

Q1512 Real Time Vector Magnetometry with Quantum Diamond Sensor

Napoom Thooppanom, Rapeephat Yodsungnoen, Natakorn Sapermsap, Sorawis Sangtawesin.

Journal article | 2025 | Journal of Physics: Conference Series | vol. 2934 | no. 1 | pp. 012022.

[DOI 10.1088/1742-6596/2934/1/012022](https://doi.org/10.1088/1742-6596/2934/1/012022) | [Source record](#) | Registered DOI metadata

Q1513 Towards robust quantum diamond magnetometry for industrial applications

Sourav Chatterjee, Shashank Kumar, S John Sharon Sandeep, Pralekh Dubey, Phani Peddibhotla.

Conference paper | 2025 | CLEO 2025 | pp. JPS200_163.

[DOI 10.1364/cleo_at.2025.jps200_163](https://doi.org/10.1364/cleo_at.2025.jps200_163) | [Source record](#) | Registered DOI metadata

Q1514 Doppler-enhanced quantum magnetometry with thermal Rydberg atoms

Shovan Kanti Barik, Silpa B S, M Venkat Ramana, Shovan Dutta, Sanjukta Roy.

Journal article | 2024 | New Journal of Physics | vol. 26 | no. 7 | pp. 073036.

[DOI 10.1088/1367-2630/ad6179](https://doi.org/10.1088/1367-2630/ad6179) | [Source record](#) | Registered DOI metadata

Q1515 Electron beam characterization via quantum coherent optical magnetometry

Nicolas DeStefano, Saeed Pegahan, Aneesh Ramaswamy, Seth Aubin, T. Averett, Alexandre Camsonne, Svetlana Malinovskaya, Eugeny E. Mikhailov et al..

Journal article | 2024 | Applied Physics Letters | vol. 125 | no. 26 | article 264001.

[DOI 10.1063/5.0234219](#) | [Source record](#) | Registered DOI metadata

Q1516 H3 diamond color centers for quantum magnetometry

E.I. Lipatov, O.I. Lyga, V.V. Chashchin, M.A. Shulepov, V.G. Vins, A.P. Yelisseyev.

Conference paper | 2024 | 2024 International Conference Laser Optics (ICLO) | pp. 437-437.

[DOI 10.1109/iclo59702.2024.10624291](#) | [Source record](#) | Registered DOI metadata

Q1517 Nanoscale covariance magnetometry with diamond quantum sensors

Nathalie de Leon.

Conference paper | 2024 | Integrated Optics: Devices, Materials, and Technologies XXVIII | pp. 22.

[DOI 10.1117/12.3008902](#) | [Source record](#) | Registered DOI metadata

Q1518 Precision magnetometry exploiting excited state quantum phase transitions

Wang Qian, Ugo Marzolino.

Journal article | 2024 | SciPost Physics | vol. 17 | no. 2 | article 043.

[DOI 10.21468/scipostphys.17.2.043](#) | [Source record](#) | Registered DOI metadata

Q1519 Quantum magnetometry using discrete-time quantum walk

Kunal Shukla, C. M. Chandrashekar.

Journal article | 2024 | Physical Review A | vol. 109 | no. 3 | article 032608.

[DOI 10.1103/physreva.109.032608](#) | [Source record](#) | Registered DOI metadata

Q1520 Quantum vector DC magnetometry via selective phase accumulation

Min Zhuang, Sijie Chen, Jiahao Huang, Chaohong Lee.

Journal article | 2024 | Science China Physics, Mechanics & Astronomy | vol. 67 | no. 10 | article 100312.

[DOI 10.1007/s11433-024-2400-1](#) | [Source record](#) | Registered DOI metadata

Q1521 A6.3 - Quantum Magnetometry for Material Testing

K. Thiemann, A. Blug, A. Bertz.

Conference paper | 2023 | Lectures | pp. 73-74.

[DOI 10.5162/smsi2023/a6.3](#) | [Source record](#) | Registered DOI metadata

Q1522 Quantum magnetometry for space

C. Deans, T. Valenzuela, M. G. Bason.

Conference paper | 2023 | International Conference on Space Optics — ICSO 2022 | pp. 248.

[DOI 10.1117/12.2691360](#) | [Source record](#) | Registered DOI metadata

Q1523 Quantum-Enhanced Magnetometry at Optimal Number Density

Charikleia Troullinou, Vito Giovanni Lucivero, Morgan W. Mitchell.

Journal article | 2023 | Physical Review Letters | vol. 131 | no. 13 | article 133602.

[DOI 10.1103/physrevlett.131.133602](#) | [Source record](#) | Registered DOI metadata

Q1524 Quantum Sensing Explained | NIST

Author metadata not supplied.

Institutional resource | Undated | NIST.

[Official resource](#) | [Source record](#) | Official page retrieved

D28T03 Atomic clocks and interferometry

Clocks and interferometers exploit controlled phase accumulation for precise measurement. Stability, systematic shifts, and entangled probe preparation set limits.

AQUIRE

Fine subcategories: Clock stability; atom interferometers; squeezing; systematic errors.

Primary catalog resources: 8. Related topic assignments are searchable in the HTML.

Q1525 Detectability of post-Newtonian classical and quantum gravity via quantum clock interferometry

Eyuri Wakakuwa.

Journal article | 2026 | Physical Review D | vol. 113 | no. 8 | article 086008.

[DOI 10.1103/hjfx-rfj](#) | [Source record](#) | Registered DOI metadata

Q1526 Gravitational time dilation in quantum clock interferometry with entangled multi-photon states and quantum memories

Mustafa Gündoğan, Roy Barzel, Dennis Rätzel.

Journal article | 2026 | Quantum Science and Technology | vol. 11 | no. 4 | pp. 045047.

[DOI 10.1088/2058-9565/aea2c3](#) | [Source record](#) | Registered DOI metadata

Q1527 Atomic clock interferometry using optical tweezers

Ilan Meltzer, Yoav Sagi.

Journal article | 2024 | Physical Review A | vol. 110 | no. 3 | article 032602.

[DOI 10.1103/physreva.110.032602](#) | [Source record](#) | Registered DOI metadata

Q1528 Finite pulse-time effects in long-baseline quantum clock interferometry

Gregor Janson, Alexander Friedrich, Richard Lopp.

Journal article | 2024 | AVS Quantum Science | vol. 6 | no. 2 | article 024403.

[DOI 10.1116/5.0178230](#) | [Source record](#) | Registered DOI metadata

Q1529 Enhancing strontium clock atom interferometry using quantum optimal control

Zilin Chen, Garrett Louie, Yiping Wang, Tejas Deshpande, Tim Kovachy.

Journal article | 2023 | Physical Review A | vol. 107 | no. 6 | article 063302.

[DOI 10.1103/physreva.107.063302](#) | [Source record](#) | Registered DOI metadata

Q1530 Quantum clock interferometry and violations of the equivalence principle

Enno Giese, Fabio Di Pumpo, Christian Ufrecht, Alexander Friedrich, Wolfgang Schleich, William Unruh.

Posted content | 2022 | Morressier.

[DOI 10.26226/m.6275705d66d5dcf63a3115f7](#) | [Source record](#) | Registered DOI metadata

Q1531 Quantum navigation with multi-axis atomic interferometry and hybrid

Yueyang Zou, Mouine Abidi, Philipp Barbey, Ashwin Rajagopalan, Christian Schubert, Matthias Gersemann, Dennis Schlippert, Sven Abend et al..

Posted content | 2022 | Copernicus GmbH.

[DOI 10.5194/egusphere-egu22-5697](#) | [Source record](#) | Registered DOI metadata

Q1532 Quantum Variational Optimization of Ramsey Interferometry and Atomic Clocks

Raphael Kaubruegger, Denis V. Vasilyev, Marius Schulte, Klemens Hammerer, Peter Zoller.

Journal article | 2021 | Physical Review X | vol. 11 | no. 4 | article 041045.

[DOI 10.1103/physrevx.11.041045](#) | [Source record](#) | Registered DOI metadata

D28T04 Quantum imaging and illumination

Quantum imaging and illumination use correlations for specified detection or imaging tasks. Advantages depend on the noise, receiver, and comparison model.

Fine subcategories: Correlation imaging; illumination receivers; resolution; noise; classical comparisons.

Primary catalog resources: 8. Related topic assignments are searchable in the HTML.

Q1533 Quantum illumination for scanning imaging under amplitude and phase noise

Jing Gao, Zhiming Qing, Jing Wang, He Jiang, Ming Zhang, Wenzheng Zhu, Hao Li, Shuangyin Huang et al..

Journal article | 2026 | Advanced Imaging | vol. 3 | no. 4 | pp. B00004.

[DOI 10.3788/ai.2026.10006](#) | [Source record](#) | Registered DOI metadata

Fine tags: noise

Q1534 High-fidelity single-pixel imaging through scattering media using quantum-state encoded illumination

Cecilia Z. C. Yu, Weiming Song, Keng C. Chou.

Journal article | 2025 | Optics Letters | vol. 50 | no. 8 | pp. 2594.

[DOI 10.1364/ol.544494](#) | [Source record](#) | Registered DOI metadata

Q1535 Using cyclic Hadamard masks for single-pixel quantum imaging under entangled photon illumination

Shuhang Bie, Xiaoxi Tong, Ziyang Lv, Hanyu Ye, Tun Cao.

Journal article | 2025 | Science Advances | vol. 11 | no. 34 | article eadw4799.

[DOI 10.1126/sciadv.adw4799](#) | [Source record](#) | Registered DOI metadata

Q1536 Quantum-illumination-inspired active single-pixel imaging with structured illumination

Tiantian Zhang, Zhiyuan Ye, Hai-Bo Wang, Jun Xiong.

Journal article | 2021 | Applied Optics | vol. 60 | no. 32 | pp. 10151.

[DOI 10.1364/ao.438642](#) | [Source record](#) | Registered DOI metadata

Q1537 Full-Field Imaging With Quantum Illumination

T. Gregory, P.-A. Moreau, E. Toninelli, M.J. Padgett.

Conference paper | 2020 | 2020 Photonics North (PN) | pp. 1-1.

[DOI 10.1109/pn50013.2020.9166939](#) | [Source record](#) | Registered DOI metadata

Q1538 Imaging through noise with quantum illumination

T. Gregory, P.-A. Moreau, E. Toninelli, M. J. Padgett.

Journal article | 2020 | Science Advances | vol. 6 | no. 6 | article eaay2652.

[DOI 10.1126/sciadv.aay2652](#) | [Source record](#) | Registered DOI metadata

Fine tags: noise

Q1539 Simultaneous multicolour imaging using quantum dot structured illumination microscopy

HUI ZENG, HUAIDONG YANG, GUOXUAN LIU, SICHUN ZHANG, XINRONG ZHANG, YINXIN ZHANG.

Journal article | 2020 | Journal of Microscopy | vol. 277 | no. 1 | pp. 32-41.

[DOI 10.1111/jmi.12862](#) | [Source record](#) | Registered DOI metadata

Q1540 Single-photon quantum imaging via single-photon illumination

Jia-Zhi Yang, Ming-Fei Li, Xiao-Xiao Chen, Wen-Kai Yu, An-Ning Zhang.

Journal article | 2020 | Applied Physics Letters | vol. 117 | no. 21 | article 214001.

[DOI 10.1063/5.0021214](#) | [Source record](#) | Registered DOI metadata

D28T05 Quantum sensing with error correction

Error corrected sensing seeks to protect useful signal accumulation while rejecting noise. The signal and noise operators must satisfy relevant conditions.

Fine subcategories: Logical sensors; noise filtering; Hamiltonian conditions; ancilla assistance.

Primary catalog resources: 3. Related topic assignments are searchable in the HTML.

Q1541 Quantum sensing with Rydberg atoms enhanced by quantum error correction

Stanisław Kurzyna, Bartosz Niewelt, Mateusz Mazelanik, Wojciech Wasilewski, Rafał Demkowicz-Dobrzański, Michał Parniak.

Conference paper | 2025 | Frontiers in Optics + Laser Science 2025 (FiO, LS) | pp. FTu6C.4.

[DOI 10.1364/fio.2025.ftu6c.4](https://doi.org/10.1364/fio.2025.ftu6c.4) | [Source record](#) | Registered DOI metadata

Q1542 Distributed quantum sensing enhanced by continuous-variable error correction

Quntao Zhuang, John Preskill, Liang Jiang.

Journal article | 2020 | New Journal of Physics | vol. 22 | no. 2 | pp. 022001.

[DOI 10.1088/1367-2630/ab7257](https://doi.org/10.1088/1367-2630/ab7257) | [Source record](#) | Registered DOI metadata

Q1543 Spatial noise filtering through error correction for quantum sensing

David Layden, Paola Cappellaro.

Journal article | 2018 | npj Quantum Information | vol. 4 | no. 1 | article 30.

[DOI 10.1038/s41534-018-0082-2](https://doi.org/10.1038/s41534-018-0082-2) | [Source record](#) | Registered DOI metadata

Fine tags: noise filtering

D29 Specialized scientific applications

Scientific applications must specify the observable or decision task and account for preparation and readout. These categories organize exploratory work without claiming universal practical advantage.

Prerequisites: Relevant scientific model and quantum algorithm prerequisites.

Assessment focus: Observable definition, model error, preparation, readout, and domain validation.

D29T01 Quantum algorithms for high energy physics

High energy applications include field simulation and selected analysis tasks. Physical observable, truncation, and classical verification define the computational problem.

Fine subcategories: Field theory; scattering; event analysis; neutrinos; gauge dynamics.

Primary catalog resources: 5. Related topic assignments are searchable in the HTML.

Q1544 Quantum Computing for High Energy Density Science

Lawrence Livermore National Laboratory (LLNL), Livermore, CA (United States), Yujin Cho, USDOE National Nuclear Security Administration (NNSA), Kristi Beck, Zachary Espley.

Report | 2026 | Office of Scientific and Technical Information (OSTI).

[DOI 10.2172/17019150](#) | [Source record](#) | Registered DOI metadata

Q1545 Quantum computing for energy correlators

Kyle Lee, Francesco Turro, Xiaojun Yao.

Journal article | 2025 | Physical Review D | vol. 111 | no. 5 | article 054514.

[DOI 10.1103/physrevd.111.054514](#) | [Source record](#) | Registered DOI metadata

Q1546 Quantum Computing and High Energy Physics

Prasanth Shyamsundar.

Conference paper | 2024 | Quantum Computing and High Energy Physics.

[DOI 10.2172/2426500](#) | [Source record](#) | Registered DOI metadata

Q1547 Hunting Quantum Gravity with Analogs: The Case of High-Energy Particle Physics

Paolo Castorina, Alfredo Iorio, Helmut Satz.

Journal article | 2022 | Universe | vol. 8 | no. 9 | pp. 482.

[DOI 10.3390/universe8090482](#) | [Source record](#) | Registered DOI metadata

Q1548 Quantum Systems for Enhanced High Energy Particle Physics Detectors

M. Doser, E. Auffray, F.M. Brunbauer, I. Frank, H. Hillemanns, G. Orlandini, G. Kornakov.

Journal article | 2022 | Frontiers in Physics | vol. 10 | article 887738.

[DOI 10.3389/fphy.2022.887738](#) | [Source record](#) | Registered DOI metadata

D29T02 Quantum algorithms for nuclear physics

Nuclear applications target states, reactions, and observables of interacting particles. Model spaces and controlled physical approximations accompany circuit errors.

Fine subcategories: Nuclear structure; reactions; few body systems; interactions; spectroscopy.

Primary catalog resources: 7. Related topic assignments are searchable in the HTML.

Q1549 Nuclear Physics in the Era of Quantum Computing and Quantum Machine Learning

José-Enrique García-Ramos, Álvaro Sáiz, José M. Arias, Lucas Lamata, Pedro Pérez-Fernández.

Journal article | 2024 | Advanced Quantum Technologies | vol. 8 | no. 12 | article 2300219.

[DOI 10.1002/qute.202300219](#) | [Source record](#) | Registered DOI metadata

Q1550 Quantum computing for nuclear physics

Martin J. Savage.

Journal article | 2024 | EPJ Web of Conferences | vol. 296 | pp. 01025.

[DOI 10.1051/epjconf/202429601025](https://doi.org/10.1051/epjconf/202429601025) | [Source record](#) | Registered DOI metadata

Q1551 Quantum computing based high-energy nuclear physics

Tian-Yin Li, Hong-Xi Xing, Dan-Bo Zhang, Key Laboratory of Atomic and Subatomic Structure and Quantum Control (Ministry of Education), Institute of Quantum Matter, South China Normal University, Guangzhou 510006, China, Guangdong Provincial Key Laboratory of Nuclear Science, Institute of Quantum Matter, South China Normal University, Guangzhou 510006, China, Guangdong-Hong Kong Joint Laboratory of Quantum Matter, Southern Nuclear Science Computing Center, South China Normal University, Guangzhou 510006, China, Key Laboratory of Atomic and Subatomic Structure and Quantum Control (Ministry of Education), Guangdong Basic Research Center of Excellence for Structure and Fundamental Interactions of Matter, School of Physics, South China Normal University, Guangzhou 510006, China.

Journal article | 2023 | Acta Physica Sinica | vol. 72 | no. 20 | pp. 200303.

[DOI 10.7498/aps.72.20230907](https://doi.org/10.7498/aps.72.20230907) | [Source record](#) | Registered DOI metadata

Q1552 Quantum Computing for Quantum Dynamics in Nuclear Physics [Slides]

Alessandro Baroni.

Report | 2022 | Office of Scientific and Technical Information (OSTI).

[DOI 10.2172/1846882](https://doi.org/10.2172/1846882) | [Source record](#) | Registered DOI metadata

Q1553 Selected topics of quantum computing for nuclear physics*

Dan-Bo Zhang, Hongxi Xing, Hui Yan, Enke Wang, Shi-Liang Zhu.

Journal article | 2021 | Chinese Physics B | vol. 30 | no. 2 | pp. 020306.

[DOI 10.1088/1674-1056/abd761](https://doi.org/10.1088/1674-1056/abd761) | [Source record](#) | Registered DOI metadata

Q1554 Quantum Computing for Theoretical Nuclear Physics, A White Paper prepared for the U.S. Department of Energy, Office of Science, Office of Nuclear Physics

Joseph Carlson, David Dean, Morten Hjorth-Jensen, David Kaplan, John Preskill, Kenneth Roche, Martin Savage, Matthias Troyer.

Report | 2018 | Office of Scientific and Technical Information (OSTI).

[DOI 10.2172/1631143](https://doi.org/10.2172/1631143) | [Source record](#) | Registered DOI metadata

Q1555 Quantum algorithms for computational nuclear physics

Jakub Věšník.

Journal article | 2015 | EPJ Web of Conferences | vol. 100 | pp. 01008.

[DOI 10.1051/epjconf/201510001008](https://doi.org/10.1051/epjconf/201510001008) | [Source record](#) | Registered DOI metadata

D29T03 Quantum algorithms for numerical integration

Integration algorithms often build on amplitude estimation. The input oracle and required precision determine whether a query benefit survives end to end accounting.

Fine subcategories: Quadrature; Monte Carlo; dimensionality; oracle access; precision.

Primary catalog resources: 3. Related topic assignments are searchable in the HTML.

Q1556 Spectral quantum algorithm for numerical differentiation and integration

Jordan Cioni, Fabio Semperlotti.

Journal article | 2026 | Physical Review Applied | vol. 26 | no. 1 | article 014063.

[DOI 10.1103/t8kt-5f2n](https://doi.org/10.1103/t8kt-5f2n) | [Source record](#) | Registered DOI metadata

Q1557 A general quantum algorithm for numerical integration

Guoqiang Shu, Zheng Shan, Jinchun Xu, Jie Zhao, Shuya Wang.

Journal article | 2024 | Scientific Reports | vol. 14 | no. 1 | article 10432.

[DOI 10.1038/s41598-024-61010-9](https://doi.org/10.1038/s41598-024-61010-9) | [Source record](#) | Registered DOI metadata

Q1558 Quantum Coin Method for Numerical Integration

N. H. Shimada, T. Hachisuka.

Journal article | 2020 | Computer Graphics Forum | vol. 39 | no. 6 | pp. 243-257.

[DOI 10.1111/cgf.14015](#) | [Source record](#) | Registered DOI metadata

D29T04 Quantum algorithms for engineering

Engineering applications connect quantum algorithms to discretized models and inverse tasks. Conditioning and output extraction can dominate the workflow.

Fine subcategories: Finite elements; fluid models; differential equations; inverse problems; discretization.

Primary catalog resources: 8. Related topic assignments are searchable in the HTML.

Q1559 Variational quantum algorithms for computational fluid dynamics

Dieter Jaksch.

Posted content | 2025 | Cassyni.

[DOI 10.52843/cassyni.qx2xbp](#) | [Source record](#) | Registered DOI metadata

Q1560 A hybrid quantum-classical framework for computational fluid dynamics

Chuang-Chao Ye, Ning-Bo An, Teng-Yang Ma, Meng-Han Dou, Wen Bai, De-Jun Sun, Zhao-Yun Chen, Guo-Ping Guo.

Journal article | 2024 | Physics of Fluids | vol. 36 | no. 12 | article 127111.

[DOI 10.1063/5.0238193](#) | [Source record](#) | Registered DOI metadata

Q1561 Computational Fluid Dynamics on Quantum Computers

Madhava Syamlal, Carter Copen, Masashi Takahashi, Benjamin Hall.

Conference paper | 2024 | AIAA AVIATION FORUM AND ASCEND 2024.

[DOI 10.2514/6.2024-3534](#) | [Source record](#) | Registered DOI metadata

Q1562 Quantum-inspired framework for computational fluid dynamics

Raghavendra Dheeraj Peddinti, Stefano Pisoni, Alessandro Marini, Philippe Lott, Henrique Argentieri, Egor Tiunov, Leandro Aolita.

Journal article | 2024 | Communications Physics | vol. 7 | no. 1 | article 135.

[DOI 10.1038/s42005-024-01623-8](#) | [Source record](#) | Registered DOI metadata

Q1563 Variational Quantum Algorithms for Computational Fluid Dynamics

Dieter Jaksch, Peyman Givi, Andrew J. Daley, Thomas Rung.

Journal article | 2023 | AIAA Journal | vol. 61 | no. 5 | pp. 1885-1894.

[DOI 10.2514/1.j062426](#) | [Source record](#) | Registered DOI metadata

Q1564 Opportunities for quantum computation in computational fluid dynamics

Yan Li.

Conference paper | 2022 | 2nd International Conference on Applied Mathematics, Modelling, and Intelligent Computing (CAMMIC 2022) | pp. 205.

[DOI 10.1117/12.2639351](#) | [Source record](#) | Registered DOI metadata

Q1565 Parallel evaluation of quantum algorithms for computational fluid dynamics

René Steijl, George N. Barakos.

Journal article | 2018 | Computers & Fluids | vol. 173 | pp. 22-28.

[DOI 10.1016/j.compfluid.2018.03.080](#) | [Source record](#) | Registered DOI metadata

Q1566 Quantum Fluid Dynamics and Quantum Computational Fluid Dynamics

C. T. Lin, J. K. Kuo, T. H. Yen.

Journal article | 2009 | Journal of Computational and Theoretical Nanoscience | vol. 6 | no. 5 | pp. 1090-1108.

[DOI 10.1166/jctn.2009.1149](#) | [Source record](#) | Registered DOI metadata

D29T05 Quantum algorithms for biology and drug discovery

Biological and drug discovery proposals often depend on quantum chemistry subproblems. Chemical accuracy and downstream biological validation should be assessed separately.

Fine subcategories: Molecular energies; docking proposals; chemistry pipelines; validation; biological relevance.

Primary catalog resources: 8. Related topic assignments are searchable in the HTML.

Q1567 Quantum Computing and Drug Discovery: Current Trends and Future

Yashraj Mukherjee, Praveen Pathak, Shweta Sinha.

Journal article | 2026 | International Journal For Multidisciplinary Research | vol. 8 | no. 1 | article 65289.

[DOI 10.36948/ijfmr.2026.v08i01.65289](https://doi.org/10.36948/ijfmr.2026.v08i01.65289) | [Source record](#) | Registered DOI metadata

Q1568 Quantum Computing in Drug Discovery

Nilesh Ingale Nilesh Ingale.

Journal article | 2026 | International Journal of Scientific Research in Engineering and Management | vol. 10 | no. 9.

[DOI 10.55041/ijcsrem.istcesd038](https://doi.org/10.55041/ijcsrem.istcesd038) | [Source record](#) | Registered DOI metadata

Q1569 Quantum Computing in Drug Discovery Techniques, Challenges, and Emerging Opportunities

Virendra S. Gomase, Arjun P. Ghatule, Rupali Sharma, Suchita P. Dhamane.

Journal article | 2026 | Current Drug Discovery Technologies | vol. 23 | no. 4 | article e15701638371707.

[DOI 10.2174/0115701638371707250729040426](https://doi.org/10.2174/0115701638371707250729040426) | [Source record](#) | Registered DOI metadata

Q1570 Quantum Computing: Revolutionizing Drug Discovery and Battery Development

G. Deepalakshmi, S.K. Sunori, K.V. Padmavathi, P. Balaramesh, M. Kathiravan, V. Venkatesh.

Book chapter | 2026 | Advanced Transportation Systems in the Field of Engineering and Technology | pp. 249-252.

[DOI 10.1201/9781003598312-79](https://doi.org/10.1201/9781003598312-79) | [Source record](#) | Registered DOI metadata

Q1571 Quantum computing applications in drug discovery

Jing Li, Leyi Wei, Henry H Y Tong, Quan Zou.

Journal article | 2026 | Briefings in Bioinformatics | vol. 27 | no. 3 | article bbag274.

[DOI 10.1093/bib/bbag274](https://doi.org/10.1093/bib/bbag274) | [Source record](#) | Registered DOI metadata

Q1572 Quantum computing applications in drug discovery: Unraveling complex molecular interactions

Anagha Deepak Kulkarni, Tejasvini Rahul Katkar, Smita Jadhav, Pranali Anandrao Jadhav.

Conference paper | 2026 | AIP Conference Proceedings | vol. 3445 | pp. 030122.

[DOI 10.1063/5.0327515](https://doi.org/10.1063/5.0327515) | [Source record](#) | Registered DOI metadata

Q1573 Quantum computing in drug discovery

Harshraj Gadgail, Rajendra Rewatkar, Nishant Jumde, Mrinal Manker, Pratik Nagre, Sujal Zade.

Journal article | 2026 | Frontiers in Drug Discovery | vol. 6 | article 1815176.

[DOI 10.3389/fddsv.2026.1815176](https://doi.org/10.3389/fddsv.2026.1815176) | [Source record](#) | Registered DOI metadata

Q1574 Quantum computing-driven evolution of drug discovery paradigms

Lu Han, Zhipeng Ke, Liying Liu, Liang Cao, Zhenzhong Wang, Xinzhuang Zhang, Wenxia Zhou, Wei Xiao.

Journal article | 2026 | Chinese Science Bulletin.

[DOI 10.1360/csb-2026-0450](https://doi.org/10.1360/csb-2026-0450) | [Source record](#) | Registered DOI metadata

D30 Education research practice and ecosystem

Research practice includes learning, reproducing results, documenting limitations, and judging evidence. Roadmaps and standards describe plans or interfaces rather than proof that a device meets them.

Prerequisites: Basic quantum computing and critical reading.

Assessment focus: Source quality, environment records, fair baselines, versioning, and full system cost.

D30T01 Quantum computing education

Education resources teach prerequisites, conceptual models, and executable experiments. Curriculum scope and assessment quality determine how to use them.

Fine subcategories: Prerequisites; laboratories; assessments; visual explanations; curriculum design.

Primary catalog resources: 8. Related topic assignments are searchable in the HTML.

Q1575 Quantum-Ready Education: Preparing Students for Post-Quantum Computing Societies

Wai Yie Leong.

Conference paper | 2026 | 2026 15th International Conference on Educational and Information Technology (ICEIT) | pp. 926-930.

[DOI 10.1109/iceit68991.2026.11521622](https://doi.org/10.1109/iceit68991.2026.11521622) | [Source record](#) | Registered DOI metadata

Q1576 BETTER VISUAL METAPHORS FOR QUANTUM COMPUTING EDUCATION

Barry Fagin.

Conference paper | 2025 | EDULEARN Proceedings | vol. 1 | pp. 8543-8552.

[DOI 10.21125/edulearn.2025.2222](https://doi.org/10.21125/edulearn.2025.2222) | [Source record](#) | Registered DOI metadata

Q1577 Enhancing Quantum Computing Education Through Virtual Reality

Yu Yan, Jeannie S. Lee, Kan Chen, Nicholas H.L. Wong.

Conference paper | 2025 | Proceedings of the 2025 20th ACM SIGGRAPH International Conference on Virtual-Reality Continuum and its Applications in Industry | pp. 1-2.

[DOI 10.1145/3779232.3779282](https://doi.org/10.1145/3779232.3779282) | [Source record](#) | Registered DOI metadata

Q1578 Enhancing Quantum Literacy in Secondary Education Through Quantum Computing and Quantum Key Distribution

Aspasia V. Oikonomou, Ilias K. Savvas, Omiros Iatrellis.

Journal article | 2025 | Education Sciences | vol. 15 | no. 9 | pp. 1167.

[DOI 10.3390/educsci15091167](https://doi.org/10.3390/educsci15091167) | [Source record](#) | Registered DOI metadata

Q1579 Integrating Quantum Computing into STEM Education via Personalized Learning

Oluwatoyin Akinbi, Mozhgan N. Entekhabi, Hongmei Chi.

Conference paper | 2025 | 2025 Interdisciplinary Conference on Electrics and Computer (INTCEC) | pp. 1-7.

[DOI 10.1109/intcec65580.2025.11256059](https://doi.org/10.1109/intcec65580.2025.11256059) | [Source record](#) | Registered DOI metadata

Q1580 Pedagogical Approach for Quantum Computing in Engineering Education

Francisco Orts.

Journal article | 2025 | Computer Applications in Engineering Education | vol. 33 | no. 2 | article e70004.

[DOI 10.1002/cae.70004](https://doi.org/10.1002/cae.70004) | [Source record](#) | Registered DOI metadata

Q1581 GitHub - NVIDIA/cuda-q-academic: This repo contains CUDA-Q Academic materials, including self-paced Jupyter notebook modules for building and optimizing hybrid quantum-classical algorithms using CUDA-Q. · GitHub

Author metadata not supplied.

Software repository | Undated | NVIDIA.

[Official resource](#) | [Source record](#) | Official page retrieved

Q1582 Part III Quantum Computation | Centre for Quantum Information and Foundations

Author metadata not supplied.

Course | Undated | University of Cambridge.

[Official resource](#) | [Source record](#) | Official page retrieved

D30T02 Quantum computing surveys and roadmaps

Surveys and roadmaps organize a fast moving field and identify bottlenecks. Roadmap milestones are plans rather than demonstrated capabilities.

Fine subcategories: Hardware surveys; algorithm surveys; milestones; uncertainty; scaling bottlenecks.

Primary catalog resources: 8. Related topic assignments are searchable in the HTML.

Q1583 Quantum Computing in the NISQ era and beyond

John Preskill.

Journal article | 2018 | Quantum | vol. 2 | pp. 79 | article 79.

[DOI 10.22331/q-2018-08-06-79](#) | [Source record](#) | Registered DOI metadata

Q1584 Classical Intelligence For Quantum Technologies: An Evidence Roadmap For AI-Enabled Quantum-Computing Workflows

Amar Singh, Vinod Kumar Shukla, Sahraoui Dhelim, Chatter Singh.

Journal article | 2026 | Advanced Quantum Technologies | vol. 9 | no. 10 | article e70495.

[DOI 10.1002/qute.70495](#) | [Source record](#) | Registered DOI metadata

Q1585 Impact of Quantum Computing on Asymmetric Cryptography Infrastructures: Prospective Study and Post-Quantum Transition Roadmap

João Manuel Marques Lucas, Carlos Caleiro, António Leonardo Gonçalves, Laercio Cruvinel Júnior.

Conference paper | 2026 | 2026 6th International Conference on Electrical, Computer and Energy Technologies (ICECET) | pp. 1-6.

[DOI 10.1109/icecet65726.2026.11633187](#) | [Source record](#) | Registered DOI metadata

Q1586 Quantum Computing and Information Technology: A Roadmap for the Next Decade

Author metadata not supplied.

Book chapter | 2026 | Advanced Studies in Multidisciplinary Research and Innovation (ASMRI).

[DOI 10.59646/745/21](#) | [Source record](#) | Registered DOI metadata

Q1587 Roadmap on quantum thermodynamics

Steve Campbell, Irene D'Amico, Mario A Ciampini, Janet Anders, Natalia Ares, Simone Artini, Alexia Auffèves, Lindsay Bassman Oftelie et al..

Journal article | 2026 | Quantum Science and Technology | vol. 11 | no. 1 | pp. 012501.

[DOI 10.1088/2058-9565/ae1e27](#) | [Source record](#) | Registered DOI metadata

Q1588 Quantum Computing Explained | NIST

Author metadata not supplied.

Institutional resource | Undated | NIST.

[Official resource](#) | [Source record](#) | Official page retrieved

Q1589 Quantum Information Science (QIS... | U.S. DOE Office of Science(SC)

Author metadata not supplied.

Institutional resource | Undated | US Department of Energy.

[Official resource](#) | [Source record](#) | Official page retrieved

Q1590 Quantum information science | NIST

Author metadata not supplied.

Institutional resource | Undated | NIST.

[Official resource](#) | [Source record](#) | Official page retrieved

D30T03 Quantum standards and interoperability

Standards and interfaces support consistent terminology and tool interoperability. The applicable version and scope should be recorded when used.

Fine subcategories: Terminology; interfaces; reference models; benchmarking definitions; interoperability.

Primary catalog resources: 1. Related topic assignments are searchable in the HTML.

Q1591 Cryogenic RF Calibrations and Standards for Quantum Computing

Peter Hopkins, Lafe Spietz, Adam Sirois, Manuel Castellanos-Beltran, Nathan FlowersJacobs, Elyse McEntee Wei, Chris Long, Dylan Williams et al..

Conference paper | 2026 | 2026 United States National Committee of URSI National Radio Science Meeting (USNC-URSI NRSM) | pp. 235-235.

[DOI 10.23919/nrsm68586.2026.11550799](#) | [Source record](#) | Registered DOI metadata

D30T04 Quantum reproducibility and research methodology

Reproducible research records environments, circuits, parameters, datasets, and uncertainty. Fair benchmarks also preserve strong classical baselines and complete cost accounting.

Fine subcategories: Experiment records; seeds; environments; baselines; statistical reporting.

Primary catalog resources: 1. Related topic assignments are searchable in the HTML.

Q1592 Reproducibility in Quantum Computing

Samudra Dasgupta, Travis S. Humble.

Conference paper | 2021 | 2021 IEEE Computer Society Annual Symposium on VLSI (ISVLSI) | pp. 458-461.

[DOI 10.1109/isvlsi51109.2021.00090](#) | [Source record](#) | Registered DOI metadata

D30T05 Quantum economics and sustainability

Economics and sustainability studies evaluate full system cost and energy use. Cooling, classical support, runtime, and utilization matter alongside quantum device operation.

Fine subcategories: Cooling energy; runtime cost; total system accounting; scaling; economic assumptions.

Primary catalog resources: 8. Related topic assignments are searchable in the HTML.

Q1593 Quantum Computing for a Sustainable Future: Transforming Energy Efficiency and Climate Solutions

Sreevatsa Bellary, Ananya Hadadi Raghavendra.

Book chapter | 2026 | Studies in Big Data | pp. 55-76.

[DOI 10.1007/978-3-032-00586-1_3](#) | [Source record](#) | Registered DOI metadata

Q1594 Application of Quantum Computing in the Energy Industry, Decarbonization, and Sustainability

Christoph Capellaro, Rafael Martín-Cuevas, Guzmán Calleja, Pascal Halfmann, Shivam Sharma.

Book chapter | 2025 | Quantum Technology Applications, Impact, and Future Challenges | pp. 9-36.

[DOI 10.1201/9781003537243-2](#) | [Source record](#) | Registered DOI metadata

Q1595 Enhancing e-commerce logistics efficiency and sustainability via quantum computing and artificial intelligence-based quantum hybrid models

Muhammad Khan, Farhan Amin, Minhaj Ud Din, Muhammad Ali Abid, Isabel de la Torre, Elisabeth Caro Montero, Irene Delgado Noya.

Journal article | 2025 | The Journal of Supercomputing | vol. 81 | no. 15 | article 1455.

[DOI 10.1007/s11227-025-07959-4](https://doi.org/10.1007/s11227-025-07959-4) | [Source record](#) | Registered DOI metadata

Q1596 Power Consumption and Energy Efficiency of Quantum Computing Platforms in High Performance Computing Integration

Xiaolong Deng, Martin Schulz, Laura Schulz.

Book chapter | 2025 | Lecture Notes in Computer Science | pp. 325-337.

[DOI 10.1007/978-3-031-85700-3_24](https://doi.org/10.1007/978-3-031-85700-3_24) | [Source record](#) | Registered DOI metadata

Q1597 Quantum computing: Impact on energy efficiency and sustainability

Vaishali Sood, Rishi Pal Chauhan.

Journal article | 2024 | Expert Systems with Applications | vol. 255 | pp. 124401 | article 124401.

[DOI 10.1016/j.eswa.2024.124401](https://doi.org/10.1016/j.eswa.2024.124401) | [Source record](#) | Registered DOI metadata

Q1598 Towards Greener Power Grids Quantum Computing Solutions for Energy Efficiency

Kuppam Mohan Babu, M. Bhaskaraiah, Kasaram Roja, T. Chandraiah.

Book chapter | 2024 | Advances in Computational Intelligence and Robotics | pp. 392-406.

[DOI 10.4018/979-8-3693-4001-1.ch026](https://doi.org/10.4018/979-8-3693-4001-1.ch026) | [Source record](#) | Registered DOI metadata

Q1599 Enhancing Cannabis Extraction Efficiency and Sustainability through Quantum Computing: A Review

Mokhlesur R. M, Tahmid C. A, Hassan S, Zubaer M, Awang M, Hasan M.

Journal article | 2023 | Oriental Journal Of Chemistry | vol. 39 | no. 6 | pp. 1419-1436.

[DOI 10.13005/ojc/390604](https://doi.org/10.13005/ojc/390604) | [Source record](#) | Registered DOI metadata

Q1600 Is quantum computing green? An estimate for an energy-efficiency quantum advantage

Daniel Jaschke, Simone Montangero.

Journal article | 2023 | Quantum Science and Technology | vol. 8 | no. 2 | pp. 025001.

[DOI 10.1088/2058-9565/acae3e](https://doi.org/10.1088/2058-9565/acae3e) | [Source record](#) | Registered DOI metadata

Catalog field definitions

Field	Meaning
ID	Stable identifier shared across all editions
Topic	Primary title screened or curated subject assignment
Fine tag	Supported lexical tag or topic level only
Year	Published or issued metadata year; no invented dates
Venue	Journal, proceedings, book container, or official provider
Conference	Supplied event or proceedings venue
DOI	Scholarly identifier from retrieved metadata
Source record	Crossref record or official page
Verification	Registered metadata or successful official page retrieval

The complete author lists, related topic assignments, access notes, ISBNs when supplied, and retrieval fields are available in the HTML and JSON. Long author lists are abbreviated in this reading edition. The source record should be consulted before relying on a bibliographic detail.

Acknowledgments

OpenAI 6.1 sol was used to obtain and organize the initial resource collection and to assist with taxonomy development, documentation, and repository preparation. Source records support bibliographic identity; AI assistance does not imply publisher endorsement or universal full-text verification.